

Curvature, support certificates, and barrier complexity of conic lifts

Abstract

A conic lift writes a convex body as the linear image of an affine slice of a product of cones, turning linear optimization over the body into a conic program. Mainly for Euclidean balls and their products, we ask how small the cone factors, dual certificate ranks, and barrier parameters of lifts can be, and how they bound distances in barrier metrics. If an s -dimensional compact convex body has a C^2 boundary patch with strictly positive curvature at some point, every exact lift by proper cones of dimensions m_i that are definable in an o-minimal structure (for example, semialgebraic) satisfies $s - 1 \leq \sum_i \max\{m_i - 2, 0\}$. Fix a finite list of simple symmetric cones, each usable repeatedly, and put $B = \max a(r - 1) > 0$, where a is the Peirce constant and r the Jordan rank; for example, $B = m - 2$ for the m -dimensional Lorentz cone and $B = r - 1$ for real $r \times r$ PSD matrices. Ranks and barriers are computed after replacing the product of cones by its smallest face containing the affine slice. A certificate for a direction v is a dual element that represents the slack $1 - \langle x, v \rangle$ on the whole affine slice. For the s -dimensional ball, $s \geq 2$, the least total Jordan rank of a certificate, maximized over directions and then minimized over lifts, is $\lceil (s - 1)/B \rceil$, without smoothness assumptions on the lift; requiring boundary points and certificates to be C^1 in the direction can increase it. For every such lift, the log-determinant barrier restricted to the affine slice has parameter at least $\lceil (s - 1)/B \rceil$. We determine the least such parameter in several cases, and the least parameter of any barrier for several concrete formulations; for some remaining cases, and for nonsymmetric cones, we prove only bounds. A weighted norm tree represents a ball by nested Euclidean-norm constraints along a rooted tree, with a barrier weight at each node. For such a tree and a fixed unit objective, the distance in the barrier metric from a fixed interior point to the ϵ -accurate points is $\sqrt{K} \log(1/\epsilon) + O(\log \log(1/\epsilon))$, where K is the total weight of the nodes whose subtree meets the support of the objective plus half the remaining weight; for the central path from the analytic center, K becomes the total weight. In the column-packed PSD formulation of a product of balls, the ball coordinates fill the columns of matrices W_ℓ , and auxiliary matrices S_ℓ satisfy one linear equation per ball on their diagonals. Let D be the residuals $S_\ell - W_\ell^T W_\ell$ at a feasible point and D^0 their value at the barrier-minimizing auxiliary point over the same projected point. If $\mu D^0 \preceq D \preceq L D^0$ ($0 < \mu \leq 1 \leq L$), the Hessians H_D and H_0 at these two points, after eliminating the auxiliary directions, satisfy $L^{-1} H_0 \preceq H_D \preceq \mu^{-1} H_0$, and this is sharp. We also give exact resource tradeoffs, approximation results with controlled cone factors, and exact compilation of input data, and compare Newton-step and query costs; distance bounds imply work bounds only under explicit assumptions on output and per-round work.

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1 Introduction

Many convex bodies in optimization are the linear image of an affine slice of a product of standard cones, such as Lorentz cones or cones of positive semidefinite (PSD) matrices. Such an exact conic lift turns linear optimization over the body into a conic program that interior-point methods can solve. The lifts of one body differ in the number and dimensions of the cone factors, the ranks of the dual certificates that prove a bound on a linear objective, the barrier parameter on the affine slice, the data that a Newton step needs, and the outputs that are inexpensive to produce; none of these resources determines the others. We ask how small each resource can be over a specified class of lifts, and how the resources bound distances in barrier metrics and computational work.

Euclidean balls and their products are the main examples; their lifts already show certificates shared between factors, boundary choices that cannot be made globally smooth, and different barrier metrics. We also treat ℓ_p norm balls, matrix balls, balance slices, power and entropy cones, and exact compilers, which map given input data to an exact representation in a prescribed format. Every exact formula is stated with the class of lifts over which it is optimized; a lower bound for one representation need not hold for every representation of its projection.

1.1 Main results

Let $C \subset \mathbb{R}^s$ be a compact convex body with the origin in its interior. An *exact lift* of C is a triple $\mathcal{L} = (K, A, \pi)$ with $C = \pi(K \cap A)$, where $K = \prod_i K_i$ is a finite product of proper cones, A is an affine subspace, and π is affine. Ranks and barriers are computed after the reduction of Lemma 2.1, which eliminates free variables and replaces K by the smallest face containing the feasible points; the result is a relative-Slater lift. For $v \in \mathbb{S}^{s-1}$, a *genuine certificate* of the support slack $1 - \langle x, v \rangle$ of the ball $\mathbb{B}_2^s$ is a dual element $Y = (Y_i) \in K^*$ with $\langle X, Y \rangle = 1 - \langle \pi(X), v \rangle$ for every $X \in A$, that is, on the whole affine slice and not only at selected boundary points. These certificates form the nonempty *certificate fiber* $\mathcal{D}_{\mathcal{L}}(v)$. For a simple symmetric cone, r is its Jordan rank and a its Peirce constant; $\text{rank}_j Y$ is the Jordan rank of an element Y .

Curvature and cone dimension. Suppose that C has a C^2 boundary patch with strictly positive curvature at some point, and that $\mathcal{L}$ is an exact lift by proper cones of real dimensions m_i that are definable in a fixed o-minimal structure (for example, semialgebraic). In the mixed second derivative of the slack identity $1 - \langle x, y \rangle = \sum_i \langle X_i, Y_i \rangle$ at a boundary point, each factor of dimension m contributes rank at most $\max\{m - 2, 0\}$, so $s - 1 \leq \sum_i \max\{m_i - 2, 0\}$. For simple symmetric cones, at some boundary point the primal ranks p_i and the complementary dual ranks q_i satisfy $s - 1 \leq \sum_i a_i p_i q_i$ (Theorem 3.2). Definable local selections suffice; no globally smooth selection is needed. For a ball of dimension $s \geq 2$ and an integer cap $d \geq 3$ on the cone dimensions, the least number of factors and the least total cone dimension over exact definable lifts are therefore

$$k = \left\lceil \frac{s-1}{d-2} \right\rceil, \quad M = s-1 + 2k,$$

and classical norm trees attain both (Corollary 3.4).

Certificate rank. Fix a finite dictionary $\mathcal{K}$ of simple symmetric cones; a lift through $\mathcal{K}$ may use any number of copies of each member, and nonnegative rays. Put $B = \max_{K \in \mathcal{K}} a(K)(r(K) - 1) > 0$; for example, $B = r - 1$ for real PSD matrices of order r , $B = m - 2$ for the Lorentz cone Q_m of

dimension m , and $B = 16$ for the Albert cone. For every $s \geq 2$, Theorem 4.1 proves

$$\inf_{\mathcal{L}} \max_{v \in \mathbb{S}^{s-1}} \min_{Y \in \mathcal{D}_{\mathcal{L}}(v)} \sum_i \text{rank}_J Y_i = \left\lceil \frac{s-1}{B} \right\rceil, \quad (1.1)$$

where the infimum is over all exact finite relative-Slater affine lifts of $\mathbb{B}_2^s$ through $\mathcal{K}$. The inner minimum runs over the whole certificate fiber, not over a certificate selected by an algorithm, and no smoothness of the lift is assumed. For each fixed lift, the lower bound holds for every certificate at every direction in a dense open set whose complement has surface measure zero. A construction from Peirce perspectives (Lemma 4.2) attains the bound in every simple family, including the Albert cone. The lift is fixed before the direction; if it may depend on v , the value is one (4.11). Section 5 treats products of balls.

Global selection. If instead the boundary points of the lift and their certificates must be C^1 functions on the whole sphere of directions, equality in a local bound forces a map into a product of support orbits or of boundaries of cone bases, and covering and cohomology arguments make some resources larger. For example, the least worst-case certificate rank becomes $\kappa_{\mathcal{K}}(s)$, which is one if $s = B + 1$ and $\mathcal{K}$ contains Q_{B+2} , and $\lceil s/B \rceil$ otherwise (Theorem 7.7). We also determine the least Lorentz resources under global selection (Theorem 6.2) and prove rigidity theorems for products (Sections 6–12); those of Section 12 assume a stronger hypothesis on entire boundary fibers, not smooth selection.

Barrier parameters. We distinguish the standard product barrier on the ambient cone, its parameter $\nu_{\text{std}, \text{slice}}$ after restriction to the affine slice (8.1), and the least parameter ϑ_{opt} of any self-concordant barrier on a fixed affine domain. The restricted parameter is at least the total nullity $\sum_i (r_i - \text{rank}_J X_i)$ at every feasible X (Lemma 8.1); recession directions give a similar bound (Lemma 8.2). With the rank theorem, this gives $\nu_{\text{std}, \text{slice}} \geq \lceil (s-1)/B \rceil$ for every lift of $\mathbb{B}_2^s$ and the exact minimum under global selection (Proposition 8.3). Without selection assumptions, we determine the minimum over lifts in several cases (Proposition 8.3, Theorems 8.4 and 8.6). For dictionaries of Hermitian PSD cones over one field up to a fixed order, or of Lorentz cones up to a fixed dimension, the remaining single-ball cases are $s = qB + 1$ with $q \geq B + 2$, where we prove only $q \leq \inf_{\mathcal{L}} \nu_{\text{std}, \text{slice}} \leq q + 1$ (8.6). Section 18 likewise proves only bounds for nonsymmetric cones. We compute ϑ_{opt} for grouped, column-packed, and balance slices (Theorems 9.4, 9.6, and 15.1), and the restricted standard parameter of every norm tree (Theorem 9.1).

Movement in barrier metrics. For a barrier F , d_F is the distance in the Riemannian metric of its Hessian. A step of local norm at most $R < 1$ moves at most $\log(1/(1-R))$, so lower bounds on d_F bound the number of steps (Corollary 23.2). If a certificate (s_i) represents the objective gap on the whole slice of a symmetric-cone lift, then in the barrier $-\sum_i \alpha_i \log \det x_i$ with $\alpha_i \geq 1$ the distance from a fixed reference point x^0 to the points with gap at most ϵ is at least $\sqrt{Q_\alpha} [\log(\Delta_\alpha/\epsilon)]_+$, where $Q_\alpha = \sum_i \alpha_i \text{rank}_J s_i$ and $\Delta_\alpha > 0$ is explicit and depends on x^0 (Theorem 23.1); with (1.1), this gives bounds in terms of B (Corollary 23.3). A classical bound in the primal–dual metric of an arbitrary logarithmically homogeneous barrier measures the distance from a point on the central path to the strictly feasible primal–dual pairs with gap at most ϵ (Theorem 24.2).

A norm tree represents a ball by a rooted tree: each internal node v has a variable t_v that bounds the Euclidean norm of its children’s values, the leaves carry the ball coordinates w , and the root variable equals one. The barrier sums the Lorentz log-determinant barriers with weights

$\omega_v \geq 1$. Fix a unit leaf objective c , and call an internal node v active if its subtree contains a leaf j with $c_j \neq 0$. For the target $1 - c^T w \leq \epsilon$, Theorem 26.1 gives, from any fixed interior reference point,

$$d(\epsilon) = \sqrt{K_c} \log(1/\epsilon) + O(\log \log(1/\epsilon)), \quad K_c = \sum_{v \text{ active}} \omega_v + \frac{1}{2} \sum_{v \text{ inactive}} \omega_v, \quad (1.2)$$

with the tree, weights, and objective fixed as $\epsilon \rightarrow 0$. The central path from the analytic center has coefficient $\sqrt{\sum_v \omega_v}$ (Corollary 26.2). Thus inactive subtrees add to the least distance although their certificate blocks vanish, and they make the central path asymptotically longer than the shortest path by the factor $(\sum_v \omega_v / K_c)^{1/2}$.

Newton systems of PSD packings. In the column-packed formulation of a product of balls, the ball coordinates are placed in the columns of matrices W_ℓ , each constrained by $\begin{pmatrix} S_\ell & W_\ell^T \\ W_\ell & I \end{pmatrix} \succeq 0$, with one linear allocation equation per ball on the diagonal entries of the S_ℓ (Section 29). Fix the projected point. At the fiber center, the minimizer of the barrier over the auxiliary variables, the residuals $D_\ell^0 = S_\ell - W_\ell^T W_\ell$ are diagonal and the projected Newton system equals that of the grouped formulation (Proposition 29.1). Let H_0 and H_D be the projected Hessians, after elimination of the auxiliary directions, at the fiber center and at another feasible auxiliary point with residuals D . If $\mu D^0 \preceq D \preceq L D^0$, where $0 < \mu \leq 1 \leq L$, Theorem 29.3 proves

$$L^{-1} H_0 \preceq H_D \preceq \mu^{-1} H_0, \quad \kappa(H_0^{-1/2} H_D H_0^{-1/2}) \leq L/\mu. \quad (1.3)$$

A feasible example with two balls attains the condition factor (Example 29.4). Exact allocations are essential: for a relative residual error $\delta < 1$, the condition bound $(1 + \delta)/(1 - \delta)$ becomes its square without them (Corollary 29.5). The comparison neither determines the right-hand side of the reduced Newton system nor constructs a cheap preconditioner.

Computational costs. Section 27 gives exact integer tradeoffs among ambient coordinates, free coordinates, and restricted barrier parameters. For the grouped, norm-tree, and centered packed ball formulations without dense additional affine coupling, sparse elimination solves each Newton equation exactly with linearly many field operations. Hence, for an outer method that accepts any direction satisfying a stated Newton residual condition and outputs the full direction in every round, a quantum Newton subroutine gives no polynomial saving in the dimension when classical elimination has the same data access and arithmetic is exact (Section 29). In the query model, we separate the cost of a normalized quantum state of a projected solution from the costs of readable values, full vectors, reusable exact compilers, and scale updates after fresh data. On one explicit family with n hidden signs, the amplitude state of the projected optimizer needs one query, whereas for every fixed $0 < \epsilon < 1/8$ any explicit ϵ -optimal output needs $\Theta_\epsilon(n)$ queries (Theorem 30.2). Our work lower bound (Theorem 28.1) assumes that every round costs at least a stated amount of new work; this does not follow from the access model, and a query lower bound for one solve cannot simply be multiplied by the number of steps that a distance bound forces.

1.2 Relation to the literature

Conic lifts and slack factorizations are the framework of Gouveia, Parrilo, and Thomas [43, Theorem 2.4]; see also the survey Fawzi et al. [37]. Fixed-size PSD factor counts [33], PSD rank and bounds from algebraic boundaries [34, 36], and semidefinite extension degree, which minimizes the largest PSD block while allowing finite products [8], measure related but different resources. Fawzi's obstruction for second-order cone (SOC) lifts and Saunderson's theorem on short chains of faces

use neighborliness to exclude finite product lifts in their settings [32, 80]. Kummer’s bounds for balls concern direct spectrahedral descriptions [58], not projections of products of cones. Positive curvature does not obstruct SOC representability: Scheiderer [81, Theorem 1.2] proves that compact convex semialgebraic sets with Nash-smooth boundary and strictly positive curvature everywhere have SOC lifts, with an arbitrary finite number of PSD blocks of order two. That existence theorem does not determine the least number of factors, the minimax certificate rank, or the minima under global selection.

The Jordan classification, Peirce spaces, quadratic representations, and support manifolds are classical [31, 45, 73], and factorization algorithms for symmetric cones are studied in [82]. Recursive norm representations by small Lorentz cones are also classical [13, Section 2]. We use standard covering, bundle, and cohomology tools [48, 83]; our contributions are the quantitative inequalities, the quantifiers over certificate fibers, the rigidity consequences, and the attaining constructions.

Self-concordant barrier theory is the analytic framework [70]. Optimality of the standard barriers on symmetric cones in the ambient space is due to Güler–Tuncel and Cardoso–Vieira [20, 47]; self-scaled barriers have been classified [49]. Conic extension, affine restriction, and projective transformation of barriers belong to an existing theory [55], so the distinction between ambient and restricted parameters is not new; we compute particular restricted and intrinsic parameters. For nonsymmetric cones we use established constructions [22, 35, 53]. The parameter bound for the universal barrier [60] and practical methods for natural cones [24] show why a small parameter should be separated from the cost of evaluating derivatives.

Nesterov–Todd and Nesterov–Nemirovski compared central paths with Riemannian distances in barrier metrics [69, 72]. The unpublished companion *The Cost of Following the Central Path* [21] already contains the weighted exposed-minor estimate, several movement bounds that depend on the distribution of the objective weights, and the standard-barrier calculations for grouped, packed, and norm-tree formulations; we reproduce the proofs needed here. New here are the combination with the certificate-rank theorem, the general exposure profile for affine PSD slices, and the weighted-tree coefficient (1.2). The general distinction between central-path length and shortest distance is not new.

The computational arguments use search, oracle interrogation, approximate counting, and information bounds [14, 18, 66, 67, 84]; sparse cone augmentation [27]; and elimination from a supplied tree decomposition [39]. The relevant algorithms for comparison are quantum interior-point methods and their end-to-end resource analyses [7, 26]. Recent iterative refinement improves their high-precision behavior [62]; we do not claim that all such methods retain the earlier dependence on inverse accuracy. The first-power improvement in (1.3) uses elementary trace inequalities and the allocation equations; it is not a new general principle about matrix order.

To our knowledge, the following results are new, in the models stated above: the selection-free minimax rank (1.1), the weighted-tree coefficient (1.2) with its active and inactive nodes, and the sharp Hessian comparison (1.3) for feasible allocations. The claim concerns these formulas and their quantifiers, not the general methods used to prove them.

1.3 Conventions

Table 1 lists the model of each main result, and Table 2 collects the recurring notation. Symbols such as p, q, r_i, L also have local meanings, defined where they are introduced; in particular, $r_i = m_i - 2$ in Sections 6 and 14 and Appendix C.

Table 1: Main results by model. Each cited statement gives its full hypotheses.

Model and resource	Conclusion and scope
Exact definable lifts; cone dimensions	Curvature bound; capped minima for balls (Theorem 3.2, Corollary 3.4).
Symmetric-cone lifts; certificate rank	$\lceil (s-1)/B \rceil$ without global selection (Theorem 4.1).
Globally C^1 selections (a stronger model)	Exact Lorentz and rank minima (Theorems 6.2, 7.7, 7.8).
Restricted standard barrier; any barrier on a fixed domain	Lower bounds, exact minima in the stated cases, and concrete optima; the range in (8.6) remains open (Sections 8, 9, 15).
Fixed barrier metric; distance to a target	Exposed-minor bounds, tree coefficient $\sqrt{K_c}$, primal-dual bound (Theorems 23.1, 26.1, 24.2).
Feasible PSD packing; projected Hessian	Grouped system at the fiber center; sharp factor L/μ (Proposition 29.1, Theorem 29.3).
Specified access, output, and per-round work	Conditional work bounds, state/readout separations, compiler costs (Sections 28–31); we do not embed the update batches of Theorem 29.7 in the Newton iterates of one fixed conic program.

Table 2: Recurring notation and conventions.

Notation	Meaning
$\mathcal{L} = (K, A, \pi)$	Exact affine lift; $K = \prod_i K_i$, reduced as in Lemma 2.1.
$\mathcal{D}_{\mathcal{L}}(v)$	Positive certificates representing $1 - \langle \pi(X), v \rangle$ for every $X \in A$, including infeasible points of the slice.
$s, s_a; m_i$	Dimensions of the projected balls; real dimensions of the cone spaces.
$r_i, a_i; p_i, q_i$	Jordan rank and Peirce constant of a simple factor; complementary primal and dual ranks at a contact. For real PSD matrices, rank_J is the matrix rank.
$B; a_i \lfloor r_i^2/4 \rfloor$	Largest mixed curvature per unit of dual rank over the dictionary; largest mixed curvature of one factor. These capacities differ.
$\nu_{\text{std}, \text{slice}}$	Least parameter of the specified standard log-determinant barrier on its effective affine slice; see (8.1).
$\vartheta_{\text{opt}}(\Omega); \nu_{\text{LH}}(C)$	Infimum over all self-concordant barriers on a fixed affine domain; infimum over logarithmically homogeneous barriers on a fixed cone.
d_F, ϵ	Distance in the Hessian metric of the displayed barrier, over paths in the stated domain with the endpoint constraints of the relevant theorem; objective gap or output accuracy.
PSD order; asymptotic notation	$H \preceq G$ means that $G - H$ is positive semidefinite. Each result states its constants and whether the instance is fixed or dimension and accuracy vary jointly.

Part I

Local curvature and certificate fibers

2 Exact lifts and local conventions

Throughout, cone-dimension and matrix-order caps are integers.

2.1 Exact lifts and genuine certificates

A *proper cone* is a closed, convex, pointed cone with nonempty interior in its finite-dimensional real ambient space. Its dual is $K^* = \{y : \langle x, y \rangle \geq 0 \text{ for all } x \in K\}$. Let $C \subset \mathbb{R}^s$ be a compact convex body with nonempty interior and $0 \in \text{int } C$. Its polar body is $C^\circ = \{v \in \mathbb{R}^s : \langle x, v \rangle \leq 1 \text{ for every } x \in C\}$. An exact lift is a triple

$$\mathcal{L} = (K, A, \pi), \quad C = \pi(K \cap A),$$

where $K = \prod_{i=1}^k K_i$, A is an affine subspace, and $\pi : A \rightarrow \mathbb{R}^s$ is affine. We do not assume $K \cap A$ is compact. A lift is *proper* if $A \cap \text{int } K \neq \emptyset$. For relative Slater feasibility we use the minimal-face reduction of the following lemma.

Lemma 2.1 (Reduction and certificate normalization). *Unrestricted free variables in a finite affine lift of a compact body can be eliminated without changing its cone factors. The cone can then be replaced by the smallest face containing all feasible points, within the span of that face; the resulting lift is proper. For a proper lift and every $v \in \partial C^\circ$, the set*

$$\mathcal{D}_{\mathcal{L}}(v) = \{Y \in K^* : \langle X, Y \rangle = 1 - \langle \pi(X), v \rangle \text{ for every } X \in A\} \quad (2.1)$$

is nonempty. Thus certificates represent the support slack on all of A , including points outside K .

Proof. Write the original affine equalities in cone coordinates X and free coordinates z . A free direction that preserves these equalities while fixing X must have zero output direction: otherwise its entire line projects into the compact body. Consequently the output depends affinely only on X , on the affine subspace of X for which the equalities admit some z . Linear elimination gives the claimed representation.

Choose $X_0 \in \text{ri}(K \cap A)$, and let F be the minimal face of K containing X_0 . For every $X \in K \cap A$, the relative-interior property gives $X_0 - \epsilon(X - X_0) \in K \cap A$ for some $\epsilon > 0$. The face property therefore gives $X \in F$. Conversely $F \cap A \subset K \cap A$. The point X_0 belongs to $\text{ri } F$, so replacing K by F and A by $A \cap \text{span } F$ gives Slater feasibility in $\text{span } F$. Faces of a product are products of faces.

For the last assertion write $A = \{X : MX = b\}$ and extend the affine output as $\pi(X) = TX + t$. Maximizing $\langle v, TX + t \rangle$ over $K \cap A$ has value one. Slater conic duality gives a multiplier λ with $Y = M^*\lambda - T^*v \in K^*$ and $\langle b, \lambda \rangle = 1 - \langle t, v \rangle$. Hence for every $X \in A$, $\langle X, Y \rangle = \langle b, \lambda \rangle - \langle TX, v \rangle = 1 - \langle \pi(X), v \rangle$. This is the duality step in the proper-lift direction of [43, Theorem 2.4]. $\square$

We call the elements of $\mathcal{D}_{\mathcal{L}}(v)$ the *genuine certificates* of v , and $\mathcal{D}_{\mathcal{L}}(v)$ its *certificate fiber*.

For symmetric cones, every face is the cone of squares of a Peirce-one subalgebra, and Jordan ranks in a face agree with ambient ranks. Throughout, ranks, determinants, and barriers of a relative-Slater lift are computed *after* this reduction. Certificates then lie in the dual of the reduced face and exist even if the original presentation fails Slater's condition. A face of a dictionary factor need not belong to the dictionary, but its capacities never exceed those of the factor.

The ball $C = \mathbb{B}_2^s = \{x : \|x\|_2 \leq 1\}$ is its own polar, so its supports are the unit vectors $v \in \mathbb{S}^{s-1}$. Define

$$r_{\mathcal{L}}(v) = \min_{Y \in \mathcal{D}_{\mathcal{L}}(v)} \sum_i \text{rank}_J Y_i, \quad R(\mathcal{L}) = \max_{v \in \mathbb{S}^{s-1}} r_{\mathcal{L}}(v). \quad (2.2)$$

Both extrema exist because they are taken over nonempty subsets of a finite set of integers; compactness of the certificate fiber is not needed. No certificate is zero, because the support slack is not identically zero on C ; this also gives the lower bound one in the interval case $s = 1$, treated separately.

2.2 Regularity obtained from the lift

We call data *definable* if they are definable, allowing real parameters, in one fixed o-minimal expansion of the ordered real field. Only the bound for general proper cones below needs this generality. All symmetric-cone data and selections used for the rank theorem are semialgebraic. Definable choice and finite differentiable stratification are standard [85]; in the semialgebraic setting see [16].

Lemma 2.2 (Generic regular selections). *For a proper definable lift, primal lifts of boundary points and certificates of supports $v \in \partial C^\circ$ can be chosen definably. On a definable C^2 boundary patch, their compositions with the definable normalized outward-normal contact map are C^1 in definable local coordinates on a dense relatively open subset of the patch. For a symmetric-cone lift of a Euclidean ball, certificates can also attain $r_{\mathcal{L}}(v)$, and the common regular subset can be chosen semialgebraic, with a lower-dimensional complement in $\mathbb{S}^{s-1}$.*

Proof. The primal fiber and the fiber (2.1) are nonempty closed definable sets; the identity on A amounts to finitely many affine equalities. Their minimum-Euclidean-norm points are unique, since both fibers are convex, and their graphs are definable by quantified formulas. For minimum Jordan rank, each condition $\sum_i \text{rank}_J Y_i = j$ is semialgebraic by spectral polynomial equations and inequalities. Partition the support sphere according to the least realized j and apply semialgebraic choice in each resulting nonempty fiber. This choice need not minimize norm and need not be globally continuous.

Finite C^1 cell decomposition, refined for all the maps at once, makes them C^1 on each full-dimensional cell. The union of the interiors of those cells is relatively open and dense; its complement has smaller dimension. In the semialgebraic case this complement has surface measure zero. A definable C^2 boundary patch and its normalized normal map are eligible for the same decomposition. Faces used in reduction are definable: for $X_0 \in \text{ri } F$, $F = \{X \in K : \exists \epsilon > 0, X_0 - \epsilon X \in K\}$. $\square$

3 Local curvature and the size of cone factors

A *contact* is a pair (x, v) with $x \in \partial C$, $v \in \partial C^\circ$, and $\langle x, v \rangle = 1$.

Let $x : U \rightarrow \partial C$ be a definable C^2 local parametrization, where $U \subset \mathbb{R}^{s-1}$, and let $n(u)$ be its outward unit normal. Set

$$y(u) = \frac{n(u)}{\langle x(u), n(u) \rangle}.$$

The denominator is positive because $0 \in \text{int } C$, and $y(u)$ lies on ∂C° . As in Lemma 2.2, select a primal lift $X(u) = (X_i(u))_i$ of $x(u)$ and a certificate $Y(u) = (Y_i(u))_i \in \mathcal{D}_{\mathcal{L}}(y(u))$. The whole-slice identity gives

$$1 - \langle x(u), y(v) \rangle = \sum_i \langle X_i(u), Y_i(v) \rangle. \quad (3.1)$$

Each summand is nonnegative and vanishes on the diagonal. The negative of the mixed second derivative at the diagonal point $u = v = 0$ is the bilinear form

$$H = Dx(0)^* Dy(0) = \frac{Dx(0)^* Dn(0)}{\langle x(0), n(0) \rangle}. \quad (3.2)$$

At a strictly positively curved point it has rank $s - 1$.

Lemma 3.1 (Universal local bound). *Let $K \subset E$ be a proper cone, $m = \dim E$, and let $X : U \rightarrow K$, $Y : U \rightarrow K^*$ be C^1 with $\langle X(u), Y(u) \rangle = 0$. The bilinear form $-\langle DX(0)[h], DY(0)[\ell] \rangle$ has rank at most $m - 2$ if $X(0), Y(0) \neq 0$, and rank zero if either is zero. In particular rays contribute no mixed curvature.*

Proof. Put $a = X(0)$ and $b = Y(0)$. When both are nonzero, nonnegativity of $\langle X(th), b \rangle$ for positive and negative t shows $DX(0)[h] \in b^\perp$. Similarly $DY(0)[\ell] \in a^\perp$. The pairing from $b^\perp$ to $(a^\perp)^*$ has kernel $b^\perp \cap \text{span}\{a\} = \text{span}\{a\}$, since $\langle a, b \rangle = 0$. Its rank is $(m - 1) - 1 = m - 2$. Pullback by the derivatives cannot increase rank. If $a = 0$, a two-sided derivative of X lies in $K \cap (-K) = \{0\}$; the case $b = 0$ is identical. $\square$

Theorem 3.2 (Curvature bounds). *Suppose a compact s -dimensional body has a relatively open C^2 boundary patch with strictly positive curvature at some point, and an exact lift by a finite product of definable proper cones of dimensions m_i . Then*

$$s - 1 \leq \sum_i \max\{m_i - 2, 0\}. \quad (3.3)$$

If the factors are simple symmetric cones of ranks r_i and Peirce constants a_i , one can choose a contact at which the complementary primal and certificate Jordan ranks p_i, q_i satisfy

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i \left\lfloor \frac{r_i^2}{4} \right\rfloor. \quad (3.4)$$

A ray has $r_i = 1$, $a_i = 0$ in this notation.

We first prove the sharper local bound for symmetric cones. In a Euclidean Jordan algebra V , write $x \circ y$ for its product, e for its identity, and $\langle x, y \rangle = \text{tr}(x \circ y)$ for the trace inner product. The cone K is the cone of squares, and $x \succeq 0$ means $x \in K$. A Jordan frame $c_1, \dots, c_r$ gives the orthogonal Peirce decomposition $V = \bigoplus_{i \leq j} V_{ij}$, where $V_{ii} = \mathbb{R}c_i$ and $\dim V_{ij} = a$ for $i < j$ in a simple algebra. The support idempotent of $x \succeq 0$ is the sum of the frame idempotents with positive eigenvalues; their number is $\text{rank}_J x$.

Lemma 3.3 (Mixed Peirce channels). *Let $X, Y : U \rightarrow K$ be C^1 and complementary: $\langle X(u), Y(u) \rangle = 0$. If $X(0)$ and $Y(0)$ have ranks p and q , then $p + q \leq r$, and $-\langle DX(0)[h], DY(0)[\ell] \rangle$ is a positive semidefinite symmetric bilinear form of rank at most apq . Neither locally constant ranks nor strict complementarity is required.*

Proof. Positive orthogonal elements are Jordan complementary and diagonalize in a common Jordan frame [31, Chapters III–IV]. Thus

$$X(0) = \sum_{i \in I} \lambda_i c_i, \quad Y(0) = \sum_{j \in J} \mu_j c_j, \quad \lambda_i, \mu_j > 0, \quad I \cap J = \emptyset.$$

For every positive z in the Peirce subalgebra on the complementary indices I^c , $\langle X(th), z \rangle \geq 0$ vanishes at $t = 0$, so its derivative there is zero. Since the positive cone of that subalgebra spans

it, the derivative of X has zero projection onto it. Likewise the derivative of Y vanishes on the J^c subalgebra. The only possible overlap of the two derivatives is therefore $\bigoplus_{i \in I, j \in J} V_{ij}$, of dimension apq .

Differentiate $X(u) \circ Y(u) = 0$. Its V_{ij} component is $\mu_j(DX[h])_{ij} + \lambda_i(DY[h])_{ij} = 0$, because multiplication by either endpoint idempotent acts by one half on V_{ij} . Hence

$$-\langle DX[h], DY[\ell] \rangle = \sum_{i \in I, j \in J} \frac{\mu_j}{\lambda_i} \langle (DX[h])_{ij}, (DY[\ell])_{ij} \rangle. \quad (3.5)$$

This is the required Gram form. If I or J is empty, the form is zero, as also follows from the pointed-vertex argument of Lemma 3.1. $\square$

Proof of Theorem 3.2. Reduce the lift by Lemma 2.1. Curvature is strictly positive on an open subpatch. There choose a contact in the common regular set given by Lemma 2.2. Differentiating (3.1) once in each independent variable and using rank subadditivity with Lemma 3.1 gives (3.3). A reduced face has dimension no larger than its original factor, so the original dimensions give valid bounds. For symmetric cones apply Lemma 3.3 instead. The inequality $pq \leq \lfloor r^2/4 \rfloor$ follows from $p + q \leq r$. Faces of a simple symmetric cone have the same Peirce constant at smaller rank, except that rank-one faces are rays. For the Albert cone its rank-two faces are spin factors with Peirce constant eight. Their capacities $a \lfloor r^2/4 \rfloor$ likewise cannot exceed that of the original factor. $\square$

The bound apq is attained locally. The inner derivations $4[L_z, L_{c_i}]$, $z \in V_{ij}$, independently rotate the support directions i and j , and their exponentials are Jordan automorphisms. Applied simultaneously to $X(0), Y(0)$, their derivatives in V_{ij} are $\lambda_i z, -\mu_j z$. Thus all apq channels in (3.5) can be realized. This local statement does not claim that a specified collection of channels integrates into an exact lift.

Write $Q_m = \{(t, z) \in \mathbb{R} \times \mathbb{R}^{m-1} : t \geq \|z\|_2\}$ for the Lorentz cone of total cone-space dimension m .

Corollary 3.4 (Least factor count and dimension for a ball under a dimension cap). *Fix $s \geq 2$ and an integer $d \geq 3$. Among exact lifts of $\mathbb{B}_2^s$ by arbitrary definable proper cone factors of dimension at most d , the minimum number of factors and the minimum total cone-space dimension are*

$$k_{\min} = \left\lceil \frac{s-1}{d-2} \right\rceil, \quad M_{\min} = s-1 + 2k_{\min}.$$

Both minima are attained by the same Lorentz norm tree, constructed in the proof. Among symmetric cone lifts under this dimension cap, the least total ambient Jordan rank is $2k_{\min}$, again attained by that lift.

Proof. Let k_+ count factors of dimension at least two and k_1 count rays. Theorem 3.2 implies $s-1 \leq k_+(d-2)$ and $M \geq s-1 + 2k_+ + k_1$, proving the first two lower bounds. For symmetric factors only non-ray irreducible factors can contribute curvature, and each has rank at least two. Decomposing reducible factors cannot increase their dimensions or change total rank, so total rank is at least $2k_{\min}$.

For attainment we use a variable-arity form of the recursive norm representation in [13, Section 2, equation (5)]. Put $k = k_{\min}$ and write $s-1 = b_1 + \dots + b_k$, with $1 \leq b_i \leq d-2$. Form a rooted tree whose internal nodes are a chain. Node $i < k$ has b_i leaf children and the next internal node as one additional child; the last node has $b_k + 1$ leaf children. It has s leaves. Assign the coordinates of $x \in \mathbb{R}^s$ to those leaves and an auxiliary scalar t_i to each internal node, and impose

$$t_i \geq \|\text{child values of node } i\|_2 \quad \text{in } Q_{b_i+2}, \quad t_1 = 1.$$

Recursive elimination of the scalars gives exactly $\|x\|_2 \leq 1$. Strict feasibility holds at $x = 0$ by choosing the nonroot scalars positive and successively smaller. The construction uses k rank-two factors and total dimension $\sum_i (b_i + 2) = s - 1 + 2k$, as required. $\square$

The total ambient Jordan rank is the parameter of the standard product log-determinant barrier on the ambient cone and remains valid after restriction to the affine slice. It need not be the least parameter of the restricted barrier, and it gives no lower bound for a barrier defined intrinsically on the projected ball. These distinctions matter when curvature bounds are used to compare formulations.

4 The exact minimum support-certificate rank

We compute the least value of $R(\mathcal{L})$ in (2.2) over lifts of a ball built from a fixed family of symmetric cones. Fix a finite dictionary $\mathcal{K}$ of simple symmetric cones. A lift through $\mathcal{K}$ may use any finite number of copies of each member and nonnegative rays; the dictionary is fixed, not the list of factors of an individual lift. For a non-ray dictionary member let r and a be its rank and Peirce constant. Put

$$B = \max_{K \in \mathcal{K}} a(K)(r(K) - 1) > 0, \quad q_B(s) = \left\lceil \frac{s-1}{B} \right\rceil. \quad (4.1)$$

We call B the *dictionary capacity*; it is the largest *primitive capacity* $a(r-1)$ of a dictionary member. It bounds mixed curvature *per unit of dual rank* and differs from the largest mixed-curvature capacity $a\lfloor r^2/4 \rfloor$ of one factor (Theorem 3.2): for real PSD matrices of order four the two capacities are three and four.

Theorem 4.1 (Least worst-case certificate rank). *Let $s \geq 2$. Over all exact finite relative-Slater affine lifts of $\mathbb{B}_2^s$ through $\mathcal{K}$,*

$$\inf_{\mathcal{L}} R(\mathcal{L}) = q_B(s). \quad (4.2)$$

For each fixed lift there is a dense open semialgebraic subset $U \subset \mathbb{S}^{s-1}$, whose complement has surface measure zero, such that

$$\sum_i \text{rank}_J Y_i \geq q_B(s) \quad \text{for every } v \in U \text{ and every } Y \in \mathcal{D}_{\mathcal{L}}(v).$$

An explicit lift attains minimum certificate rank $q_B(s)$ at every support except one pole, where its minimum rank is one. Consequently

$$\inf_{\mathcal{L}} \text{ess inf}_{v \in \mathbb{S}^{s-1}} r_{\mathcal{L}}(v) = q_B(s). \quad (4.3)$$

Lower bound. Use Lemma 2.2 to choose a primal lift $X(u)$ of u and a minimum-total-rank certificate $Y(v)$ of $1 - \langle u, v \rangle$. On a common dense open regular subset of the sphere these are C^1 and satisfy

$$\sum_i \langle X_i(u), Y_i(v) \rangle = 1 - \langle u, v \rangle.$$

At a contact $u = v$, the negative mixed derivative of this identity is the sphere metric, of rank $s - 1$. With $p_i = \text{rank}_J X_i(u)$ and $q_i = \text{rank}_J Y_i(u)$ there, Lemma 3.3 gives

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - 1) q_i \leq B \sum_i q_i. \quad (4.4)$$

The middle inequality uses $p_i \leq r_i - 1$ whenever $q_i > 0$; when $q_i = 0$ both terms vanish. Rays contribute zero curvature. Reduced proper faces have no larger $a_i(r_i - 1)$ than the original dictionary factor. Since the selected certificate has minimum total rank, (4.4) implies the asserted bound for *all* certificates at that support, and hence at every support in the regular set. The same bound gives both lower bounds in (4.2) and (4.3). $\square$

4.1 A uniform Peirce-perspective construction

The matching upper bound uses one construction valid in every simple Euclidean Jordan algebra. For $x \in V$ put $L_x y = x \circ y$ and $P(x) = 2L_x^2 - L_{x^2}$, the quadratic representation. For an idempotent c , $P(c)$ is orthogonal projection onto $V(c, 1)$. We use the standard facts that $P(x)K \subset K$ and that $P(x)$ is a cone automorphism when x is invertible. These hold in every Euclidean Jordan algebra; see [31, Chapters II–IV] and the Peirce and quadratic representation discussion in [45, pp. 1554–1555]. We also need a rank consequence. Approximate any x by invertible elements x_n . Cone automorphisms preserve extreme rays, so $P(x_n)c$ has rank one when c is primitive. The closedness of the set of nonnegative elements of rank at most one gives

$$P(x)c \succeq 0, \quad \text{rank}_J(P(x)c) \leq 1. \quad (4.5)$$

This argument does not use associative matrix multiplication.

Lemma 4.2 (Primitive Peirce perspective). *Let V be simple of rank $r \geq 2$, let c be primitive, and set $d = e - c$. The half-Peirce space $W = V(c, 1/2)$ has dimension $a(r - 1)$; give it the norm induced by the trace inner product, $\|w\|^2 = \langle w, w \rangle$. For $w \in W$,*

$$X(\delta, w, z) = \delta c + \sqrt{2}w + zd \succeq 0 \iff \delta, z \geq 0, \quad \delta z \geq \|w\|^2, \quad (4.6)$$

$$\det X(\delta, w, z) = z^{r-2}(\delta z - \|w\|^2). \quad (4.7)$$

For $g \in W$ and $A > 0$ the element

$$Y = AP \left(c - \frac{g}{\sqrt{2}A} \right) c = Ac - \frac{g}{\sqrt{2}} + \frac{P(g)c}{2A} \quad (4.8)$$

is positive of rank one. Its component in $V(d, 1)$ has trace $\|g\|^2/(4A)$.

Proof. Peirce multiplication gives $w^2 = tc + u$, where $u \in V(d, 1)$ and

$$t = \langle w^2, c \rangle = \langle w, w \circ c \rangle = \frac{\|w\|^2}{2}.$$

The definition of P and $c \circ w = w/2$ give $P(w)c = u$. Its trace is $\text{tr } u = \text{tr } w^2 - t = t$. By (4.5), for $w \neq 0$ we have $u = tk$ for a primitive idempotent $k \in V(d, 1)$. Thus $w^2 = t(c + k)$. Spectral calculus shows that w lies in the Peirce-one subalgebra with identity $c + k$: the supports of w and w^2 coincide. Since $c \circ w = w/2$, we have $k \circ w = w/2$. The span of c, k, w is therefore a rank-two Jordan subalgebra, orthogonal to $d - k$, and in this subalgebra the two eigenvalues of $\delta c + \sqrt{2}w + zk$ have sum $\delta + z$ and product $\delta z - 2t = \delta z - \|w\|^2$. The remaining $r - 2$ eigenvalues are all z . This proves both displayed claims for $w \neq 0$. For $w = 0$ they follow from the diagonal element $\delta c + zd$.

Expanding the quadratic representation by its definition proves (4.8). The first expression and (4.5) show positivity and rank at most one; its c -coefficient $A > 0$ makes the rank exactly one. The calculation above with $w = g$ gives $\text{tr } P(g)c = \|g\|^2/2$, proving the trace formula. $\square$

Attainment in Theorem 4.1. Choose a dictionary member attaining B , and use its notation from Lemma 4.2. Put $q = q_B(s)$ and partition the first $s - 1$ real coordinates into q nonempty groups, each of size at most B . Isometrically embed each group into a subspace $W_G \subset W$. Write a ball point as $x = (w_1, \dots, w_q, b)$ and impose

$$X_G = (1 - b)c + \sqrt{2}w_G + z_G d \succeq 0, \quad \sum_{G=1}^q z_G = 1 + b. \quad (4.9)$$

The variables z_G are auxiliary and $w_G \in W_G$. If the constraints hold, Lemma 4.2 gives $\sum_G \|w_G\|^2 \leq (1 - b) \sum_G z_G = 1 - b^2$, hence $x \in \mathbb{B}_2^s$. Conversely, if $b < 1$ and $x \in \mathbb{B}_2^s$, choose $z_G \geq \|w_G\|^2 / (1 - b)$ with sum $1 + b$, which is possible because $\sum_G \|w_G\|^2 / (1 - b) \leq 1 + b$. At $b = 1$ all $w_G = 0$ and any nonnegative z_G of sum two works. Thus the projection is exact. At $x = 0$ with $z_G = 1/q$, all blocks are positive definite, which proves Slater feasibility.

Let $v = (g_1, \dots, g_q, \eta) \in \mathbb{S}^{s-1}$, with $g_G \in W_G$. Every certificate admits the Peirce decomposition

$$Y_G = A_G c + u_G + H_G, \quad u_G \in W, \quad H_G \in V(d, 1),$$

and positivity implies $A_G \geq 0$ and $H_G \succeq 0$. Comparing coefficients on the whole affine slice (4.9) gives

$$\text{proj}_{W_G} u_G = -g_G / \sqrt{2}, \quad \text{tr } H_G = \gamma = \frac{1 - \eta}{2}, \quad \sum_G A_G = \frac{1 + \eta}{2}. \quad (4.10)$$

Indeed, varying the z_G with fixed sum makes their trace coefficients equal; the constant and b coefficients give the two remaining scalar identities. When $q = 1$ these two comparisons give them directly. Conversely, these conditions imply the whole-slice identity.

If $\eta < 1$, define

$$A_G = \frac{\|g_G\|^2}{2(1 - \eta)}.$$

For $g_G \neq 0$, take the rank-one completion in (4.8); its $V(d, 1)$ component has trace $\|g_G\|^2 / (4A_G) = \gamma$. For $g_G = 0$, take $Y_G = \gamma k_G$ with any primitive $k_G \in V(d, 1)$. The identity $\sum_G \|g_G\|^2 = 1 - \eta^2$ gives the prescribed sum of the A_G . Thus every block has rank one and the total rank is q . Conversely, $\gamma > 0$ forces every block of every certificate to be nonzero, so the minimum total rank is exactly q . This includes the south pole $\eta = -1$.

At the north pole $\eta = 1$, positivity and $\text{tr } H_G = 0$ give $H_G = 0$. Positivity then gives $u_G = 0$: a positive element orthogonal to d belongs to the face $\mathbb{R}_+ c$. Concentrating $\sum_G A_G = 1$ in one block gives total rank one, which is minimal. If $W_G \neq W$, a certificate may have extra half-Peirce components orthogonal to W_G . Our construction sets them to zero, while the positive trace γ of the $V(d, 1)$ component still forces each block to be nonzero. We do not claim that certificates are unique. The exceptional pole has surface measure zero for $s \geq 2$, proving (4.3) and finishing the theorem. $\square$

4.2 Examples and the order of quantifiers

The primitive capacity $a(r - 1)$ is $r - 1$, $2(r - 1)$, and $4(r - 1)$ for real, complex, and quaternionic Hermitian PSD matrices of order r , and $m - 2$ for a Lorentz cone Q_m . For the Albert cone $H_+^3(\mathbb{O})$ it is $8(3 - 1) = 16$. Consequently, lifts by any finite number of Albert factors give the exact value $\lceil (s - 1)/16 \rceil$. The proof uses neither an associative representation of the Albert algebra nor an embedding into a real PSD cone.

In Theorem 4.1 the lift is fixed before the support (the objective); reversing this order can change the value:

$$\inf_{\mathcal{L}} \sup_v r_{\mathcal{L}}(v) = q_B(s), \quad \sup_v \inf_{\mathcal{L}} r_{\mathcal{L}}(v) = 1. \quad (4.11)$$

For the second identity rotate the construction so that the prescribed v is its north pole; the first is Theorem 4.1. The universal lower bound one follows from the nonzero support slack. Likewise, the theorem minimizes over the full certificate fiber at each support, not over certificates *selected in advance* by an algorithm. For $s = 1$, whenever a non-ray factor is available, choose orthogonal primitive idempotents c, k and the section $X = tc + (1 - t)k$, $0 \leq t \leq 1$, with output $2t - 1$. Its two support slacks have the rank-one certificates $2k$ and $2c$, so the minimum is one; the formula $\lceil (s - 1)/B \rceil = 0$ does not apply.

5 Products: certificates, compression, and private curvature

Write $C = \prod_{a=1}^h \mathbb{B}_2^{s_a}$, with $s_a \geq 2$, and $M = \prod_a \mathbb{S}^{s_a-1}$. The pure support rows of C are the slacks $1 - \langle x_a, v \rangle$, $v \in \mathbb{S}^{s_a-1}$. A pure-row certificate represents such a slack on the entire slice of the original lift, so its pairing has zero coefficients on the projected coordinates of the other balls. This is stronger than an identity on a selected *boundary sheet*, a set of primal lifts of boundary points.

5.1 The support-join identity

Idempotents are ordered by their faces; the join $c \vee d$ is the support of $c + d$. In a product of simple factors, joins and Jordan ranks are taken factor by factor.

Lemma 5.1 (Compression measures new support). *For an idempotent c , put $f = e - c$. For $y \succeq 0$, let $t = \text{supp } P(f)y$. Then*

$$c \vee \text{supp } y = c + t, \quad \text{rank}_J P(f)y = \text{rank}_J(c \vee \text{supp } y) - \text{rank}_J c.$$

For positive weights, $\text{supp}(\sum_a \lambda_a y_a) = \bigvee_a \text{supp } y_a$.

Proof. We have $t \leq f$. Set $g = f - t$. Self-adjointness of $P(f)$ gives $\langle g, y \rangle = \langle g, P(f)y \rangle = 0$. Positive orthogonality implies $\text{supp } y \perp g$, hence $c \vee \text{supp } y \leq e - g = c + t$. Conversely let $k = e - (c \vee \text{supp } y)$. Then $k \leq f$ and $\langle k, P(f)y \rangle = \langle k, y \rangle = 0$, so $k \perp t$. Consequently both c and t belong to $V(e - k, 1)$, proving the reverse inclusion. Ranks add on orthogonal idempotents. Finally a positive element is orthogonal to a positive sum precisely when it is orthogonal to every summand; their complementary faces therefore agree. This proves the sum identity. $\square$

Theorem 5.2 (Additive existence of genuine certificates). *For every finite exact relative-Slater affine lift of C through the symmetric-cone dictionary of Section 4, there are directions $u_a \in \mathbb{S}^{s_a-1}$ and genuine certificates Y^a of the pure rows $1 - \langle x_a, u_a \rangle$ such that*

$$Q := \sum_i \text{rank}_J \left(\sum_a \lambda_a Y_i^a \right) \geq \sum_a q_B(s_a) \quad (\lambda_a > 0). \quad (5.1)$$

Every feasible primal tuple over the simultaneous contact $(u_1, \dots, u_h) \in M$ has total Jordan nullity at least Q . No regularity of primal or dual selections is assumed.

Proof. Choose the rows successively. Suppose the first $a - 1$ certificates have been chosen, and in factor i let $c_i = \bigvee_{j < a} \text{supp } Y_i^j$, $f_i = e_i - c_i$. Fix the coordinates of the previous balls at their chosen contacts. Their slacks vanish on the whole remaining cylinder. Nonnegativity of each block pairing therefore puts every feasible tuple on this cylinder in $\prod_i V_i(f_i, 1)_+$. Intersect the restricted affine slice with $\prod_i V_i(f_i, 1)$; this preserves its feasible set and its projection. Fix all later projected coordinates, say at zero. The resulting affine lift projects exactly onto the a th ball.

Compress every original pure-row certificate Y by $P(f_i)$. Its pairing with every tuple on this slice remains $1 - \langle x_a, v \rangle$. Reduce this restricted lift further to its minimal face $V_i(k_i, 1)_+$, where $k_i \leq f_i$. The further compressed tuple $P(k_i)Y_i$ is a certificate on the relative-Slater restricted slice. Face reduction cannot increase the primitive capacity. By the dense-open lower bound of Theorem 4.1, there is a support u_a at which every such further compression has rank at least $q_B(s_a)$. Slater duality on the original slice gives a certificate Y^a of the original lift for this support. Compression cannot increase rank: by Lemma 5.1 its rank is the Jordan-rank increment of a support join, which is at most the rank of the added support (equivalently use the usual face-lattice rank inequality). Alternatively apply a spectral decomposition of y and (4.5) termwise to $P(k)y$. Thus $\sum_i \text{rank}_J P(f_i)Y_i^a \geq q_B(s_a)$.

Lemma 5.1 identifies this sum with the increase of $\sum_i \text{rank}_J c_i$. Summing over the rows proves (5.1). Every final primal tuple is orthogonal to each chosen certificate, hence to their positive aggregate. Complementary ranks sum to at most the ambient rank, proving the nullity assertion. $\square$

The proof compresses certificates of the original lift because a certificate of a restricted slice need not extend to that lift. The theorem gives only the existence of pure-row certificates with high-rank positive aggregates. It gives no lower bound on the rank of every certificate of the aggregate objective, or on the minimum rank in its full fiber. Separate copies of (4.9) give aggregate certificates of rank at most $\sum_a q_B(s_a)$ at every support, which confirms the scale of (5.1) without reversing those quantifiers.

5.2 Private directions in Hermitian factors

For Hermitian PSD factors, a different argument gives a pointwise bound. It counts, for each ball, the *private* directions of its certificate range: those not used by the other balls' certificates. Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, $\delta = \dim_{\mathbb{R}} \mathbb{F}$, and use the real trace pairing. Suppose that C^1 maps satisfy the full-row identities

$$1 - \langle x_a, v \rangle = \sum_i \langle X_i(x), Y_i^a(v) \rangle, \quad x \in M, \quad v \in \mathbb{S}^{s_a-1}. \quad (5.2)$$

The maps X_i are shared by all rows; Y_i^a depends only on its own row argument.

Theorem 5.3 (Private curvature). *Suppose the factors in (5.2) belong to $H_+^{r_i}(\mathbb{F})$, $r_i \leq R$, for an integer $R \geq 2$, and put $B = \delta(R - 1)$. At every simultaneous contact $x \in M$, with supports $v = x_a$, write $p_i = \text{rank } X_i$, $U_{ia} = \text{Ran } Y_i^a(x_a)$, $U_{i,-a} = \sum_{b \neq a} U_{ib}$, and*

$$d_{ia} = \dim_{\mathbb{F}} U_{ia} / (U_{ia} \cap U_{i,-a}).$$

Then

$$s_a - 1 \leq \sum_i \delta p_i d_{ia}, \quad (5.3)$$

$$\sum_a d_{ia} \leq \text{rank} \left(\sum_a \lambda_a Y_i^a(x_a) \right) \leq \dim_{\mathbb{F}} \ker X_i, \quad \lambda_a > 0. \quad (5.4)$$

In particular, both total aggregate rank and total primal nullity are at least $\sum_a \lceil (s_a - 1)/B \rceil \geq h$.

Proof. Complementarity gives $X_i Y_i^a(x_a) = 0$. Its cylindrical form, with x_b varying and $a \neq b$, gives $(D_b X_i) Y_i^a = 0$. Decompose the underlying vector space as $\text{Ran } X_i \oplus \ker X_i$. Two-sided positivity forces the kernel–kernel compression of $D X_i$ to vanish, as in Lemma 3.3. The only part entering the row- a mixed pairing is its off-diagonal map from $\ker X_i$ to $\text{Ran } X_i$. This map annihilates $U_{i,-a}$. The corresponding derivative of Y_i^a has its other endpoint in U_{ia} ; the derivative pairing therefore factors through a real space of dimension $\delta p_i d_{ia}$. Differentiating the row identity gives the sphere metric, proving (5.3) by rank subadditivity.

Choose a complement E_{ia} of $U_{ia} \cap U_{i,-a}$ in U_{ia} . These complements are jointly independent: a relation $\sum_a w_a = 0$, $w_a \in E_{ia}$, puts each w_a in $U_{ia} \cap U_{i,-a}$, hence makes it zero. Their total dimension is therefore at most $\dim_{\mathbb{F}} \sum_a U_{ia}$, which is the aggregate rank by positivity. This sum lies in $\ker X_i$, proving (5.4). When $d_{ia} > 0$, $p_i \leq R - 1$. Thus (5.3) gives $\sum_i d_{ia} \geq \lceil (s_a - 1)/B \rceil$. Sum this inequality in a . $\square$

Inequality (5.3), stated for each ball or *source*, says more than a bound on total rank: it records which source uses each available kernel direction. It requires differentiability only at the contact under consideration. Definable selections of an affine lift are differentiable at generic contacts, but not necessarily at all of them.

Example 5.4 (One factor can serve every source). Let $p = \max_a \lceil s_a/\delta \rceil$, and embed each $\mathbb{R}^{s_a}$ isometrically into $\mathbb{F}^p$. Write their images as columns w_a of W . The single-factor lift

$$Z = \begin{pmatrix} S & W^* \\ W & I_p \end{pmatrix} \succeq 0, \quad S \in H^h(\mathbb{F}), \quad S_{aa} = 1 \quad (5.5)$$

projects exactly onto C . Indeed its Schur complement $D = S - W^*W$ is positive, giving $\|w_a\| \leq 1$; conversely choose $S = W^*W + \text{Diag}(1 - \|w_a\|^2)$. At an extreme point of the product, D has zero diagonal, hence is zero, so $S = W^*W$. This boundary sheet is polynomial and has nullity exactly h . For the embedded support $\tilde{v}$, put

$$Y^a(v) = \frac{1}{2}(e_a, -\tilde{v})(e_a, -\tilde{v})^*.$$

On the entire affine slice, its pairing with Z is $\frac{1}{2}(1 + \|\tilde{v}\|^2 - 2 \text{Re}(w_a^* \tilde{v})) = 1 - \langle x_a, v \rangle$. These are genuine rank-one row certificates. Their ranges are independent because their first h coordinates are the distinct e_a . Thus total aggregate rank and nullity are both exactly h , although the lift has only one cone factor.

Part II

Global regularity and affine barriers

6 Global regularity changes the minimal lift resources

The generic selectors used above are guaranteed to be differentiable only on a dense open set. Requiring C^1 factor maps on the entire sphere is a stronger condition. We determine its exact cost in the number, total capacity, and total dimension of Lorentz factors in the full-row identities (5.2), including factors that vanish on open sets.

6.1 Lorentz phases and pointwise no-sharing

We use the Euclidean self-dual pairing on $Q_m = \{(t, z) : t \geq \|z\|\}$. A fixed rescaling converts it to the Jordan trace pairing and changes no rank or resource count. Suppose (5.2) has globally C^1 factors in Q_{m_i} , $m_i \geq 3$, and put $r_i = m_i - 2$, the capacity of factor i ; we call i a *label*. At a contact point define

$$H_i^a(x)(u, v) = -\langle D_a X_i(x)[u], D Y_i^a(x_a)[v] \rangle.$$

Then $\sum_i H_i^a$ is the sphere metric, of rank $p_a = s_a - 1$. If X_i or Y_i^a is zero at the contact, so is its derivative, by pointedness, and $H_i^a = 0$. If one factor is interior, the other is zero. Otherwise, on a neighborhood where both factors remain nonzero, complementarity gives

$$X_i = \alpha_i(1, q_i^a(x_a)), \quad Y_i^a = \beta_i^a(1, -q_i^a), \quad q_i^a \in \mathbb{S}^{r_i}, \quad \alpha_i, \beta_i^a > 0.$$

Complementarity holds for every x with the same x_a (the cylinder identity), so the primal phase q_i^a depends only on x_a there. Differentiation cancels every term containing a derivative of the amplitudes, and

$$H_i^a = \alpha_i \beta_i^a (D q_i^a)^* D q_i^a \succeq 0, \quad \text{rank } H_i^a \leq r_i. \quad (6.1)$$

Label i is *active* for source a at x if $H_i^a(x) \neq 0$. Then the primal phase is locally constant in every other source direction, so at each point a label is active for at most one source (*pointwise no-sharing*):

$$H_i^a(x) \neq 0 \implies H_i^b(x) = 0 \quad (b \neq a). \quad (6.2)$$

A label may be active for different sources at different points.

6.2 A covering argument with top cohomology classes

The lower bounds use three standard topological facts, with real cohomology coefficients. First, if open sets U_a cover a space M and classes $u_a \in H^*(M)$ satisfy $u_a|_{U_a} = 0$, the relative cup product implies $\prod_a u_a = 0$: lift each class to $H^*(M, U_a)$ and multiply to get a class in $H^*(M, \bigcup_a U_a) = 0$. The same argument applies to any subfamily that covers. Second, by the Künneth formula, on a product of spheres the product of any subset of the pulled-back top classes is nonzero. Third, a proper open subset of a connected closed oriented p -manifold has zero ordinary cohomology in degree p when every component is noncompact; in particular this holds for every proper open subset of $\mathbb{S}^p$. See [48, Sections 3.2–3.3].

Lemma 6.1 (Saturated labels have a proper phase domain). *Fix a source a with $p_a \geq 2$ and a compact set $E \subset M$ on which the labels active for a are exactly those in J , with $\sum_{i \in J} r_i = p_a$ and*

$r_i < p_a$ for every $i \in J$. There is an open neighborhood P of $\pi_a(E)$ in $\mathbb{S}^{p_a}$ such that the dual phases give a local C^1 diffeomorphism

$$(q_i^a)_{i \in J} : P \longrightarrow \prod_{i \in J} \mathbb{S}^{r_i}.$$

Moreover, $P \neq \mathbb{S}^{p_a}$, so the pulled-back top class of the source sphere vanishes on $\pi_a^{-1}(P)$.

Proof. At a point of E , every active primal and dual factor is nonzero. Fix all other primal coordinates and vary x_a nearby. By continuity the primal factor stays nonzero, so the dual factor stays on the nonzero boundary. Hence its normalized phase is defined on an open neighborhood of x_a . The equality $\sum_{i \in J} r_i = p_a$, together with (6.1), makes the joint phase derivative invertible. Invertibility is open. The union of these phase domains, restricted to their common invertibility locus, provides P .

If P were the whole sphere, compactness would make this local diffeomorphism a finite covering of the connected target. If any $r_i = 1$, the target has infinite fundamental group, which is incompatible with a finite covering by the simply connected $\mathbb{S}^{p_a}$. Otherwise the target is simply connected, so the covering is a diffeomorphism. But the target has at least two factors, since $r_i < p_a$, so its intermediate cohomology differs from that of a sphere. Hence P is proper, and the top-cohomology fact above completes the proof. $\square$

Theorem 6.2 (Exact smooth Lorentz resources). *Fix an integer $c \geq 1$ and allow Lorentz dimensions $3 \leq m_i \leq c + 2$. For full-row factorizations (5.2) with globally C^1 primal and dual maps, define*

$$h_a = \begin{cases} 1, & p_a \leq c, \\ \lceil (p_a + 1)/c \rceil, & p_a > c, \end{cases} \quad \tau_a = \begin{cases} p_a, & p_a \leq c, \\ p_a + 1, & p_a > c. \end{cases}$$

Let L be the number of non-ray factors, $T = \sum_i (m_i - 2)$ their total capacity, and $D = \sum_i m_i$ their total ambient dimension. Their exact minima, attained simultaneously, are

$$L_{\min} = \sum_a h_a, \quad T_{\min} = \sum_a \tau_a, \quad D_{\min} = \sum_a (\tau_a + 2h_a). \quad (6.3)$$

The minima are attained by Slater affine lifts with genuine whole-slice certificates. Additional ray factors do not lower them.

Proof. Write $C_a = \sum_{i \text{ active for } a} r_i$, $N_a = \#\{i \text{ active for } a\}$. The metric identity and (6.2) give

$$C_a \geq p_a, \quad N_a \geq \lceil p_a/c \rceil, \quad \sum_a C_a \leq T, \quad \sum_a N_a \leq L.$$

Rays have $H_i^a = 0$ and are omitted.

Let $A = \{a : p_a > c\}$. If $T < \sum_a (p_a + \mathbf{1}_{a \in A})$, at every point some $a \in A$ has $C_a = p_a$. Thus the sets $E_a = \{C_a = p_a\}$, $a \in A$, cover M . They are closed: each active locus is open, and $C_a \geq p_a$. Partition E_a by its exact set of active labels. This is a finite clopen partition into compact sets $E_{a,J}$: active labels remain active nearby, and an additional active label would increase C_a . Lemma 6.1 supplies a proper phase neighborhood $P_{a,J}$ for each compact piece. Choose pairwise disjoint open neighborhoods $W_{a,J}$ of the pieces, small enough to lie in $\pi_a^{-1}(P_{a,J})$. The source top class vanishes on $U_a = \coprod_J W_{a,J}$. These sets cover M , so their relative cup product vanishes, contradicting Künneth. Hence $T \geq \sum_a \tau_a$.

The pointwise count already gives $N_a \geq h_a$ except for $A_0 = \{a : p_a > c, c \mid p_a\}$. If $L < \sum_a h_a$, at every point some $a \in A_0$ has exactly p_a/c active labels. These closed sets cover M . Partition them by their active labels; on each piece every label must have capacity c , and their capacities sum

to p_a . The same phase and relative cup argument gives a contradiction. Thus $L \geq \sum_a h_a$. Since $D = T + 2L$, this also proves the dimension lower bound.

For attainment, if $p_a \leq c$ use the direct trace section $X_a = (1, x_a) \in Q_{p_a+2}$, with certificate $(1, -v)$ in its row and zero in other rows. Otherwise partition all $s_a = p_a + 1$ coordinates into h_a nonempty groups of size at most c , and impose

$$A_G(t_G, x_G) = \left(\frac{1+t_G}{2}, \frac{1-t_G}{2}, x_G \right) \in Q_{|G|+2}, \quad \sum_G t_G = 1. \quad (6.4)$$

This is equivalent to $t_G \geq \|x_G\|^2$, hence projects exactly onto the ball and is Slater at $x = 0$ with all $t_G > 0$. On the boundary choose $t_G = \|x_G\|^2$. The polynomial certificate

$$B_G(v) = \left(\frac{1 + \|v_G\|^2}{2}, -\frac{1 - \|v_G\|^2}{2}, -v_G \right)$$

has pairing $(t_G + \|v_G\|^2)/2 - \langle x_G, v_G \rangle$. Summing gives $1 - \langle x, v \rangle$ on the entire slice. Each group contributes one factor and capacity $|G|$. Separate factors for different sources attain all three claimed minima. $\square$

For comparison, among finite exact relative-Slater affine lifts without a global C^1 requirement, generic definable selectors and the pointwise argument give the exact minima

$$L_0 = \sum_a \lceil p_a/c \rceil, \quad T_0 = \sum_a p_a, \quad D_0 = T_0 + 2L_0. \quad (6.5)$$

Independent norm trees attain them by the construction in the proof of Corollary 3.4: partition p_a into $\lceil p_a/c \rceil$ positive integers at most c ; a rooted tree with one internal node with $k+1$ children for each part k has $1+p_a = s_a$ leaves. Pointwise no-sharing at a common generic regular contact proves the matching lower bounds. Such a contact lies in the primal regular locus and in the inverse images, under each source projection, of the dual regular loci; by definable selection this finite intersection is dense and open. The comparison makes no claim about arbitrary factor maps. Thus global regularity adds one capacity unit for each source with $p_a > c$, and one factor precisely for those sources with $p_a > c$ and $c \mid p_a$. These are formulation resources, not intrinsic barrier parameters of the projected body: the standard parameter $2L$ of a product of L Lorentz factors can decrease on the affine slice, as examined below.

6.3 Why local and joint identities are weaker

If only a positive weighted sum of the full labelled rows must factor, fewer factors can suffice.

Proposition 6.3 (Joint weighted rows have a different sharp count). *For two balls of common dimension $s \geq 3$ and positive weights $\lambda_1 + \lambda_2 = 1$, the kernel*

$$S_\lambda(x, z) = \sum_{a=1}^2 \lambda_a (1 - \langle x_a, z_a \rangle)$$

on $(\mathbb{S}^{s-1})^2$ has globally C^1 factorizations over copies of Q_3 , and the least number of factors in them is $2s - 1$. The full labelled rows (5.2) require $2s$ factors.

Proof. Mixed differentiation of S_λ at a contact gives a metric $g_\lambda = \sum_i H_i$ of rank $n = 2(s-1)$, so $L \geq n$. If $L = n$, every H_i has rank one everywhere and both factors of each label are nonzero. The phases $q_i : M \rightarrow \mathbb{S}^1$ lift to C^1 functions $\theta_i : M \rightarrow \mathbb{R}$ since M is simply connected. The identity $g_\lambda = \sum_i \alpha_i \beta_i d\theta_i^2$ makes $(\theta_1, \dots, \theta_n) : M \rightarrow \mathbb{R}^n$ a local diffeomorphism. Its image would be both nonempty open and compact, which is impossible. Thus $L \geq n+1 = 2s-1$.

For the upper bound put $Z(x) = (\sqrt{\lambda_1}x_1, \sqrt{\lambda_2}x_2) \in \mathbb{S}^{2s-1}$. Choose a unit vector q supported in the first coordinate block, and set

$$c(x) = 1 - \langle q, Z(x) \rangle > 0, \quad f(x) = \frac{Z(x) - \langle q, Z(x) \rangle q}{c(x)} \in q^\perp \cong \mathbb{R}^{2s-1}.$$

Direct expansion of squared norms gives the stereographic identity

$$1 - \langle Z(x), Z(z) \rangle = \frac{c(x)c(z)}{2} \|f(x) - f(z)\|^2.$$

For $U(t) = ((1+t^2)/2, (1-t^2)/2, t)$ and $V(t) = ((1+t^2)/2, -(1-t^2)/2, -t)$, both vectors are in Q_3 and $\langle U(t), V(u) \rangle = (t-u)^2/2$. Thus $c(x)U(f_j(x))$, $c(z)V(f_j(z))$, $1 \leq j \leq 2s-1$, give the required smooth factors. The full-row count follows from Theorem 6.2 with $c = 1$. $\square$

For $k \geq 2$ equal balls, the same lower-bound argument gives $k(s-1) + 1$ factors. Pairing the sources and using the preceding construction on each pair, with one coordinate-square construction on any remaining source, gives $ks - \lfloor k/2 \rfloor$ factors. We do not determine the exact count for $k \geq 3$. Both bounds concern only the contact kernel; neither construction supplies the full labelled rows of an affine lift. The phase argument alone cannot close the gap, because a product of spheres of total dimension n embeds in $\mathbb{R}^{n+1}$. To see this, embed a sphere as an orientable hypersurface, then observe that the boundary of a small tubular neighborhood of $N^m \subset \mathbb{R}^{m+1} \subset \mathbb{R}^{m+p+1}$ is $N \times \mathbb{S}^p$; the added normal bundle is trivial. Induction proves the assertion.

Remark 6.4 (Switching must pass through vertex sets with interior). Call a label *productive* for a source if it is active for that source at some point. A Lorentz label productive for two sources must have a primal vertex set $\{x : X_i(x) = 0\}$ with nonempty interior. Indeed, choose dual phase patches for both sources on which their derivatives are nonzero. Their images each contain more than one point, so smaller patches can be chosen with disjoint images. On the product of these patches, cylindrical complementarity would force a nonzero primal phase to equal both images. Thus the primal factor is zero on that open cylinder. Moreover, the nonzero locus of each productive dual factor cannot be dense. Otherwise, the identity saying that the primal derivative in every unrelated source direction is radial extends by continuity to all points; this excludes productivity for the second source. In particular, if the factor maps are analytic, or every primal vertex set has empty interior, no label is productive for two sources. The proof of Theorem 6.2 does not need either extra hypothesis.

Example 6.5 (A codimension-two singular set removes the extra cost of smoothness). In a full binary norm tree for $\mathbb{B}_2^s$, let $t_v(x) = \|x_{S_v}\|$ for each internal node v , where S_v is its set of descendant leaves; leaves carry their signed coordinates and the root value is one. At each internal node with children v_0, v_1 take

$$A_v(x) = (t_v(x), t_{v_0}(x), t_{v_1}(x)), \quad B_v(z) = (t_v(z), -t_{v_0}(z), -t_{v_1}(z)).$$

Their pairings telescope to $1 - \langle x, z \rangle$. The dual factors are genuine certificates of the tree slice, since the same telescoping works when the primal internal variables are free. Boundary fibers are

unique: the nonnegative squared cone slacks sum to $1 - \|x\|^2 = 0$, forcing every internal norm to its minimum. Both factor maps are globally Lipschitz and semialgebraic, and analytic away from the sets $x_{S_v} = 0$. Every nonroot internal node has at least two leaves, so these sets have codimension at least two in the boundary sphere when nonempty. Nevertheless, the tree uses only $s - 1$ copies of Q_3 , whereas for $s \geq 3$ Theorem 6.2 requires s copies for globally C^1 factors. Thus unique boundary fibers, Lipschitz regularity, and analyticity on a dense open set do not imply the conclusion of Theorem 6.2.

The tree also has a unique genuine dual certificate for every support $z \in \mathbb{S}^{s-1}$. Write an arbitrary dual block at node v as $(\alpha_v, \beta_{v,0}, \beta_{v,1}) \in Q_3$. Matching coefficients on the entire affine slice forces

$$\alpha_{\text{root}} = 1, \quad \beta_{v,j} = \begin{cases} -z_\ell, & \text{if child } v_j \text{ is leaf } \ell, \\ -\alpha_{v_j}, & \text{if child } v_j \text{ is internal.} \end{cases}$$

Induction from the leaves, using $\alpha_v \geq (\beta_{v,0}^2 + \beta_{v,1}^2)^{1/2}$, gives $\alpha_v \geq \|z_{S_v}\|$. At the root this is equality because $\|z\| = 1$. Equality then propagates down the tree: the head (first coordinate) of each child block is nonnegative and at least its subtree norm, so equality in the sum of their squares forces equality for each child separately. This also handles a zero subtree, whose head and descendant coordinates must all vanish. Thus the unique tuple is exactly $B_v(z)$ above. Every nonzero block has Jordan rank one; a zero block has rank zero.

Corollary 6.6 (Exact worst pure-row certificate rank over real PSD2). *For $C = \prod_{a=1}^h \mathbb{B}_2^{s_a}$, $s_a \geq 2$, fix any positive weights $\lambda_1, \dots, \lambda_h$. Let $\mathcal{D}_{\mathcal{L},a}(u_a)$ denote the full fiber of genuine certificates of $1 - \langle x_a, u_a \rangle$ on the affine slice of a lift $\mathcal{L}$. Among finite exact relative-Slater lifts over real 2×2 positive semidefinite cones and optional rays,*

$$\inf_{\mathcal{L}} \sup_{u \in M} \sup_{\substack{Y^a \in \mathcal{D}_{\mathcal{L},a}(u_a) \\ 1 \leq a \leq h}} \sum_i \text{rank} \left(\sum_a \lambda_a Y_i^a \right) = \sum_a (s_a - 1). \quad (6.6)$$

Proof. The lower bound is Theorem 5.2 with $B = 1$. For the upper bound take independent binary norm trees, with $s_a - 1$ factors for source a . The linear isomorphism

$$(t, z_1, z_2) \mapsto \begin{pmatrix} t + z_1 & z_2 \\ z_2 & t - z_1 \end{pmatrix}$$

identifies Q_3 with the real PSD2 cone; multiplying the dual image by $1/2$ preserves the pairing and does not change rank.

Any genuine pure-row certificate for source a vanishes on all factors belonging to other sources. Indeed evaluate its identity at a feasible tuple with source a at its support contact and every other source at a Slater point of its tree. The slack is zero, and each nonnegative block pairing must therefore vanish. A positive dual element orthogonal to an interior primal element is zero. What remains on source a is its unique tree certificate, by the coefficient argument above. Consequently every positive pure-row aggregate has rank at most $\sum_a (s_a - 1)$. Equality holds on the dense open support set where every internal subtree norm is positive. This proves the upper bound and hence the formula. $\square$

The inner supremum in (6.6) ranges only over genuine *pure-row* certificates; the corollary neither minimizes over nor describes all certificates of the aggregate objective.

Local saturation alone gives an exact linear-algebra statement. If positive semidefinite contact forms H_i sum to a positive metric H and $\sum_i \text{rank } H_i = \text{rank } H$, then $P_i = H^{-1/2} H_i H^{-1/2}$ are mutually orthogonal projections. Indeed write $P_i = Z_i Z_i^*$ with Z_i of full column rank; concatenating the Z_i gives a square matrix Z with $ZZ^* = I$, hence $Z^*Z = I$. This decomposes the tangent metric at the contact orthogonally but says nothing about unused parts of the cone, higher derivatives, or the global affine slice. The global arguments above use compactness and the full two-point identities in addition to this local rank equality.

Proposition 6.7 (Everywhere differentiability already forbids saturation). *Assume the Lorentz full-row factors are everywhere Fréchet differentiable, without continuity of their derivatives, and $p_a \geq 2$. If $r_i \leq c < \max_a p_a$ for an integer $c \geq 1$, then*

$$T \geq \sum_a p_a + 1.$$

For one source this gives the same exact resource minima as Theorem 6.2. For several sources it gives the additional count $L \geq \sum_a \lceil p_a/c \rceil + 1$ whenever every p_a is divisible by c . We do not claim that, under differentiability alone, the extra cost is additive over sources.

Proof. All pointwise phase and no-sharing calculations remain valid with pointwise derivatives. Suppose $T = \sum_a p_a =: n$. Sum the row slacks and aggregate their dual factors. At every contact, equality in the rank bound forces every factor to be nonzero and to use its full capacity. Its primal phase is therefore defined everywhere. Their product $q : M \rightarrow \prod_i \mathbb{S}^{r_i}$ is differentiable with invertible derivative, because its kernel is annihilated by the positive contact metric. The finite-dimensional local inversion theorem for everywhere differentiable maps with everywhere nonsingular derivative [78] makes q a local homeomorphism. Compactness makes it a finite covering. Since M is simply connected, circle factors are impossible; with all $r_i \geq 2$ the covering is a homeomorphism. Consequently $\prod_a (1 + t^{p_a}) = \prod_i (1 + t^{r_i})$. The smallest exponent j and its coefficient determine the multiplicity of $1 + t^j$; cancel that power and repeat. Thus the multisets of exponents agree, contradicting the cap. This proves the strict capacity bound. If every p_a is divisible by c , the baseline count L_0 satisfies $cL_0 = \sum_a p_a$, so $T \leq cL$ forces $L \geq L_0 + 1$. For one source, combine the capacity and count bounds with the grouped polynomial construction. If $c \nmid p$, its factor count is already $\lceil (p+1)/c \rceil$ by rounding; if $c \mid p$, use the extra factor. $\square$

A smooth example shows why the proofs use the cover argument instead of analytic continuation. Put $h(t) = e^{-1/t^2}$ for $t > 0$ and $h(t) = 0$ for $t \leq 0$, and use U from Proposition 6.3. For distinct sources a, b , the factors

$$A(x) = h((x_a)_1)U((x_b)_1), \quad B^a(v) = h(-v_1)U(0)$$

are smooth and nonnegative Lorentz-valued, with $\langle A(x), B^a(x_a) \rangle = 0$ on the entire cylinder. The dual is not identically zero, yet the primal phase varies in source b wherever $(x_a)_1 > 0$. Hence the identity that the primal derivative is radial in other source directions, which holds where the dual factor is nonzero, cannot in general be extended across an open set where the dual factor vanishes. The example refutes only this proof step; it is not a full-row factorization with too few factors.

7 Support-orbit rigidity and selected ranks

We now allow general symmetric cones with globally C^1 factor maps. If the local curvature bound holds with equality, the supports of the primal factors form a finite covering of a product of

idempotent manifolds (support orbits), and only one positive-capacity factor remains; for a ball it is a Lorentz cone. We then determine the least worst-case total rank of the chosen dual factors for a ball and for products of balls.

7.1 Saturation and support orbits

For a simple Euclidean Jordan algebra V of rank r and Peirce constant a , write $\mathcal{I}_p(V)$ for its manifold of rank- p idempotents. Its dimension is $ap(r-p)$ and its tangent space at an idempotent c is $V(c, 1/2)$. These are classical facts about idempotent manifolds [31, 73].

Theorem 7.1 (Global saturation). *Let $C \subset \mathbb{R}^{n+1}$ be a compact convex body, $n \geq 1$, containing the origin in its interior, and suppose $P = \partial C$ and $D = \partial C^\circ$ are strictly convex C^1 hypersurfaces. Suppose the full slack has a finite globally labelled factorization*

$$1 - \langle x, z \rangle = \sum_i \langle X_i(x), Y_i(z) \rangle, \quad (x, z) \in P \times D,$$

where $X_i : P \rightarrow K_i$ and $Y_i : D \rightarrow K_i$ are C^1 and K_i is a simple symmetric cone with rank r_i and Peirce constant a_i . Put $c_i = a_i \lfloor r_i^2/4 \rfloor$, with $c_i = 0$ for rays. If $\sum_i c_i = n$, then for each positive-capacity factor the Jordan ranks of $X_i(x)$ and $Y_i(z)$ at contact pairs are constant and balanced, equal to p_i and $r_i - p_i$ with $\{p_i, r_i - p_i\} = \{\lfloor r_i/2 \rfloor, \lceil r_i/2 \rceil\}$, and the joint support map

$$P \longrightarrow \prod_{i:c_i>0} \mathcal{I}_{p_i}(V_i), \quad x \longmapsto (\text{supp } X_i(x))_i$$

is a finite covering. There is exactly one positive-capacity factor. It is the Lorentz cone Q_{n+2} , except that for $n = 2$ the topological argument does not exclude $\mathbb{S}_+^3$.

Proof. Let z be the unique polar contact of x . The contact correspondence is a homeomorphism; no differentiability of that correspondence is needed. The tangent spaces are $T_x P = z^\perp$ and $T_z D = x^\perp$. Their Euclidean pairing is nondegenerate: its left kernel consists of vectors in $z^\perp \cap \text{span}\{x\}$, which is zero because $\langle x, z \rangle = 1$. Mixed differentiation in independent boundary charts gives

$$\langle u, v \rangle = \sum_i H_i(u, v), \quad H_i(u, v) = -\langle dX_i(x)[u], dY_i(z)[v] \rangle.$$

The cross-Peirce argument of Lemma 3.3, which for its rank bound uses only the two independent derivative images, gives

$$\text{rank } H_i \leq a_i p_i q_i \leq c_i, \quad p_i = \text{rank}_J X_i(x), \quad q_i = \text{rank}_J Y_i(z).$$

Nondegeneracy and $\sum_i c_i = n$ force equality throughout. Hence, for every positive-capacity factor, $p_i + q_i = r_i$, the ranks are balanced, and $\text{rank } H_i = c_i$. Ranks are lower semicontinuous along the contact graph. Since the two ranks sum to r_i , each is locally constant, hence constant by connectedness.

In the following support-differential argument, all indices satisfy $c_i > 0$; zero-capacity factors have $H_i = 0$ and play no role. Let $e_i(x) = \text{supp } X_i(x)$. This is C^1 : locally the zero spectral cluster is separated from the positive spectrum, and a smooth scalar cutoff equal to zero on the former and one on the latter gives e_i by spectral functional calculus. Differentiating $e_i \circ e_i = e_i$ and $X_i \circ e_i = X_i$ shows that

$$de_i[u] \in V_i(e_i, 1/2), \quad L_{X_i} de_i[u] = \frac{1}{2} \text{proj}_{V_i(e_i, 1/2)} dX_i[u].$$

Here L_{X_i} is invertible on the half-Peirce space: in a frame adapted to the positive support it acts on a cross channel by $\lambda_j/2 > 0$. Consequently $de_i[u] = 0$ precisely when the cross-Peirce component of $dX_i[u]$ vanishes. In particular

$$c_i = \text{rank } H_i \leq \text{rank } de_i \leq c_i.$$

If every de_i kills u , every $H_i(u, v)$ is zero, so $u = 0$. The joint support differential is therefore injective, and its domain and codomain have the same dimension n . The joint map is a local C^1 diffeomorphism. Its image is open and, by compactness, closed in the connected target. It is surjective and proper, hence a finite covering.

It remains to identify the target. For $n \geq 2$, the source $P \cong S^n$ is simply connected, so this is the universal cover of the target. There can be no circle factor, since it would make the target fundamental group infinite. The remaining factors have compact simply connected universal covers. The universal cover of their product is the product of those covers. If there were at least two positive-dimensional factors, the mod-two top cohomology class of one factor, tensored with the unit classes of the others, would give nonzero cohomology in a degree strictly between zero and n . The Künneth formula contradicts the cohomology of S^n . Thus there is one factor. For $n = 1$, the same conclusion follows directly from the sum of the positive integer dimensions.

The simple-algebra classification now leaves the following cases. A spin factor has primitive-idempotent orbit S^{m-2} and therefore gives Q_{n+2} . For $H_r(\mathbb{R})$ the orbit is the real Grassmannian $\text{Gr}_p(\mathbb{R}^{p+q})$, $p + q = r$. At $r = 3$ its balanced orbit is $\mathbb{RP}^2$, covered by S^2 . At $r \geq 4$, balanced ranks satisfy $p, q \geq 2$. The oriented Grassmannian is simply connected by the exact sequence of

$$SO(p) \times SO(q) \longrightarrow SO(p+q) \longrightarrow \text{Gr}_p^+(\mathbb{R}^{p+q}) :$$

the induced map on fundamental groups onto $\pi_1(SO(p+q)) = \mathbb{Z}_2$ is surjective. Its kernel is nonzero: use the diagonal element when $p, q \geq 3$, and an even nonzero winding in an $SO(2)$ factor otherwise. Exactness therefore gives nonzero π_2 of the oriented Grassmannian. Its dimension $pq \geq 4$ precludes a sphere.

The complex Grassmannian is simply connected by the fibration with isotropy $U(p) \times U(q)$ in $U(p+q)$. The map on π_1 is addition $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, so its kernel gives nonzero π_2 of the Grassmannian. For $r \geq 3$ its dimension $2pq \geq 4$ precludes a sphere. For quaternionic Grassmannians, the groups $Sp(k)$ are simply connected and $\pi_3(Sp(k)) = \mathbb{Z}$, with standard inclusions preserving a generator. These facts follow inductively from $Sp(k-1) \rightarrow Sp(k) \rightarrow S^{4k-1}$ and $Sp(1) = S^3$. The map on π_3 of the isotropy inclusion is again addition $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, giving nonzero π_4 of the Grassmannian. Its dimension $4pq \geq 8$ precludes a sphere. The classical-group and Grassmannian fibrations used here are described in [48, Examples 4.53–4.55]. Finally the balanced idempotent orbit of $H_3(\mathbb{O})$ is the Cayley plane: rank-one idempotents are its points and complementation identifies the rank-two orbit with it [9, Section 3.4]. Its cells have dimensions 0, 8, 16, so $H^8(-; \mathbb{Z}) = \mathbb{Z}$; see [48, Example 4.47]. It is not a sphere. Order-two real, complex, and quaternionic Hermitian algebras are already spin factors. This exhausts the classification. $\square$

7.2 Eliminating the real order-three exception

For a ball, the full slack identity rules out the case $\mathbb{S}_+^3$, $n = 2$, left open by Theorem 7.1. The argument is elementary and needs no classification of linear determinantal representations. The next example shows that a purely local argument does not suffice.

Example 7.2 (Local constant rank need not extend globally). The affine pencil

$$L(x) = \begin{pmatrix} 1+x_3 & x_1 & x_2 \\ x_1 & 1-x_3 & 0 \\ x_2 & 0 & 1-x_3 \end{pmatrix}, \quad \det L(x) = (1-x_3)(1-\|x\|^2),$$

is positive semidefinite of rank two on $S^2 \setminus \{(0,0,1)\}$. Indeed, its lower 2×2 block is positive definite and its Schur complement is zero there. Its kernel line is

$$\text{span} \left(1, -\frac{x_1}{1-x_3}, -\frac{x_2}{1-x_3} \right),$$

so its kernel map is a local diffeomorphism onto an affine chart of $\mathbb{RP}^2$ by stereographic coordinates. At the north pole, $L(0,0,1) = \text{diag}(2,0,0)$ has rank one. Thus an affine pencil can have constant rank on an open subset of the sphere without having constant rank on the whole sphere or a common kernel. Proposition 7.5 below excludes the global slack factorization by using the full C^1 factor maps and finite affine gluing.

Lemma 7.3 (Injective kernels of an affine sphere pencil). *Let $s \geq 2$ and $X(x) = C_0 + \sum_{j=1}^s x_j C_j$ be a real symmetric matrix pencil, positive semidefinite of nullity one for every $x \in S^{s-1}$. If its kernels do not share a nonzero vector, its kernel-line map is injective.*

Proof. Suppose distinct x, y have the same kernel line, containing $k \neq 0$. The affine scalar function $f(v) = k^T X(v)k = a + b \cdot v$ is nonnegative on the unit sphere and vanishes at x and y . Nonnegativity says $a \geq \|b\|$. If $b \neq 0$, a zero is possible only when $a = \|b\|$ and $v = -b/\|b\|$, giving only one zero. Thus $a = b = 0$. For every v , positivity and $k^T X(v)k = 0$ imply $X(v)k = 0$, contradicting the absence of a common nonzero kernel vector. $\square$

Lemma 7.4 (Finite affine selection on a sphere). *Let $n \geq 2$ and let $F : S^n \rightarrow W$ be C^1 , with W finite dimensional. Suppose a dense open subset is the union of finitely many nonempty open sets U_i , and $F|_{U_i} = L_i|_{U_i}$ for ambient affine maps $L_i : \mathbb{R}^{n+1} \rightarrow W$. Then F is the restriction of one affine map.*

Proof. Merge sets with identical L_i and put $E_i = \overline{U_i}$. Continuity of values and tangent derivatives gives $j^1 F = j^1(L_i|_{S^n})$ on E_i . If distinct affine maps have the same tangent first jet at $z \in S^n$, their difference is $M(z \cdot x - 1)$ for some nonzero $M \in W$. Such a difference has matching first jet at no other sphere point: matching derivatives force the other normal line to be $\mathbb{R}z$, and matching values exclude $-z$. Thus each intersection $E_i \cap E_j$ lies in at most one point. Remove the resulting finite set Z . The connected space $S^n \setminus Z$ is partitioned into the disjoint relatively closed sets $E_i \setminus Z$; each is also relatively open because the partition is finite. Only one is nonempty. But the closure of a nonempty open U_i cannot be finite, so there was only one distinct affine map. Density finishes the proof. $\square$

Proposition 7.5 (No real order-three saturation for a ball). *There is no factorization*

$$1 - x \cdot y = \text{tr}(X(x)Y(y)) + \sum_{j=1}^q a_j(x)b_j(y), \quad x, y \in S^2,$$

with $X, Y : S^2 \rightarrow \mathbb{S}_+^3$ of class C^1 , continuous nonnegative a_j, b_j , finite q , constant complementary ranks 2, 1, and a rank-one support map that is a local diffeomorphism onto $\mathbb{RP}^2$.

Proof. Exchange X, Y if necessary so that Y has rank one. Every summand vanishes at $y = x$, so $a_j(x)b_j(x) = 0$. For each of the finitely many continuous nonnegative functions $f = a_j, b_j$, the set $\{f > 0\} \cup \text{int}\{f = 0\}$ is dense open. Intersect these sets and partition the result by their finitely many positivity/zero patterns. On each nonempty open pattern set U , for each j at least one of $a_j|_U, b_j|_U$ is identically zero. Therefore all ray terms vanish for every pair $(x, y) \in U \times U$.

The rank-one support map of Y sends U onto a projective open set. Consequently the matrices $Y(y)$, $y \in U$, span $\mathbb{S}^3$: an annihilating symmetric matrix would have a quadratic form zero on an open set of projective directions, and hence zero identically. Choose six $y_\ell \in U$ whose matrices are a basis. The equations

$$\text{tr}(X(x)Y(y_\ell)) = 1 - x \cdot y_\ell, \quad x \in U,$$

force $X|_U$ to be an ambient affine pencil. There are only finitely many pattern sets, so Lemma 7.4 makes X one affine pencil on the whole sphere.

Complementarity identifies the kernel line of $X(x)$ with the support line of $Y(x)$. This map is a finite covering $S^2 \rightarrow \mathbb{RP}^2$, hence the universal double cover. Its image contains all lines, so its kernels have no common nonzero vector. But Lemma 7.3 makes the map injective, contradicting its two-point fibers. $\square$

Corollary 7.6 (Ball saturation classification). *A finite globally bi- C^1 symmetric-cone factorization of the full slack of $\mathbb{B}_2^s$, $s \geq 2$, satisfying $\sum_i a_i \lfloor r_i^2/4 \rfloor = s - 1$ has exactly one positive-capacity factor, namely Q_{s+1} . Remaining factors, if any, are rays.*

Proof. Apply Theorem 7.1 and exclude its sole additional case by Proposition 7.5. $\square$

7.3 The least selected certificate rank for a fixed dictionary

Let $\mathcal{K}$ be a finite repeatable dictionary as in Section 4, with the capacity B defined there, and define

$$\kappa_{\mathcal{K}}(s) = \begin{cases} 1, & s = B + 1 \text{ and the dictionary contains } Q_{B+2}, \\ \lceil s/B \rceil, & \text{otherwise.} \end{cases} \quad (7.1)$$

Here “contains” means up to linear cone isomorphism. The formula also gives one when $s \leq B$. A dictionary of Hermitian matrices over a fixed field $\mathbb{F}$ with order cap R has $B = \delta(R - 1)$, and the exceptional branch occurs exactly when $R = 2$ and $s = \delta + 1$.

Theorem 7.7 (Least worst-case rank of globally selected certificates). *For globally bi- C^1 full-slack factorizations of $\mathbb{B}_2^s$ through a fixed finite repeatable symmetric-cone dictionary, the least possible value of $\sup_v \sum_i \text{rank}_J Y_i(v)$ is $\kappa_{\mathcal{K}}(s)$. The upper bound is attained by a relative-Slater affine lift with globally smooth boundary selections and genuine whole-slice certificates.*

Proof. At every contact, with primal and dual ranks p_i, q_i ,

$$s - 1 \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - q_i) q_i \leq B \sum_i q_i.$$

Rays contribute zero to the first two sums and cannot help attain equality. Unless $s - 1 = qB$ for an integer $q \geq 1$, the lower bound already rounds to $\lceil s/B \rceil$. In the remaining case suppose $Q(v) = \sum_i q_i(v) \leq q$ everywhere. Equality forces $Q \equiv q$; lower semicontinuity makes each q_i constant. Each nonzero dual factor has $q_i = 1$, $a_i(r_i - 1) = B$, and $p_i = r_i - 1$. Its primitive support orbit has dimension B . The support-differential argument above gives a finite covering from $\mathbb{S}^{qB}$ to the product of these q primitive-idempotent orbits.

If $q > 1$, a circle factor would give the target infinite fundamental group. Otherwise the compact universal covers of the positive-dimensional target factors have nonzero intermediate product cohomology, ruling out a sphere source. If $q = 1$, the simple-algebra classification identifies the primitive orbit as a sphere for spin factors, real projective space for real Hermitian factors, complex or quaternionic projective space in the other classical cases, and the Cayley plane in the exceptional case. Complex and quaternionic projective spaces have intermediate cohomology unless their matrix order is two; the Cayley plane has its nonzero middle cohomology. These order-two cases are spin factors. For real order $R \geq 3$ the remaining map is the double covering $\mathbb{S}^{R-1} \rightarrow \mathbb{RP}^{R-1}$. All other dual factors vanish identically. Pairing the primal factor map with a basis of rank-one dual matrices $Y(v)$ makes it an ambient affine pencil, since rank-one matrices with directions in a projective open set span the symmetric matrices. Its kernel-line map has no common nonzero kernel and hence is injective by Lemma 7.3, contradicting the double cover. Thus $Q \equiv q$ is possible only in the exceptional case of (7.1): $q = 1$ and the spin factor Q_{B+2} . In every other case $Q(v) \geq q + 1$ at some v .

For attainment outside the exceptional branch, choose a dictionary member attaining B and partition all s coordinates into $g = \lceil s/B \rceil$ nonempty groups embedded in its primitive half-Peirce space. With $d = e - c$ as in Lemma 4.2, impose

$$X_G = t_G c + \sqrt{2} w_G + d \succeq 0, \quad \sum_G t_G = 1. \quad (7.2)$$

These constraints say $t_G \geq \|w_G\|^2$, so the projection is the ball and the slice is strictly feasible. On the boundary the constraints force $t_G = \|w_G\|^2$. At support $v = (g_G)_G$ the rank-one completion

$$Y_G = \frac{1}{2}c - \frac{g_G}{\sqrt{2}} + P(g_G)c = \frac{1}{2}P(c - \sqrt{2}g_G)c$$

has pairing $t_G/2 - \langle w_G, g_G \rangle + \|g_G\|^2/2$. It is nonzero of rank one even when $g_G = 0$. Summing over G gives the slack identity on the whole slice, so the Y_G are genuine certificates of total rank g at every support. For the exceptional branch use the direct Lorentz trace section, whose boundary primal and dual factors have rank one. This proves attainment. $\square$

Theorem 7.8 (Sequential smooth ranks). *Suppose the full rows of $\prod_a \mathbb{B}_2^{s_a}$ have globally bi- C^1 factors through $\mathcal{K}$. There is a simultaneous contact u such that, for every choice of positive weights,*

$$\sum_i \text{rank}_J \left(\sum_a \lambda_a Y_i^a(u_a) \right) \geq \sum_a \kappa_{\mathcal{K}}(s_a). \quad (7.3)$$

The same bound holds for total primal nullity there. These bounds are sharp over globally selected factorizations, and their upper constructions are actual affine lifts with genuine certificates.

Proof. Use the successive joins c_i from Theorem 5.2. After fixing earlier primal coordinates at contact, cylindrical complementarity puts the primal factors in $V_i(e_i - c_i, 1)_+$. Fix future primal coordinates arbitrarily and apply the next one-ball row to the compressed dual $P(e_i - c_i)Y_i^a$. The compressed maps remain globally C^1 and satisfy the exact full two-point row identity. Lemma 5.1 identifies their total rank with the next support increment.

To apply Theorem 7.7, note that proper faces have strictly smaller primitive capacity than their parent simple non-ray cone: classical faces have smaller order, spin faces are rays, and proper non-ray faces of the Albert cone have capacity eight rather than sixteen. Thus each remaining face has capacity at most B . If equality is needed in the rank chain of that theorem, the face must be

an original dictionary cone attaining B . In particular, a newly created proper face cannot introduce a matching spin exception at capacity B . The same proof therefore gives at least $\kappa_{\mathcal{H}}(s_a)$ new rank at some next contact. Iteration and Lemma 5.1 prove the bound; complementarity gives nullity. Separate copies of (7.2) and the exceptional direct sections give equality at every contact. $\square$

In particular, under an integer real-dimension cap $d \geq 3$ on arbitrary symmetric factors, the least possible worst-case aggregate rank of the chosen dual factors is

$$\sum_a h_d(s_a), \quad h_d(s) = \begin{cases} 1, & d \geq s + 1, \\ \lceil s/(d-2) \rceil, & d < s + 1. \end{cases}$$

Indeed the primitive capacity is at most $d - 2$, with equality only for a spin factor of dimension d . The equality analysis in Theorem 7.7 gives the lower bound, and separate grouped Lorentz sections attain it. This result concerns only the chosen globally C^1 selections. It does not extend to the full certificate fibers of arbitrary affine lifts: for $d = 3$ and $s \geq 3$, the nonsmooth norm tree of Example 6.5 has, for each support, a certificate of total Jordan rank at most $s - 1 < h_3(s) = s$.

The bound can be refined using the faces that occur. At stage a of sequential compression, suppose the non-ray faces have ranks R_{ia} and Peirce constants α_{ia} . Define

$$g_a = \min \left\{ \sum_i q_i : q_i \in \{0, \dots, R_{ia}\}, \quad \sum_i \alpha_{ia} q_i (R_{ia} - q_i) \geq s_a - 1 \right\}.$$

The mixed-channel inequality gives at least g_a new non-ray rank at every next contact. Hence one can replace the summand in (7.3) by $\max\{g_a, \kappa_{\mathcal{H}}(s_a)\}$. This bound depends on the faces of the given factorization; we do not claim it is exact. For at most L faces of a common Peirce constant α and order at most R , a fixed total dual rank $Q = uL + t$, $0 \leq t < L$, has capacity at most

$$\alpha[(L-t)u(R-u) + t(u+1)(R-u-1)].$$

To prove it, pad with zero ranks and increase every order to R . If two ranks differ by at least two, transferring one unit from the larger to the smaller increases their sum of $q(R-q)$. Iteration balances all ranks to u or $u+1$, proving the formula. Inverting this finite envelope gives a convenient explicit lower bound on g_a .

Corollary 7.9 (Dimension-capped smooth symmetric-cone resources). *Consider globally bi- C^1 full-slack factorizations of $\mathbb{B}_2^s$ over simple symmetric cones of real dimension at most an integer $d \geq 3$, with optional rays. Let L be the number of non-ray factors, D the total ambient dimension, and ρ the total ambient Jordan rank. If $d < s + 1$, the exact minima of these quantities, attained simultaneously, are*

$$L_{\min} = \left\lceil \frac{s}{d-2} \right\rceil, \quad D_{\min} = s + 2L_{\min}, \quad \rho_{\min} = 2L_{\min}.$$

If $d \geq s + 1$, the exact triple is $(1, s + 1, 2)$. The standard ambient product barrier has parameter ρ .

Proof. For every simple non-ray cone, $a[r^2/4] \leq m - 2$, with equality only for rank-two spin factors; substitution of $m = r + ar(r-1)/2$ verifies strictness for $r \geq 3$. Write $T = \sum_{i \text{ non-ray}} (m_i - 2)$. Local curvature gives $T \geq s - 1$. If equality holds, all factors are spin and Corollary 7.6 forces the single factor Q_{s+1} , which exceeds the cap when $d < s + 1$. Thus $T \geq s$ in that case. Since $T \leq (d-2)L$, $D \geq T + 2L$, and $\rho \geq 2L$, the displayed lower bounds follow. The grouped Lorentz construction attains them simultaneously. In the other case local curvature gives $T \geq s - 1$, $L \geq 1$, and hence $D \geq s + 1$, $\rho \geq 2$; the direct Lorentz section attains equality. $\square$

8 The standard barrier after affine restriction

The rank of a certificate concerns a single support direction. A barrier must control every direction in its domain, including directions inside a projection fiber. We now make this distinction quantitative.

8.1 The parameter and two boundary tests

Write $L = X^\circ + T$ for the effective affine slice of a product of Euclidean Jordan algebras, after minimal-face reduction, and set $\Omega = L \cap \text{int } K$. “Effective” means that redundant variables that do not change the cone factors have been removed. For the standard barrier

$$F(X) = - \sum_i \log \det X_i,$$

its least gradient parameter on this slice is

$$\nu_{\text{std}, \text{slice}} = \sup_{X \in \Omega} \sup_{0 \neq H \in T} \frac{DF(X)[H]^2}{D^2F(X)[H, H]}. \quad (8.1)$$

The restricted Hessian is positive definite. The standard third-derivative self-concordance inequality is preserved under affine restriction, so (8.1) is exactly the least self-concordant-barrier parameter of this function. It is not the infimum over all barriers on Ω . The ambient Jordan rank is an upper bound for $\nu_{\text{std}, \text{slice}}$ but need not equal it.

Lemma 8.1 (Boundary order). *For every $\bar{X} \in L \cap K$,*

$$\nu_{\text{std}, \text{slice}} \geq \sum_i (r_i - \text{rank}_J \bar{X}_i). \quad (8.2)$$

In particular, the bound holds at every point of every fiber over the boundary, not only at a chosen boundary selection.

Proof. Let $X(t) = (1-t)\bar{X} + tX^\circ$, $0 < t < 1$, and put $z_i = r_i - \text{rank}_J \bar{X}_i$. Apply the positive invertible quadratic representation $P((X_i^\circ)^{-1/2})$. It sends X_i° to the unit, preserves rank, and changes the determinant by a positive constant. Diagonalizing the transformed $\bar{X}_i$ therefore gives

$$\det_i X_i(t) = c_i t^{z_i} \prod_{j=1}^{r_i - z_i} ((1-t)\lambda_{ij} + t), \quad c_i > 0, \quad \lambda_{ij} > 0.$$

If $z = \sum_i z_i$, the restriction $f(t) = F(X(t))$ satisfies $f' = -z/t + O(1)$ and $f'' = z/t^2 + O(1)$. For $z > 0$, the quotient $(f')^2/f''$ tends to z ; for $z = 0$ the claim is immediate. These statements use Jordan eigenvalues and apply also to the Albert algebra. $\square$

Lemma 8.2 (Recession and complementary boundary order). *Suppose $D \neq 0$ is a recession direction of $L \cap K$. In each factor let $c_i = \text{supp } D_i$, $f_i = e_i - c_i$, and let $C_i = P(f_i)\bar{X}_i$, where $\bar{X} \in L \cap K$. Define*

$$r = \sum_i \text{rank}_J D_i, \quad z = \sum_i (\text{rank } f_i - \text{rank}_{V_i(f_i, 1)} C_i).$$

Then $\nu_{\text{std}, \text{slice}} \geq r + z$.

Proof. The two-dimensional affine family $X(\sigma, t) = (1 - \sigma)\bar{X} + \sigma X^\circ + tD$ is interior for $0 < \sigma < 1$ and $t > 0$. Put $A(\sigma) = (1 - \sigma)\bar{X} + \sigma X^\circ$. The Jordan Schur determinant formula [45], applied to the Peirce decomposition for c_i , gives

$$\det_i(A_i(\sigma) + tD_i) = t^{r'_i} \det_{c_i} D_i \det_{f_i}(P(f_i)A_i(\sigma)) [1 + O_\sigma(t^{-1})], \quad r'_i = \text{rank}_J D_i. \quad (8.3)$$

For a zero D_i the formula means the unchanged determinant and $r'_i = 0$. The Schur correction is quadratic in the mixed Peirce part and linear in the inverse of the c_i part, hence is $O_\sigma(t^{-1})$. Alternatively, the determinant is a polynomial in (σ, t) whose leading coefficient is the positive coefficient displayed in (8.3). On every compact subinterval of $0 < \sigma < 1$, the expansion may consequently be differentiated twice, including after multiplication of each t derivative by t .

Write $f(\sigma, t) = F(X(\sigma, t))$ and $g(\sigma) = -\sum_i \log \det_{f_i}(P(f_i)A_i(\sigma))$, omitting rank-zero factors. As $t \rightarrow \infty$ with σ fixed,

$$t f_t \rightarrow -r, \quad t^2 f_{tt} \rightarrow r, \quad f_\sigma \rightarrow g', \quad f_{\sigma\sigma} \rightarrow g'', \quad t f_{t\sigma} \rightarrow 0.$$

The compression $P(f_i)X_i^\circ$ of the Slater point is positive definite in $V_i(f_i, 1)$ whenever this algebra has positive rank. The determinant computation of Lemma 8.1, applied to g , gives $\sigma g' \rightarrow -z$ and $\sigma^2 g'' \rightarrow z$ as $\sigma \downarrow 0$. At each point test the affine direction $H = tD + \sigma(X^\circ - \bar{X})$. Its gradient tends, in the iterated limit ($t \rightarrow \infty$ first, then $\sigma \downarrow 0$), to $-(r + z)$ and its Hessian quadratic form to $r + z$. Equation (8.1) therefore gives $\nu_{\text{std}, \text{slice}} \geq r + z$. The derivative expansions are needed here: a bounded remainder in the barrier value alone would not justify the conclusion. $\square$

Proposition 8.3 (General lower bound and minimum under global selection). *For a finite repeatable symmetric-cone dictionary with $B > 0$, every relative-Slater affine lift of $\mathbb{B}_2^s$ satisfies*

$$\left\lceil \frac{s-1}{B} \right\rceil \leq \nu_{\text{std}, \text{slice}}. \quad (8.4)$$

There is a bounded lift with parameter $\kappa_{\mathcal{K}}(s)$ from (7.1). Among lifts of $\prod_a \mathbb{B}_2^{s_a}$ that admit globally bi- C^1 full-row selections (5.2) with genuine certificates, the minimum of $\nu_{\text{std}, \text{slice}}$ is exactly $\sum_a \kappa_{\mathcal{K}}(s_a)$. For a single ball, every unbounded lift satisfies the stronger bound $\nu_{\text{std}, \text{slice}} \geq 1 + \lceil (s-1)/B \rceil$.

Proof. The dense-open rank bound for full certificate fibers in Theorem 4.1 and complementarity force a boundary tuple with nullity at least $\lceil (s-1)/B \rceil$; apply Lemma 8.1. For the upper bound use (7.2). Its determinant is $t_G - \|w_G\|^2$, so its barrier is

$$F = -\sum_{G=1}^g \log(t_G - \|w_G\|^2), \quad \sum_G t_G = 1. \quad (8.5)$$

Each summand has parameter one on its paraboloid domain. Indeed, if $h = t - \|w\|^2$, then $D^2(-\log h)[H, H] = (Dh[H]/h)^2 + 2\|H_w\|^2/h$. The third-derivative inequality follows by restriction of the rank-two Lorentz log determinant with one perspective coordinate fixed. By Cauchy-Schwarz, the sum has parameter at most g . At any sphere point set $t_G = \|w_G\|^2$. Each fixed-tail block then has nullity one, so Lemma 8.1 gives $\nu_{\text{std}, \text{slice}} \geq g$, and hence equality. For the direct Lorentz section $t = 1$ of Q_{s+1} , $F = -\log(1 - \|x\|^2)$ has parameter one: its squared dual gradient norm is $2\|x\|^2/(1 - \|x\|^2)$, with supremum one. Products give the stated upper bound, and Theorem 7.8 gives the matching nullity lower bound.

For the last assertion, a nonempty closed unbounded convex slice has a nonzero recession direction $D \in K \cap T$; compactness of its projection implies $\pi D = 0$. Choose a generic support

certificate Y of rank at least $q = \lceil (s-1)/B \rceil$ and any tuple $\bar{X}$ over that support. The whole-slice identity gives $\langle D, Y \rangle = 0$, so Y_i lies in the face complementary to $\text{supp } D_i$. Its complementarity with $\bar{X}_i$ implies that the compressed nullity z in Lemma 8.2 is at least q . Since $\sum_i \text{rank}_J D_i \geq 1$, that lemma proves the assertion. $\square$

Consequently, when B does not divide $s-1$, the minimum over all lifts of a single ball is exact without selection hypotheses: both bounds equal $\lceil s/B \rceil$. The same holds for Hermitian order caps with $B = a(R-1)$, where $a = \delta = \dim_{\mathbb{R}} \mathbb{F}$, and for Lorentz dimension caps with $B = d-2$. Proposition 8.3 leaves open bounded lifts with $s = qB + 1$. Theorems 8.4 and 8.6 settle $q = 1$ and, for Hermitian and Lorentz caps, $2 \leq q \leq B+1$; for these caps the remaining range is (8.6), and for other dictionaries bounded lifts with $q \geq 2$ remain open. If every symmetric cone of real dimension at most d is permitted, the maximum primitive capacity is $B = d-2$, attained by Q_d . Hence, over lifts with smooth global selections, the minimum standard-barrier parameter is $\lceil s/(d-2) \rceil$ for $3 \leq d < s+1$, and one for $d \geq s+1$. For several source balls these minima add. This statement allows all symmetric cone types; higher-rank types cannot improve the optimum.

8.2 An exact result for $s = B+1$ without selection hypotheses

Theorem 8.4 (One-channel rigidity). *Let $s = B+1$. The infimum of $\nu_{\text{std}, \text{slice}}$ over finite affine lifts through $\mathcal{K}$ equals one if the dictionary contains Q_{B+2} , and equals two otherwise. In the latter case the corresponding infimum for $(\mathbb{B}_2^{B+1})^b$ equals $2b$. Moreover, at some simultaneous contact there are pure-row certificates, one per ball, whose positive aggregates all have total Jordan rank at least $2b$.*

Proof. Suppose first that one lift has $\nu_{\text{std}, \text{slice}} < 2$. The integral boundary orders in Lemma 8.1 imply that every boundary tuple has nullity at most one. Thus every member of every normalized genuine certificate fiber $\mathcal{D}(v)$ has rank at most one. Such a member is nonzero, since its slack is not identically zero. The positive midpoint of two elements on different product-cone rays has rank at least two. Convexity therefore puts $\mathcal{D}(v)$ on one ray. The whole-slice identity fixes its scalar, making $\mathcal{D}(v) = \{Y(v)\}$.

A Slater point provides the uniform normalization

$$0 < 1 - \|\pi X^\circ\| \leq \langle X^\circ, Y(v) \rangle \leq 1 + \|\pi X^\circ\|.$$

Strict positivity of X° bounds $Y(v)$ above and away from zero. Since Y is bounded with closed graph, it is continuous. As the sphere is connected, the factor in which $Y(v)$ is nonzero does not depend on v . Projectivization gives a continuous injection $\mathbb{S}^B \rightarrow \mathcal{P}(V_j)$ into that factor's primitive-ray manifold: if the rays agree, evaluating the genuine identities at a lift of the ball's center fixes their scalar, and the identities then fix v .

The target dimension is $a_j(r_j - 1) \leq B$. Invariance of dimension excludes a smaller dimension, and invariance of domain makes the map a homeomorphism in equal dimension. The primitive-ray spaces are $\mathbb{RP}^{r-1}$, $\mathbb{CP}^{r-1}$, $\mathbb{HP}^{r-1}$, spheres for spin factors, and $\mathbb{OP}^2$ for the Albert algebra. The classification and cohomology used in Section 7 show that the only sphere cases are the spin factors, including the order-two Hermitian cones, which are Lorentz cones. Thus a value below two requires Q_{B+2} in the dictionary. Its direct section attains one; otherwise the two-group construction (7.2) attains two.

For the product assertion apply sequential compression as in Section 5. At the next source, compress the entire original certificate fiber to the faces complementary to the previously chosen support join. The resulting families are compact, convex, and have closed graphs: the original fibers are uniformly bounded by the same Slater normalization, and compression is linear. Their genuine

row identities on the remaining contact cylinder ensure that zero is not in any compressed fiber. If every compressed certificate had total rank at most one, the preceding singleton and primitive-ray argument would again give a spin factor of capacity B in one of the remaining faces. A proper face of a simple factor has strictly smaller primitive capacity than that factor; hence no such face can arise when the original dictionary does not contain Q_{B+2} . Some compressed certificate must therefore have rank at least two. The support-join compression identity of Section 5 adds at least two new rank units at each source. The final join, and every positive aggregate of the chosen certificates, has rank at least $2b$. All tuples at that simultaneous contact are complementary to this join, so the boundary-order test gives $\nu_{\text{std,slice}} \geq 2b$. Separate two-group constructions attain equality. $\square$

For example, an Albert-cone dictionary has $B = 16$, and the minimum of $\nu_{\text{std,slice}}$ for the 17-dimensional ball is two. The theorem concerns the optimum over lifts; it does not say that every lift using a matching spin factor has parameter one.

8.3 Arbitrarily many factors under a wide cap

For Hermitian order caps and Lorentz dimension caps, we settle the case $s - 1 = qB$ with $q \geq 2$ and $B \geq q - 1$, again without selection hypotheses or a bound on the number of factors.

Lemma 8.5 (A face-codimension budget). *Suppose a bounded relative-Slater affine slice $P = L \cap K$ projects onto $\mathbb{B}_2^s$. For any $X \in P$ over the boundary of the ball, let E_X be the linear span of the minimal face of K containing X . Then $\text{codim } E_X \geq s$.*

Proof. Write $L = X^\circ + T$. Since P is bounded and nonempty, $T \cap K = \{0\}$; also $0 \notin L$, since otherwise P would be a bounded cone with nonempty relative interior projecting onto a ball. On $\hat{L} = \text{span}\{X^\circ\} + T$ define height $\ell(tX^\circ + H) = t$. A cone point of zero height is zero. There are no cone points of negative height: otherwise adding a suitable positive multiple of X° gives a nonzero cone point of height zero; pointedness excludes cancellation to zero. Hence $\hat{L} \cap K$ is precisely the cone over P . The projection homogenizes to a linear surjection $\hat{\pi} : \hat{L} \rightarrow \mathbb{R}^{s+1}$ whose cone image is Q_{s+1} . If $k = \dim \ker \hat{\pi}$, then $\dim \hat{L} = s + 1 + k$. Every $H \in \hat{L} \cap E_X$ permits two-sided small feasible perturbations of X , because $X \in \text{ri}(K \cap E_X)$. Their images lie in the line spanned by the target extreme ray. Thus $\dim(\hat{L} \cap E_X) \leq k + 1$. Grassmann's dimension inequality gives $k + 1 \geq s + 1 + k - \text{codim } E_X$, as required. $\square$

Theorem 8.6 (Wide-cap rigidity). *Fix one of two dictionaries: Hermitian cones over one field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ of order at most R , with $B = a(R - 1)$; or Lorentz cones of dimension at most d , with $B = d - 2$. Consider lifts through any finite number of rays and of factors from this dictionary. If $q \geq 2$ and $B \geq q - 1$, the minimum of $\nu_{\text{std,slice}}$ for affine lifts of $\mathbb{B}_2^{qB+1}$ is $q + 1$.*

Proof. Unbounded lifts were settled by Proposition 8.3. For a bounded lift suppose $\nu_{\text{std,slice}} < q + 1$. All boundary tuples then have integer total nullity at most q . In a Hermitian block of order $r \leq R$ and nullity m , the minimal-face codimension is

$$\left\lceil r + \frac{ar(r-1)}{2} \right\rceil - \left\lceil r - m + \frac{a(r-m)(r-m-1)}{2} \right\rceil = m + \frac{am(2r-m-1)}{2} \leq (B+1)m.$$

A Lorentz block has codimension $d_i - 1 \leq B + 1$ at a nonzero boundary point, nullity one there, and codimension $d_i \leq 2(B + 1)$ at its vertex, nullity two there. Rays obey the same bound. Hence $\text{codim } E_X \leq (B + 1) \text{nul } X$. Lemma 8.5 rules out nullity at most $q - 1$, since $(q - 1)(B + 1) < qB + 1$ under the stated hypothesis. Every tuple in every boundary fiber therefore has nullity exactly q .

Each compact boundary fiber is a singleton. Indeed, from a relative-interior point of a nontrivial fiber take a maximal nonzero affine chord. Its endpoint must leave the relative interior of the minimal cone face, and thus acquire an additional null eigenvalue. The resulting nullity exceeds q , a contradiction. Compactness makes the unique boundary lift $X(v)$ continuous.

With p_i and q_i the Jordan ranks of complementary primal and certificate blocks, as in (4.4), the curvature chain holds with equality on a dense open set of supports:

$$qB \leq \sum_i a_i p_i q_i \leq \sum_i a_i (r_i - q_i) q_i \leq B \sum_i q_i \leq qB.$$

Hence exactly q cap-attaining blocks have corank one, and all other blocks are interior. Call the set of corank-one blocks at a generic point its active pattern. At a limit of points with one pattern, those zero eigenvalues persist, and constant total nullity excludes any additional zero eigenvalue or new active block. The closures of the regions with distinct patterns are therefore disjoint, and their finite union covers the connected sphere, so one pattern occurs everywhere.

The corresponding kernel lines define a continuous map from $\mathbb{S}^{qB}$ to $(\mathbb{F}^{R-1})^q$, or to $(\mathbb{S}^B)^q$ in the Lorentz case. It is injective. Indeed, equal lines put the two tuples in the relative interior of the same product face. Any certificate exposing one projected point annihilates that face, so the genuine slack identity makes the other projected point identical. Source and target are compact connected manifolds of the same dimension, so invariance of domain would make the map a homeomorphism. The target has nonzero intermediate mod-two cohomology, whereas the sphere does not. This contradiction proves the lower bound. The grouped construction with $q + 1$ groups attains it. $\square$

The proof also gives a statement that does not depend on the cap width: a compact lift with $s - 1 = qB$, $q \geq 2$, cannot have total nullity exactly q at every point of every boundary fiber. Such constant nullity would force singleton fibers and hence the same impossible kernel-line injection.

8.4 Properties of a hypothetical better bounded lift

For a fixed Hermitian order cap or Lorentz dimension cap, the single-ball cases not resolved by the preceding results are those with

$$s = qB + 1, \quad q \geq B + 2, \quad q \leq \inf_{\text{affine lifts}} \nu_{\text{std,slice}} \leq q + 1. \quad (8.6)$$

By Proposition 8.3, any lift with $\nu_{\text{std,slice}} < q + 1$ must be bounded. The next proposition derives necessary properties of such a lift; none of the preceding theorems depends on resolving this interval.

Proposition 8.7 (Certificate collapse and fixed generic labels). *Fix a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, an order cap $R \geq 2$, and $B = a(R - 1)$, where $a = \dim_{\mathbb{R}} \mathbb{F}$. Consider a bounded affine lift through finitely many Hermitian cones of orders at most R and optional rays, in the divisible case $s - 1 = qB$, $q \geq 2$. Suppose $\nu_{\text{std,slice}} < q + 1$. For $v \in \mathbb{S}^{qB}$ let $\mathcal{P}(v)$ and $\mathcal{D}(v)$ be its full primal and normalized certificate fibers. Define*

$$E_i(v) = \sum_{Y \in \mathcal{D}(v)} \text{Ran}_{\mathbb{F}} Y_i, \quad Z = \left\{ v : \sum_i \dim_{\mathbb{F}} E_i(v) \leq q - 1 \right\},$$

counting a nonzero ray certificate as dimension one. Then Z is nonempty and semialgebraic, and $\dim Z \leq qB - B$. On a dense open semialgebraic set U , both fibers are singletons, strict complementarity holds, and exactly q full-order blocks are of corank one. If $B \geq 2$, one and the same set of q block labels is active on all of U . Thus changing generic label patterns cannot produce such a lift; what we cannot exclude is that this fixed pattern degenerates over Z .

Proof. The boundary-order bound gives nullity at most q at every primal boundary tuple, so every certificate has rank at most q . A finite positive average spans all the spaces $E_i(v)$, hence $\sum_i \dim E_i(v) \leq q$. Semialgebraicity follows by expressing the spanning condition with finitely many certificates (the ambient dimension bounds their number) and eliminating quantifiers.

Choose a dense open generic set where the full-fiber minimum-rank theorem holds. All certificates there have rank exactly q , and all primal tuples have nullity exactly q . The compact primal chord argument in Theorem 8.6 gives a singleton primal fiber. The dual fiber lies in a product face of rank at most q ; if it contained a nontrivial maximal chord, an endpoint would leave that face's relative interior and have rank below q . Thus it too is a singleton. On a further dense open smooth locus, equality in the curvature chain gives exactly q full-order corank-one blocks and strict complementarity.

On each smooth stratum of Z of dimension t , choose smooth primal and certificate selections after further stratification. Differentiating the slack on two copies of this stratum gives its positive definite tangent metric, of rank t . The certificate ranks sum to at most $q - 1$, so the mixed Peirce estimate gives $t \leq B(q - 1)$. This proves the dimension bound, including for the semialgebraic closure $\overline{Z}$.

For any support v , compactness gives a limit of generic primal tuples. Its nullity is at least q , because nullity cannot drop in a limit, and at most q by the parameter hypothesis. If $v \notin Z$, an averaged certificate of rank q forces every primal in that fiber to have nullity q . The fiber is consequently a singleton. This singleton is the limit of every approaching sequence of generic tuples. Passing to a subsequence of fixed generic labels shows that it has exactly that pattern: an additional zero eigenvalue would exceed nullity q . Its inactive blocks are positive definite. Compactness then provides a neighborhood of v in which no generic tuple has a new active label; since each generic tuple has exactly q labels, its pattern is fixed.

If Z were empty, this argument would give a continuous singleton lift and one fixed pattern on the whole sphere. The kernel-line injection and projective-product contradiction in Theorem 8.6 would apply. Hence $Z \neq \emptyset$. If $B \geq 2$, the complement of the closed semialgebraic set $\overline{Z}$, of codimension at least two, is connected. For example, semialgebraic triangulation and a small general-position perturbation of a path avoid a subcomplex of codimension at least two. The local pattern just proved is constant on this complement. Every nonempty open generic pattern region meets the complement because $\overline{Z}$ has empty interior. There can therefore be only one such pattern. $\square$

For Lorentz cones of dimension at most $d \geq 3$, set $B = d - 2$ and retain the same boundedness and parameter hypotheses, with optional rays. The Lorentz version replaces E_i by the support face generated by the whole certificate fiber, and its dimension count by Jordan rank. Its rank-one orbit has dimension at most B . The same proof gives $\dim Z \leq qB - B$ and, for $B \geq 2$, the same fixed-label conclusion. When $B = 1$, the exceptional set may have codimension one and the connected-complement argument does not apply.

We record one precise topological fact about the exceptional fibers. The contact incidence space

$$\Gamma = \{(v, X, Y) : v \in \mathbb{S}^{qB}, X \in \mathcal{P}(v), Y \in \mathcal{D}(v)\}$$

is compact: primal compactness is assumed and the Slater normalization uniformly bounds all dual fibers. The fiber of $\Gamma \rightarrow \mathbb{S}^{qB}$ is exactly $\mathcal{P}(v) \times \mathcal{D}(v)$, since every primal is complementary to every genuine certificate. This is a nonempty compact convex set. The Vietoris–Begle theorem for compact spaces with acyclic fibers [12] implies

$$\check{H}^k(\Gamma; \mathbb{Z}/2) \cong \check{H}^k(\mathbb{S}^{qB}; \mathbb{Z}/2). \quad (8.7)$$

Thus the incidence space has no cohomology beyond that of the sphere. The difficulty is that the generic kernel-line map need not extend over the rank-deficient fibers. Equation (8.7) gives neither such an extension nor its degree, and we do not claim that one exists.

8.5 Counterexamples to stronger fiberwise conclusions

Example 8.8 (Boundary selections can escape over a compact target). Put $\delta = 1 - x_1$ and impose three positive semidefinite blocks:

$$\begin{pmatrix} 1+x_1 & x_2 \\ x_2 & \delta \end{pmatrix} \succeq 0, \quad \begin{pmatrix} \delta & x_2 \\ x_2 & u \end{pmatrix} \succeq 0, \quad \begin{pmatrix} \delta & u \\ u & z-1 \end{pmatrix} \succeq 0.$$

The first block describes exactly $\mathbb{B}_2^2$. For every point of this ball with $\delta > 0$, the remaining blocks are feasible precisely when $u \geq x_2^2/\delta$ and $z \geq 1 + u^2/\delta$. At $(x_1, x_2) = (1, 0)$, they instead require $u = 0$ and $z \geq 1$. Thus the projection is the entire closed disk. The choice $x = 0, u = 1, z = 3$ makes all three blocks positive definite, so the lift has Slater feasibility. On the boundary $x_1 = 1 - \delta$, $x_2^2 = 2\delta - \delta^2$, every feasible completion satisfies

$$u \geq 2 - \delta, \quad z \geq 1 + \frac{(2 - \delta)^2}{\delta} \longrightarrow \infty \quad (\delta \downarrow 0).$$

Consequently no boundary selection is locally bounded at $(1, 0)$. Compactness of the projected set and Slater feasibility therefore do not imply bounded boundary selections; any compactness assumption on the lifted boundary needs separate justification.

Lemma 8.9 (A compact base removes one gradient unit). *Let F be a k -logarithmically homogeneous self-concordant barrier on a pointed cone, with positive definite Hessian on its span. On a nonempty base $\ell(X) = 1$, where ℓ is nonnegative on the cone, its restricted gradient parameter is at most $k - 1$.*

Proof. Write $g = \nabla F$, $H = \nabla^2 F$. Homogeneity gives $HX = -g$ and $X^T H X = k$. The squared dual norm of the gradient on $\ker \ell$ is its orthogonal projection in the inverse-Hessian norm:

$$\min_{\alpha} (g - \alpha \ell)^T H^{-1} (g - \alpha \ell) = k - \frac{1}{\ell^T H^{-1} \ell}.$$

The unit Dikin ellipsoid lies in the cone; since $\ell(X) = 1$ and ℓ is nonnegative there, $\ell^T H^{-1} \ell \leq 1$. This proves the claim. Compactness of the base is a convenient sufficient condition for a strictly positive height; the argument only needs the conditions stated above. $\square$

Example 8.10 (Small minimum nullity, larger global parameter). The Lorentz case of the perspective construction in Section 4 shows that the standard barrier parameter can exceed both the minimum fiber nullity and the maximum certificate rank; all three are computed exactly. Partition $\mathbb{R}^{s-1}$ into q nonempty coordinate groups and write $x = (a, b)$. The lift is

$$W_G = \left(\frac{1-b+z_G}{2}, \frac{1-b-z_G}{2}, a_G \right) \in Q_{|G|+2}, \quad \sum_G z_G = 1+b. \quad (8.8)$$

Equivalently, $1-b \geq 0$, $z_G \geq 0$, and $\|a_G\|^2 \leq (1-b)z_G$. Summing them shows that the projection lies in the ball. Conversely, every ball point lifts by distributing the nonnegative residual $1-b^2 - \|a\|^2$ among the z_G . All variables are bounded and $a = b = 0$, $z_G = 1/q$ is strictly feasible.

At every boundary point with $b < 1$, every inequality is an equality, $z_G = \|a_G\|^2/(1-b)$, and each block is a nonzero extreme ray. At the north pole, the fiber is $z_G \geq 0$, $\sum_G z_G = 2$. Its relative

interior again has one null eigenvalue per block, whereas a vertex has one such block and $q - 1$ zero blocks. Consequently

$$\min_{X \in \mathcal{P}(v)} \text{nul } X = q \quad \text{for every } v, \quad \max_{X \in \mathcal{P}(\text{north})} \text{nul } X = 2q - 1.$$

For completeness, at support $v = (c, \eta)$ the entire certificate fiber has blocks

$$V_G = (A_G + \gamma, A_G - \gamma, -c_G), \quad \gamma = (1 - \eta)/2, \quad \sum_G A_G = (1 + \eta)/2, \quad A_G \geq 0, \quad 4A_G\gamma \geq \|c_G\|^2.$$

Coefficient matching on the entire affine slice proves these conditions necessary and sufficient. For $\eta < 1$, all the inequalities are equalities and the certificate is unique with every block nonzero of rank one. At $\eta = 1$ it is the simplex $A_G \geq 0$, $\sum_G A_G = 1$. Thus the maximum certificate rank is q at every support, although its minimum at the north pole is one.

The restricted standard barrier is $F = -\sum_G \log((1 - b)z_G - \|a_G\|^2)$, with exact parameter $2q - 1$. For the upper bound homogenize 1 to τ in (8.8); the resulting cone has a compact $\tau = 1$ base and its restricted product barrier is $2q$ -homogeneous. Apply Lemma 8.9. The north-fiber vertex and Lemma 8.1 give the matching lower bound. For balls with different q_a , separate copies of this lift give minimum fiber nullity $\sum_a q_a$, maximum rank $\sum_a q_a$ of aggregate certificates with strictly positive source weights, and standard parameter $\sum_a (2q_a - 1)$. Constant terms cannot move between source certificates: a source affine slack $\mu_a - \lambda_a v_a^T x_a$ is nonnegative only if $\mu_a \geq \lambda_a$, and equality of the total constants forces equality sourcewise.

This example refutes an extra-unit lower bound stated with the quantifier “every primal completion at some support”: every support has a completion of nullity only q . The barrier lower bound of Lemma 8.1 needs only one completion of high nullity, such as the north-fiber vertex.

Two simpler examples show that convexity of the fibers alone does not yield a global kernel-line map into a product of projective spaces. In a two-dimensional $\mathbb{F}$ -subspace U , the set $\{Y \succeq 0 : \text{Ran } Y \subset U, \text{tr } Y = 1\}$ contains all primitive rays in U , normalized to trace one, and their rank-two mixtures. At its scalar center there is no distinguished eigenline. Adding $q - 2$ fixed rank-one blocks keeps the total rank at most q . For a primal example, encode the simplex $\sum_{j=0}^q p_j = 1$, $p_j \geq 0$, using

$$X_0 = \text{diag}(p_0, p_1), \quad X_j = \text{diag}(1, p_{j+1}), \quad 1 \leq j \leq q - 1,$$

and project to $\sigma = \sum_{j=2}^q p_j \in [0, 1]$. The fiber at $\sigma = 0$ has $p_0 = t$, $p_1 = 1 - t$: its first kernel changes between two lines at the endpoints and disappears in between. Its nullity is $q - 1$ inside the fiber and q at either endpoint. The standard barrier is $-\sum_{j=0}^q \log p_j$, with exact parameter q by Lemmas 8.9 and 8.1. This is a lift of an interval, not of a ball in the critical dimension. It shows that a proof for the unresolved range (8.6) must use more than boundedness, convexity, and a nullity bound.

9 Exact parameters of concrete affine formulations

This section computes barrier parameters for three formulations of balls, each on its own affine domain: norm trees, grouped slices, and column packing. The norm tree uses the fewest small cone factors, but its restricted standard barrier often has a larger parameter than that of the grouped formulation. For the grouped and column-packed slices we also prove lower bounds for every self-concordant barrier on the same domain, including barriers with arbitrary mixed derivatives. For an open convex domain Ω in its affine hull, the optimal, or intrinsic, parameter $\vartheta_{\text{opt}}(\Omega)$ is the

infimum of the parameters of all self-concordant barriers with positive definite Hessian on that affine hull; the formulation stays fixed.

These standard-parameter calculations also appear in the unpublished companion manuscript *The Cost of Following the Central Path* [21]. We reproduce them with proofs so that this paper is self-contained, and use them to study arbitrary barriers on a fixed affine domain and to compare resources across lifts. We do not claim the shared calculations and constructions as new.

9.1 Norm trees: the children of the root determine the standard parameter

A norm tree for $\mathbb{B}_2^N$ uses a rooted tree T with $N \geq 2$ leaves, one per coordinate, and L internal nodes, each with $c_v \geq 2$ children and one Lorentz cone. At an internal node v , let t_v be its axial variable and let y_v collect the values of its children: a leaf contributes a projected coordinate, and an internal child w contributes t_w . The linked cone is defined by $t_v > \|y_v\|$ for every internal node v . Its root slice, with $t_r = 1$, is denoted by Ω_T and projects onto $\text{int } \mathbb{B}_2^N$. Put

$$q_v = t_v^2 - \|y_v\|^2, \quad \Phi_T = - \sum_v \log q_v, \quad F_T = \Phi_T|_{t_r=1}.$$

The positive-sheet condition $t_v > 0$ is part of the domain.

Theorem 9.1 (Standard parameter of a norm tree). *The least self-concordance parameter of the restricted standard barrier F_T is*

$$\nu(F_T) = \begin{cases} 2L - 2, & \text{every root child is internal,} \\ 2L - 1, & \text{at least one root child is a leaf.} \end{cases} \quad (9.1)$$

For a forest of independent root-fixed trees, these parameters add.

Proof. Affine restriction of the standard Lorentz barriers gives standard self-concordance. The Hessian is positive definite, since every nonzero linked direction changes a cone coordinate. Before fixing $t = t_r$, Φ_T is logarithmically homogeneous of degree $2L$. With $H = D^2\Phi_T$ at the point z and e_t the root covector, homogeneity gives $H z = -D\Phi_T$ and $\|D\Phi_T\|_{H^{-1}}^2 = 2L$. Orthogonal projection onto the slice tangent space yields

$$\|DF_T\|_{D^2F_T, *}^2 = 2L - \frac{t^2}{e_t^T H^{-1} e_t}, \quad \nu(F_T) = 2L - \frac{1}{\sup \beta_T}, \quad \beta_T = \frac{e_t^T H^{-1} e_t}{t^2}. \quad (9.2)$$

The notation on the left means the dual norm induced by the restricted Hessian. The unit Dikin ellipsoid is contained in the linked cone, which lies in $t > 0$. Consequently $e_t^T H^{-1} e_t \leq t^2$: otherwise a vector of local norm less than one could move t to zero.

If a root child is a leaf, set $t = 1$, that leaf to $\sqrt{1 - \epsilon}$, and the other root leaves to zero. Scale every internal child subtree from a fixed interior point by ϵ^2 . Then $q_r = \epsilon - O(\epsilon^4) > 0$. The direction with $h_t = 1$, $h_{\text{leaf}} = 1$, and all other components zero has

$$D^2\Phi_T[h, h] = \frac{4(1 - \sqrt{1 - \epsilon})^2}{q_r^2} \rightarrow 1.$$

The descendant terms vanish in this direction. Since $(e_t^T H^{-1} e_t)^{-1} = \min_{h_t=1} h^T H h$, the preceding limit and the Dikin bound prove $\sup \beta_T = 1$.

Suppose instead that every root child is internal, with axial values $a_i > 0$. Eliminate its strict descendants from the Hessian. The effective axial curvature of child i is at least a_i^{-2} by Dikin

containment in that child's linked cone. Increasing these curvatures can only increase the final root Schur complement. It therefore suffices to consider

$$-\log(t^2 - \sum_i a_i^2) - \sum_i \log a_i.$$

Normalize $t = 1$, and put $r = \sum_i a_i^2$, $q = 1 - r$, and $Z = \sum_i a_i^4 / (q + 2a_i^2)$. The child Hessian is the diagonal matrix with entries $2/q + a_i^{-2}$ plus $4aa^T/q^2$. Its cross row with the root is $-4a^T/q^2$. The rank-one inverse formula gives the root Schur complement

$$S = \frac{2(1+r) - 16Z/(q+4Z)}{q^2}. \quad (9.3)$$

Since $x/(q+2x)$ increases for $x \geq 0$, $Z \leq r^2/(q+2r) = r^2/(1+r)$. The inequality $S \geq 2$ is equivalent to $r(3-r) \geq 4(2-r)Z$; substituting this upper bound leaves $3r(1-r)^2/(1+r) \geq 0$. Hence $\beta_T \leq 1/2$. Conversely, scale all child subtrees by $\epsilon \downarrow 0$. The direction changing only the root has squared local norm $2(1+r)/(1-r)^2 \rightarrow 2$. The same variational formula proves $\sup \beta_T = 1/2$. Equation (9.2) now proves both cases. On a Cartesian product, squared dual gradient norms add; their suprema can be approached independently, proving the forest claim. $\square$

Corollary 9.2 (The best capped norm tree). *Fix an integer Lorentz dimension cap $d \geq 3$, and set*

$$A = d - 2, \quad E = N - 1, \quad L_0 = \lceil E/A \rceil, \quad \sigma = L_0 A - E.$$

Define $\chi = 1$ if $L_0 \geq 3$ and $\sigma \geq \max\{0, d - L_0\}$, and define $\chi = 0$ otherwise. Among all norm trees with $c_v + 1 \leq d$,

$$\min_T \nu(F_T) = 2L_0 - 1 - \chi. \quad (9.4)$$

Proof. The tree identity $\sum_v (c_v - 1) = E$, with $1 \leq c_v - 1 \leq A$, gives $L \geq L_0$. Any list of L_0 increments in this interval with sum E is realized by a chain of internal nodes, as in Corollary 3.4. For an all-internal root of arity c , one must additionally have $2 \leq c \leq \min(d-1, L_0-1)$. The remaining $L_0 - 1$ increments must sum to $E - (c-1)$. Such increments exist exactly when

$$L_0 \geq 3, \quad E \leq (L_0 - 1)A + \min(d-1, L_0-1) - 1.$$

The lower bound on their sum causes no further constraint, since $E \geq L_0$ and the root increment may be chosen no larger than $E - L_0 + 1$. Partitioning the remaining internal nodes into c nonempty chains realizes the required descendant forest. The displayed inequality is precisely the condition defining χ . Trees with more than L_0 internal nodes cannot do better: even with an all-internal root, $L_0 + 1$ internal nodes give $2(L_0 + 1) - 2 > 2L_0 - 1$. $\square$

The next proposition bounds every self-concordant barrier on the root-fixed domain, not only the standard one.

Proposition 9.3 (Intrinsic norm-tree bounds). *Every self-concordant barrier on Ω_T has parameter at least L . Every logarithmically homogeneous self-concordant barrier on the unfixed linked cone has degree at least $L + 1$. For a forest of independent root-fixed trees, every self-concordant barrier has parameter at least the total number of internal nodes.*

Proof. Choose positive leaf values with squared sum one and set t_v^* equal to the norm of the leaves below v . Each $z_v^* = (t_v^*, y_v^*)$ is a nonzero boundary vector. Intersect the linked affine slice with the product of two-planes

$$z_v = \alpha_v(t_v^*, y_v^*) + \beta_v(t_v^*, -y_v^*).$$

Cone membership is exactly $\alpha_v, \beta_v \geq 0$; its determinant is $4(t_v^*)^2 \alpha_v \beta_v$. Root fixing and linking become

$$\alpha_r + \beta_r = 1, \quad \alpha_w + \beta_w = \alpha_v - \beta_v \quad (w \text{ an internal child of } v). \quad (9.5)$$

The L variables β_v are affine coordinates: these equations determine every α_v recursively. Small positive values of all β_v give strictly positive α_v , so the section has dimension L and meets the interior of the linked slice. Nonnegativity and the equations bound all coordinates by one. At $\beta = 0$, all $\alpha_v = 1$, and exactly the L independent inequalities $\beta_v \geq 0$ are active. Every relative boundary point of this polytope lies on a cone boundary. Restriction of any barrier on Ω_T is therefore a barrier on this polytope. The facet bound [70, Proposition 2.3.6] at this vertex, where L linearly independent facet inequalities are active, proves the claim. Product sections prove the forest extension.

The corresponding independent-facet principle for logarithmically homogeneous barriers on polyhedral cones, with one additional parameter unit at a nonzero boundary point, is due to Hildebrand [53, Theorem 6.1]. The tree section above gives the following direct argument for this application. For the conic claim let the logarithmic degree be θ . Lemma 8.9 gives a root-slice parameter at most $\theta - 1$. The first claim gives $\theta - 1 \geq L$. Logarithmic homogeneity is essential to this last argument. $\square$

Thus $\vartheta_{\text{opt}}(\Omega_T)$ lies between L and the value in (9.1). We do not determine it for general trees; in particular, we do not claim that the standard barrier is optimal.

A natural attempt to lower the standard parameter adds $\log t_w$ for every non-root internal node w , removing one axial logarithm per link. This already fails for the smallest such tree, a root with a leaf child x and an internal child a whose children are leaves y and z . The function

$$\tilde{F}(t, x, a, y, z) = -\log(t^2 - x^2 - a^2) - \log(a^2 - y^2 - z^2) + \log a$$

has logarithmic degree three. At the interior point $(3, 0, 1, 0, 0)$, in direction $(0, 0, 1, 1, 0)$, its derivatives are $D^2 \tilde{F} = 53/16$ and $D^3 \tilde{F} = -441/32$. The squared self-concordance inequality fails, since $(441/32)^2 - 4(53/16)^3 = 45604/1024 > 0$. Hence this modification does not give a self-concordant barrier.

9.2 Grouped fixed-tail slices: coupled barriers cannot lower the parameter

For each ball a , partition its coordinates into nonempty groups $G \in \mathcal{G}_a$, and put $H = \sum_a |\mathcal{G}_a|$. Consider

$$\Omega_{\mathcal{G}} = \left\{ (u, w) : u_{aG} > \|w_{aG}\|^2, \quad \sum_{G \in \mathcal{G}_a} u_{aG} = 1 \text{ for every } a \right\}. \quad (9.6)$$

Its projection is the product of the open balls. The Lorentz realization uses $((1+u)/2, (1-u)/2, w) \in \text{int } Q_{|G|+2}$. The real PSD realization uses $\begin{pmatrix} u & w^T \\ w & I \end{pmatrix} \succ 0$. More generally, fixing the tail $z = 1$ in Lemma 4.2 gives the same domain in every simple Jordan family, for group size at most $a(r-1)$. In all these cases the restricted determinant is $u - \|w\|^2$.

Theorem 9.4 (Intrinsic grouped parameter). *The domain in (9.6) satisfies*

$$\vartheta_{\text{opt}}(\Omega_{\mathcal{G}}) = H. \quad (9.7)$$

The restricted standard barrier $F_{\mathcal{G}} = -\sum_{a,G} \log(u_{aG} - \|w_{aG}\|^2)$ attains this value.

Proof. For $f(u, w) = -\log q$, $q = u - \|w\|^2$, consider an affine line and write $A = q'/q$, $C = 2\|w'\|^2/q \geq 0$. Then

$$f'' = A^2 + C, \quad f''' = -2A^3 - 3AC, \quad 4(A^2 + C)^3 - (2A^3 + 3AC)^2 = 3A^2C^2 + 4C^3 \geq 0.$$

The Hessian is positive definite, and $(D^2f)^{-1}Df = (-q, 0)$ gives squared dual gradient norm one. Thus each summand is a one-self-concordant barrier; addition and affine restriction give the upper bound H .

For the lower bound choose positive σ_{aG} summing to one for each a , and a unit vector e_{aG} in each nonempty group. The affine section $u_{aG} = \sigma_{aG}$, $w_{aG} = t_{aG}e_{aG}$ is exactly the open box

$$\prod_{a,G} (-\sqrt{\sigma_{aG}}, \sqrt{\sigma_{aG}}).$$

Its boundary lies on the boundary of Ω_G . An arbitrary barrier, including one coupling all groups, restricts to a barrier on this H -dimensional box. The classical facet bound [70, Proposition 2.3.6] yields $\nu \geq H$. $\square$

For comparison, the product of k positive-dimensional balls itself has $\vartheta_{\text{opt}} = k$. The upper bound follows by restricting the one-parameter paraboloid barriers to $u = 1$ and adding; the lower bound follows from a k -dimensional box section. Consequently the optimal parameter of the grouped domain exceeds that of its projection by $H - k$. Under a Lorentz dimension cap d with $3 \leq d < N + 1$, the grouped formulation of a single ball in $\mathbb{R}^N$ uses $h = \lceil N/(d - 2) \rceil$ groups. In the notation of Corollary 9.2, put $\zeta = \mathbf{1}_{\{d-2|N-1\}}$, so $h = L_0 + \zeta$. The exact comparison is

$$\min_T \nu(F_T) - \nu(F_G) = L_0 - 1 - \chi - \zeta \geq 0. \quad (9.8)$$

Here $L_0 \geq 2$; when $L_0 = 2$, $\chi = 0$, and when $L_0 \geq 3$ the last inequality is immediate. The grouped lift uses at most one more factor than the smallest capped tree, while its parameter can be asymptotically half the tree's. If $d \geq N + 1$, the direct Lorentz slice already has parameter one.

The same box section also gives the lower bound H for $u_{aG} > \|w_{aG}\|_p^p$, $1 < p < \infty$, with the same allocation equations. The matching upper bound just proved uses the quadratic case $p = 2$; it is not asserted for other exponents.

9.3 Bounded-fiber projection and one-block column packing

A lower bound on the barrier parameter of a projected body implies one for the lift only under hypotheses on the projection; counting lift variables is not enough. The next lemma, a form of the classical exact partial-minimization theorem [22, Theorem 5.2.1], gives such hypotheses.

Lemma 9.5 (Parameter monotonicity under bounded-fiber projection). *Let $Q \subset E \times V$ be a closed convex set with nonempty interior, and suppose every nonempty fiber $Q_x = \{y : (x, y) \in Q\}$ is bounded. Suppose $C = \pi_E Q$ contains no line. If F is a self-concordant barrier on $\text{int } Q$ with positive definite Hessian and parameter ν , then*

$$\phi(x) = \min\{F(x, y) : (x, y) \in \text{int } Q\}$$

is a self-concordant barrier with positive definite Hessian and parameter at most ν on $\text{int } C$. The same statement holds in affine hulls after eliminating affine equations. In particular, $\vartheta_{\text{opt}}(\text{int } Q) \geq \vartheta_{\text{opt}}(\text{int } C)$.

Proof. First, inverse images in Q of bounded subsets of E are bounded. Otherwise choose $(x_j, y_j) \in Q$ with x_j bounded and $\|y_j\| \rightarrow \infty$. Normalize their differences from a fixed $q_0 \in Q$ and take a subsequence converging to a vertical unit vector $(0, v)$. Closedness and convexity imply $q_0 + t(0, v) \in Q$ for every $t \geq 0$, contradicting the bounded fiber through q_0 . This argument also proves that C is closed, by lifting a convergent sequence in C and extracting a subsequence. The standard relative-interior identity for linear images of convex sets gives $\pi_E(\text{int } Q) = \text{int } C$.

For each interior x , the fiber is bounded and its boundary lies on ∂Q . Barrier divergence and strict convexity therefore give a unique interior minimizer $y(x)$. The equation $F_y = 0$ and the implicit-function theorem show that $y(x)$ is C^2 . For a direction h , set $\hat{h} = (h, Dy(x)h)$. Differentiating stationarity gives Hessian orthogonality to every vertical direction, and hence

$$\begin{aligned} D\phi[h] &= DF[\hat{h}], \\ D^2\phi[h, h] &= D^2F[\hat{h}, \hat{h}] = \min_k D^2F[(h, k), (h, k)], \\ D^3\phi[h, h, h] &= D^3F[\hat{h}, \hat{h}, \hat{h}]. \end{aligned}$$

In the third identity the acceleration term is vertical and its Hessian pairing with $\hat{h}$ vanishes. The positive definite Schur complement proves nondegeneracy; the two defining derivative inequalities for F prove the same inequalities for ϕ . Finally, as x_j tends to a finite boundary point of C , the minimizers remain bounded by the first paragraph. Any cluster point lies on ∂Q , since an interior cluster point would project to an interior point of C . Barrier divergence therefore implies $\phi(x_j) \rightarrow +\infty$. This completes the proof. $\square$

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, let $p, b \geq 1$, and choose nonzero real linear subspaces $E_a \subset \mathbb{F}^p$. With $W = [w_1 \cdots w_b]$, $w_a \in E_a$, define

$$\Omega_{\text{pack}} = \left\{ (S, W) : S \in H^b(\mathbb{F}), S_{aa} = 1, \begin{pmatrix} S & W^* \\ W & I_p \end{pmatrix} \succ 0 \right\}. \quad (9.9)$$

Determinants for quaternionic Hermitian matrices mean Jordan determinants, and traces in the calculations below mean real traces.

Theorem 9.6 (Intrinsic column-packing parameter). *For the single-factor domain in (9.9),*

$$\vartheta_{\text{opt}}(\Omega_{\text{pack}}) = b. \quad (9.10)$$

*The restricted standard barrier $F_{\text{pack}} = -\log \det(S - W^*W)$ attains this value. This holds without any relation between the number of columns b and the row dimension p .*

Proof. With the Schur complement $D = S - W^*W$, the projection of Ω_{pack} onto W is exactly $\prod_a \{w_a \in E_a : \|w_a\| < 1\}$. Indeed its diagonal entries are $d_a = 1 - \|w_a\|^2$; conversely, set $S = W^*W + \text{Diag}(d_a)$. For fixed W , the closed set of feasible S satisfies $|D_{ac}|^2 \leq d_a d_c \leq 1$ and is bounded. The whole closed slice is bounded as well: $\|w_a\| \leq 1$, and $S \succeq 0$ with diagonal one implies $|S_{ac}| \leq 1$. It has relative interior, for example $S = I_b$, $W = 0$. Lemma 9.5 and the product-of-balls parameter established above show that every barrier has $\nu \geq b$.

For the upper bound work first on the unrestricted matrix epigraph $S - W^*W \succ 0$. It is an affine slice of $H_{++}^{p+b}(\mathbb{F})$, so its log-determinant is standard self-concordant. An affine shear at any base point sets its W coordinate to zero while preserving the Schur complement. At the resulting point $(D, 0)$,

$$DF[H, K] = -\text{tr}(D^{-1}H), \quad D^2F[(H, K), (H, K)] = \text{tr}(D^{-1}HD^{-1}H) + 2\text{tr}(D^{-1}K^*K).$$

The squared dual norm of this gradient is b , attained by $(H, K) = (-D, 0)$. Therefore this is a b -self-concordant barrier; the diagonal and column-subspace restrictions cannot increase its parameter. The lower bound proves equality. $\square$

For the standard barrier the partial minimizer is explicit. Hadamard's determinant inequality, with equality exactly at a diagonal positive matrix, gives

$$\min_{S: S_{aa}=1} F_{\text{pack}}(S, W) = - \sum_{a=1}^b \log(1 - \|w_a\|^2).$$

Thus column packing reduces the ambient parameter $p + b$ to b on the fixed slice, and no barrier there, coupled or not, does better. In contrast, the optimal parameter of a grouped slice may be much larger than the number of balls. The factor count (here one), the ambient rank, and the optimal slice parameter measure different costs.

Example 9.7 (Repeated boundary blocks do not bound arbitrary barriers). For $m \geq 2$, the diagonal affine slice

$$\{((1/m, x/m), \dots, (1/m, x/m)) : x \in \text{int } \mathbb{B}_2^s\} \subset Q_{s+1}^m$$

is affinely isomorphic to one ball. All m blocks reach nonzero boundary points simultaneously, and polynomial slack factors are $A_j(x) = (1, x)/m$, $B_j(z) = (1, -z)$. The restricted standard product barrier is $-m \log(1 - \|x\|^2)$ plus a constant, with exact parameter m ; its squared dual gradient norm approaches m as $\|x\| \rightarrow 1$. Yet the slice admits the one-parameter barrier $-\log(1 - \|x\|^2)$. The active conormals coincide after restriction. Consequently simultaneous boundary multiplicity alone does not give a lower bound for arbitrary barriers; the polyhedral sections or the projection argument above provide the missing geometric information.

These results concern the parameters of self-concordant barriers on specified formulations. They do not assert that every algorithm takes a number of iterations proportional to the square root of the parameter, and they do not bound the conditioning or the arithmetic cost of a Newton solve.

10 Topology of saturated contact factorizations

We extend the support-orbit covering of Section 7 to arbitrary proper cones. The targets are then the boundaries of compact cone bases. These need not be smooth a priori, but saturation makes them C^1 hypersurfaces.

10.1 Normalized cone bases and the joint covering

Let P, D be compact connected C^1 manifolds without boundary, both of dimension $n \geq 1$. Suppose $\zeta : P \rightarrow D$ is continuous and

$$s(x, z) = \sum_i \langle A_i(x), B_i(z) \rangle, \quad A_i : P \rightarrow K_i, \quad B_i : D \rightarrow K_i^*,$$

where the maps are C^1 , the cones are proper, and $s(x, \zeta(x)) = 0$. Assume that the contact pairing

$$H_x(v, w) = - \sum_i \langle DA_i(x)[v], DB_i(\zeta(x))[w] \rangle \quad (10.1)$$

is nondegenerate between $T_x P$ and $T_{\zeta(x)} D$. No differentiability of ζ is required. Write $c_i = \max\{\dim K_i - 2, 0\}$ and $I_+ = \{i : c_i > 0\}$.

Theorem 10.1 (Joint covering at capacity equality). *If $\sum_i c_i = n$, choose any $\ell_i \in \text{int } K_i^*$ for $i \in I_+$. Then A_i is nowhere zero, the boundary*

$$J_i = \partial\{a \in K_i : \ell_i(a) = 1\}$$

is a C^1 embedded hypersurface diffeomorphic to $\mathbb{S}^{c_i}$, and

$$p : P \longrightarrow \prod_{i \in I_+} J_i, \quad p_i(x) = \frac{A_i(x)}{\ell_i(A_i(x))},$$

is a finite C^1 covering. If P is simply connected, every $c_i \geq 2$ and the covering has one sheet. In particular, if $P \cong \mathbb{S}^n$, there is exactly one positive capacity, equal to n . If $P \cong \prod_{a=1}^b \mathbb{S}^{p_a}$ with $p_a \geq 2$, the multiset of positive capacities is exactly $\{p_1, \dots, p_b\}$.

Proof. At each contact every cone pairing is nonnegative and zero, so its partial derivatives in each variable vanish:

$$DA_i(T_x P) \subset B_i(\zeta(x))^\perp, \quad DB_i(T_{\zeta(x)} D) \subset A_i(x)^\perp. \quad (10.2)$$

Lemma 3.1 applies with independent primal and dual tangent vectors. Nondegeneracy and rank subadditivity force every mixed channel to have rank c_i . For $c_i > 0$ this excludes the vertices $A_i(x) = 0$ and $B_i(\zeta(x)) = 0$, because a two-sided derivative of a map into a pointed cone is zero at the vertex. Complementarity puts $p_i(x)$ in J_i .

For $a = A_i(x)$ and $b = B_i(\zeta(x))$, the channel factors through $b^\perp/\mathbb{R}a$, a space of dimension c_i . The quotient derivative has rank c_i , since the channel does. Direct differentiation shows

$$Dp_i(x)[v] = 0 \iff DA_i(x)[v] \in \mathbb{R}A_i(x).$$

Thus Dp_i has constant rank c_i . A priori, the base boundary J_i is only a topological sphere. The constant-rank theorem supplies a local c_i -dimensional transversal on which p_i is a regular embedding. Invariance of domain makes this embedded patch relatively open in J_i . Consequently $p_i(P)$ is open and compact in the connected J_i , and is all of J_i . The regular patches give its C^1 embedded structure. Radial projection from an interior point of the base is a C^1 diffeomorphism to $\mathbb{S}^{c_i}$: a supporting normal has strictly positive pairing with the radial direction, so that direction is never tangent.

If $v \in \bigcap_{i \in I_+} \ker Dp_i$, the preceding kernel formula and (10.2) give $H_x(v, w) = 0$ for every w ; zero-capacity channels also vanish. Hence $v = 0$. Source and target of p have equal dimensions, so p is a local diffeomorphism. Compactness and connectedness make it a surjective finite covering.

If P is simply connected, $c_i \neq 1$: a submersion $p_i : P \rightarrow J_i \cong \mathbb{S}^1$ would lift to a real-valued function, whose maximum contradicts submersivity. Thus the target is simply connected and the cover has one sheet. A product of two positive-dimensional spheres has intermediate cohomology, which proves the single-sphere assertion. More generally, the graded indecomposable quotient $H^+/(H^+)^2$ of rational cohomology has one generator in the dimension of each sphere. Comparing this quotient on the source and target proves the multiset assertion. $\square$

The nondegeneracy hypothesis holds for the slack of a convex body with C^1 primal and polar boundaries: at a contact (x, y) , where $\langle x, y \rangle = 1$, the independent tangent spaces are $y^\perp$ and $x^\perp$, and their Euclidean pairing is nondegenerate because its first-slot kernel would lie in $y^\perp \cap \mathbb{R}x = \{0\}$. Thus the theorem applies without differentiating the supporting-polarity map. It also applies to the contact kernels of weighted products of balls, where the covering need not split into maps of individual source coordinates. The theorem does not identify a saturated cone factor as a Lorentz cone; it determines only its normalized base boundary, up to C^1 diffeomorphism, and the capacity profile.

Corollary 10.2 (Smooth ball lifts with arbitrary proper cones). *Let $N \geq 3$ be the dimension of the ball and $d \geq 3$ an integer. Allow arbitrary proper cone factors of dimension at most d . Over exact ball lifts with global C^1 selections of primal factors and genuine dual certificates, the minimum number of factors, total cone dimension, and parameter of a barrier on the full ambient cone product are attained simultaneously and equal*

$$\begin{cases} (h, N + 2h, 2h), & h = \lceil N/(d-2) \rceil, \quad d < N + 1, \\ (1, N + 1, 2), & d \geq N + 1. \end{cases}$$

The parameter minimum is over all barriers on the ambient product, including coupled barriers and, in particular, coupled logarithmically homogeneous ones.

Proof. If $d < N + 1$, every positive capacity is smaller than $N - 1$. The local bound gives $\sum_i c_i \geq N - 1$, and by Theorem 10.1 equality would require a single positive capacity $N - 1$. Hence $\sum_i c_i \geq N$. If there are k such factors, $k \geq h$ and their dimensions sum to at least $N + 2k$. Every proper cone of dimension at least two has a two-dimensional linear section through its interior that is isomorphic to $\mathbb{R}_+^2$. Restricting any ambient barrier to the product of these sections gives an orthant barrier, whose parameter is at least $2k$ [70, Proposition 2.3.6]. Extra factors cannot reduce these bounds. For $d \geq N + 1$, the local bound already gives total dimension at least $N + 1$, and the same section argument gives parameter at least two. The grouped Lorentz lifts in Theorem 6.2, and the direct Lorentz lift in the second case, attain all bounds. $\square$

The next lemma localizes the covering argument for use when the active factor labels change over the sphere. It also shows that nonsmooth cone boundaries do not remove the topological obstruction.

Lemma 10.3 (Proper normalized-phase domains). *Let $p \geq 2$, let $0 < c_i < p$ with $\sum_i c_i = p$, and let J_i be the boundaries of compact convex bodies of dimensions $c_i + 1$. Suppose $U \subset \mathbb{S}^p$ is open and $q_i : U \rightarrow J_i$ are C^1 as ambient maps. If the combined derivative of $q = (q_i)_i$ is injective everywhere, then $U \neq \mathbb{S}^p$. In particular the top class of $\mathbb{S}^p$ restricts to zero on U . The same holds on the inverse image of U under any coordinate projection from a product of spheres.*

Proof. A map into J_i has derivative rank at most c_i : rank $c_i + 1$ would make its image contain an ambient open set. Therefore joint injectivity forces each component to have rank c_i . The transversal and invariance-of-domain argument above gives regular charts on their images. If U were the whole sphere, every image would be all of J_i and q would be a finite covering of their product. A circle factor is impossible, since a finite covering by the simply connected $\mathbb{S}^p$ has a target with finite fundamental group. Without circle factors, the product is simply connected and its intermediate cohomology contradicts that of the sphere. Finally every proper open subset of a connected closed oriented manifold has zero ordinary top cohomology, by noncompact-manifold duality, component by component [48, Section 3.3]. $\square$

11 Capacity equality for products of balls

We extend the single-ball support-orbit obstruction to products of balls, where each ball contributes one full row identity. Equality in the capacity bound forces exactly one spin (Lorentz) factor for each ball, used only by that ball's row, even when scalar ray factors are present. The proof combines the topology of the joint support map with the fact that each dual row depends on only one source coordinate.

Theorem 11.1 (Product saturation with scalar rays). *Let $M = \prod_{a=1}^h S^{p_a}$, where $p_a \geq 1$, and $n = \sum_a p_a$. Suppose the full row identities*

$$1 - x_a \cdot z = \sum_i \langle X_i(x), Y_i^a(z) \rangle + \sum_j \alpha_j(x) \beta_j^a(z), \quad x \in M, \quad z \in S^{p_a}, \quad (11.1)$$

hold with finitely many globally labelled C^1 factors. The maps X_i, Y_i^a take values in simple non-ray symmetric cones K_i , of Jordan rank r_i , Peirce constant a_i , and real dimension m_i ; the scalar factors are nonnegative. Put $c_i = a_i \lfloor r_i^2/4 \rfloor$. Then

$$\sum_i c_i \geq n.$$

If equality holds, there are exactly h non-ray factors, and after relabelling

$$K_a \cong Q_{p_a+2}, \quad Y_a^d \equiv 0 \quad (d \neq a).$$

Conversely, separate direct Lorentz sections, one for each ball, attain equality. Thus the non-ray factors at equality have total dimension $n + 2h$ and total Jordan rank $2h$. If this profile fails, then $\sum_i c_i \geq n + 1$.

The first lemma determines the topology of the joint support map. As in Section 7, $\mathcal{I}_k(V)$ is the orbit of rank- k idempotents of a simple Euclidean Jordan algebra V .

Lemma 11.2 (Product support covering). *Assume equality in Theorem 11.1. Then the positive-capacity factors have constant, balanced, complementary ranks. Their support maps, complemented when necessary, form a finite covering*

$$\Sigma : M \longrightarrow \prod_i \mathcal{I}_{\lfloor r_i/2 \rfloor}(V_i). \quad (11.2)$$

The topology of this covering leaves only two possible non-spin factors, $\mathbb{S}_+^3$ and $\mathbb{S}_+^4$, whose orbits have universal covers S^2 and $S^2 \times S^2$. Each of these target S^2 factors is assigned to a distinct S^2 factor of M : the restriction of Σ to that source factor, lifted to the universal cover and projected onto the target S^2 , has degree ± 1 , and the restriction to every other source S^2 has degree zero.

Proof. At contact, put $Y_i(x) = \sum_a Y_i^a(x_a)$. Termwise nonnegativity gives $X_i(x) \circ Y_i(x) = 0$. A nonnegative differentiable scalar function has zero derivative at each zero. Hence each scalar product in (11.1) contributes zero to the mixed derivative at contact. Summing the rows and differentiating in the primal and dual variables independently gives the nondegenerate product-sphere metric. The Peirce rank bound yields

$$n \leq \sum_i a_i u_i v_i \leq \sum_i c_i, \quad u_i = \text{rank}_J X_i(x), \quad v_i = \text{rank}_J Y_i(x).$$

Equality forces $u_i + v_i = r_i$ and balanced ranks everywhere. Lower semicontinuity makes each rank constant. The support-differential calculation in Theorem 7.1 now applies without change: its joint differential is injective and has equal-dimensional domain and codomain. Compactness therefore gives the finite covering (11.2).

Lift this covering to universal covers. The lift is a diffeomorphism. The source universal cover is $\mathbb{R}^t \times \prod_{p_a \geq 2} S^{p_a}$, where t counts source circles. Each spin orbit is a sphere; each non-spin classical orbit is a balanced Grassmannian. The Albert orbit is $\mathbb{O}P^2$. We exclude all non-spin possibilities except $\mathbb{S}_+^3$ and $\mathbb{S}_+^4$.

Every positive-degree class in the mod-two cohomology of a product of spheres has square zero. In a complex Grassmannian of order at least three, the degree-two Schubert class instead satisfies $\sigma_1^2 = \sigma_2 + \sigma_{1,1} \neq 0$, omitting terms outside its rectangle. The same relation holds with degrees doubled for quaternionic Grassmannians; see [79, Section 2] for the Schubert ring and its quaternionic counterpart. These spaces are simply connected, so the nonzero square is also present in the universal cover. For the Albert orbit, $H^*(\mathbb{O}P^2; \mathbb{F}_2) = \mathbb{F}_2[u]/(u^3)$, $|u| = 8$, gives the same obstruction [48, Example 4.47].

For a balanced real Grassmannian of order $r \geq 6$, both ranks $k, r - k$ are at least three. Its simply connected oriented cover has

$$\pi_2(\mathrm{Gr}_k^+(\mathbb{R}^r)) = \ker((\mathbb{Z}/2)^2 \xrightarrow{+} \mathbb{Z}/2) = \mathbb{Z}/2.$$

This follows from the fibration with fiber $SO(k) \times SO(r - k)$ in $SO(r)$; the same exact sequence proves simple connectivity. Hurewicz then gives $H_2 = \mathbb{Z}/2$, which remains a direct summand of H_2 of the universal-cover product. The source has torsion-free H_2 .

At real order five the oriented cover is the complex quadric $Q^3 \subset \mathbb{C}P^4$: an oriented orthonormal pair (u, v) maps to $[u + iv]$, identifying its plane with an isotropic complex line. Here $H^2(Q^3; \mathbb{Q})$ is one-dimensional. Indeed the same homogeneous fibration has fundamental-group map $\mathbb{Z} \times \mathbb{Z}/2 \rightarrow \mathbb{Z}/2$, $(b, e) \mapsto b + e$; its kernel is $\mathbb{Z}$, so Hurewicz computes $H_2 = \mathbb{Z}$. The hyperplane class h satisfies $\int_{Q^3} h^3 = 2$, the degree of the quadric, and hence $h^2 \neq 0$. Any square-zero degree-two class on a product containing this quadric restricts to zero on that factor. Consequently square-zero classes cannot span the product's degree-two rational cohomology. They do span degree two for a product of spheres, a contradiction. This excludes real order five without a torsion calculation in its integral cohomology ring.

The remaining non-spin possibilities are real orders three and four, whose oriented covers are S^2 and $\mathrm{Gr}_2^+(\mathbb{R}^4) \cong S^2 \times S^2$. These identifications, and the classical-group fibrations used above, are compatible with the Grassmannian descriptions in [48, Examples 4.53–4.55]. All remaining universal-cover factors are therefore spheres or contractible real lines. The indecomposable quotient of their positive-degree cohomology records one generator in each sphere dimension. It follows that the target sphere dimensions, with multiplicity, equal the source dimensions at least two. Circle multiplicities agree because the finite covering induces an inclusion of finite index on fundamental groups, whose free-abelian ranks count circles.

Finally, in the integral cohomology of a product of simply connected spheres, let $u_1, \dots, u_b$ be the degree-two sphere generators. Then

$$\left(\sum_j b_j u_j \right)^2 = 2 \sum_{j < k} b_j b_k u_j u_k.$$

The primitive square-zero degree-two classes are exactly the $\pm u_j$. The ring isomorphism induced by the universal-cover diffeomorphism thus permutes these classes up to sign. Restricting to a source factor $S^{p_a} = S^2$ gives the claimed degree assignment. Changing the other fixed coordinates gives homotopic restrictions in the universal cover. Hence these degrees hold, up to orientation choices, on every a -sphere, the copy of S^{p_a} in M on which only x_a varies. $\square$

Lemma 11.3 (The two real exceptions violate the row identities). *Under the hypotheses of Lemma 11.2, neither $\mathbb{S}_+^3$ nor $\mathbb{S}_+^4$ occurs.*

Proof. Fix a row d , a point x_d , and a real matrix block i , and let $W = \mathrm{Ran} Y_i^d(x_d)$. Row d vanishes at $z = x_d$ on the whole cylinder of points x with this d th coordinate, so complementarity gives

$$W \subseteq \ker X_i(x) \quad \text{whenever } x_d \text{ is fixed.} \quad (11.3)$$

Suppose first that i is an order-four block. Its primal support has rank two, and the two S^2 rulings of its oriented Grassmannian are assigned to distinct source factors a, b . If $W \neq 0$ for a row $d \neq a$, vary only x_a . If $\dim W = 2$, the primal plane is constant. If $\dim W = 1$, the plane lies in $W^\perp$, a fixed three-space. The inclusion

$$\mathrm{Gr}_2^+(W^\perp) \cong S^2 \longrightarrow \mathrm{Gr}_2^+(\mathbb{R}^4) \cong S_+^2 \times S_-^2$$

has degree ± 1 in both projections. To see this, represent an oriented plane by a unit simple bivector and take its self-dual and anti-self-dual parts. For planes in a fixed three-space, the two parts are related by an orthogonal map. Every map factoring through this inclusion therefore has its two degrees equal up to sign. This contradicts the degrees $(\pm 1, 0)$ on the assigned a -sphere. Thus $Y_i^d \equiv 0$ for every $d \neq a$. Applying the same argument to the assigned b -sphere makes all rows zero, contradicting the positive complementary dual rank. No order-four block remains.

For an order-three block assigned to source a , the primal rank is one or two. A nonzero W for $d \neq a$ either fixes its kernel line (primal rank two), or confines its range line to $\mathbb{RP}(W^\perp)$ of dimension at most one (primal rank one). The lifted map on the assigned S^2 then has degree zero, contradicting Lemma 11.2. Hence this block is used only by row a .

We next show that no other non-ray block is used by row a . Every other order-three block has a distinct assigned S^2 , so the preceding argument applies. Each remaining block is spin. Its sphere target can be assigned to a source sphere of the same dimension with nonzero restriction degree: in each degree at least two choose a nonzero term in the determinant of the invertible matrix on cohomological indecomposables of the universal covers. For circles use the invertible rational matrix induced on fundamental groups. These assignments are bijections; the degree-two assignments agree with the signed permutation already proved, so none of these blocks is assigned to a . Fixing row a and varying such an assigned source, a nonzero dual factor would force the spin primal support to be constant. Indeed, a rank-two cone has only one primitive idempotent orthogonal to any fixed nonzero complementary element. This contradicts its nonzero degree or winding. All other non-ray factors in row a vanish.

Fix all primal coordinates except x_a . Row a is now a factorization of the S^2 ball slack through one $\mathbb{S}_+^3$ block and finitely many scalar rays. At each contact the ray mixed derivatives vanish, so this block carries the full two-dimensional mixed metric. Its projective support map is a local diffeomorphism and its ranks are complementary one and two. Proposition 7.5 excludes precisely this factorization, exchanging its two arguments if necessary. Thus no order-three block remains either. $\square$

Proof of Theorem 11.1. Lemma 11.2 gives the capacity inequality and covering. Its remaining conclusions and Lemma 11.3 leave only spin factors. The universal-cover cohomology and fundamental-group counts give exactly one target sphere for each source sphere, of the same dimension. The determinant assignment in the preceding proof matches each target to one source with nonzero restriction degree or winding. No splitting of the covering as a product map is required.

Let block i be assigned to source a , and fix a different row d . If $Y_i^d(x_a) \neq 0$, complementarity makes the primitive support of $X_i(x)$ constant as x_a varies, by the uniqueness of the complementary primitive ray in a spin cone. Its assigned degree would then be zero. Hence $Y_i^d \equiv 0$ for $d \neq a$, as asserted. The direct separate Lorentz representations give the converse. Since all capacities are integers, failure of the equality profile gives $\sum_i c_i \geq n + 1$. $\square$

If the scalar ray terms vanish identically, a shorter argument shows more: the joint support map at equality is injective. If two points have the same primal supports, the dual rows at each point annihilate the primal factors at the other, and the full row identities force all their coordinates to

agree. Scalar rays break this argument, which is why the proof above keeps a covering map until it has used the row identities on the cylinders.

For Hermitian factors over one fixed field, equality is possible only with order-two blocks, and every source sphere then has dimension 1, 2, or 4 over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, respectively. Over all symmetric cones, spin factors of any size are allowed, but a cap $m_i \leq d$ makes equality impossible if some source ball has $p_a + 2 > d$. Both statements concern globally C^1 full-row factorizations, not lifts whose selections exist only on generic patches.

The Grassmannian topology and its cohomology are classical. Our contribution is to combine them with the full row identities to classify capacity equality and exclude sharing of a factor between rows, also with finitely many scalar ray terms. Aubrun, La Piana, and Müller-Hermes [6, Theorem 1] classify cones for which every positive linear self-map factors through a finite direct sum of Lorentz cones. Their hypothesis concerns all positive linear maps; ours concerns one globally C^1 full-row slack factorization at equality in its mixed-curvature capacity bound. Our theorem does not assert their Lorentz-factorization property for general cones.

12 Rigidity from entire boundary fibers

The preceding saturation theorems assume globally C^1 primal and dual selections on the contact set. This section obtains a global support map without any selection. If the total nullity takes one common value at every point of every fiber of the projection over the product M of the boundary spheres, then all points of a fiber have the same factor supports, so the lift itself defines a support map on M . We prove the resulting rigidity theorem for products of balls. Its hypothesis concerns entire fibers, including points that no chosen regular selection reaches.

Theorem 12.1 (Constant-nullity rigidity). *Let $C = \prod_{a=1}^h \mathbb{B}^{s_a}$, where $s_a \geq 2$, and put $M = \prod_a \mathbb{S}^{p_a}$, $p_a = s_a - 1$, and $n = \sum_a p_a$. Consider an exact semialgebraic affine lift in relative-Slater form,*

$$C = \pi(\mathcal{L} \cap K), \quad K = \prod_i K_i,$$

where the K_i are simple symmetric cones, rays are allowed, and $\dim K_i \leq d$ with $d \geq 3$. Free variables have been eliminated as in Lemma 2.1. Suppose the full preimage

$$E = \{X \in \mathcal{L} \cap K : \pi(X) \in M\}$$

is compact. If an integer $L \geq 1$ satisfies

$$n = L(d - 2), \quad Z(X) := \sum_i (r_i - \text{rank}_J X_i) = L \quad (X \in E), \quad (12.1)$$

then

$$\{p_a : 1 \leq a \leq h\} = \underbrace{\{d - 2, \dots, d - 2\}}_{L \text{ copies}} \quad (12.2)$$

as multisets. Exactly L cone factors have positive nullity on E ; each is Q_d and has nullity one. All other factors are interior on E , and their dual coordinates vanish in every genuine pure-row certificate there. The profile (12.2) is attained by the separate direct Lorentz lift.

Proof. Projection $E \rightarrow M$ is a continuous surjection with convex, hence connected, fibers. It follows that E is connected. Indeed, a separation of the compact set E into two nonempty clopen subsets

would give two disjoint compact images covering M : a connected fiber must lie on one side of the separation. This contradicts connectedness of M .

Each nullity $z_i(X) = r_i - \text{rank}_J X_i$ is upper semicontinuous and integer-valued. Near any $X \in E$, all z_i are at most their values at X . Their constant sum forces equality term by term. Thus each z_i is locally constant, and connectedness makes it constant on E . If X, X' belong to one fiber, their midpoint belongs to the same fiber. In a symmetric cone the support of a positive sum is the join of the supports. Equality of the three ranks therefore implies

$$\text{supp } X_i = \text{supp } X'_i \quad \text{for every } i.$$

The support map is continuous on a fixed-rank stratum. Since a continuous surjection from a compact space to a Hausdorff space is a quotient map, these support maps descend to continuous maps on M .

Next choose primal and genuine pure-row dual selections on a common dense open semialgebraic contact set; Lemma 2.2 supplies such selections with locally constant ranks. Summing the pure rows gives the joint kernel $\sum_a (1 - \langle x_a, v_a \rangle)$, whose mixed contact form is the product metric of rank n . Write u_i and v_i for the primal and aggregate dual Jordan ranks at one such contact. The Peirce bound gives

$$\begin{aligned} n &\leq \sum_{i:r_i \geq 2} a_i u_i v_i \leq \sum_{i:r_i \geq 2} a_i (r_i - 1) z_i \leq \sum_{i:r_i \geq 2} (m_i - 2) z_i \\ &\leq (d - 2) \sum_{i:r_i \geq 2} z_i \leq (d - 2) Z = L(d - 2) = n, \end{aligned} \tag{12.3}$$

where $m_i = \dim K_i$. For the second inequality, complementarity gives $v_i \leq z_i$ and a nonzero primal rank is at most $r_i - 1$ whenever $z_i > 0$; if the primal vanishes, its mixed contribution is zero. The third inequality follows from

$$m_i - 2 - a_i(r_i - 1) = (r_i - 2) \left(1 + \frac{a_i(r_i - 1)}{2} \right).$$

Equality throughout (12.3) implies that rays have zero nullity and every positive-nullity non-ray factor has rank two, dimension d , and primal and dual ranks one. Consequently $z_i = 1$ in exactly L factors. Constancy of each z_i extends this conclusion to all of E . The other factors have interior primal coordinates; complementarity forces their dual coordinate to vanish for every genuine pure supporting row at every point of M .

The supports of the L positive-nullity Lorentz factors, descended to M , define

$$\Sigma : M \longrightarrow (\mathbb{S}^{d-2})^L.$$

This map is injective. If $\Sigma(x) = \Sigma(x')$, take any genuine certificate for the row $1 - \langle \pi_a(X), x_a \rangle$ and evaluate it at any lift over x' . Equal support rays in these L factors make their pairings zero, and all other dual coordinates are zero. Hence $1 - \langle x'_a, x_a \rangle = 0$ for every a , so $x' = x$. Source and target have equal dimension. Invariance of domain makes the image open, and compactness makes it closed. The target is connected; thus Σ is a homeomorphism. In the mod-two cohomology ring of a product of positive-dimensional spheres, the quotient of the positive-degree ideal by its square has one generator in each sphere dimension, with multiplicity. A graded ring isomorphism therefore gives (12.2), including any degree-one generators. Conversely, under that profile the lift $(1, x_a) \in Q_d$ has compact singleton fibers and one unit of nullity per source, as asserted. $\square$

For a single s -dimensional ball with $d < s + 1$ and $s - 1 = L(d - 2)$, Theorem 12.1 shows that hypothesis (12.1) cannot hold. For products of balls it is a sharp classification: only the profile of separate direct Lorentz factors is possible. The theorem assumes no continuous choice of primal lift or dual certificate.

Definition 12.2 (Accessibility by regular contacts). A point $X \in E$ is *contact-accessible* if there are contact points $x^{(j)} \rightarrow \pi(X)$ and local C^1 primal and genuine pure-row dual contact selections near $x^{(j)}$, on constant-rank strata, whose primal values $X^{(j)}$ satisfy $X^{(j)} \rightarrow X$.

Corollary 12.3 (Accessible equality cases). *Retain the lift and compactness assumptions of Theorem 12.1 and suppose $n = L(d - 2)$. If every $X \in E$ is contact-accessible and the restricted standard barrier satisfies $\nu_{\text{std},\text{slice}} \leq L$, then the profile (12.2) is necessary.*

Proof. Lemma 8.1 gives $Z(X) \leq \nu_{\text{std},\text{slice}} \leq L$ at every $X \in E$. At each approximating regular contact the curvature chain (12.3), without its final equalities, gives $Z(X^{(j)}) \geq L$. Upper semicontinuity then yields $Z(X) \geq \limsup_j Z(X^{(j)}) \geq L$. Thus $Z = L$ throughout E , and Theorem 12.1 applies. $\square$

A bound on the standard barrier parameter gives only the upper bound on nullity, not accessibility or the reverse inequality at fiber points where the projection is singular. A single global section controls the nullity only on its own image. Hence the corollary does not close the gaps between the lower and upper bounds on $\nu_{\text{std},\text{slice}}$ for arbitrary affine lifts in Section 8.

Part III

Formulation families and barrier comparisons

13 Whole rows, minimum dimension, and affine normalization

The differential bounds above allow one slack row to be split across many factors. We first study the more restrictive *whole-row* model, in which the source bodies are divided into groups and each group is assigned to one cone that factors all support rows of its bodies. There the linear rank of the slack functions bounds the cone dimension, the dimension of a contact face bounds the largest exposed-face dimension, and one cone attains both bounds, without any regularity assumption. We then prove two results that remain valid when rows split: rigidity at minimum total dimension, and a dimension bound for projections of a single normalized base of a product cone. Finally, we characterize when one level set of a linear functional normalizes a lift without changing its projection, so that every barrier on the lift induces a barrier on the body with no larger parameter.

13.1 Contact faces and exact grouping

Let $C_a \subset \mathbb{R}^{n_a}$ be compact full-dimensional convex bodies with $0 \in \text{int } C_a$, for $a \in [k]$. Write

$$E_a = \text{ext } C_a, \quad P_a = \text{ext } C_a^\circ, \quad F_a(z) = \{x \in C_a : \langle x, z \rangle = 1\}.$$

Define the maximum contact-face dimension and its codimension by

$$e_a = \max_{z \in P_a} \dim F_a(z), \quad \lambda_a = n_a - e_a.$$

Here face dimension is affine dimension, so $1 \leq \lambda_a \leq n_a$. The maximum exists because the nonempty set of possible dimensions is a finite set of integers. For a proper cone K define

$$f(K) = \max_{0 \neq y \in K^*} \dim \text{span}(K \cap y^\perp).$$

Thus $f(K)$ is the largest dimension of the linear span of a proper exposed face of K ; the zero face is allowed.

Theorem 13.1 (Whole-row dimension and face bounds). *Fix a nonempty group $G \subseteq [k]$. Suppose arbitrary maps*

$$A : \prod_{a \in G} E_a \longrightarrow K, \quad B^a : P_a \longrightarrow K^*$$

into a proper m -dimensional cone satisfy

$$\langle A(x), B^a(z) \rangle = 1 - \langle x_a, z \rangle \quad (a \in G, x \in \prod_{b \in G} E_b, z \in P_a). \quad (13.1)$$

Put $N_G = \sum_{a \in G} n_a$. Then

$$m \geq N_G + 1, \quad f(K) \geq N_G + 1 - \min_{a \in G} \lambda_a. \quad (13.2)$$

Both bounds are attained by the shared homogenization cone

$$H_G = \{(t, (y_a)_{a \in G}) : t \geq 0, y_a \in tC_a \text{ for all } a \in G\}. \quad (13.3)$$

No continuity or definability of the factor maps is required.

Proof. The sets E_a and P_a affinely span their respective spaces: their convex hulls are C_a and C_a° . Consequently the functions $1 - \langle x_a, z \rangle$, indexed by all a, z , span the constant and every coordinate of x . These $N_G + 1$ functions are linearly independent on $\prod_{a \in G} E_a$: fix all coordinates but one and use its affine span, successively for each block. By (13.1), their span has dimension at most m . This proves the first bound.

Fix a, z and restrict x to the contact cylinder

$$(F_a(z) \cap E_a) \times \prod_{b \in G \setminus \{a\}} E_b.$$

Its image under A lies in $K \cap B^a(z)^\perp$. The vector $B^a(z)$ is nonzero, since its slack is not identically zero. The extreme points of the exposed face $F_a(z)$ are precisely $F_a(z) \cap E_a$, and they affinely span that face. Thus the restricted affine functions have rank

$$1 + \dim F_a(z) + \sum_{b \in G \setminus \{a\}} n_b.$$

They are restrictions of linear functionals of $A(x)$, so this quantity cannot exceed $\dim \text{span}(K \cap B^a(z)^\perp) \leq f(K)$. Maximizing over a, z proves the second bound.

The cone H_G is proper and has dimension $N_G + 1$. The maps $A(x) = (1, x)$ and $B^a(z) = (1, 0, \dots, -z, \dots, 0)$ give the required factorization. To compute all its exposed faces, write $h_a(w) = \max_{x \in C_a} \langle x, w \rangle$. Its dual is

$$H_G^* = \{(\alpha, (w_a)_a) : \alpha \geq \sum_a h_a(-w_a)\}.$$

An interior dual point exposes only zero. At a nonzero dual boundary point let $J = \{a : w_a \neq 0\}$, which is nonempty. Nonnegativity of each support slack shows that the complementary face has dimension

$$1 + \sum_{a \notin J} n_a + \sum_{a \in J} \dim \arg \min_{x \in C_a} \langle x, w_a \rangle. \quad (13.4)$$

Every face $F_a(z)$ with z on the boundary of C_a° is an intersection of faces $F_a(z')$ with $z' \in P_a$. Indeed, express z as a finite convex combination of extreme points of the polar; equality in its supporting inequality forces equality in each inequality having positive weight. Each block $a \in J$ in (13.4) therefore contributes at most $n_a - \lambda_a$. A single block in J with smallest λ_a , using a normal attaining e_a , attains the claimed maximum. $\square$

By a *whole-row partition* we mean a partition $\mathcal{G}$ of $[k]$ with one cone assigned to each group, factoring all of that group's rows as in (13.1). In particular, no row is split between different cones. Allowing a group's primal map to depend on other source coordinates does not weaken the lower bound: fix those coordinates and apply the theorem.

Corollary 13.2 (Exact grouping under two caps). *Let $D \geq 1$ and $F \geq 0$ be integers, and allow all proper cones with dimension at most D and $f(K) \leq F$. A group G is feasible in the whole-row model if and only if*

$$\sum_{a \in G} n_a \leq D - 1, \quad \sum_{a \in G} n_a - \min_{a \in G} \lambda_a \leq F - 1. \quad (13.5)$$

The minimum factor count is the minimum number of parts satisfying these two inequalities. For k identical bodies with dimension n and contact codimension λ , put

$$Q = \min \left\{ \left\lfloor \frac{D-1}{n} \right\rfloor, \left\lfloor \frac{F+\lambda-1}{n} \right\rfloor \right\}.$$

If $Q \geq 1$, the exact factor count is $\lceil k/Q \rceil$; if $Q = 0$, no whole-row partition is feasible.

Proof. Necessity is (13.2), and H_G proves sufficiency for every feasible group. The identical-body inequalities reduce to $|G| \leq Q$. $\square$

For strictly convex bodies, $\lambda_a = n_a$. For polytopes, $\lambda_a = 1$, since some extreme polar normal exposes a facet. For Euclidean balls of arbitrary dimensions, the theorem gives exactly

$$\dim H_G = 1 + \sum_{a \in G} n_a, \quad f(H_G) = 1 + \sum_{a \in G} n_a - \min_{a \in G} n_a.$$

Thus the face bound counts the full coordinate spaces of the other bodies in the group, not only tangent directions detected by curvature.

Proposition 13.3 (Complexity of heterogeneous grouping). *For Euclidean balls with dimensions given as positive integers, deciding whether the whole-row grouping problem has a partition with at most a specified number of groups is strongly NP-complete. This holds already when $F = D - 1$.*

Proof. Under $F = D - 1$, the second inequality in (13.5) follows from the first because $\min_{a \in G} n_a \geq 1$. Consider a 3-PARTITION instance with integers $b_1, \dots, b_{3q}$, $B/4 < b_a < B/2$, and $\sum_a b_a = qB$ [40, Problem SP15]. Set $n_a = b_a$, $D = B + 1$, $F = B$, and ask for at most q groups. Their total dimension is qB , so every feasible group must have dimension exactly B and there must be exactly q groups. The size inequalities force exactly three items in each group. This is precisely the original 3-PARTITION question. The reduction preserves polynomial bounds on all integers, so on the polynomially bounded instances used for strong NP-completeness even explicit ball descriptions have polynomial size. A partition is a polynomially checkable certificate for the two integer inequalities, which also proves membership in NP. The hardness comes from the classical packing problem, not from testing a fixed group. $\square$

13.2 Rigidity at the absolute dimension minimum

The next result applies to arbitrary compact bodies and allows every row to split. A lift of minimum total dimension is, up to a linear isomorphism, the homogenization cone of the body, so it cannot improve any linear-isomorphism invariant.

Theorem 13.4 (Minimum-dimensional rigidity). *Let $C \subset \mathbb{R}^N$, $N \geq 1$, be a compact full-dimensional convex body. Every exact lift over a proper cone of total dimension M , allowing affine slices, affine projections, and free variables, has $M \geq N + 1$. At equality, after the reduction of Lemma 2.1, an invertible linear map carries the cone to*

$$\text{hom}(C) = \{(t, tx) : t \geq 0, x \in C\}$$

and its feasible slice to $\{1\} \times C$. The reduced projection is an affine isomorphism on that slice.

The same cone-isomorphism conclusion holds for any factorization of the full slack operator through a proper cone of dimension $N + 1$, without regularity assumptions on its factor maps. In particular, it holds when that cone is a product and rows split across its factors.

Proof. First consider an exact lift. Eliminate free variables and reduce to the minimal face, of dimension $M' \leq M$, with a Slater affine slice A . Its affine output has full rank on the direction space of A , so $\dim A \geq N$. If $M' \leq N$, these facts force A to be the entire cone space. Its image is an affine translate of a cone, which cannot be a full-dimensional bounded body. Thus $M' \geq N + 1$.

When $M = N + 1$, no face reduction is possible, and A must be an N -dimensional affine hyperplane: the full-space case gives the same contradiction. The output map on A is an affine

isomorphism, so $S = K \cap A$ is compact. The hyperplane does not contain zero, again because otherwise its feasible set and output would be translated cones. Write $A = \{z : \ell(z) = 1\}$. Fix $w \in A \cap \text{int } K$. If nonzero $v \in K$ satisfies $\ell(v) = 0$, then $w + tv \in S$ for all $t \geq 0$, contradicting compactness. If $\ell(v) < 0$, the nonzero vector $v - \ell(v)w \in K \cap \ker \ell$ gives the same contradiction; pointedness ensures this vector is nonzero. Therefore ℓ is strictly positive on $K \setminus \{0\}$, and S is a compact base. The affine output on A extends to a linear map $\tilde{\pi}$ on the cone space because A is a hyperplane not containing zero. The map $Tz = (\ell(z), \tilde{\pi}(z))$ is invertible: its kernel lies in the direction space of A , where the output is injective. Writing each nonzero $z \in K$ uniquely as $z = \ell(z)s$, $s \in S$, proves both claims.

For the factorization assertion, translate C so $0 \in \text{int } C$. The canonical vectors $U(x) = (1, x)$ for $x \in \text{ext } C$ and $V(z) = (1, -z)$ for $z \in \text{ext } C^\circ$ each span $\mathbb{R}^{N+1}$. The slack has rank $N + 1$, by the affine-function argument in Theorem 13.1. If $A(x), B(z)$ give another factorization in that dimension, their images span as well. A linear relation among the $U(x)$ is equivalent, by pairing with the spanning dual family, to the same relation among the $A(x)$. Hence an invertible T satisfies $A = TU$ and $B = T^{-\top}V$. Taking conic hulls gives

$$T \text{hom}(C) \subseteq K, \quad T^{-\top} \text{hom}(C)^* \subseteq K^*.$$

The extreme points of the body and its polar generate their respective compact bases, so these are the full homogenization cones. Dualizing the second inclusion reverses the first, and proves equality. $\square$

The rows in Theorem 13.1 are exactly the slack rows of $\prod_{a \in G} C_a$ indexed by the extreme points of its polar. Indeed, its polar is $\{(z_a) : \sum_a h_a(z_a) \leq 1\}$. A nonzero extreme point must have one supported coordinate block, since two positive support-function contributions give a nontrivial convex decomposition into single-block points. Within that block it must be an extreme point of C_a° . Conversely each such single-block point is extreme. Thus equality $m = N_G + 1$ in that theorem forces K to be linearly isomorphic to H_G . For a fixed partition into g groups, every whole-row formulation of total dimension $\sum_a n_a + g$ therefore uses precisely those shared homogenization cones up to linear changes of coordinates. Barrier parameters are preserved by such changes. In particular, the exact shared spectral-cone values of Theorem 17.3, including its Euclidean vector-ball case, apply to every such minimum-dimensional formulation.

Corollary 13.5 (Connected extreme sets under a strict factor cap). *Suppose $\text{ext } C$ is connected. A full slack factorization through $K = \prod_{i=1}^L K_i$, with every K_i nonzero and proper, can have total dimension $N + 1$ only when $L = 1$. Consequently, if $\dim K_i \leq d < N + 1$, then*

$$M \geq N + 2, \quad L \geq \max \left\{ 2, \left\lceil \frac{N + 2}{d} \right\rceil \right\}. \quad (13.6)$$

These conclusions also hold for exact lifts after the reduction of Lemma 2.1, and allow arbitrary row splitting.

Proof. Normalize extreme rays by any strictly positive linear functional. For a product of L nonzero proper cones, the extreme rays are supported in exactly one factor; the L factor-supported pieces are nonempty and both open and closed in the normalized extreme-ray space. For $\text{hom}(C)$ that space is homeomorphic to $\text{ext } C$. Theorem 13.4 therefore excludes $L > 1$ at equality. A single factor under the strict cap cannot supply dimension $N + 1$. Integrality gives $M \geq N + 2$, and $M \leq Ld$ gives the count bound. For an unreduced exact lift with $M = N + 1$, the theorem already ensures that no reduction occurs. $\square$

The corollary excludes only $M = N + 1$; it does not force one extra coordinate per factor. The balance slices in Theorem 15.1 have connected extreme sets and $M = N + 2$ for arbitrarily many essential factors, so $M \geq N + L$ fails in general.

13.3 What one normalization equation does imply

Proposition 13.6 (An additive bound for one normalized base). *Let $K = \prod_{i=1}^L K_i$, where K_i is nonzero and proper of dimension m_i , and choose $\ell_i \in \text{int } K_i^*$. Let*

$$B = \{z \in K : \sum_i \ell_i(z_i) = 1\}, \quad C = \pi(B),$$

where π is affine and $\dim \text{aff } C = N$. If $\text{ext } C$ is connected, then $\sum_i m_i \geq N + L$. No irredundancy assumption is needed.

Proof. Let B_i be the compact normalized base of factor i , embedded in the product with all other coordinates zero, and put $C_i = \pi(B_i)$. Then $C = \text{conv } \bigcup_i C_i$, and $\dim \text{aff } C_i \leq m_i - 1$. Every extreme point of C belongs to some C_i , because any finite convex representation of an extreme point uses only that point. The nonempty sets $E_i = C_i \cap \text{ext } C$ form a finite relatively closed cover of the connected extreme set. Their intersection graph is connected: a graph separation would give a separation into two disjoint nonempty relatively closed finite unions. Along a spanning tree, the affine hull of each newly added E_i intersects the previous affine hull. The dimension of their combined affine hull is therefore at most the sum of their dimensions. With $I = \{i : E_i \neq \emptyset\}$ this gives

$$N = \dim \text{aff}(\text{ext } C) \leq \sum_{i \in I} \dim \text{aff } E_i \leq \sum_{i \in I} (m_i - 1) \leq \sum_{i=1}^L (m_i - 1).$$

The final inequality uses $m_i \geq 1$. Rearrangement proves the result. $\square$

A second affine equation can create extreme points with nonzero coordinates in several factors. The sets C_i in the proof then need not cover the extreme points of the smaller slice, so the proposition does not contradict the balance-slice examples.

13.4 Recession normalization and induced barriers

The projection lemma in Section 9 requires bounded fibers. The next criterion states exactly when one level set of a linear functional provides them without changing the projection. All interiors in the statement are relative to affine hulls.

Proposition 13.7 (A criterion for barrier-preserving normalization). *Let D be a nonempty closed convex set containing no affine line, and let an affine map π satisfy $\pi(D) = C$, where C is a compact convex body. Write $R = \text{rec } D$ for its recession cone. If $R \neq \{0\}$, choose a linear functional a strictly positive on $R \setminus \{0\}$, and define*

$$m_a(x) = \min\{a(z) : z \in D, \pi(z) = x\}.$$

Every minimum is finite and attained. For any real M ,

$$\pi(D \cap \{a = M\}) = C \iff M \geq \sup_{x \in C} m_a(x). \quad (13.7)$$

Whenever this holds, the level section is compact. If the supremum is finite, every self-concordant barrier on $\text{ri } D$ induces a barrier on $\text{int } C$ with no larger parameter. If $R = \{0\}$, the same parameter conclusion holds without taking a level section.

Proof. Compactness of C implies $R \subseteq \ker \pi_0$, where π_0 is the linear part of π . The cone R is pointed because D contains no line, so a strictly positive a exists. Every nonempty sublevel $D \cap \{a \leq b\}$ is bounded. Otherwise normalize differences of an unbounded sequence from a fixed point of D and pass to a unit limit r . Closedness and convexity imply $r \in R$, while boundedness above of a on the sequence gives $a(r) \leq 0$, a contradiction. Each fiber is closed; minimizing a on a compact nonempty sublevel proves attainment and finiteness of $m_a(x)$.

Fix $0 \neq r_0 \in R$. Adding a nonnegative multiple of r_0 to a minimizer in a fiber preserves that fiber and reaches every larger a -value. This proves (13.7). The level section is closed and bounded by the same sublevel argument. If the supremum is finite, choose M greater than it and greater than $a(z^0)$ for some $z^0 \in \text{ri } D$. Adding a suitable multiple of r_0 to z^0 gives a point of the level in $\text{ri } D$. Indeed, a relative ball around z^0 remains in D after translating by a recession direction. Thus the affine restriction of the original barrier is a barrier on the relative interior of the compact level section, with the same parameter bound. Lemma 9.5 then gives a barrier on C . If $R = \{0\}$, the closed convex set D is bounded by the same normalized-sequence argument, and that lemma applies directly. $\square$

Corollary 13.8 (Arbitrary line-free lifts of a simple-vertex polytope). *Let C be an N -dimensional compact polytope with a simple vertex. Every self-concordant barrier on the relative interior of every closed line-free affine lift of C has parameter at least N , even if its fibers are unbounded. The minimum over all such lifts and barriers is N .*

Proof. Choose a lift $z_v \in D$ of each vertex v of C . For a convex combination $x = \sum_v \theta_v v$, its lifted combination belongs to D and therefore

$$m_a(x) \leq \sum_v \theta_v a(z_v) \leq \max_v a(z_v).$$

By Proposition 13.7, the barrier induces a barrier on C with no larger parameter. At a simple vertex, the classical independent-facet lower bound gives $\nu \geq N$ [70, Proposition 2.3.6]. The universal barrier on C attains the bound N [60, Theorem 2]; it may also be used on the canonical homogenization slice, which is affinely isomorphic to C . $\square$

The simple-vertex hypothesis is essential to this argument. Choosing N independent normals at a nonsimple vertex does not reproduce its tangent cone. At e_1 of the three-dimensional cross-polytope, the four active normals are $(1, \pm 1, \pm 1)$. The three with sign pairs $(+, +), (+, -), (-, +)$ are independent, but $(-1, -1, -1)$ satisfies their three strict tangent inequalities and violates the fourth. The cited simple-vertex lower bound cannot be applied after deleting that facet.

Example 13.9 (Compact projection need not admit affine normalization). Let D be the positive-semidefinite lift of the disk in Example 8.8, with coordinates (x_1, x_2, u, z) . Its recession cone is exactly the positive z -ray. Indeed, compactness of the projected disk makes every recession direction have zero x -coordinates. The second block then requires its u -coordinate to be nonnegative, while the last homogeneous block $\begin{pmatrix} 0 & \dot{u} \\ \dot{u} & \dot{z} \end{pmatrix}$ forces $\dot{u} = 0$ and $\dot{z} \geq 0$. Conversely every such direction is a recession direction. If an affine subspace removed this ray from a nonempty section of D , that closed convex section would have zero recession cone and hence be bounded. It could not project onto the entire disk, since every completion of the boundary points approaching $(1, 0)$ in Example 8.8 has $z \rightarrow \infty$. Thus no affine restriction that preserves the projection removes the recession ray, even for this semialgebraic Slater lift over three 2×2 positive-semidefinite cones. The example rules out only normalization by an affine restriction; it does not show that projecting a general lift with unbounded fibers can increase the barrier parameter.

The slack-factorization framework used in this section is classical [43]. The minimum-dimension rigidity theorem is an elementary consequence of it, and the normalized-base dimension bound uses only a finite connected cover. The barrier bounds in Corollary 13.8 are also classical; the corollary adds that the lower bound holds on any given closed line-free lift, by Proposition 13.7 and Lemma 9.5. The whole-row grouping formulas apply only to the whole-row model; they do not claim optimality among lifts in which rows may split.

14 Complementary faces control sharing between source rows

Unlike Section 13, we now allow every row to split over several blocks, but require globally C^1 factor maps for the full slack. Then the exposed-face dimensions of a block limit how many source rows can be active in it, that is, have a nonzero mixed derivative there. Globally, the exact Lorentz bounds of Theorem 6.2 extend to every cone whose nonzero proper exposed faces are rays, even with nonsmooth boundaries.

Let $M = (\mathbb{S}^p)^k$, $p = s - 1 \geq 2$, and suppose

$$1 - \langle x_a, z \rangle = \sum_{i=1}^L \langle A_i(x), B_i^a(z) \rangle, \quad x \in M, \quad z \in \mathbb{S}^p, \quad a \in [k], \quad (14.1)$$

where $A_i : M \rightarrow K_i$ and $B_i^a : \mathbb{S}^p \rightarrow K_i^*$ are C^1 and the cones are proper of dimensions $3 \leq m_i \leq d$, for an integer $d \geq 3$. Set

$$r_i = m_i - 2, \quad c = d - 2, \quad f_i = \max_{\substack{0 \neq b \in \partial K_i^* \\ K_i \cap b^\perp \neq \{0\}}} \dim(K_i \cap b^\perp), \quad \bar{f}_i = \min(f_i, k).$$

We call r_i the capacity of block i ; block i is active on row a at x when $t_i^a(x) > 0$, with t_i^a as in Theorem 14.1. Here f_i is a maximum of face dimensions, not of numbers of extreme rays; below it may be replaced by any upper bound on the dimensions of the nonzero exposed faces met by the dual factors. Ray factors may be added: their mixed forms vanish, so they contribute nothing to the bounds.

Theorem 14.1 (Local face packing). *At $x \in M$ put*

$$H_i^a(u, v) = -\langle D_a A_i(x)[u], DB_i^a(x_a)[v] \rangle, \quad t_i^a = \text{rank } H_i^a, \quad I_i = \{a : t_i^a > 0\}.$$

Then $t_i^a \leq r_i$, $\sum_i t_i^a \geq p$, and

$$\sum_{a \in I_i} t_i^a \leq r_i, \quad 1 + \sum_{b \in I_i \setminus \{a\}} t_i^b \leq \dim(K_i \cap B_i^a(x_a)^\perp) \quad (a \in I_i). \quad (14.2)$$

In particular $|I_i| \leq \bar{f}_i$.

Proof. Contact and nonnegativity make every pairing $\langle A_i(x), B_i^a(x_a) \rangle$ zero. On the cylinder $x_a = u$, the function $z \mapsto \langle A_i(x), B_i^a(z) \rangle$ has a minimum at u . Consequently $\langle A_i(x), DB_i^a(u)[v] \rangle = 0$ on that entire cylinder. Differentiating in a different source direction, with the dual derivative held fixed, gives

$$\langle D_b A_i(x)[w], DB_i^a(x_a)[v] \rangle = 0 \quad (b \neq a).$$

This uses only first derivatives. The ray-quotient contact argument of Lemma 3.1 gives $t_i^a \leq m_i - 2$; at a zero factor its derivative is zero by pointedness. Mixed differentiation of the full row gives $\sum_i H_i^a = g_{\mathbb{S}^p}$, hence $\sum_i t_i^a \geq p$.

Fix one block with an active row, write $A = A_i(x) \neq 0$, and let $\Lambda_a(v)$ be the functional $w \mapsto \langle v, DB_i^a(x_a)[w] \rangle$. Choose $E_a \subset \text{im } DA_i$ of dimension t_i^a on which Λ_a is injective. The identities just proved imply $\Lambda_a A = 0$ and $\Lambda_a E_b = 0$ for $b \neq a$. Applying each Λ_a to a linear dependence proves that

$$\mathbb{R}A \oplus \bigoplus_{a \in I_i} E_a$$

is a direct sum. Fix any active row a_0 and put $b = B_i^{a_0}(x_{a_0}) \neq 0$. The nonnegative function $x' \mapsto \langle A_i(x'), b \rangle$ vanishes at x , so $\text{im } DA_i(x) \subset b^\perp$. Since $A \in b^\perp$, the entire direct sum lies in this hyperplane of dimension $m_i - 1$. Thus $1 + \sum_a t_i^a \leq m_i - 1$, proving the first inequality. When there is no active row that inequality is immediate. On the contact cylinder for row a , the primal factor lies in $F_a = K_i \cap B_i^a(x_a)^\perp$. Thus $E_b \subset \text{span } F_a$ for $b \neq a$, and $A \in F_a$. Their direct sum proves the second inequality. Since each active rank is at least one, it also gives $|I_i| \leq f_i$. $\square$

Theorem 14.2 (Global face budgets). *Define*

$$h = \begin{cases} \lceil (p+1)/c \rceil, & c < p, \\ 1, & c \geq p, \end{cases} \quad \tau = \begin{cases} p+1, & c < p, \\ p, & c \geq p, \end{cases} \quad R = \sum_i r_i, \quad K = kp + \mathbf{1}_{\{c < p\}}.$$

Every factorization (14.1) satisfies

$$\sum_i \bar{f}_i \geq kh, \quad \sum_i \bar{f}_i r_i \geq k\tau, \quad R \geq K. \quad (14.3)$$

If $f_i \leq f$ for a common $f \geq 1$, set

$$R_* = \max\{\lceil k\tau/f \rceil, K\}, \quad L_* = \max\{\lceil kh/f \rceil, \lceil R_*/c \rceil\}.$$

Then

$$L \geq L_*, \quad R \geq R_*, \quad D := \sum_i m_i \geq R_* + 2L_*. \quad (14.4)$$

Every self-concordant barrier on the ambient product has parameter at least $2L_*$, even if it couples the factors or is not logarithmically homogeneous.

Proof. Write $N_a(x) = |\{i : t_i^a(x) > 0\}|$ and $C_a(x) = \sum_{i: t_i^a(x) > 0} r_i$. Pointwise, $N_a \geq \lceil p/c \rceil$ and $C_a \geq p$. Counting incidences gives

$$\sum_a N_a \leq \sum_i \bar{f}_i, \quad \sum_a C_a \leq \sum_i \bar{f}_i r_i.$$

Summing (14.2) gives $kp \leq R$. To exclude equality when $c < p$, aggregate the labelled dual maps:

$$\widehat{B}_i(z_1, \dots, z_k) = \sum_a B_i^a(z_a), \quad \sum_i \langle A_i(x), \widehat{B}_i(z) \rangle = k - \sum_a \langle x_a, z_a \rangle.$$

These are C^1 cone-valued maps on M , with contact map $z = x$ and nondegenerate mixed pairing equal to the product spherical metric. At $R = kp$, Theorem 10.1 therefore forces the positive capacities to be exactly k copies of p . This contradicts $r_i \leq c < p$. Integrality gives $R \geq kp + 1$ in this case. Zero-capacity ray factors do not affect this argument. These observations prove the last bound in (14.3), and all its bounds when $c \geq p$.

Suppose $c < p$. If $\sum_i \bar{f}_i r_i < k(p+1)$, every point belongs to some $Z_a = \{C_a = p\}$. Each Z_a is closed because activity is open and $C_a \geq p$ everywhere. For a label set J , let $E_{a,J}$ be the set of points of Z_a where the active labels for row a are exactly J . These finitely many sets are clopen in

Z_a , hence compact: an active label stays active nearby, while a new one would raise C_a above p . On such a piece $\sum_{i \in J} r_i = p$. Normalize the nonzero dual factors B_i^a using an interior primal vector to obtain maps into the boundaries of compact dual-cone bases. These maps are defined near the projection of $E_{a,J}$ onto its source sphere. Indeed fix the other source coordinates at any point of the piece; nearby contact primal factors remain nonzero and force the corresponding dual factors to remain on the nonzero boundary.

The joint derivative of these normalized dual maps is injective at every projected point. If a tangent vector is killed by all normalized derivatives, each unnormalized derivative is radial and therefore annihilates every primal contact derivative. The identity $\sum_i H_i^a = g_{\mathbb{S}^p}$ then kills that tangent vector itself. Injectivity persists on an open neighborhood. Lemma 10.3 shows that this neighborhood is a proper open subset of $\mathbb{S}^p$, since every $r_i < p$ and their sum is p .

For a fixed a , choose pairwise disjoint open neighborhoods of the finitely many compact pieces $E_{a,J}$, each inside the inverse image of its proper source neighborhood. The pulled-back top class u_a of the a th source sphere vanishes on their union U_a . The U_a cover M , so the relative cup-product argument of Section 6 contradicts $u_1 \cdots u_k \neq 0$. Thus $\sum_i \bar{f}_i r_i \geq k(p+1)$. For the bound on $\sum_i \bar{f}_i$, if $c \nmid p$ the pointwise count already equals h . Otherwise write $p = tc$. If $\sum_i \bar{f}_i < k(t+1)$, then M is covered by the loci where some source has exactly t active labels. Every label there must have capacity c , and the same compact-piece argument applies. This proves the count bound.

Uniform face caps and $R \geq K$ imply $R \geq R_*$ and $L \geq \lceil kh/f \rceil$. Since $R \leq cL$, they also imply $L \geq \lceil R_*/c \rceil$. The dimension bound follows from $D = R + 2L$. Finally each proper cone of dimension at least two has a two-dimensional linear section through an interior point that is a pointed wedge. Its relative boundary lies in the cone boundary. The product sections form a $2L$ -orthant. Restriction of any ambient barrier and the orthant lower bound give $\nu \geq 2L \geq 2L_*$. $\square$

Corollary 14.3 (Exact minima for ray-exposed cones). *Suppose every nonzero proper exposed face of each K_i is a ray. Over all dictionaries of such cones of dimension at most d , the minima are attained simultaneously and equal*

$$L = kh, \quad R = k\tau, \quad D = k\tau + 2kh, \quad \nu_{\text{ambient}} = 2kh,$$

where ν_{ambient} is the least barrier parameter on the ambient product. They are attained by Lorentz cones and the grouped constructions of Theorem 6.2. A fixed dictionary need not contain these attaining cones. We do not claim exact values for $f > 1$.

The face inequalities also limit a single shared block. If block i is active on $q_i \geq 2$ rows and $T_i = \sum_a t_i^a$, summing its face inequalities gives

$$(q_i - 1)T_i \leq q_i(f_i - 1), \quad T_i \leq \min\{r_i, 2(f_i - 1)\}. \quad (14.5)$$

Thus a small exposed face can further restrict a block's total mixed rank when it serves several rows; sharing never increases that rank beyond its capacity r_i . For a real PSD factor whose contact dual has nullity h_{ia} , the face inequality reads $1 + \sum_{b \neq a} t_i^b \leq h_{ia}(h_{ia} + 1)/2$; the sharper one-row bound of Lemma 3.3 holds at the same time.

14.1 A sharp model for shared scalar channels

The next cone shows how split rows can share one block. Define

$$P_q = \{(t, u, w) \in \mathbb{R} \times \mathbb{R}^q \times \mathbb{R}^q : t \geq 0, w_a \geq 0, tw_a \geq u_a^2\}.$$

This closed proper cone has dimension $2q + 1$. The polynomial factors

$$A(v) = (1, v, v^{\circ 2}), \quad B^a(z) = (z^2/2, -ze_a, e_a/2) \quad (14.6)$$

give $\langle A(v), B^a(z) \rangle = (v_a - z)^2/2$ on $[-1, 1]^q \times [-1, 1]$. Dual feasibility follows directly by completing a square when $t > 0$ and by continuity when $t = 0$.

Proposition 14.4 (The shared-square cone). *The cone P_q has maximum nonzero proper exposed-face dimension $2q - 1$. Its dimension $2q + 1$ and this face dimension are the smallest possible for a one-block factorization of all the kernels in (14.6). Its optimal self-concordant barrier parameter is $q + 1$, attained by*

$$F_q(t, u, w) = - \sum_{a=1}^q \log(tw_a - u_a^2) + (q - 1) \log t. \quad (14.7)$$

For a partition of k source balls into g groups of sizes q_ℓ , using these cones coordinate by coordinate gives an analytic factorization of the full slack with sg factors of total dimension $s(2k + g)$, maximum block dimension $2 \max q_\ell + 1$, maximum exposed-face dimension $2 \max q_\ell - 1$, and exact ambient parameter $s(k + g)$.

Proof. The dual cone consists of $\gamma \geq 0$, $\beta_a = 0$ when $\gamma_a = 0$, and $\alpha \geq \sum_{\gamma_a > 0} \beta_a^2/(4\gamma_a)$. Let $J = \{a : \gamma_a > 0\}$. In the equality case for a nonzero dual vector, J is nonempty and the contact equations fix $u_a = -t\beta_a/(2\gamma_a)$ and $w_a = t\beta_a^2/(4\gamma_a^2)$ for $a \in J$. The other coordinates range over a copy of $P_{q-|J|}$, so the face dimension is $1 + 2(q - |J|)$. In the strict case $t = 0$, hence $u = 0$; only w_a for $a \notin J$ remain, giving dimension $q - |J|$. These cases give the maximum $2q - 1$, including $q = 1$.

The row functions span the linearly independent family $1, v_a, v_a^2$ of size $2q + 1$, proving the dimension lower bound. On a contact cylinder $v_a = z$, the other rows linearly recover $1, v_b, v_b^2$ for $b \neq a$; a nonzero constant is also recovered from a different dual argument in row a . The contact face must therefore span at least $1 + 2(q - 1)$ dimensions.

The cone P_q is the PSD-completable partial-matrix cone on a star: the center diagonal is t , its edge entries are u_a , and the leaf diagonals are w_a . For $t > 0$ its maximum-determinant completion has determinant $t \prod_a (w_a - u_a^2/t)$. The classical completion barrier proved in Section 17 consequently is (14.7), of parameter $q + 1$. Equivalently it is the Legendre barrier of the sparse arrow-matrix log determinant. Restricting any barrier to $u = 0$ gives a $(q + 1)$ -orthant with boundary on the original boundary, so the parameter is optimal against arbitrary barriers.

For source a in a group use $v_b = x_{b,j}$ and $z = z_j$ in coordinate j . Summing gives $\frac{1}{2} \sum_j (x_{a,j} - z_j)^2 = 1 - \langle x_a, z \rangle$ on the unit spheres. Summing dimensions and parameters gives the stated counts; the product orthant sections prove the ambient lower bound even for coupled barriers. $\square$

For comparison, the rank-one real PSD factors $A(v) = (1, v)(1, v)^T$ and $B^a(z) = (e_a - ze_0)(e_a - ze_0)^T/2$ give the same scalar kernels, but expose a PSD face of dimension $q(q + 1)/2$. A whole-row version uses $v = (1, x_1, \dots, x_q)$ and sums those dual outer products over the s coordinates of a row; its dual rank is s , its nullity is $1 + (q - 1)s$, and its contact face has dimension $\binom{2+(q-1)s}{2}$. These PSD constructions are not claimed to be optimal among homogeneous cones.

The homogenization of a group of q whole ball rows, by contrast, has dimension $qs + 1$ and maximum exposed-face dimension $1 + (q - 1)s$ by Theorem 13.1. Its exact ambient parameter is $q + 1$, as the vector-row specialization of Theorem 17.3. For k rows in g groups the ambient parameter is $k + g$, while fixing each scale to one gives exactly $-\sum_a \log(1 - \|x_a\|^2)$ with intrinsic parameter k , independently of grouping. Thus factor sharing, exposed-face dimension, and the barrier parameter on the actual slice measure different properties of a formulation.

The correspondence between full slack factorizations and cone lifts is classical [43]. Our bounds are the direct-sum inequalities (14.2) across contact cylinders and the global budgets (14.3); both assume a global C^1 choice of factors for every labelled row. The barriers in the examples are classical spectral and PSD-completion barriers. Neither the face bounds nor a small ambient count alone implies an unrestricted algorithmic speedup.

15 Balance slices with sharp dimension and barrier bounds

By Corollary 13.5, a lift of an N -dimensional body with a connected set of extreme points, through factors of dimension less than $N + 1$, has total dimension at least $N + 2$. We show that connectivity forces only this one extra coordinate, not one per factor: slices of products of arbitrarily many symmetric cones attain the bound. We also compute their optimal barrier parameters exactly. Comparing two such families shows that factor count and total cone dimension do not determine the barrier parameter.

Let V_i , $1 \leq i \leq L$, be simple Euclidean Jordan algebras of rank $r_i \geq 2$, with cones of squares K_i , identities e_i , and Jordan frames $(c_{ij})_{j=1}^{r_i}$. Here $L \geq 2$. Choose a nonzero off-diagonal Peirce element $A_i \in V_i(c_{i1}, c_{i2})$, and put

$$\ell_i(X_i) = \langle A_i, X_i \rangle, \quad M = \sum_i \dim V_i, \quad \rho = \sum_i r_i.$$

Define the balance cone and its trace-one base by

$$\widehat{Z} = \left\{ X \in \prod_i K_i : \sum_i \ell_i(X_i) = 0 \right\}, \quad Z = \left\{ X \in \widehat{Z} : \sum_i \operatorname{tr} X_i = 1 \right\}. \quad (15.1)$$

All interiors and barriers below are taken in the respective affine hulls.

Theorem 15.1 (Balance slices). *The set Z is a compact convex body of dimension $M - 2$, with a path-connected extreme-point set. Each displayed factor is indecomposable and appears in a nonconstant supporting slack, and its membership constraint is essential: without $X_i \in K_i$ the feasible set is unbounded. For arbitrary self-concordant barriers, including barriers coupling all blocks,*

$$\vartheta_{\text{opt}}(\operatorname{ri} \widehat{Z}) = \rho, \quad \vartheta_{\text{opt}}(\operatorname{ri} Z) = \rho - 1. \quad (15.2)$$

The product Jordan log-determinant, restricted to these domains, attains both values. Every closed convex affine lift of Z with nonempty bounded fibers has intrinsic barrier parameter at least $\rho - 1$. The factors may be any mixture of classical Hermitian, spin, and Albert algebras.

The elementary minimal-face criterion used below is classical in finite dimensions; see also the general injectivity formulation of Henrion, Kružík, and Weis [52, Theorem 2.6]. The ambient symmetric-cone barrier theory is standard [31, 47]. The theorem contributes the explicit balance construction, its connected extreme set, and the exact parameters on its two sections; it contains no new general extremality or barrier theorem.

Proof. The point $X_i^0 = e_i/\rho$ is strictly feasible. The total trace and balance functionals are independent, so $\dim Z = M - 2$. Positivity and fixed total trace imply compactness.

We first classify and connect the extreme points. If a point X belongs to an affine section $\{X \in K : \mathcal{A}X = b\}$, it is extreme precisely when

$$\ker \mathcal{A} \cap \operatorname{span} F_K(X) = \{0\}. \quad (15.3)$$

Indeed, X is relatively interior to its minimal face, so a nonzero vector in this intersection gives a feasible segment through X . Conversely, the endpoints of any such segment lie in the minimal face. A rank- $q \geq 2$ face in a simple Jordan cone has dimension $q + a_i q(q-1)/2 \geq 3$, where a_i is the Peirce multiplicity. There are only two affine equations in (15.1). Thus each extreme point has one or two nonzero blocks, each on a primitive ray. Normalize its primitive idempotents to trace one, write $h_i(P) = \ell_i(P)$, and call this value the height of P . The constraint columns are $(1, h_i(P))$. The exact possibilities are consequently

- one block P with $h_i(P) = 0$;
- two blocks wP and $(1-w)Q$ with $h_i(P)h_j(Q) < 0$, where $w = -h_j(Q)/(h_i(P) - h_j(Q))$.

In the second case $i \neq j$. These conditions also suffice by (15.3).

We next describe the sign regions of the height h_i . By a Jordan automorphism and positive rescaling, its defining element becomes $c_{i1} - c_{i2}$; this follows from the spectral theorem and transitivity on Jordan frames. For a classical factor $H_r(\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, primitive idempotents are $uu^*/\|u\|^2$, and their height is proportional to

$$\frac{|u_1|^2 - |u_2|^2}{\|u\|^2}.$$

The positive region, in the chart $u_1 = 1$, is $\{u_2 : |u_2| < 1\} \times \mathbb{F}^{r-2}$ and is path-connected; the negative region is obtained by interchanging coordinates. If $r \geq 3$, the zero region is path-connected: for $u_1 \neq 0$ it is $\mathbb{S}^{\dim_{\mathbb{R}} \mathbb{F}-1} \times \mathbb{F}^{r-2}$, and every such point can be joined, by letting a third coordinate tend to infinity, to a fixed ray in the subspace $u_1 = u_2 = 0$. That subspace has connected projective space (or is one point). This also joins the two chart components when $\mathbb{F} = \mathbb{R}$. For $r = 2$ the zero region is $\mathbb{S}^{\dim_{\mathbb{R}} \mathbb{F}-1}$; only the real case has two points. Every zero ray admits a path entering either strict sign region: vary $|u_2|$ in the first chart, or introduce a first or second coordinate at a ray with $u_1 = u_2 = 0$. For a spin factor, primitive idempotents form a sphere and height is a nonzero linear coordinate. Its sign regions are hemispheres, its zero region is an equator, and the same assertions hold, with a two-point equator in the three-dimensional cone.

We check the Albert case directly in coordinates, not by analogy with associative projective spaces. The chart of the octonionic projective plane with first diagonal coordinate positive is parametrized by $(x, y) \in \mathbb{O}^2$. Its primitive idempotent is

$$P(x, y) = \frac{1}{1 + |x|^2 + |y|^2} \begin{pmatrix} 1 & \bar{x} & \bar{y} \\ x & |x|^2 & x\bar{y} \\ y & y\bar{x} & |y|^2 \end{pmatrix}, \quad h(P) = \frac{1 - |x|^2}{1 + |x|^2 + |y|^2}. \quad (15.4)$$

These are the usual reduced homogeneous coordinates [51, Section 3]; the primitive-idempotent model is described in [9]. The two octonions x, y generate an associative subalgebra, which directly verifies $P^2 = P$ and $\text{tr } P = 1$. The positive region is $\mathbb{B}^8 \times \mathbb{R}^8$, and permutation of the first two frame idempotents identifies the negative region with it. The zero region in this chart is $\mathbb{S}^7 \times \mathbb{R}^8$. Outside the chart, zero height forces both first diagonal coordinates to vanish, leaving only c_{i3} : positivity makes their Peirce off-diagonal components vanish as well. A path (x, ty_0) with $|x| = |y_0| = 1$ tends to c_{i3} as $t \rightarrow \infty$. After reparametrization this is a continuous path on a closed interval, so the entire zero region is path-connected. Varying $|x|$ gives both signs near every finite zero point; keeping $|x| < 1$ or $|x| > 1$ and letting $|y| \rightarrow \infty$ gives such paths at c_{i3} .

These facts give paths of extreme points of Z as follows. From any two-block extreme point, move its positive-height ray within the positive region to a path ending at height zero, while holding the negative-height ray fixed. The displayed balance weights converge to one and zero, so this joins

the point to a one-block extreme point. For prescribed zero-height rays P and Q in different blocks, choose continuous paths P_t and Q_t such that $P_0 = P$, $Q_1 = Q$, $h(P_t) > 0$ for $t > 0$, and $h(Q_t) < 0$ for $t < 1$. The positive path may be held constant after entering its sign region, and similarly for the negative path. For $0 < t < 1$, the unique balanced two-block extreme point formed from P_t and Q_t tends to P as $t \rightarrow 0$ and to Q as $t \rightarrow 1$. Finally, any two zero rays in one block can be connected through a zero ray in another block. This last step uses $L \geq 2$ and also handles each two-point zero region. Thus all extreme points lie in one path component.

We next compute barrier parameters. Let $\Phi(X) = -\sum_i \log \det X_i$. The frame-diagonal section of $\hat{Z}$ is $\mathbb{R}_+^\rho$, because each A_i is off diagonal. It meets $\text{ri } \hat{Z}$, and its boundary belongs to $\partial \hat{Z}$. To see the latter assertion, a vanished frame coordinate is detected by a nonnegative primitive functional, which is strictly positive at X^0 ; it is therefore a nonconstant supporting functional on the balance section. Its trace-one section is a simplex of dimension $\rho - 1$ whose relative boundary lies in that of Z . Restriction of any barrier to these sections preserves its parameter. The classical orthant and simplex lower bounds [70, Section 2.3.4 and Proposition 2.3.6] give respectively ρ and $\rho - 1$, even for coupled barriers.

The standard barrier gives the upper bound ρ on $\hat{Z}$. For the sharper normalized bound, let $g = \nabla \Phi$, $H = \nabla^2 \Phi$, and let a be the total-trace covector. Jordan identities give

$$H^{-1}g = -X, \quad g^T H^{-1}g = \rho, \quad a^T H^{-1}a = \sum_i \text{tr}(X_i^2) \leq \left(\sum_i \text{tr } X_i \right)^2.$$

Here $H_i = P(X_i^{-1})$ and $H_i^{-1} = P(X_i)$, including in the Albert algebra. On the trace-one hyperplane the squared dual norm of the restricted gradient is exactly

$$\rho - \frac{(a^T H^{-1}g)^2}{a^T H^{-1}a} = \rho - \frac{1}{\sum_i \text{tr}(X_i^2)} \leq \rho - 1. \quad (15.5)$$

Restriction to the balance tangent can only decrease this norm. Self-concordance and boundary divergence are preserved under affine restriction, which proves (15.2). Lemma 9.5 proves the assertion for other closed lifts with bounded fibers; no restriction on cross-block derivatives of their barriers is imposed.

Finally, simplicity makes each factor indecomposable. If membership in K_i is removed, choose $j \neq i$ and set $X_i = -Te_i/r_i$, $X_j = (T+1)e_j/r_j$, with all other blocks zero. This is an unbounded family satisfying both affine equations and all remaining cone constraints. Each factor also supports a nonconstant nonnegative slack: $X \mapsto \langle c_{i2}, X_i \rangle$ vanishes at the feasible one-block point c_{i1} and is positive at X^0 . $\square$

Corollary 15.2 (Exact minima under a strict dimension cap). *Suppose the factors in (15.1) all have dimension d , and put $N = dL - 2$. Among full-slack factorizations of Z through products of arbitrary nonzero proper cones of dimension at most d , the minimum factor count and minimum total cone dimension are*

$$L_{\min} = L, \quad M_{\min} = N + 2 = dL.$$

These statements permit row splitting and impose no regularity on the factor maps.

Proof. For a competing product, ordinary slack rank gives $M' \geq N + 1$. Equality would identify that product with the homogenization cone of Z by Theorem 13.4. The projectivized extreme rays of a nontrivial product form a disconnected union, whereas those of the homogenization are homeomorphic to the connected set of extreme points of Z . Equality therefore requires a single factor. This violates the cap, since $d < N + 1 = dL - 1$. Consequently $M' \geq N + 2 = dL$ and

$L' \geq M'/d \geq L$. The displayed relative-Slater slice gives an attaining full-slack factorization by Lemma 2.1. $\square$

Corollary 15.3 (An alternative balance functional for Hermitian factors). *For L copies of $H_r(\mathbb{F})_+$, $r \geq 2$, the preceding theorem and corollary remain valid with $A_i = \mathbf{1}\mathbf{1}^* - I_r$ instead of an off-diagonal Peirce element. Writing $\delta = \dim_{\mathbb{R}} \mathbb{F}$, the zero-height primitive rays in one block form $\mathbb{S}^{\delta(r-1)-1}$.*

Proof. This matrix has zero diagonal and signature $(1, r-1)$, with eigenvalues $r-1, -1, \dots, -1$. The Slater, barrier, and essentiality proofs therefore apply unchanged. In eigenvector coordinates (a, b) , the height is $((r-1)|a|^2 - \|b\|^2)/(|a|^2 + \|b\|^2)$. A zero ray has the unique representative $(1, \sqrt{r-1}u)$ with $\|u\| = 1$, proving the sphere assertion. Every strict-sign ray can be moved to a zero ray while retaining its sign until the endpoint, by scaling the two components; if one component vanishes, first introduce it. Two prescribed zero rays in distinct blocks admit an explicit extreme path. For their fixed unit vectors u , replace the representative by $(1, \lambda(c)\sqrt{r-1}u)$, where

$$\lambda(c)^2 = \frac{r-1-c}{(r-1)(1+c)}, \quad -1 < c < r-1.$$

Fix $0 < c_0 < 1$ and assign heights tc_0 and $-(1-t)c_0$. Their balance weights are $1-t$ and t . This connects the two prescribed one-block points. A second block again connects the two components when $\mathbb{F} = \mathbb{R}$, $r = 2$. Connectedness and the cap conclusion follow exactly as above. $\square$

Take $d = r + \delta r(r-1)/2$, the common dimension of a classical factor and of the spin factor Q_d . The balance slices built from L classical factors and from Q_d^L both have body dimension $dL-2$, minimum factor count L under the cap d , and minimum total dimension dL . Their optimal intrinsic barrier parameters are $rL-1$ and $2L-1$. The bodies differ, so the comparison shows only that factor count and total dimension do not determine the intrinsic barrier parameter; it does not construct a Lorentz lift of the classical body. For Albert factors the same formulas give $N = 27L-2$, $M_{\min} = 27L$, and intrinsic parameter $3L-1$.

The sharp results above combine classical minimal-face, Jordan-barrier, and simplex arguments with the explicit connectivity proof and minimum-dimension rigidity. We claim no priority for the alternative balance functional. Relating these parameters to intrinsic path length requires an additional metric calculation, treated separately below.

Example 15.4 (A failed coupled logarithmic cancellation). In two three-dimensional Lorentz factors write $q_i = t_i^2 - x_i^2 - y_i^2$, and impose $x_1 + x_2 = 0$ and $t_1 + t_2 = 1$. The candidate $\Psi = -\log q_1 - \log q_2 + \log(q_1 + q_2)$ cancels one logarithmic order where q_1 and q_2 both vanish, but is not self-concordant. At $(t_1, t_2, x_1, x_2, y_1, y_2) = (1/3, 2/3, 0, 0, 0, 1/3)$, the feasible direction varying only y_2 gives $D^2\Psi = 13/4$ and $D^3\Psi = 25$. Thus $(D^3\Psi)^2 > 4(D^2\Psi)^3$. These derivatives follow directly from $q_1 = 1/9$ and $q_2 = 1/3 - 2h/3 - h^2$ along the line $y_2 = 1/3 + h$. This illustrates why the parameter proof above uses the restricted Hessian metric and a separate lower-bound section, rather than the orders of boundary singularities alone.

16 Norm balls and weighted power cones

The local curvature bound needs only a curved boundary patch, not rotational symmetry. Hence the minimum factor count and total cone dimension found for Euclidean balls hold for every ℓ_p ball with $1 < p < \infty$. The minima under global C^1 selections also carry over, but their proof treats the primal and polar boundaries as separate C^1 manifolds; this matters at points with a zero coordinate. The last subsection bounds the capacity $\sum_i (d_i - 2)_+$ of exact lifts of generalized Euclidean power cones.

16.1 Exact counts and the dimensions of count-optimal lifts

Write

$$B_p^N = \{x \in \mathbb{R}^N : \sum_j |x_j|^p \leq 1\}, \quad K_{p,m} = \{(t, u) \in \mathbb{R} \times \mathbb{R}^{m-1} : t \geq \|u\|_p\}.$$

Fix an o-minimal expansion in which the fixed powers used below are definable; $\mathbb{R}_{\text{exp}}$ suffices. For rational exponents, all cones and lifts can be taken semialgebraic.

Theorem 16.1 (Dimension-capped ℓ_p balls). *Let $1 < p < \infty$, $N \geq 2$, and let $d \geq 3$ be an integer. Among exact lifts of B_p^N through arbitrary definable proper cones of dimensions at most d , with arbitrary affine equations, projections, and free variables, the minimum number of factors and minimum total cone dimension are simultaneously*

$$k_* = \left\lceil \frac{N-1}{d-2} \right\rceil, \quad M_* = N-1 + 2k_*. \quad (16.1)$$

Every such lift has at least k_ factors of dimension at least three, even if arbitrarily many rays are allowed.*

Proof. On the boundary patch where every coordinate is nonzero, $f(x) = \sum_j |x_j|^p$ is smooth and

$$D^2 f(x) = p(p-1) \text{diag}(|x_j|^{p-2}) \succ 0.$$

Its restriction to the tangent space, divided by $\|Df(x)\|$, is the positive second fundamental form. After the reductions of Lemma 2.1, Theorem 3.2 gives $N-1 \leq \sum_i (s_i - 2)_+$, where $s_i \leq d$ are the reduced factor dimensions. If k_+ factors have $s_i \geq 3$, then $k_+ \geq k_*$ and their dimensions sum to at least $N-1 + 2k_+$. The original dimensions and the total factor count are no smaller.

For attainment, partition $N-1 = b_1 + \dots + b_{k_*}$ with $1 \leq b_i \leq d-2$. Build a chain of internal tree nodes: node $i < k_*$ has b_i new leaf children and node $i+1$; the last node has $b_{k_*} + 1$ leaf children. Node i requires its scalar to dominate the p -norm of its children, using K_{p, b_i+2} , and the root scalar is one. There are N leaves. Induction from the leaves shows that a node scalar dominates the p -norm of its descendant leaves. Conversely, assigning exactly those subtree norms makes every internal constraint feasible whenever $\|x\|_p \leq 1$. The projection is therefore exactly B_p^N , and the total dimension is $\sum_i (b_i + 2) = N-1 + 2k_*$. For a Slater point, set all leaves to zero and choose strictly positive internal scalars that decrease from the root.

The cones $K_{p,m}$ are closed, pointed, full-dimensional and convex. For $p = a/b$ rational, nonnegative auxiliary variables satisfying $z_j^b = |u_j|^a$, $w^b = t^a$, $t \geq 0$ and $\sum_j z_j \leq w$ give a semialgebraic description after projection. This also handles an even denominator without changing the meaning of the power. $\square$

The tree construction is prior work: the three-dimensional version is Krokmal and Soberanis [57, Proposition 1]; see also Blanco and Martínez-Antón [15] for representations by specified cone families. What Theorem 16.1 adds is the matching lower bound over arbitrary definable proper factors and affine lifts, and the minimum total dimension. The polyhedral endpoints $p = 1, \infty$ are excluded: their boundaries have no curved patch. As in Corollary 10.2, restricting to interior two-dimensional sections of the positive-capacity factors shows that every barrier on the ambient cone product has parameter at least $2k_*$. This bound is attained for $p = 2$; we make no claim for other p .

Proposition 16.2 (Capacity excess of lifts with the minimum count). *Under the hypotheses of Theorem 16.1, set $n = N-1$, $c = d-2$, and $\sigma = k_*c - n$. Suppose an exact lift uses exactly k_**

original factors, of dimensions $m_i \leq d$, and let s_i be their dimensions after minimal-face reduction. Then every $s_i \geq 3$ and

$$\Delta = \sum_i (s_i - 2) - n \in \{0, \dots, \sigma\}, \quad \sum_i s_i = n + 2k_* + \Delta. \quad (16.2)$$

Every Δ in this range occurs for a p -norm tree lift in which each cone is its own minimal face. If c divides n , every count-optimal lift has $s_i = m_i = d$ for all i .

Proof. If one reduced factor had dimension at most two, the remaining capacity would be at most $(k_* - 1)c < n$, a contradiction. The lower curvature bound and the dimension cap now give (16.2). For any desired Δ , start with the preceding chain and choose integers $0 \leq r_i \leq c - b_i$ with $\sum_i r_i = \Delta$. Replace K_{p,b_i+2} by K_{p,b_i+2+r_i} and fix the added norm coordinates to zero. This preserves the projection and strict feasibility, so each enlarged cone is still its own minimal face. If $\sigma = 0$, equality in $\sum_i (s_i - 2) \leq k_*c$ forces every $s_i = m_i = d$. $\square$

When $\Delta = 0$, the local curvature argument gives more than the dimension count. On a patch where every coordinate is nonzero, the supporting-polarity map is smooth, so the generic C^2 selections can be written in a common contact chart. For the selections $A_i(u), B_i(v)$ of factor i , put $H_i = -DA_i^*DB_i$. Each function $(u, v) \mapsto \langle A_i(u), B_i(v) \rangle$ is nonnegative and zero when $u = v$. Its Hessian at a diagonal point is positive semidefinite and annihilates (h, h) ; thus H_i is symmetric positive semidefinite. The total $H = \sum_i H_i$ is positive definite. Since $\Delta = 0$, equality holds in the rank count, so $\text{rank } H_i = s_i - 2$, and the linear-algebra argument in Section 6 gives

$$P_i = H^{-1/2} H_i H^{-1/2}, \quad P_i^2 = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_i P_i = I. \quad (16.3)$$

Moreover, both quotient derivatives into $B_i^\perp/\mathbb{R}A_i$ and $A_i^\perp/\mathbb{R}B_i$ are onto: otherwise the mixed rank would be smaller than $s_i - 2$. These are local statements; they do not determine the full cones.

Example 16.3 (A saturated lift with a non-strictly-convex factor). For B_p^3 , set

$$K_1 = K_{p,3}, \quad K'_2 = \{(s, t, z) \in K_{p,3} : t \geq 0\}.$$

The two constraints

$$(t, x_1, x_2) \in K_1, \quad (1, t, x_3) \in K'_2$$

project exactly onto B_p^3 : the first already implies $t \geq 0$, so the extra halfspace in the second is redundant on the affine lift. At $x = 0$ and $0 < t < 1$ both constraints are strict. Thus neither factor reduces to a smaller face, and their total capacity is $2 = 3 - 1$: the lift is saturated. Nevertheless K'_2 has the two-dimensional exposed face $\{(s, 0, z) : s \geq |z|\}$, whereas every proper nonzero face of $K_{p,3}$ is a ray, so the two factors are not linearly isomorphic. At every contact with $(x_1, x_2) \neq 0$ the added halfspace is inactive, so near the contact K'_2 coincides with $K_{p,3}$. Hence saturation alone neither determines the global factors nor justifies applying to every factor the p -dependent lower bound on the barrier parameter that Theorem 18.2 proves for norm-cone factors.

16.2 Global selections and the Young construction

Let $q = p/(p-1)$. The two boundaries ∂B_p^N and ∂B_q^N are C^1 and strictly convex. In this subsection, a global factorization consists of C^1 primal selections on the first boundary and C^1 normalized genuine dual certificates on the second that represent $1 - \langle x, y \rangle$ for all x, y on the two boundaries. Each selection is differentiated in the coordinates of its own boundary.

Theorem 16.4 (Minimum resources for ℓ_p balls under global C^1 selections). *Let $N \geq 3$ and let $d \geq 3$ be an integer. Consider exact lifts through arbitrary proper factors of dimension at most d that admit these global selections. Over these lifts, the capacity $S = \sum_i (\dim K_i - 2)_+$, the factor count k , and the total cone dimension M have the simultaneous minima*

$$(S, k, M) = \begin{cases} (N, h, N + 2h), & h = \lceil N/(d-2) \rceil, \quad d < N + 1, \\ (N - 1, 1, N + 1), & d \geq N + 1. \end{cases} \quad (16.4)$$

The lower bound does not require definability of the cones or selections.

Proof. At a contact $\langle x, y \rangle = 1$ the tangent spaces are $y^\perp$ and $x^\perp$. Their Euclidean pairing is nondegenerate, since its first kernel is $y^\perp \cap \mathbb{R}x = \{0\}$. Consequently $S \geq N - 1$ by Lemma 3.1. If equality holds, Theorem 10.1 allows exactly one positive-capacity factor, of dimension $N + 1$: both boundaries are C^1 diffeomorphic to $\mathbb{S}^{N-1}$. The assumption $d < N + 1$ excludes this case, so integrality gives $S \geq N$. Counting positive-capacity factors then gives $k \geq h$ and $M \geq N + 2h$. For $d \geq N + 1$, the direct lift $K_{p,N+1}$ with primal and dual maps $(1, x), (1, -y)$ attains the second line.

For the first line define, for $r \geq 1$,

$$\mathcal{P}_{p,r} = \{(u, v, w) \in \mathbb{R}_+^2 \times \mathbb{R}^r : uv^{p-1} \geq \|w\|_p^p\}. \quad (16.5)$$

This is the closed perspective epigraph of $\|w\|_p^p$, hence a proper cone of dimension $r + 2$. At $v = 0$ the definition gives $w = 0$ and $u \geq 0$, exactly the closure of its part with $v > 0$. Partition the coordinates into h nonempty groups G of sizes at most $d - 2$, and impose

$$(u_G, v_G, w_G) \in \mathcal{P}_{p,|G|}, \quad v_G = 1, \quad \sum_G u_G = 1, \quad x_G = w_G. \quad (16.6)$$

The projection is B_p^N , and $x = 0, u_G = 1/h$ is a Slater point. On the boundary the primal fiber is unique, with $u_G = \|x_G\|_p^p$. Define

$$A_G(x) = (\|x_G\|_p^p, 1, x_G), \quad B_G(y) = \left(\frac{1}{p}, \frac{\|y_G\|_q^q}{q}, -y_G \right). \quad (16.7)$$

For $v > 0$, Young's inequality gives

$$\frac{u}{p} + \frac{v\|y_G\|_q^q}{q} - \langle w, y_G \rangle \geq v \left(\frac{\|w/v\|_p^p}{p} + \frac{\|y_G\|_q^q}{q} - \langle w/v, y_G \rangle \right) \geq 0.$$

Continuity includes $v = 0$, so $B_G(y)$ belongs to the dual cone. Its entries are the multiplier $1/p$ of $\sum_G u_G = 1$, the multiplier $\|y_G\|_q^q/q$ of $v_G = 1$, and the negated objective $-y_G$ on w_G . Their objective value is $1/p + \sum_G \|y_G\|_q^q/q = 1$. Thus these are genuine certificates on the whole affine slice. They give

$$\sum_G \langle A_G(x), B_G(y) \rangle = 1 - \langle x, y \rangle \quad (x \in \partial B_p^N, y \in \partial B_q^N).$$

Both maps are globally C^1 because $p, q > 1$. Their total capacity, count and dimension are $N, h, N + 2h$, respectively. $\square$

The scalar case $r = 1$ of (16.5) is the power cone $P_{1/p} = \{(u, v, w) : u, v \geq 0, u^{1/p}v^{1/q} \geq |w|\}$. The corresponding standard p -norm model is recorded in MOSEK ApS [65, Section 4.2.2]. We do not claim the construction or the use of Young's inequality as new. Theorem 16.4 proves that it

uses the minimum resources under the stated global regularity condition. For $d = 3$ the minimum is exactly N factors, also when only three-dimensional power cones are allowed.

The selections in (16.7) are both C^2 only when $p = 2$. If $1 < p < 2$, the primal map fails to be C^2 at coordinate zeros and the dual map is C^2 ; for $p > 2$ the roles reverse. This does not affect the preceding proof, which does not differentiate the polarity map $y_j = \text{sign}(x_j)|x_j|^{p-1}$. When $d < N + 1$, every barrier on the ambient cone product has parameter at least $2h$, by the product orthant-section argument of Corollary 10.2. This bound is attained for $p = 2$; we make no such claim for the nonsymmetric perspective cones. Every self-concordant barrier on the reduced affine domain of (16.6) has parameter at least h : fix positive u_G with sum one, set all but one coordinate per group to zero, and vary the remaining coordinate in $(-u_G^{1/p}, u_G^{1/p})$. This gives an h -dimensional open box through the domain interior whose boundary lies on the domain boundary; restriction to this box and the lower bound for boxes prove the claim. In the Euclidean case the grouped barrier of Section 9 attains h ; for $p \neq 2$ we make no such claim.

Proposition 16.5 (The two-dimensional power-cone exception). *For $N = 2$, one arbitrary three-dimensional proper cone suffices for B_p^2 , with global C^1 primal and dual selections. Within the dictionary of three-dimensional power cones, the minimum total factor count is one for $p = 2$ and two for $p \neq 2$, even without requiring global selections.*

Proof. The direct $K_{p,3}$ lift proves the first assertion, and the two scalar Young factors give the upper bound in the power dictionary. A one-factor power-cone lift has total dimension three; by Theorem 13.4 its cone would be linearly isomorphic to $K_{p,3}$. We show that this is impossible unless $p = 2$.

For P_α , $0 < \alpha < 1$, the only projective boundary rays with possibly non-smooth second derivatives are its two axial rays. In the chart $u = 1$ at the first, the boundary is $v = |w|^{1/(1-\alpha)}$; in the chart $v = 1$ at the second it is $u = |w|^{1/\alpha}$. Elsewhere the boundary is smooth with positive curvature. If $\alpha \neq 1/2$, exactly one of these exponents is below two, so exactly one boundary ray is not C^2 . For $K_{p,3}$ the projective section is the p -unit circle. Near $(1, 0)$ it has graph

$$x_1 = (1 - |x_2|^p)^{1/p} = 1 - \frac{|x_2|^p}{p} + o(|x_2|^p),$$

and the same holds at its other three coordinate points. It therefore has exactly four non- C^2 rays when $p < 2$, and none when $p > 2$. An invertible linear cone map induces smooth projective coordinate changes and preserves this count. This excludes $\alpha \neq 1/2$. For $\alpha = 1/2$, the section is a nonsingular conic with nonzero second fundamental form everywhere. When $p > 2$ the four coordinate points instead have zero second fundamental form, and this vanishing is preserved by projective transformations, as follows by differentiating a homogeneous supporting equation along its tangent. When $p < 2$ the C^2 obstruction already suffices. At $p = 2$ both cones are Lorentz cones, which proves the remaining case. $\square$

16.3 A curvature bound for generalized Euclidean power cones

Let $\alpha_j > 0$, $\sum_{j=1}^m \alpha_j = 1$, and set

$$g(x) = \prod_{j=1}^m x_j^{\alpha_j}, \quad C_{\alpha,k} = \{(x, z) \in \mathbb{R}_+^m \times \mathbb{R}^k : \|z\|_2 \leq g(x)\}.$$

Here g is extended continuously to the whole nonnegative orthant. It is concave, so $C_{\alpha,k}$ is a proper convex cone. It is definable in $\mathbb{R}_{\text{exp}}$, and semialgebraic for rational weights.

Theorem 16.6 (Power-cone curvature capacity). *Let $m, k \geq 1$ and $m + k \geq 3$. Every exact affine lift of $C_{\alpha, k}$ by definable proper cones of dimensions d_i satisfies*

$$\sum_i (d_i - 2)_+ \geq m + k - 2. \quad (16.8)$$

In particular, a lift using three-dimensional Lorentz cones and any number of rays requires at least $m + k - 2$ Lorentz factors. The corresponding bound, with right-hand side $m - 1$ in (16.8), holds for the nonnegative geometric-mean cone $\{(x, t) : x \geq 0, 0 \leq t \leq g(x)\}$ when $m \geq 2$.

Proof. Intersect the cone with $\sum_j x_j = 1$. This is a compact convex body of dimension $m + k - 1$ in its affine hull: x ranges over the simplex and $\|z\| \leq g(x)$ is uniformly bounded. Restricting an exact lift by this affine equation introduces no cone factors. On the patch $x_j > 0, z \neq 0$, the boundary is $\Phi(x, z) = 0$, where $\Phi(x, z) = \|z\| - g(x)$. Write $z = g(x)\omega$ with $\|\omega\| = 1$. A tangent vector (h, ξ) obeys

$$\sum_j h_j = 0, \quad \langle \omega, \xi \rangle = Dg(x)[h].$$

The Hessian on such tangent vectors is

$$D^2\Phi[(h, \xi), (h, \xi)] = -D^2g(x)[h, h] + \frac{\|\xi\|_2^2 - \langle \omega, \xi \rangle^2}{g(x)}. \quad (16.9)$$

Direct differentiation gives the weighted variance identity

$$-D^2g(x)[h, h] = g(x) \left\{ \sum_j \alpha_j \left(\frac{h_j}{x_j} \right)^2 - \left(\sum_j \alpha_j \frac{h_j}{x_j} \right)^2 \right\}.$$

It is nonnegative and vanishes exactly when h is proportional to x . On $\sum_j h_j = 0$, this forces $h = 0$. If also the second term of (16.9) vanishes, then ξ is proportional to ω ; the tangent equation with $h = 0$ forces $\xi = 0$. Thus the boundary patch has strictly positive second fundamental form of rank $m + k - 2$. Theorem 3.2, applied in the affine hull of this compact section, gives (16.8). Minimal-face reduction can only reduce the displayed dimensions. For the nonnegative scalar cone the positive upper graph $t = g(x)$ has exactly the same Hessian on the simplex tangent; its section has dimension m and curved boundary dimension $m - 1$. The case $m = 1$ in the vector statement reduces to the ordinary Euclidean ball section. $\square$

Theorem 16.6 is a lower bound on dimensions, not an exact factor count for every rational weight. Wang [87, Theorem 3.6] proves the scalar bound $m - 1$ for simple mediated-set representations, with separate denominator-dependent bounds; his Theorem 3.15 gives $\lceil m/2 \rceil$ for arbitrary lifts by three-dimensional second-order cones. The proof above allows arbitrary affine maps into the cone coordinates, auxiliary equations, and free variables, and also gives the term $k - 1$ from the Euclidean output z . Blanco and Martínez-Antón [15, Definition 1 and Theorem 14] relate extended representations and mediated graphs; their constructive maps select triples of mediated variables. We do not claim priority for the scalar inequality and do not extend the denominator-dependent bounds beyond the representation classes in which they are proved. Theorem 16.6 gives a self-contained geometric proof in the affine-lift model of this paper.

17 Matrix balls and sparse completion cones

This section compares two lifts of products of spectral-norm matrix balls in which one cone factor serves several blocks. The first places several blocks under one scale variable, which reduces the number of cone coordinates and the ambient barrier parameter. The second uses a positive-semidefinite completion cone, which keeps a sparse matrix structure. After the scale or the diagonal blocks are fixed, both constructions have exactly the same barrier on the product of matrix balls. The spectral and completion barriers used below are classical [3, 24]. Our results are factorizations of the full slack rows $1 - \langle X_a, Z \rangle$ by these cones, sharp bounds on cone and face dimensions, and the equality of the two reduced Newton systems after this fixing.

17.1 Mixed contact rank with split rows

For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ let $\delta = \dim_{\mathbb{R}} \mathbb{F}$ and use the real Frobenius pairing $\langle X, Z \rangle = \text{Re tr}(X^* Z)$. The extreme points of the spectral-norm ball in $\mathbb{F}^{r \times c}$, $1 \leq r \leq c$, are the coisometries

$$\mathcal{E}_{r,c} = \{X : XX^* = I_r\};$$

those of its nuclear-norm polar are $\mathcal{Z}_{r,c} = \{uv^* : \|u\| = \|v\| = 1\}$. These descriptions follow from singular-value decomposition: a contraction with a singular value below one is a nontrivial average of contractions, whereas equality on every row forces any such averaging at a coisometry to be trivial. Similarly, the nuclear unit ball is the convex hull of the rank-one unit matrices, which are exposed by suitable contractions.

Consider k blocks $X_a \in \mathcal{E}_{r_a, c_a}$ over fields $\mathbb{F}_a$, with $\delta_a = \dim_{\mathbb{R}} \mathbb{F}_a$. Suppose the full slack rows $1 - \langle X_a, Z \rangle$ have a globally C^1 factorization with dual maps labelled by the row:

$$1 - \langle X_a, Z \rangle = \sum_i \langle A_i(X_1, \dots, X_k), B_i^a(Z) \rangle, \quad A_i \in K_i, \quad B_i^a \in K_i^*. \quad (17.1)$$

Here the nonray proper cones have real dimensions $m_i \geq 3$; ray factors contribute no mixed rank. Put

$$q_i = m_i - 2, \quad f_i = \max_{0 \neq y \in K_i^*} \dim(K_i \cap y^\perp), \quad \bar{f}_i = \min(f_i, k), \quad \kappa_a = \delta_a c_a - 1.$$

All dimensions are real.

Proposition 17.1 (Rank and face bounds at spectral contacts). *At every contact $\langle X, Z \rangle = 1$, the mixed pairing $(H, \dot{Z}) \mapsto -\langle H, \dot{Z} \rangle$ on $T_X \mathcal{E}_{r,c} \times T_Z \mathcal{Z}_{r,c}$ has rank $\delta c - 1$. At a simultaneous contact $\langle X_a, Z_a \rangle = 1$ for all a , let*

$$t_i^a = \text{rank}[(H, \dot{Z}) \mapsto -\langle D_a A_i[H], D B_i^a[\dot{Z}] \rangle], \quad I_i = \{a : t_i^a > 0\},$$

and call I_i the rows active in factor i . Then

$$t_i^a \leq q_i, \quad \sum_i t_i^a \geq \kappa_a, \quad |I_i| \leq \bar{f}_i, \quad (17.2)$$

$$\sum_{a \in I_i} t_i^a \leq q_i, \quad 1 + \sum_{b \in I_i \setminus \{a\}} t_i^b \leq \dim(K_i \cap B_i^a(Z_a)^\perp) \quad (a \in I_i). \quad (17.3)$$

If $m_i \leq d$ for an integer $d \geq 3$, these imply

$$\sum_i \bar{f}_i \geq \sum_a \left\lceil \frac{\kappa_a}{d-2} \right\rceil, \quad \sum_i q_i \geq \sum_a \kappa_a. \quad (17.4)$$

In particular, if the L nonray factors satisfy $f_i \leq f$ for a common $f \geq 1$, then

$$L \geq \max \left\{ \left\lceil \frac{1}{f} \sum_a \left\lceil \frac{\kappa_a}{d-2} \right\rceil \right\rceil, \left\lceil \frac{\sum_a \kappa_a}{d-2} \right\rceil \right\}.$$

Proof. Contact implies $Xv = u$ and $X^*u = v$ by equality in the contraction inequality. Unitary changes of bases therefore give $X = (I_r \ 0)$ and $u = v = e_1$. Write a coisometry tangent as $H = (A \ B)$ with $A^* = -A$. Since (u, v) and (uq, vq) give the same uv^* for a unit scalar $q \in \mathbb{F}$, we may assume $u^*u = 0$. The pairing $\langle H, \dot{u}e_1^* + e_1\dot{v}^* \rangle$ splits into independent parts: the entries A_{j1} , $j > 1$, paired with the off-axis differences of $\dot{u}$ and $\dot{v}$; the first row of B , paired with the tail of $\dot{v}$; and the imaginary entry A_{11} , paired with the imaginary entry $\dot{v}_1$. They have respectively $\delta(r-1)$, $\delta(c-r)$, and $\delta-1$ real dimensions, and the pairing on each part is nondegenerate. Their sum is $\delta c - 1$. This argument includes real scalar rows, where the rank can be zero. In the quaternionic case all pairings are real Euclidean pairings, so cyclic rearrangements of quaternionic traces are not required.

The face argument does not need nondegeneracy of the full spectral pairing. At contact, each summand in (17.1) vanishes. If either factor is zero, its derivative is zero because a differentiable map into a pointed cone has zero derivative at the vertex. Otherwise Lemma 3.1 bounds its mixed rank by $m_i - 2$. Mixed differentiation and rank subadditivity give $\sum_i t_i^a \geq \kappa_a$.

Fix i and an active row a . On the contact cylinder, where X_a stays in contact with Z_a and the other blocks vary, nonnegativity gives

$$\langle D_b A_i[H_b], DB_i^a[\dot{Z}] \rangle = 0 \quad (b \neq a), \quad \langle A_i, DB_i^a[\dot{Z}] \rangle = 0.$$

Indeed, at every point of that cylinder the summand, as a function of Z , has minimum zero at Z_a ; its vanishing Z -derivative is the second identity, and differentiating it along X_b gives the first. Choose $E_a \subseteq \text{im } D_a A_i$ of dimension t_i^a on which pairing with DB_i^a is injective. These identities show that $\mathbb{R}A_i \oplus \bigoplus_{a \in I_i} E_a$ is a direct sum. For any fixed active row a_0 , the nonzero vector $b = B_i^{a_0}(Z_{a_0})$ annihilates all of $\text{im } DA_i$: the nonnegative function $X' \mapsto \langle A_i(X'), b \rangle$ has value zero at the contact X . It also annihilates A_i , so the direct sum lies in $b^\perp$ and $1 + \sum_a t_i^a \leq m_i - 1$. If there is no active row, the total rank bound is immediate. Moreover, A_i and every E_b with $b \neq a$ lie in the span of the exposed face $K_i \cap B_i^a(Z_a)^\perp$. This proves (17.3) and $|I_i| \leq f_i$. Finally, each row a is active in at least $\lceil \kappa_a/(d-2) \rceil$ factors. Counting these incidences and summing the mixed-rank bounds proves (17.4). Since $\sum_i q_i \leq (d-2)L$, the count bound follows. $\square$

These are local necessary conditions for split rows; they include no global topological bound, and we do not claim that every bound is attained. The mixed rank is $\delta c - 1$, not the dimension of the polar extreme manifold: some polar directions move within the contact set and do not enter the mixed pairing.

17.2 Whole-row grouping and exact spectral parameters

We return to the whole-row model of Section 13. Here take $\mathbb{F}_a \in \{\mathbb{R}, \mathbb{C}\}$, with $\delta_a = \dim_{\mathbb{R}} \mathbb{F}_a$, and orient each matrix with $r_a \leq c_a$. For a nonempty group G define

$$S_G = \{(t, (X_a)_{a \in G}) : \|X_a\|_{\text{op}} \leq t\}, \quad N_G = \sum_{a \in G} \delta_a r_a c_a, \quad R_G = \sum_{a \in G} r_a, \quad \lambda_a = \delta_a(r_a + c_a - 1).$$

The dual cone is $\{(\alpha, (Z_a)) : \alpha \geq \sum_a \|Z_a\|_*\}$, under the real pairing. Thus the polynomial maps

$$A_G(X) = (1, (X_a)_{a \in G}), \quad B_G^a(Z) = (1, (0, \dots, -Z, \dots, 0))$$

represent the entire row $1 - \langle X_a, Z \rangle$.

Corollary 17.2 (Sharp whole-row spectral caps). *In the whole-row model, a group G can be carried by one proper cone of dimension at most D and maximal proper exposed-face dimension at most F if and only if*

$$N_G \leq D - 1, \quad N_G - \min_{a \in G} \lambda_a \leq F - 1. \quad (17.5)$$

The minimum factor count is the minimum number of groups in a partition of the rows satisfying these inequalities; the cones $\mathcal{S}_G$ attain it. For k identical $r \times c$ rows over a field of real dimension δ , one group carries at most

$$Q = \min \left\{ \left\lfloor \frac{D-1}{\delta rc} \right\rfloor, \left\lfloor \frac{F + \delta(r+c-1) - 1}{\delta rc} \right\rfloor \right\}.$$

If $Q \geq 1$, the exact count is $\lceil k/Q \rceil$.

Proof. For a rank-one polar support, contact fixes a 1×1 corner of X_a and sets both off-diagonal blocks to zero; the remaining contraction has $(r_a - 1)(c_a - 1)$ entries over $\mathbb{F}_a$. Its affine face dimension is therefore $\delta_a(r_a - 1)(c_a - 1)$, with codimension λ_a . Theorem 13.1 gives necessity. For sufficiency, the dimension of $\mathcal{S}_G$ is $1 + N_G$. More generally, a dual boundary support with nonzero blocks J of ranks h_a exposes a face of dimension

$$1 + \sum_{a \notin J} \delta_a r_a c_a + \sum_{a \in J} \delta_a (r_a - h_a)(c_a - h_a).$$

To see this, equality in spectral/nuclear duality fixes the supported singular-vector corner to $-tI_{h_a}$, sets the off-diagonal blocks to zero, and leaves an arbitrary contraction in the complementary rectangle. The scale contributes one dimension. Each supported block loses $\delta_a h_a(r_a + c_a - h_a)$ coordinates; this is minimized at $h_a = 1$. Consequently the maximal proper exposed-face dimension is $1 + N_G - \min_a \lambda_a$, proving sufficiency and the count formula. $\square$

For rows of different sizes, computing the minimum partition is strongly NP-hard: for real vector rows ($r_a = 1$) and a nonbinding face cap $F \geq D - 1$, it is bin packing with item sizes c_a and bin capacity $D - 1$. Since 3-PARTITION is NP-hard with polynomially bounded numbers, the matrix dimensions stay polynomial. This hardness concerns whole-row grouping, not split-row factorizations.

Theorem 17.3 (Shared spectral cone and slice). *For a partition $\mathcal{G}$ into g nonempty groups, put $R = \sum_a r_a$ and $N = \sum_a \delta_a r_a c_a$. The product of the shared cones has real dimension $N + g$ and*

$$\vartheta_{\text{opt}} \left(\text{int} \prod_{G \in \mathcal{G}} \mathcal{S}_G \right) = R + g, \quad \vartheta_{\text{opt}} \left(\prod_a \{X_a : \|X_a\|_{\text{op}} < 1\} \right) = R. \quad (17.6)$$

Both lower bounds hold for arbitrary, possibly coupled, self-concordant barriers. The attaining barriers are respectively the sum over groups of

$$F_G(t, X) = - \sum_{a \in G} \log \det(t^2 I_{r_a} - X_a X_a^*) + (R_G - 1) \log t \quad (17.7)$$

and its fixed-scale restriction

$$\Phi(X) = - \sum_a \log \det(I_{r_a} - X_a X_a^*). \quad (17.8)$$

Proof. The classical spectral-cone barrier is $-\log t - \log \det(tI - XX^*/t)$, with parameter one plus the number of rows [24, Section 4.2]; the same formula for complex matrices is recorded in Coey [25, Section 2.6.1]. Apply it to the block-diagonal matrix with blocks X_a ; real blocks can be embedded in a complex matrix without changing singular values. This proves self-concordance of (17.7) and its degree $R_G + 1$ of logarithmic homogeneity.

For the lower bound we use Nesterov's recession bound, which does not require logarithmic homogeneity. For an interior point x , recession directions p_j , and positive numbers a_j, b_j satisfying $x - \sum_j a_j p_j \in C$ and $x - b_j p_j \notin \text{int } C$, every self-concordant barrier of C has parameter at least $\sum_j a_j/b_j$ [35, Theorem 3.9]. We call (x, p_j, a_j, b_j) a recession certificate. Restrict the matrices of one group to their real rectangular diagonals. Writing $r = R_G$, this gives $C_r = \{(t, \xi) : t \geq \|\xi\|_\infty\}$. For $0 < \epsilon < \min(1, 2/r)$ choose

$$x = (1, (1 - \epsilon)\mathbf{1}), \quad p_0 = (1, \mathbf{1}), \quad p_j = (1, \mathbf{1} - 2e_j) \quad (1 \leq j \leq r),$$

$$a_j = b_j = \epsilon/2 \quad (j \geq 1), \quad a_0 = 1 - r\epsilon/2, \quad b_0 = 1 - \epsilon/2.$$

All p_j belong to C_r , $x = \sum_{j=0}^r a_j p_j$, and $x - b_j p_j$ is on the boundary for every j . The recession bound is therefore $r + (1 - r\epsilon/2)/(1 - \epsilon/2)$, which tends to $r + 1$. For several groups, place each recession certificate in its own direct summand and take x to be the tuple of their interior points. The resulting single recession certificate gives $\sum_G (R_G + 1)$ even for a coupled barrier. These diagonal sections meet the cone interiors, so restricting any ambient barrier to them does not increase its parameter.

For the slice, ordinary self-concordance follows by restricting the positive-definite barrier to $\begin{pmatrix} I_r & X \\ X^* & I_c \end{pmatrix}$. At a singular-value representative $X = (\text{diag}(\sigma_j) \ 0)$, the gradient is real diagonal and this subspace is invariant under the Hessian. On its j th coordinate the first two derivatives are $2\sigma_j/(1 - \sigma_j^2)$ and $2(1 + \sigma_j^2)/(1 - \sigma_j^2)^2$. Consequently

$$\|D\Phi(X)\|_{D^2\Phi(X),*}^2 = \sum_{a,j} \frac{2\sigma_j(X_a)^2}{1 + \sigma_j(X_a)^2} < R.$$

The real diagonal section is the cube $(-1, 1)^R$, whose sharp barrier lower bound is R [70, Proposition 2.3.6]. This proves the second equality, including coupled competitors. $\square$

Corollary 17.2 optimizes only the factor count and dimensions in the whole-row model. Equation (17.6) gives the exact barrier parameters of the attaining cones $\mathcal{S}_G$, not a lower bound for other, higher-dimensional cones carrying the same rows. For k identical rows and $g = \lceil k/Q \rceil$, the total dimension and parameters are $(M, \nu_{\text{ambient}}, \nu_{\text{slice}}) = (k\delta rc + g, kr + g, kr)$.

17.3 Completion cones and the clique-separator barrier

Let G be an undirected graph on n vertices; the specified entries are the diagonal and the positions of edges of G . Over $\mathbb{R}$ or $\mathbb{C}$, let $\mathcal{H}_G$ be the space of Hermitian matrices supported on those entries, with the real trace pairing. The sparse positive-semidefinite cone is $K_G = \{Y \in \mathcal{H}_G : Y \succeq 0\}$. Its dual C_G is identified with the partial Hermitian matrices that admit a positive-semidefinite completion. This set is closed: in a convergent sequence of partial matrices all diagonals are bounded, and every positive-semidefinite completion obeys $|Z_{ij}|^2 \leq Z_{ii}Z_{jj}$. A convergent subsequence of completions therefore completes the limit. Also, C_G is pointed because a matrix and its negative can both be completed only if all their diagonals, and hence all entries, vanish.

Proposition 17.4 (Classical completion barrier and its exact parameter). *For arbitrary graphs $G_1, \dots, G_l$ of orders $n_1, \dots, n_l$,*

$$\vartheta_{\text{opt}} \left(\text{int} \prod_j C_{G_j} \right) = \sum_j n_j.$$

For one graph an attaining barrier, up to a constant, is

$$F_G(S) = -\log \det Z(S), \quad (17.9)$$

where $Z(S)$ is the unique maximum-determinant positive-definite completion. If the graph is chordal, its maximal cliques $\mathcal{Q}$ and a clique forest's separators $\mathcal{T}$, counted with multiplicity, give

$$F_G(S) = -\sum_{Q \in \mathcal{Q}} \log \det S_Q + \sum_{T \in \mathcal{T}} \log \det S_T. \quad (17.10)$$

Empty separators contribute zero.

Proof. The restriction of $-\log \det Y$ to $\text{int } K_G$ is an n -logarithmically homogeneous self-concordant (“ n -LHSC”) barrier. Its signed Legendre dual

$$F_G(S) = \sup_{Y \in \text{int } K_G} \{-\langle S, Y \rangle + \log \det Y\}$$

is another n -LHSC barrier [3]. Stationarity says that Y^{-1} completes S . Equivalently, stationarity of the strictly concave completion problem says that $Z(S)^{-1}$ is sparse. Boundedness of the set of completions and existence of a positive-definite completion guarantee existence; strict concavity gives uniqueness. Since Y is sparse, $\langle S, Y \rangle = \text{tr}(Y^{-1}Y) = n$, giving $F_G = -n - \log \det Z(S)$. The diagonal section of C_G is precisely $\mathbb{R}_+^n$, and it meets the interior. Across a product it is $\mathbb{R}_+^{\sum n_j}$. The sharp orthant bound proves the stated lower bound for every coupled barrier.

For a chordal graph, eliminate leaf cliques successively. If a leaf clique meets the remainder in separator T , Schur complementation maximizes the determinant by making the residual cross block zero and multiplies the determinant of the remainder by $\det S_Q / \det S_T$. Iteration gives (17.10), the classical clique–separator formula [10, 46]. The running-intersection identity $\sum_Q |Q| - \sum_T |T| = n$ gives its degree of logarithmic homogeneity. Self-concordance of this formula follows from duality, not from a general rule for subtracting logarithmic barriers. $\square$

Proposition 17.5 (Block-star realization). *Consider a hub clique of size $s \geq 1$, and k disjoint leaf cliques of sizes $p_a \geq 1$, each joined completely to the hub. Put $\delta = 1$ over $\mathbb{R}$ and $\delta = 2$ over $\mathbb{C}$, and $h_\delta(v) = v + \delta v(v-1)/2$. The completion cone has real dimension and exact ambient parameter*

$$D = h_\delta(s) + \sum_a (\delta s p_a + h_\delta(p_a)), \quad \nu_{\text{ambient}}^{\text{opt}} = s + \sum_a p_a. \quad (17.11)$$

Fixing the hub and leaf diagonal blocks to identities gives the product $\prod_a \{X_a \in \mathbb{F}^{s \times p_a} : \|X_a\|_{\text{op}} \leq 1\}$. Its restricted completion barrier is exactly (17.8), with optimal parameter $\sum_a \min(s, p_a)$, and it represents every full slack row $1 - \langle X_a, Z \rangle$ with Z an extreme point of the nuclear-norm unit ball.

Proof. Write the specified blocks as T , U_a , W_a on the hub, cross edges, and leaves. Chordal completion is equivalent to

$$\begin{pmatrix} T & U_a \\ U_a^* & W_a \end{pmatrix} \succeq 0 \quad \text{for every } a.$$

The count of independent entries gives D , and the graph order gives the exact parameter by Proposition 17.4. All $k - 1$ clique-tree separators are the hub, so the barrier is

$$-\sum_a \log \det \begin{pmatrix} T & U_a \\ U_a^* & W_a \end{pmatrix} + (k - 1) \log \det T = -\log \det T - \sum_a \log \det (W_a - U_a^* T^{-1} U_a). \quad (17.12)$$

At $T = I_s$, $W_a = I_{p_a}$, $U_a = X_a$, the Schur complement gives exactly the stated product. Sylvester's determinant identity gives $\det(I - X_a^* X_a) = \det(I - X_a X_a^*)$, so its barrier is (17.8), using the smaller order if desired. For a polar extreme $Z = uv^*$, put $b = (-u, v)$ on the hub and leaf a and zero on all other leaves. The sparse positive-semidefinite matrix $bb^*/2$ is a dual factor, and

$$\frac{1}{2} b^* \begin{pmatrix} I_s & X_a \\ X_a^* & I_{p_a} \end{pmatrix} b = 1 - \operatorname{Re}(u^* X_a v).$$

Thus the identity-block section and these dual factors represent every slack row $1 - \langle X_a, Z \rangle$. The rank-one map depends only on Z : simultaneous phase changes in u, v leave bb^* invariant. $\square$

For scalar leaves, $p_a = 1$, the construction has $D = h_\delta(s) + k(\delta s + 1)$ and parameter $s + k$. It removes $\delta k(k - 1)/2$ real leaf-leaf coordinates compared with a dense Hermitian matrix of order $s + k$. Partitioning k leaves among g stars gives total dimension $k(\delta s + 1) + gh_\delta(s)$ and parameter $k + gs$. In the real case, factoring T once and solving against all leaf vectors evaluates (17.12) in $O(s^3 + ks^2)$ dense arithmetic and stores $O(s^2 + ks)$ scalars. These counts concern barrier evaluation; a Newton solve with additional constraints has a separate cost.

17.4 Equality of the restricted barriers and Newton systems

After fixing the shared scales or the star's diagonal blocks, all constructions have the same retained coordinates and the same function Φ in (17.8). For one oriented block, with $M = I - XX^*$, its derivatives are

$$\nabla \Phi(X) = 2M^{-1}X, \quad D^2 \Phi(X)[H] = 2M^{-1}H + 2M^{-1}(HX^* + XH^*)M^{-1}X.$$

Hence their Hessian metrics and reduced Newton systems are identical: with the same real affine map $\mathcal{A}$, right side b , and objective c , both systems use

$$\begin{pmatrix} D^2 \Phi(X) & \mathcal{A}^* \\ \mathcal{A} & 0 \end{pmatrix}, \quad \begin{pmatrix} -D\Phi(X) - \tau c - \mathcal{A}^* y \\ b - \mathcal{A}X \end{pmatrix}.$$

Their central paths and reduced condition numbers also coincide. When orientation requires transposition or adjunction, the identification is a fixed real orthogonal coordinate change (including conjugation signs in the complex case). An oracle for one reduced object therefore yields one for the other through this known coordinate change. Building it from either construction's ambient data is a different task: the ambient scales, completion coordinates, dual variables, and KKT systems differ. Neither the smaller ambient parameter nor the smaller sparse coordinate count alone implies a saving in running time or quantum queries.

18 Nonsymmetric cones: barrier parameters and evaluation costs

The dimension m of a cone and its contact capacity (at most $m - 2$) do not determine its best barrier parameter. We bound that parameter for the norm and perspective cones of Section 16. Throughout, $\nu_{\text{LH}}(C)$ denotes the infimum of the parameters of logarithmically homogeneous self-concordant (LHSC) barriers on C ; on a product, these barriers may couple the factors. The cross-ratio arguments below require logarithmic homogeneity.

18.1 An additive lower bound for fixed cone dictionaries

For $1 < p < \infty$, put $c = \min\{1/p, 1 - 1/p\}$. If $p \neq 2$, let $\gamma \in (0, 1)$ solve

$$(1 - 2c)\gamma^{1-c} + (1 - c)\gamma^{1-2c} = c, \quad h(p) = 1 + \frac{1 + \gamma^c}{1 + \gamma^{1-c}}. \quad (18.1)$$

Set $h(2) = 2$. The left side of the root equation is strictly increasing from zero to $2 - 3c > c$, so the root exists and is unique. Rearranging gives

$$h(p) - 2 = \frac{1 - 2c}{1 - c} \gamma^c. \quad (18.2)$$

Consequently $2 < h(p) < 3$ for $p \neq 2$, and $h(p) = h(q)$ when $1/p + 1/q = 1$. The lower bound $h(p)$ for a single three-dimensional power cone or norm cone is due to Hildebrand [53, Corollaries 7.1–7.2].

For a proper cone C , choose a compact affine base, a basis $e_1, \dots, e_m$ of boundary points of this base, and a point x in the relative interior of their simplex. On the projective line through e_a and x , let y_a be the other boundary point of the base and z_a the intersection with the span of the other basis vectors. Write $q_a = (e_a, x; y_a, z_a)$, using the convention $(a, b; c, d) = (a - c)(b - d)/((a - d)(b - c))$ and continuity at infinity. Hildebrand's Theorem 5.4 states, in particular, that

$$\sum_a q_a \geq 2 \implies \nu_{\text{LH}}(C) \geq \sum_a q_a. \quad (18.3)$$

Lemma 18.1 (Cross-ratio lower bound for products). *Suppose each proper cone C_i has a configuration as above with cross-ratio sum $S_i \geq 2$. Then $\nu_{\text{LH}}(\prod_i C_i) \geq \sum_i S_i$.*

Proof. Normalize each local base by $\ell_i = 1$ and the product base by $\sum_i \ell_i = 1$. Embed each e_{ia} into its own direct summand, and choose $X = (\alpha_i x_i)_i$, where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. These embedded vectors form a boundary basis, and X lies in the interior of their simplex. Along the line from an embedded e_{ia} through X and beyond, every other block remains a positive multiple of x_j . The point where the line leaves the base beyond X , and its intersection with the span of the other basis vectors, are therefore determined by block i . Projection onto block i induces a projective isomorphism on this line, sending the four defining points to their local counterparts. Thus its cross-ratio is q_{ia} , unchanged by α_i . Applying (18.3) to the product gives the result. $\square$

Theorem 18.2 (Lower bounds for products of nonsymmetric cones). *For $m_i \geq 3$ and $r_i \geq 1$,*

$$\nu_{\text{LH}} \left(\prod_{i=1}^k K_{p, m_i} \right) \geq kh(p), \quad (18.4)$$

$$\nu_{\text{LH}} \left(\prod_{i=1}^k \mathcal{P}_{p, r_i} \right) \geq kh(p). \quad (18.5)$$

At $p = 2$ both values of ν_{LH} equal $2k$.

Proof. For $1 < p < 2$, Hildebrand's norm-cone configurations have positive cross-ratio sums tending to $h(p) > 2$. Apply Lemma 18.1 and take the limit. For $p > 2$, the signed Legendre transform $F_*(s) = \sup_{x \in \text{int } C} \{-\langle s, x \rangle - F(x)\}$ preserves the LHSC parameter and has domain $\text{int } C^*$. Since $K_{p,3}^* = K_{q,3}$, the preceding case applied to the dual product proves the assertion. This dual step is necessary: for $p > 2$, Hildebrand's configurations have $\sum_a q_a < 2$, so Lemma 18.1 does not apply to them directly. In a larger norm cone, set all but two norm coordinates to zero. This linear section meets the ambient interior and is $K_{p,3}$, so barrier restriction proves (18.4).

In $\mathcal{P}_{p,r_i}$ retain one output coordinate. The resulting interior-meeting section is the scalar power cone $u^{1/p}v^{1/q} \geq |w|$. For $p \geq 2$, Hildebrand's power-cone configurations have $\sum_a q_a \geq 2$; for $p < 2$, swap u and v . Lemma 18.1 proves (18.5). At $p = 2$ every displayed cone is a Lorentz cone; the orthant lower bound and the sum of the standard Lorentz barriers both give $2k$. $\square$

The theorem gives additive lower bounds for a *fixed* product, without assuming a separable barrier. With Theorem 16.1, it shows that every lift of B_p^N by norm-cone factors of dimensions between three and d has an ambient cone with $\nu_{\text{LH}} \geq h(p) \lceil (N-1)/(d-2) \rceil$. At the minimum dimension, Theorem 13.4 gives a dictionary-independent statement: in any lift with $N+1$ cone coordinates, the lifting cone is isomorphic to $K_{p,N+1}$ and so has $\nu_{\text{LH}} \geq h(p)$.

The bound $h(p)$ per factor does not extend to arbitrary factors merely because the target is a non-Euclidean ball. For rational p , scalar power inequalities have finite Lorentz lifts [87]; applying these to $|x_j|^p \leq u_j$, $\sum_j u_j = 1$, gives an exact Lorentz-product lift of B_p^N . Each of its factors has optimal parameter two. This does not determine the best parameter over all dictionaries with a dimension cap, because this lift may use more than the minimum number of factors.

18.2 Grouped perspective cones: duality and parameter bounds

The perspective cones are known constructions [22, 41]. Their exact dual is

$$\mathcal{P}_{p,r}^* = \{(a, b, z) : a, b \geq 0, \quad a^{1/p}b^{1/q} \geq p^{-1/p}q^{-1/q}\|z\|_q\}. \quad (18.6)$$

Indeed, writing $w = vs$ reduces dual membership to $a\|s\|_p^p + b + \langle z, s \rangle \geq 0$ for every s ; the conjugate of $a\|s\|_p^p$ is $\|z\|_q^q/[q(ap)^{q-1}]$ for $a > 0$, with the boundary cases obtained by closure. Thus the map $(a, b, z) \mapsto (b, a, p^{-1/p}q^{-1/q}z)$ identifies the dual with $\mathcal{P}_{q,r}$. The section $u = v = t$ also gives

$$\nu_{\text{LH}}(\mathcal{P}_{p,r}) \geq \nu_{\text{LH}}(K_{p,r+1}). \quad (18.7)$$

The same comparison holds for products, by simultaneous restriction. A dimension- m proper cone admits an m -LHSC barrier, for example its canonical barrier [54]. Therefore

$$kh(p) \leq \nu_{\text{LH}} \left(\prod_i \mathcal{P}_{p,r_i} \right) \leq \sum_i (r_i + 2). \quad (18.8)$$

We do not claim that either bound is exact for $p \neq 2$.

We next give an explicit barrier with computable derivatives but a larger parameter. Put $\alpha = 1/p$, $\beta = 1/q$. Chares's scalar power barrier is

$$\phi_\alpha(a, v, z) = -\log(a^{2\alpha}v^{2\beta} - z^2) - \beta \log a - \alpha \log v. \quad (18.9)$$

It has parameter three [22, Theorem 3.1.1]. Since $\mathcal{P}_{p,r}$ is represented by $\sum_i a_i = u$, $(a_i, v, w_i) \in \mathcal{P}_{p,1}$, define

$$\Psi_{p,r}(u, v, w) = \min_{\sum_i a_i = u} \sum_i \phi_\alpha(a_i, v, w_i). \quad (18.10)$$

For an interior point the fiber has a nonempty interior and lies in $0 < a_i < u$. The objective is strictly convex and diverges at its boundary. The partial-minimization theorem for bounded fibers in Section 9 therefore gives a nondegenerate self-concordant barrier. Scaling the unique minimizer proves logarithmic homogeneity with parameter $3r$. This construction and the use of partial minimization are due to Chares; the formulas below give its evaluation cost.

At the minimizer, write $\phi_a(a_i, v, w_i) + \lambda = 0$ and $\sum_i a_i = u$. For a direction $(\dot{u}, \dot{v}, \dot{w})$ set

$$A_i = \phi_{aa} > 0, \quad g_i = \phi_{av}\dot{v} + \phi_{aw}\dot{w}_i, \quad S = \sum_i A_i^{-1}.$$

Differentiation gives

$$\dot{\lambda} = -\frac{\dot{u} + \sum_i g_i/A_i}{S}, \quad \dot{a}_i = -\frac{g_i + \dot{\lambda}}{A_i}, \quad \Psi_u = -\lambda, \quad \Psi_v = \sum_i \phi_v, \quad \Psi_{w_i} = \phi_w. \quad (18.11)$$

Once λ and the values a_i solving $\phi_a(a_i, v, w_i) = -\lambda$ are known, these identities evaluate a Hessian-vector product in $O(r)$ scalar operations. Equivalently, the Hessian is an arrowhead matrix plus a rank-one correction. This count excludes the cost of root finding to a given precision; we give no exact finite-arithmetic root-finding algorithm. Sparse-plus-low-rank Hessian structure for related nonsymmetric cones is known [23].

For an integer cap $d \geq 3$, set $k_g = \lceil N/(d-2) \rceil$ and $k_t = \lceil (N-1)/(d-2) \rceil$. The preceding results give the following parameters for three lifts of B_p^N : one scalar power cone per coordinate, grouped perspective cones $\mathcal{P}_{p,r}$, and the norm-cone tree from the proof of Theorem 16.1.

Representation of B_p^N	Bounds on ambient ν_{LH}	Scalar-power barrier
Coordinate power cones	$[Nh(p), 3N]$	$3N$
Grouped perspectives	$[k_g h(p), N + 2k_g]$	$3N$
Norm-cone tree	$[k_t h(p), N - 1 + 2k_t]$	$3(N - 1 + k_t)$

For the last entry, restrict a grouped barrier to $u = v = t$ in every norm-cone block; the tree has $N - 1 + k_t$ norm coordinates. The upper bounds in the middle column use barriers with parameter equal to the cone dimension, which are only known to exist; the last column uses the explicit barrier built from (18.9). These guarantees concern different barriers and cannot be combined. At $p = 2$ the exact ambient optima improve to $2N, 2k_g, 2k_t$. The generalized power cone theorem of Roy and Xiao [77] concerns Euclidean vector outputs; it gives no barrier of constant parameter for the ℓ_p vector output in (16.5).

18.3 Lower bounds for scaled characteristic barriers

For a proper cone $C \subset \mathbb{R}^m$, define its characteristic integral $Z_C(x) = \int_{C^*} \exp(-\langle x, y \rangle) dy$. Its logarithm is m -logarithmically homogeneous and, by the universal-barrier theorem, a self-concordant barrier with parameter m [60]. Among the multiples $\kappa \log Z_C$, a parameter below m requires $\kappa < 1$, and self-concordance then needs a separate proof. We call the functions $\kappa \log Z_C$, $\kappa > 0$, scaled characteristic barriers; they form the characteristic family.

Theorem 18.3 (Boundary asymptotics of the characteristic family). *Let $p > 2$ and $r \geq 2$. If $\kappa \log Z_C$ is self-concordant, its degree ν of logarithmic homogeneity satisfies*

$$C = \mathcal{P}_{p,r} : \quad \nu \geq \frac{r+2}{3/2 + (r-1)/p} > 2, \quad (18.12)$$

$$C = K_{p,r+1} : \quad \nu \geq \frac{r+1}{1 + (r-1)/p} > 2. \quad (18.13)$$

Both right sides tend to p as $r \rightarrow \infty$. For $1 < p < 2$, the same bounds, with $q = p/(p-1) > 2$ in place of p , apply to the Legendre duals of scaled characteristic barriers of the dual cone.

Proof. Write $\alpha = 1/p$, $\beta = 1/q$. Parameterize the interior of the dual in (18.6) by

$$a = \alpha s, \quad z = -sy, \quad b = s(\beta \|y\|_q^q + \rho), \quad s, \rho > 0, \quad y \in \mathbb{R}^r.$$

The absolute Jacobian is αs^{r+1} . At $x_\delta = (1 + \delta, 1, e_1)$, integration first in ρ and then in s gives

$$Z_C(x_\delta) = \alpha \Gamma(r+1) \int_{\mathbb{R}^r} [\alpha \delta + g(y)]^{-(r+1)} dy, \quad g(y) = \alpha + \beta \|y\|_q^q - y_1. \quad (18.14)$$

Young's inequality shows that g vanishes only at e_1 . Near this point, with $y_1 = 1 + \eta$,

$$g(y) = \frac{q-1}{2} \eta^2 + \beta \|y_\perp\|_q^q + O(|\eta|^3).$$

On a sufficiently small fixed neighborhood it is bounded above and below by positive multiples of $\eta^2 + \|y_\perp\|_q^q$. On compact sets outside that neighborhood it has a positive minimum, and for large y it is bounded below by a positive multiple of $\|y\|_q^q$. For derivative orders $j = 0, 1, 2, 3$, differentiation under the integral replaces the exponent $r+1$ by $r+1+j$ and multiplies by $(-\alpha)^j (r+1)_j$. The outside integrals are bounded uniformly as $\delta \downarrow 0$, since $q(r+1+j) > r$.

Inside the neighborhood substitute $\eta = \delta^{1/2} t$, $y_\perp = \delta^{1/q} z$. The resulting integrands are dominated by a constant multiple of $(1+t^2 + \|z\|_q^q)^{-(r+1+j)}$, which is integrable: the anisotropic volume exponent is $a = 1/2 + (r-1)/q < r+1$. Dominated convergence gives, with $B = r+1-a = 3/2 + (r-1)/p$,

$$\frac{d^j}{d\delta^j} Z_C(x_\delta) \sim (-1)^j (B)_j C \delta^{-B-j}, \quad 0 \leq j \leq 3, \quad (18.15)$$

for a constant $C > 0$. The coefficient ratios follow either by the beta integral in anisotropic radial coordinates or by differentiating that homogeneous model integral. Thus $f = \kappa \log Z_C(x_\delta)$ obeys $f'' \sim \kappa B \delta^{-2}$ and $f''' \sim -2\kappa B \delta^{-3}$. Univariate self-concordance forces $\kappa B \geq 1$. Since $\nu = (r+2)\kappa$, this proves the first bound.

For $K_{p,r+1}$ the dual parameterization (s, sy) , $y \in B_q^r$, has Jacobian s^r , so at $\hat{x}_\delta = (1 + \delta, e_1)$,

$$Z_C(\hat{x}_\delta) = \Gamma(r+1) \int_{B_q^r} (1 + \delta + y_1)^{-(r+1)} dy.$$

Writing $a = 1 + y_1$, the transverse section volume is $\text{vol}(B_q^{r-1})[1 - |1 - a|^q]^{(r-1)/q}$, which equals a positive constant times $a^{(r-1)/q}(1 + O(a))$ near zero. The substitution $a = \delta t$, with the same differentiated dominated-convergence argument, gives (18.15) with $B = r - (r-1)/q = 1 + (r-1)/p$. This proves the second bound. The strict inequalities are equivalent to $(r-1)(1 - 2/p) > 0$. The final statement follows from (18.6), $K_{p,r+1}^* = K_{q,r+1}$, and preservation of parameter by Legendre duality. The statement concerns Legendre duals because the Legendre dual of a scaled characteristic barrier need not be one. $\square$

The theorem applies to one family of barriers, not to all LHSC barriers, so its bounds do not replace the lower bounds in the table. For $p > 2$, the perspective bound is strictly smaller than the norm bound at the same r , since their cross-multiplied difference has the sign of $(r-1)(1/p - 1/2)$. This compares the lower bounds only, not the optimal parameters.

18.4 The three-dimensional case and explicit formulas that fail

For $K_{p,3}$, the general bounds are

$$h(p) \leq \nu_{\text{LH}}(K_{p,3}) \leq 3. \quad (18.16)$$

The value is two at $p = 2$ and three at the polyhedral endpoints $p = 1$ and $p = \infty$ [53]. We do not determine the optimal value for other p ; recent work on barrier design still treats it as unknown [75]. For $p > 2$, Theorem 18.3 gives the stronger bound $\nu \geq 3p/(p+1)$ for scaled characteristic barriers. Chares's numerical experiments suggested that this value suffices; we do not prove this.

Proposition 18.4 (Strict separation from the cross-ratio bound). *For $p > 2$, $3p/(p+1) > h(p)$. Consequently no scaled characteristic barrier on $K_{p,3}$ attains Hildebrand's bound. For $p < 2$ the analogous statement concerns Legendre duals of scaled characteristic barriers on the dual cone and the bound $3p/(2p-1)$.*

Proof. Set $s = p - 2 > 0$ and $A = \gamma^{1/p}$. Equation (18.1) becomes $A^s(sA + s + 1) = 1$, whereas $h(p) = 2 + sA/(s+1)$ and $3p/(p+1) = 2 + s/(s+3)$. The left side is increasing in A . At $A_0 = (s+1)/(s+3)$ its logarithm is $g(s) = \log(2s+3) + (s+1)\log((s+1)/(s+3))$. Here $g(0) = 0$ and, putting $z = 2/(s+3)$,

$$g'(s) = \log(1-z) + z + \frac{2z}{4-3z} =: H(z), \quad H'(z) = \frac{8-24z+24z^2-9z^3}{(1-z)(4-3z)^2} > 0 \quad (0 < z < 2/3).$$

The numerator decreases to zero at $2/3$: its derivative is $-3(8-16z+9z^2) < 0$. Since $H(0) = 0$, $g(s) > 0$ and $A < A_0$. The desired inequality follows. Duality gives the last assertion. $\square$

Several explicit barrier forms cannot close the gap in (18.16). First, a hypothetical 2-LHSC barrier F on a proper cone must be a Lorentz barrier up to a linear map and an additive constant. To see this, take h tangent to a level set of F at x , put $a = F''[h, h]$, $b = F'''[h, h, h]$, and use homogeneity to obtain

$$F''[h+tx, h+tx] = a + 2t^2, \quad F'''[h+tx, h+tx, h+tx] = b - 6at - 4t^3.$$

Self-concordance at $t = \pm\sqrt{a/2}$ forces $b = 0$. For $Q = e^{-F}$ this implies $D^3Q = 0$ on triples of vectors tangent to the level set; homogeneity of degree two gives $D^3Q[x, \cdot, \cdot] = 0$. Hence Q is quadratic. Its Hessian is positive on x and negative definite on the level-set tangent space, so its signature is $(1, m-1)$. The barrier boundary condition identifies the cone with a component of $Q \geq 0$. Thus no non-Lorentz cone has a 2-LHSC barrier; this alone does not show that the infimum exceeds two.

For a more concrete obstruction, consider, on $K_{p,3}$,

$$F = -\alpha_0 \log A - \beta_0 \log(A^{2/p} - y^2), \quad A = t^p - |x|^p.$$

If this is an LHSC barrier, an affine line reaching a generic boundary point where $A > 0$ forces $\beta_0 \geq 1$ by univariate self-concordance. Along $(t, x, y) = (1, 1-\delta, 0)$ the same argument forces $\alpha_0 + 2\beta_0/p \geq 1$. Its degree of logarithmic homogeneity therefore satisfies $\nu = p\alpha_0 + 2\beta_0 \geq p > h(p)$ for $p > 2$. For $2 < p \leq 3$ the nonzero coefficient of $|x|^p$ at $y = 0$ also precludes C^3 regularity on the interior axis $x = y = 0$. The still simpler formula

$$-a \log(t^p - |x|^p - |y|^p) - b \log t, \quad a > 0,$$

is not C^2 on that axis for $1 < p < 2$ and has zero transverse Hessian there for $p > 2$.

Adding a regular correction with bounded derivatives cannot repair the last formula. On $t = 1$, let $L(x, y) = -a \log(1 - |x|^p - |y|^p)$. For $p \leq 3$, $p \neq 2$, adding a C^3 function cannot cancel its loss of C^2 or C^3 regularity on the interior axis. For $p > 3$, put $x_\delta = (1-\delta)^{1/p}$ and $y_\delta = \delta^{1/(p-2)}$. These points are interior for small δ , and direct differentiation gives

$$L_{yy}(x_\delta, y_\delta) = ap(p-1) + o(1), \quad L_{yyy}(x_\delta, y_\delta) = ap(p-1)(p-2)\delta^{-1/(p-2)}(1 + o(1)).$$

If a correction has bounded yy and yyy derivatives along this sequence, the corrected second derivative stays bounded while the third diverges. Either convexity fails or self-concordance does. An admissible correction must therefore be singular in the transverse direction near the flat boundary; we draw no conclusion about unrestricted corrections.

Finally, we expand $h(p)$ near the endpoints. Write $d_0 = |1 - 2/p|$ and $w = W(e^{-1})$, where W is Lambert's function. Then

$$h(p) = 2 + 2wd_0 + 2w^2d_0^2 + O(d_0^3), \quad p \rightarrow 2, \quad (18.17)$$

$$h(p) = 3 - \frac{\log(2p) + 1}{p} + \frac{\log^2(2p) - \log(2p) + 1}{2p^2} + O\left(\frac{\log^3 p}{p^3}\right), \quad p \rightarrow \infty. \quad (18.18)$$

In particular $h(p) = 2 + w|p - 2| + O((p - 2)^2)$. To verify the expansions set $a = \gamma^c$ and $s = 1/c - 2$. The root equation is $a^s(1 + s(a + 1)) = 1$. After division of its logarithm by s , the analytic implicit-function theorem at $s = 0$ gives $a = w + \frac{1}{2}w(1 + w)s + O(s^2)$, proving (18.17). At $c \downarrow 0$, let $\ell = \log(c/2)$. Substitution in the root equation gives

$$\gamma = \frac{c}{2} \left[1 + \frac{3}{2}c(\ell + 1) + O(c^2\ell^2) \right], \quad h(p) = 3 + c(\ell - 1) + \frac{1}{2}c^2(\ell^2 + \ell + 1) + O(c^3|\ell|^3),$$

which proves (18.18). For $p = 1 + \varepsilon$, Hölder duality yields, with $L = \log(2/\varepsilon)$, $h(p) = 3 - \varepsilon(L + 1) + \frac{1}{2}\varepsilon^2(L^2 + L + 1) + O(\varepsilon^3L^3)$.

The scalar cross-ratio bound, the perspective cones, the scalar power barrier, and the partial-minimization theorem are from the literature. Our contributions are Lemma 18.1 and its consequences for fixed dictionaries, the lower bounds for scaled characteristic barriers, and the explicit formulas shown to fail. To the best of our knowledge, the dimension-dependent bounds (18.12)–(18.13) have not been stated for these families.

19 Exponential factors and entropy aggregation

Changing the ambient cone can lower a barrier parameter even when coupling the factors of a fixed product cannot. We compute the exact optimal parameters for products of exponential cones and for relative-entropy cones with a nonnegative output map. The barriers and lower-bound method are special cases of Fawzi and Saunderson [35]; we give the explicit certificates and closure arguments needed to compare formulations, and claim no new general theorem on optimal barriers.

19.1 A recession lower bound for products

For a closed full-dimensional convex set C , let $\mathbf{n}(C)$ be the supremum of $\sum_j a_j/b_j$ over interior points x , recession directions p_j , and positive numbers a_j, b_j satisfying

$$x - \sum_j a_j p_j \in C, \quad x - b_j p_j \notin \text{int } C. \quad (19.1)$$

We call such data a recession certificate with value $\sum_j a_j/b_j$, coefficients a_j , and thresholds b_j . Nesterov's recession theorem, in the form [35, Theorem 3.9], gives $\nu \geq \mathbf{n}(C)$ for every self-concordant barrier on C , including nonhomogeneous barriers. Certificates for separate factors combine by padding each recession direction with zeros; both conditions in (19.1) still hold, so

$$\mathbf{n}(C_1 \times \cdots \times C_k) \geq \sum_j \mathbf{n}(C_j). \quad (19.2)$$

This bound applies to every barrier on the product, including coupled ones; the sum rule for barriers gives only an upper bound.

Lemma 19.1 (Orthant-input hypographs). *Let $n, B \geq 1$ be integers and $h : \mathbb{R}_{++}^n \rightarrow \mathbb{R}^B$ be componentwise concave and homogeneous of degree one. For*

$$H_h = \overline{\{(x, z) : x > 0, z \leq h(x)\}},$$

every self-concordant barrier on H_h has parameter at least $n + B$.

Proof. This is the orthant-output specialization of Fawzi and Saunderson [35, Theorem 3.10]. We give the certificate explicitly. For $0 < \epsilon < 1$, set $v_i = \epsilon \mathbf{1} + (1 - \epsilon)e_i$ and $x_0 = \sum_i v_i = b\mathbf{1}$, where $b = 1 + (n - 1)\epsilon$. Concavity and homogeneity imply $d = h(x_0) - \sum_i h(v_i) \geq 0$ componentwise. Choose $T' > \max_j d_j$ and $T > T'$. At $y_0 = (x_0, h(x_0) - T\mathbf{1})$ use recession directions $(v_i, h(v_i))$ with coefficients 1 and thresholds b , and $(0, -Te_j)$ with coefficients $1 - T'/T$ and thresholds 1. Subtracting these directions, scaled by their coefficients, leaves $(0, d - T'\mathbf{1}) \in H_h$. Each input threshold makes an input coordinate zero; each output threshold makes an output inequality tight. Hence each threshold point is noninterior. The certificate has value $n/b + B(1 - T'/T)$. First let $T \rightarrow \infty$, then $\epsilon \downarrow 0$. $\square$

Theorem 19.2 (Exact exponential-product parameter). *Let*

$$K_{\text{exp}} = \overline{\{(u, v, w) : v > 0, w \geq ve^{u/v}\}}.$$

The smallest parameter of any self-concordant barrier on a product of R primal or dual exponential cones and P rays is $3R + P$.

Proof. Write the primal cone as the hypograph of $h(v, w) = v \log(w/v)$. Lemma 19.1 gives certificates with values tending to three. The compatible barrier

$$-\log(v \log(w/v) - u) - \log v - \log w$$

has parameter three [35]. A ray has a certificate of value one and barrier $-\log t$. Equation (19.2) and the sum of these barriers give the product assertion. The map $(a, b, c) \mapsto (a - b, -a, c)$ is a linear isomorphism from K_{exp}^* onto K_{exp} : its interior formula is $a < 0, c > -a \exp(b/a - 1)$, and its closure face $a = 0, b, c \geq 0$ maps to the primal face $v = 0, u \leq 0, w \geq 0$. $\square$

The same argument gives the optimal parameter $\sum_j (n_j + 1)$ for a product of scalar hypographs of homogeneous concave $h_j : \mathbb{R}_{++}^{n_j} \rightarrow \mathbb{R}$ whenever the compatible barriers $-\log(h_j(x) - z) - \sum_i \log x_i$ exist. This includes the usual one-sided geometric-mean hypographs. To check this directly, let $g(x) = \prod_i x_i^{\alpha_i}$, $\alpha_i > 0$, $\sum_i \alpha_i = 1$. For a direction put $u_i = h_i/x_i$ and $\bar{u} = \sum_i \alpha_i u_i$. Then $-D^2g = g \sum_i \alpha_i (u_i - \bar{u})^2$ and $D^3g = g \sum_i \alpha_i (\bar{u} + 2u_i)(u_i - \bar{u})^2$. Hence $|D^3g| \leq 3\|u\|_2(-D^2g)$, the compatibility inequality with the orthant barrier. The argument needs the output recession direction, so it gives no exact parameter for an absolute power cone. For matrix relative entropy, the published optimal parameter $2n + 1$ also adds over products; restricting the inputs to diagonal matrices preserves the recession certificates and yields $\sum_j (2n_j + 1)$ for a product of these matrix epigraphs [35].

19.2 Positive output maps and their boundary fibers

Put $\phi_i(x_i, y_i) = x_i \log(x_i/y_i)$ for positive inputs. Let N, B be positive integers and $A \in \mathbb{R}_+^{B \times N}$. The cone

$$C_A = \overline{\{(x, y, z) : x, y > 0, z \geq A\phi(x, y)\}}. \quad (19.3)$$

keeps all N input pairs and all B output coordinates as independent variables. Zero or repeated rows and zero columns are permitted. We call C_A the intrinsic cone.

Theorem 19.3 (Exact intrinsic aggregation parameter). *The optimal barrier parameter of C_A is $2N + B$, attained by*

$$F_A = - \sum_{b=1}^B \log s_b - \sum_{i=1}^N (\log x_i + \log y_i), \quad s = z - A\phi(x, y) > 0. \quad (19.4)$$

Let $J = \{i : A_{\cdot i} \neq 0\}$ (the active indices) and $n_+ = |J|$. An exact closed affine lift of C_A uses n_+ exponential cones, $2(N - n_+)$ input rays, and B output rays. The optimal parameter of this ambient product is $2N + n_+ + B$.

Proof. We first verify the compatibility inequality behind the classical barrier construction. For one pair and a direction (a, b) write $u = a/x, v = b/y$. Direct differentiation gives

$$D^2\phi = x(u - v)^2, \quad D^3\phi = -x(u - v)^2(u + 2v).$$

Therefore $|D^3\phi| \leq 3\sqrt{u^2 + v^2} D^2\phi$. The common orthant barrier has second derivative $\sum_i (u_i^2 + v_i^2)$, so multiplication by a nonnegative row of A preserves the compatibility inequality. Nesterov's compatible hypograph theorem [35, Theorem 3.3], applied to the diagonal output matrix, proves self-concordance of (19.4). Its homogeneity degree is $2N + B$; its Hessian, given below, is positive definite. Lemma 19.1 with $h(x, y) = -A\phi(x, y)$ proves the matching lower bound.

For the exact lift use $q_i \geq \phi_i(x_i, y_i)$ for $i \in J$, $(x_i, y_i) \geq 0$ for $i \notin J$, and

$$z = A_{\cdot J}q + r, \quad r \geq 0. \quad (19.5)$$

At the boundary, extend each active scalar entropy by $\bar{\phi}(0, y) = 0$ for $y \geq 0$, and $\bar{\phi}(x, 0) = +\infty$ for $x > 0$. The elementary bound $\phi(x, y) \geq -y/e$ shows that these terms are uniformly bounded below along convergent input sequences. If a point belongs to (19.3), choose an interior approximating sequence. Every active i has some $A_{bi} > 0$; boundedness of z_b and the lower bounds on all other terms give an upper bound on ϕ_i . Thus $x_i > 0, y_i = 0$ is impossible for active i . Lower semicontinuity then gives $z \geq A_{\cdot J}\bar{\phi}_J$. Conversely, every such inequality is approximated by positive inputs: replace each zero coordinate by ϵ and shift the output by the resulting change in the active entropies, which vanishes because they converge to their finite extended values. Choosing $q = \bar{\phi}_J$ and $r = z - A_{\cdot J}q$ gives every point of C_A a preimage with finite q in (19.5), so the lift is exact. The final parameter follows from Theorem 19.2. $\square$

Exactness requires rays for zero columns. With $N = B = 1$ and $A = 0$, $C_A = \mathbb{R}_+^3$ includes $(x, y, z) = (1, 0, 0)$; a finite exponential constraint $q \geq \phi(1, 0)$ cannot represent this point. Keeping an unused exponential cone gives only a nonclosed image with the right closure. Replacing it by two rays restores exactness.

For a partition of N terms into B nonempty blocks, one may instead sum the local epigraph coordinates within each block, without separate output rays. The ambient product then has optimal

parameter $3N$, and C_A has $2N + B$, a difference of $N - B$. This lift is onto C_A , including its boundary, because a nonnegative residual in each block output can be absorbed into one local epigraph coordinate. For a general positive map, passing from the lift (19.5) to the intrinsic cone lowers the parameter by n_+ , and the ratio of the usual short-step iteration bounds is

$$\sqrt{\frac{2N + n_+ + B}{2N + B}} < \sqrt{3/2}.$$

This compares parameter-based upper bounds, not actual iteration counts.

19.3 A sparse intrinsic Hessian

Let $p = (x_1, y_1, \dots, x_N, y_N)$, let J_A denote the Jacobian of $A\phi$, and put $R = \text{Diag}(s_b^{-1})$. Differentiation of (19.4) gives

$$\nabla^2 F_A = \begin{pmatrix} I & -J_A^T \\ 0 & I \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & R^2 \end{pmatrix} \begin{pmatrix} I & 0 \\ -J_A & I \end{pmatrix}, \quad (19.6)$$

where G has independent positive-definite 2×2 blocks

$$G_i = \text{Diag}(x_i^{-2}, y_i^{-2}) + \left(\sum_b A_{bi}/s_b \right) \begin{pmatrix} x_i^{-1} & -y_i^{-1} \\ -y_i^{-1} & x_i/y_i^2 \end{pmatrix}.$$

The second summand is positive semidefinite. Assembling G requires the product $A^T(s_b^{-1})_{b=1}^B$. After assembly, each solve with (19.6) uses triangular substitution, two products by A or A^T , diagonal scaling, and N two-dimensional solves. Given the point and slacks, assembly and one solve take $O(N + B + \text{nnz } A)$ arithmetic operations. This is a solve with the cone-barrier Hessian only; affine constraints add a separate Schur complement system. Sparse and low-rank Hessian formulas for ordinary relative entropy are due to Chen and Goulart [23]. He et al. [50] tailor entropy cones to positive input maps, including maps with singular images, and remove redundant determinant terms. Here the explicit output map records which coordinates remain independent, and the closure analysis shows when the elementary lift is exact.

Part IV

Approximation and exact compilation

20 Approximation with controlled cone factors

In an exact lift, a proper cone of dimension m supplies at most $\max\{m - 2, 0\}$ mixed-curvature directions. The bound extends to approximate lifts only under additional control of the factors. We give two versions: a theorem for finitely many sampled directions, which uses no derivatives, and a local theorem for smooth factorizations, in which the mixed derivative of the slack is exact. The hypotheses concern the selected factors in fixed Euclidean coordinates and are stronger than approximation of the projected convex body.

20.1 A finite sampling theorem

Write $r = N - 1 \geq 1$, fix $0 < h < 1$, and put

$$v_0 = e_N, \quad v_j = he_j + \sqrt{1 - h^2}e_N \quad (1 \leq j \leq r), \quad \tau = 1 - \sqrt{1 - h^2}.$$

Suppose proper cones $K_i \subset E_i$, of dimensions m_i , admit factors $A_i(v_j) \in K_i$, $B_i(v_j) \in K_i^*$ such that

$$\left| \sum_i \langle A_i(v_j), B_i(v_\ell) \rangle - (1 - \langle v_j, v_\ell \rangle) \right| \leq \epsilon \quad (0 \leq j, \ell \leq r). \quad (20.1)$$

Let $a_i = A_i(v_0)$, $b_i = B_i(v_0)$, and let $X_i, Y_i : \mathbb{R}^r \rightarrow E_i$ have columns $A_i(v_j) - a_i$, $B_i(v_j) - b_i$, respectively. All matrix norms below are operator norms. Define

$$E_i = \frac{\sqrt{r}(\tau + \epsilon)}{\|b_i\|} \|Y_i\| + \frac{\sqrt{r}(\tau + \epsilon)}{\|a_i\|} \|X_i\| + \frac{\epsilon \|X_i\| \|Y_i\|}{\|a_i\| \|b_i\|} \quad (20.2)$$

when $a_i, b_i \neq 0$; otherwise define $E_i = \|X_i\| \|Y_i\|$.

Theorem 20.1 (Finite samples with controlled factors). *If $4r\epsilon + \sum_i E_i < h^2$, then*

$$N - 1 \leq \sum_i (m_i - 2)_+.$$

In particular, if all factors of dimension at least three have dimension at most an integer $d \geq 3$, at least $\lceil (N - 1)/(d - 2) \rceil$ such factors are required.

Proof. Fix a block with $a, b \neq 0$. Nonnegativity of every summand gives $0 \leq \langle a, b \rangle \leq \epsilon$ and

$$\|X^*b\|, \|Y^*a\| \leq \sqrt{r}(\tau + \epsilon).$$

Indeed, each coordinate is the difference of two numbers in $[0, \tau + \epsilon]$. Set $X_0 = P_{b^\perp}X$ and $Y_0 = P_{a^\perp}Y$. Then

$$\|X^*Y - X_0^*Y_0\| \leq \frac{\|X^*b\|}{\|b\|} \|Y\| + \|X\| \frac{\|Y^*a\|}{\|a\|}. \quad (20.3)$$

For $m \geq 2$, the product $P_{b^\perp}P_{a^\perp}$ has singular values 1 with multiplicity $m - 2$, together with $\gamma = \langle a, b \rangle / (\|a\| \|b\|)$ and 0. This follows by splitting off $(\text{span}\{a, b\})^\perp$ and computing the remaining plane; the parallel case gives the same list with $\gamma = 1$. Deleting the singular direction of size γ therefore approximates $X_0^*Y_0$ by a rank at most $m - 2$ matrix with error at most $\gamma \|X\| \|Y\|$.

Equation (20.3) proves the bound E_i . For $m = 1$, $X_0 = Y_0 = 0$, so the first two terms of E_i suffice. If a base factor is zero, the zero approximant has error at most $\|X_i\|\|Y_i\| = E_i$.

The mixed difference matrix of the kernel $k(v, w) = \sum_i \langle A_i(v), B_i(w) \rangle$, with entries $k(v_j, v_\ell) - k(v_j, v_0) - k(v_0, v_\ell) + k(v_0, v_0)$, is $\sum_i X_i^* Y_i$. For the exact sphere slack it is $D_* = -h^2 I_r - \tau^2 \mathbf{1}\mathbf{1}^*$, whose smallest singular value is at least h^2 . Each entry combines four errors from (20.1), so the operator error is at most $4r\epsilon$. The sum of the block approximants has rank at most $\sum_i (m_i - 2)_+$ and lies within $4r\epsilon + \sum_i E_i$ of D_* . A singular matrix cannot lie at operator distance less than h^2 from D_* : apply it to a unit vector in its kernel. This proves the assertion. $\square$

A simpler sufficient condition is available when there are k blocks, $\|a_i\|, \|b_i\| \geq \rho > 0$, and $\|X_i\|, \|Y_i\| \leq Lh$. If $\epsilon \leq h^2$, then $\tau \leq h^2$, and it suffices that

$$\frac{4r\epsilon}{h^2} + \frac{4k\sqrt{r}Lh}{\rho} + \frac{k\epsilon L^2}{\rho^2} < 1. \quad (20.4)$$

For fixed r, k, L, ρ , the choice h of order $\epsilon^{1/3}$ makes the left side tend to zero, provided the bounds on $\|X_i\|, \|Y_i\|$ hold for the shrinking samples.

If $(1 - \eta)\mathbb{B}_2^N \subset C \subset (1 + \eta)\mathbb{B}_2^N$, $0 < \eta < 1$, choose radial boundary points $x(v) = \rho_C(v)v$ and $y(w) = \rho_{C^\circ}(w)w$. Their radial lengths lie in $[1 - \eta, 1 + \eta]$ and $[(1 + \eta)^{-1}, (1 - \eta)^{-1}]$. Consequently

$$|1 - \langle x(v), y(w) \rangle - (1 - \langle v, w \rangle)| \leq \frac{2\eta}{1 - \eta}.$$

The reduction and certificate argument of Lemma 2.1 provides a factorization of this slack from any exact affine lift of C , including at radial points that are not extreme. Thus metric accuracy gives (20.1) with $\epsilon = 2\eta/(1 - \eta)$, but not the factor bounds in (20.2) or (20.4).

20.2 Smooth near-complementarity

For $r_0 > 0$, $\delta, H \geq 0$, define

$$\psi_{r_0}(\delta, H) = \inf_{0 < s \leq r_0} \left(\frac{\delta}{s} + \frac{Hs}{2} \right). \quad (20.5)$$

Because the radius is finite, this covers $H = 0$, where the value is δ/r_0 . If $H > 0$ and $\sqrt{2\delta/H} \leq r_0$, it equals $\sqrt{2H\delta}$; for $\delta = 0$ the value is zero by a limit.

Lemma 20.2 (Quantitative near-complementarity). *Let $A : U \rightarrow K$, $B : U \rightarrow K^*$ be C^2 , where $K \subset E$ is a proper cone of dimension m and U contains the closed Euclidean ball of radius r_0 about zero. Put $a = A(0)$, $b = B(0)$, $X = DA(0)$, $Y = DB(0)$, and $\delta = \langle a, b \rangle$. Suppose $a, b \neq 0$ and let*

$$H_A = \sup_{\|u\|=1, |t| \leq r_0} |\langle D^2 A(tu)[u, u], b \rangle|, \quad H_B = \sup_{\|u\|=1, |t| \leq r_0} |\langle a, D^2 B(tu)[u, u] \rangle|.$$

Write $U_1 = \|X\|$, $V_1 = \|Y\|$ and

$$\alpha = \frac{\psi_{r_0}(\delta, H_A)}{\|b\|}, \quad \beta = \frac{\psi_{r_0}(\delta, H_B)}{\|a\|}, \quad \gamma = \frac{\delta}{\|a\|\|b\|}.$$

For $m \geq 2$, X^*Y is within

$$e = U_1\beta + V_1\alpha + U_1V_1\gamma \quad (20.6)$$

of a matrix of rank at most $m - 2$. For a ray, the zero approximant has error at most $e = \alpha\beta$. If either base factor is zero, $X^*Y = 0$.

Proof. For a unit u , the function $f(t) = \langle A(tu), b \rangle$ is nonnegative, $f(0) = \delta$, and $|f''| \leq H_A$. Taylor's inequality at $-s$ or s , with sign opposite to $f'(0)$, gives $|f'(0)| \leq \delta/s + H_A s/2$. Hence $\|X^*b\| \leq \psi_{r_0}(\delta, H_A)$, and similarly for Y^*a . Apply (20.3) and its singular-value truncation to obtain (20.6). For $m = 1$ the normal components are the entire derivatives, giving $\alpha\beta$. If $a = 0$, every two-sided derivative of A lies in $K \cap (-K) = \{0\}$, so $X = 0$; the dual case is identical. $\square$

The bound (20.6) implies the more conservative bound $U_1\beta + V_1\alpha + \alpha\beta\gamma + U_1V_1\gamma$, obtained by expanding both normal components separately. Keeping one projected factor in (20.3) removes the extra term. Both bounds are unchanged when A is multiplied by a positive constant and B by its reciprocal.

Theorem 20.3 (Smooth approximation with controlled factors). *Let $\mathbb{B}_2^N \subset C \subset (1 + \epsilon)\mathbb{B}_2^N$ and suppose a local C^2 factorization on a sphere chart $\sigma : U \rightarrow \mathbb{S}^{N-1}$ satisfies*

$$\sum_i \langle A_i(u), B_i(v) \rangle = h_C(\sigma(v)) - \langle \sigma(u), \sigma(v) \rangle, \quad D\sigma(0)^* D\sigma(0) = I_{N-1}. \quad (20.7)$$

Apply Lemma 20.2 in each factor span, of dimension s_i , with resulting error e_i . If $\sum_i e_i < 1$, then

$$N - 1 \leq \sum_i (s_i - 2)_+.$$

For a definable exact lift of C , after minimal-face reduction, such local C^2 factorizations exist on some nonempty open subset of the sphere. The condition $\sum_i e_i < 1$ is an additional hypothesis.

Proof. At the chart center, nonnegativity gives $\delta_i \geq 0$ and $\sum_i \delta_i = h_C(\sigma(0)) - 1 \leq \epsilon$. Mixed differentiation of (20.7) gives the exact identity $-\sum_i DA_i(0)^* DB_i(0) = I_{N-1}$, because the support term depends only on v . Lemma 20.2 approximates this identity by a matrix of rank at most $\sum_i (s_i - 2)_+$ with error at most $\sum_i e_i$. Error less than one makes that matrix invertible.

For the existence statement, write the reduced lift as $C = \pi\{z \in K : Mz = b_0\}$ with relative Slater feasibility, absorbing an affine output offset into the dual objective if necessary. Choose the minimum-norm feasible lift of $\sigma(u)$ and the minimum-norm optimal dual multiplier $\lambda(v)$ for maximizing $\langle \sigma(v), \pi z \rangle$. Both choices exist in closed nonempty fibers; Slater dual attainment supplies the second fiber. With $B(v) = M^* \lambda(v) - \pi^* \sigma(v)$, weak duality and attainment give (20.7). Definable choice by these unique minimizers and C^2 cell decomposition give a common open patch. We do not claim a uniform patch size or uniform derivative bounds across families of lifts. $\square$

For example, suppose there are at most k nonzero selected blocks, $\|a_i\|, \|b_i\| \geq \mu > 0$, $\|DA_i\|, \|DB_i\| \leq L$, and $H_{A,i}, H_{B,i} \leq H$, where $H > 0$ and $\sqrt{2\epsilon/H} \leq r_0$. The sum of the errors for blocks of dimension at least two is at most

$$\mathcal{E}_k(\epsilon) = \frac{2L\sqrt{2Hk\epsilon}}{\mu} + \frac{L^2\epsilon}{\mu^2}. \quad (20.8)$$

This follows from $\sum_i \sqrt{\delta_i} \leq \sqrt{k\epsilon}$. Rays add at most $2H\epsilon/\mu^2$. Zero selected factors add zero. Thus a bounded number of blocks, factor norms bounded away from zero, bounded derivatives, and a fixed chart radius give error $O(\sqrt{\epsilon})$. Conversely, a family with $\sum_i (s_i - 2)_+ < N - 1$ must have $\mathcal{E}_k(\epsilon) + 2H\epsilon/\mu^2 \geq 1$ (omit the last term when no rays occur). If the terms other than the square-root term sum to at most $1/2$, then

$$L\sqrt{H} \geq \frac{\mu}{4\sqrt{2k\epsilon}}.$$

This concerns the selected factors, not an intrinsic condition number of a Newton system. Global consequences under stronger selection hypotheses appear in Appendix C.

20.3 Why approximating the convex body is not enough

For every $\epsilon > 0$, choose a finite δ -net $\mathcal{U} \subset \mathbb{S}^{N-1}$ with $(1 - \delta^2/2)^{-1} \leq 1 + \epsilon$. The polytope

$$P = \{x : \langle u, x \rangle \leq 1 \text{ for all } u \in \mathcal{U}\}$$

contains $\mathbb{B}_2^N$. For $x \neq 0$ choose u within δ of $x/\|x\|$. Then $\langle u, x/\|x\| \rangle \geq 1 - \delta^2/2$, so $\|x\| \leq 1 + \epsilon$ for $x \in P$. One nonnegative slack per inequality gives an exact lift of P by rays only. Therefore no bound on the Hausdorff or multiplicative error alone, however small, forces $N - 1 \leq \sum_i (m_i - 2)_+$. The number of scalar inequalities is a separate resource. For lifted polyhedral approximations of the Lorentz cone, Ben-Tal and Nemirovski prove matching bounds of order $N \log(1/\epsilon)$ on this number for small ϵ [13, Theorem 1.1 and Proposition 3.1]. If such an approximation is realized with an orthant of that dimension, the optimal ambient barrier parameter equals the number of slacks. This is a property of that realization, not of every cone that lifts the same polytope.

At a fixed error, the bound can fail even for lifts without rays. For $d \geq 3$,

$$\mathbb{B}_2^{2(d-1)} \subset \mathbb{B}_2^{d-1} \times \mathbb{B}_2^{d-1} \subset \sqrt{2} \mathbb{B}_2^{2(d-1)}.$$

The outer ratio and Hausdorff distance $\sqrt{2} - 1$ are attained by pairs of unit vectors. Two Q_d factors represent the product, whereas the exact ball requires at least $\lceil (2d - 3)/(d - 2) \rceil = 3$ factors of dimension at most d .

Bounded factors and an arbitrarily small sampled slack error do not replace the hypothesis of Theorem 20.1, which controls the increments relative to the base factors. For its two-point sample ($r = 1$), fix $\alpha > 0$ and put $\beta = \tau/\alpha$. In $\mathbb{R}_+^2$ take

$$A(v_0) = B(v_0) = (\alpha, \alpha), \quad A(v_1) = (0, \alpha + \beta), \quad B(v_1) = (\alpha + \beta, 0).$$

The kernel values are $2\alpha^2, \tau + \alpha^2, \tau + \alpha^2, 0$, so the error tends to zero. Its mixed difference is nevertheless $-2\tau = -h^2 - \tau^2$, exactly the sphere value, although $(m - 2)_+ = 0$. Apply the diagonal automorphism $D = \text{diag}(\sqrt{\beta/\alpha}, \sqrt{\alpha/\beta})$ to A and D^{-1} to B . All factors become bounded as $\alpha \downarrow 0$ and the pairings stay unchanged, but the hypothesis still fails. This finite example does not by itself construct a global fixed-size approximation of a ball.

Approximate slack factorizations and their geometric approximation properties are classical. Gouveia, Parrilo, and Thomas bound inner and outer polytope approximations using the maximum column and row Euclidean norms of a finite slack-matrix error; they also obtain approximate factorizations from lifted approximations of nested polyhedra [44, Sections 2.7 and 4.2]. Our results add explicit local conditions on the factors at curved contact points under which the rank bound $\dim K_i - 2$ holds for approximate lifts. We do not claim that every metric approximation admits such factors, or that violating the conditions imposes an algorithmic cost.

21 Exact compilers and the limits of representation-specific rank

We study three ways in which an exact formulation can become smaller: several input constraints are replaced by one exact constraint; a function of source data is evaluated from a finite summary of the data; or auxiliary variables write a function as a composition of simpler functions. We give exact criteria for the first two. For the third, we show that the minimum number of exponential terms in an exact representation of a function (its signed exponential rank) does not bound the number of exponential cones needed. A *compiler* is a map from the given input data to an exact representation in a prescribed output format; each result depends on that format.

21.1 Intersections of translates

For a proper cone $K \subset \mathbb{R}^d$, write $x \leq_K y$ if $y - x \in K$, and call a the apex of the translate $a - K$. The next proposition restates, as a compilation criterion, the classical characterization of simplicial cones as the cones whose order makes $\mathbb{R}^d$ a vector lattice; see, for example, Németh [68, Introduction] and the references there.

Proposition 21.1 (One-translate compilation). *Let $d \geq 1$. The identity*

$$(a - K) \cap (b - K) = c - K$$

has a solution c for every pair $a, b \in \mathbb{R}^d$ if and only if $K = T\mathbb{R}_+^d$ for an invertible linear map T . In that case, for any finite family,

$$\bigcap_{i=1}^N (a_i - K) = Tm - K, \quad m_j = \min_i (T^{-1}a_i)_j. \quad (21.1)$$

If c must be one of the input apices a, b , a solution exists for every pair if and only if $d = 1$.

Proof. The intersection identity says precisely that c is the greatest lower bound $a \wedge b$. Negation supplies least upper bounds, so the induced order is a vector lattice. We prove in finite dimensions that K is then simplicial. Distinct extreme rays of K are disjoint in the lattice sense: a positive element below representatives of both rays must lie on both and therefore vanish. The lattice decomposition identity

$$0 \leq z \leq x + y \implies z = (z \wedge x) + (z - x)^+, \quad 0 \leq z \wedge x \leq x, \quad 0 \leq (z - x)^+ \leq y$$

shows by induction that finite sums of disjoint groups of extreme rays are disjoint. Any linear dependence among distinct extreme-ray representatives would equate two nonzero positive sums over disjoint groups, which is impossible. Thus there are at most d extreme rays. A compact base of a closed pointed cone is the convex hull of its extreme points; since K spans $\mathbb{R}^d$, there are exactly d such rays and they form a basis. This gives $K = T\mathbb{R}_+^d$. Coordinatewise minima prove the converse and (21.1).

The output apex can always be chosen among the inputs exactly when every pair is comparable. This is equivalent to $K \cup (-K) = \mathbb{R}^d$. For $d \geq 2$, the unit sphere is connected, whereas its intersections with K and $-K$ would be nonempty closed sets covering it, disjoint because K is pointed. A one-dimensional proper cone has a total order. $\square$

The output size does not grow with the number of incomparable input apices: even an antichain of N apices in a simplicial cone is summarized by d coordinate minima. For explicit classical output and coherent access to each transformed entry $(T^{-1}a_i)_j$, quantum minimum finding [28] gives the upper bound in

$$Q = O(d\sqrt{N} \log(d+1)), \quad Q = \Omega(d\sqrt{N}), \quad R = \Theta(dN). \quad (21.2)$$

Here Q and R denote bounded-error quantum and randomized entry-query complexities, respectively, and success means that all d output coordinates are correct. The oracle model includes exact retrieval and comparison of encoded entries. For the upper bound, run each minimum search with failure probability at most $1/(3d)$. For the quantum lower bound, restrict each column to all ones or to a single hidden zero. A correct output decides, for each of the d columns independently, whether it contains a zero, so the adversary direct-sum theorem applies [2, Theorem 4]. For the randomized

lower bound, take independent columns that are all ones or contain a single zero, each with probability $1/2$, and couple this input with the one obtained by changing any one column. If an algorithm succeeds on the full answer vector with probability at least $2/3$, each column must be distinguished with constant advantage; a uniform hidden zero then forces $\Omega(N)$ expected probes in each column. Summing over columns gives $\Omega(dN)$. The logarithm in the quantum upper bound makes all d searches succeed simultaneously; we do not claim a $\Theta(d\sqrt{N})$ bound. If the oracle instead returns a physical coordinate of a_i and T^{-1} is dense, forming one transformed entry can require d source queries.

With the canonical logarithmic barriers, the N input constraints $x \in a_i - K$ have barrier parameter Nd , while the compiled translate has the optimal parameter d . When all apices coincide, a radial curve along which the common transformed slack is $\alpha \mathbf{1}$ has lengths

$$\sqrt{Nd} |\log(\alpha_1/\alpha_0)| \quad \text{and} \quad \sqrt{d} |\log(\alpha_1/\alpha_0)|,$$

in the two barrier metrics, by the Hessians of $-N \sum_j \log s_j$ and $-\sum_j \log s_j$. This compares the two barriers only, not all algorithms for the underlying optimization problem.

For arbitrary proper cones, the same compilation works when the apices are radial in each factor of a product cone. If $K = \prod_{j=1}^B K_j$, $e_j \in \text{int } K_j$, and $a_i = (r_{i1}e_1, \dots, r_{iB}e_B)$, then

$$\bigcap_i (a_i - K) = \prod_{j=1}^B (r_{*j}e_j - K_j), \quad r_{*j} = \min_i r_{ij}. \quad (21.3)$$

Indeed, in each factor the translates are nested and ordered by the scalar r_{ij} . The corresponding query bounds are (21.2) with $d = B$. Given barriers of parameters ν_j on the factors, the compiled product has a barrier of parameter $\sum_j \nu_j$, whereas the sum of the N original barriers has parameter $N \sum_j \nu_j$. No symmetry of K_j is needed.

Proposition 21.2 (Lorentz-translate obstruction). *For every N , there are $a_1, \dots, a_N \in \mathbb{R}^3$ such that every finite identity*

$$\bigcap_{i=1}^N (a_i - Q_3) = \bigcap_{\ell=1}^m (c_\ell - Q_3)$$

satisfies $m \geq N$.

Proof. Choose distinct $v_i \in \mathbb{S}^1$ and $0 < \rho < R$, and set $a_i = (0, \rho v_i)$. On the plane $t = -R$, the intersection is $\bigcap_i B(\rho v_i, R)$. The point $z_i = -(R - \rho)v_i$ satisfies

$$\|z_i - \rho v_i\| = R, \quad \|z_i - \rho v_j\|^2 = (R - \rho)^2 + \rho^2 + 2\rho(R - \rho)\langle v_i, v_j \rangle < R^2 \quad (j \neq i).$$

Hence each original circle contributes an open boundary arc. An alternative translate intersects this plane in a disk. Its radius must be positive because the common intersection contains an open disk. Every point of an original open arc lies on the boundary of at least one alternative disk. Two distinct circles intersect in at most two points, so finitely many alternative circles can cover that arc only if one coincides with the original circle. Distinct original circles therefore require distinct alternative disks. $\square$

The obstruction occurs already in Jordan rank two. The example also has a formulation whose graph of variable blocks is a path: copy the three-dimensional variable for each constraint and require consecutive copies to be equal. Thus neither cone rank nor this path structure bounds the number of translates. The proposition concerns only translates of the same cone in the same space; it gives no lower bound for lifts with auxiliary variables, different cone maps, or custom barriers.

21.2 Reusable summaries of perspective data

Let $h : (0, \infty) \rightarrow \mathbb{R}$ be continuous. Its *dilation rank* is

$$R_h = \dim \text{span}\{r \mapsto h(\lambda r) : \lambda > 0\},$$

possibly infinite. For fixed source data $(w_i, c_i, d_i)_{i \leq N}$, consider the perspective sum

$$\Phi(U, V) = \sum_{i=1}^N w_i c_i U h\left(\frac{d_i}{c_i} \frac{V}{U}\right), \quad U, V, w_i, c_i, d_i > 0. \quad (21.4)$$

A reusable summary is a finite real vector computed from the source data that determines this function for every $U, V > 0$, not only at one point.

Theorem 21.3 (Continuous summary dimension). *If $R_h = R < \infty$, then Φ in (21.4) depends on the source data only through R real moments, each linear in the weights w_i . Conversely, choose any q linearly independent dilates $h(\lambda_1 r), \dots, h(\lambda_q r)$. If a continuous map $E : \mathbb{R}_{++}^q \rightarrow \mathbb{R}^m$ admits a deterministic decoder recovering $\sum_i a_i h(\lambda_i r)$ for every $r > 0$ and every $a \in \mathbb{R}_{++}^q$, then $m \geq q$. Consequently, R_h is the exact minimum dimension of a continuous summary when the positive coefficients are unrestricted; if $R_h = \infty$, no fixed finite dimension suffices.*

Proof. Choose a basis $h_1, \dots, h_R$ of the dilation space and write $h(\lambda r) = \sum_{\ell} m_{\ell}(\lambda) h_{\ell}(r)$. Then

$$M_{\ell} = \sum_i w_i c_i m_{\ell}(d_i/c_i), \quad \Phi(U, V) = U \sum_{\ell=1}^R M_{\ell} h_{\ell}(V/U).$$

The coefficient functions m_{ℓ} are continuous: choose R evaluation points giving an invertible evaluation matrix and solve for the coefficients using the continuous values $h(\lambda r_j)$.

For the lower bound, equality $E(a) = E(b)$ would give equality of the reconstructed functions, hence $a = b$ by linear independence. Thus E is injective. If $m < q$, append $q - m$ zeros to obtain a continuous injection from the open set $\mathbb{R}_{++}^q$ into $\mathbb{R}^q$. Invariance of domain [48, Theorem 2B.3] says its image is open, whereas the image lies in a proper coordinate subspace. This is impossible. $\square$

The decoder need not be continuous. Continuity of the encoder and a finite-dimensional real output are essential; they exclude encoding all the data into one real number of infinite precision. For a fixed restricted family of coefficients, the minimum dimension need not equal R_h .

Proposition 21.4 (Classical dilation classification). *The rank R_h is finite if and only if*

$$h(r) = \sum_{j=1}^J r^{\rho_j} P_j(\log r), \quad (21.5)$$

where the ρ_j are distinct complex numbers and P_j are nonzero polynomials, with conjugate terms paired when h is real-valued. The empty sum represents $h = 0$, with $R_h = 0$. Otherwise $R_h = \sum_j (\deg P_j + 1)$.

Proof. Put $f(y) = h(e^y)$. Its translation span W is finite dimensional exactly when the dilation span is. Translation acts on W by a continuous group $T(t)$ of invertible matrices: continuity follows by choosing an invertible finite evaluation matrix as above. For completeness, choose small $\delta > 0$ so that $M = \int_0^{\delta} T(t) dt$ is invertible. The identity $T(s)M = \int_s^{s+\delta} T(t) dt$ proves differentiability of T . The group law then gives $T'(s) = T(s)A$ with $A = T'(0)$, so $T(s) = e^{sA}$. Evaluation at zero shows

that f is an exponential polynomial; the Jordan form of A gives (21.5). Conversely, translates of each $e^{\rho_j y} P_j(y)$ lie in the span of $y^\ell e^{\rho_j y}$ for $0 \leq \ell \leq \deg P_j$. These functions are linearly independent. Each occurs in the cyclic translation span: on this finite-dimensional span the differential operator acts with one Jordan chain of length $\deg P_j + 1$ for each distinct ρ_j , and a polynomial in that operator isolates each chain. This also gives the stated rank. $\square$

The translation-space classification is due to Anselone–Korevaar [4]; see also Engert’s extension and historical account [30]. We include an elementary proof via continuous one-parameter groups to make all assumptions explicit. For example, $-\log r$ has rank two, r^α has rank one, and e^r has infinite rank. The latter follows from independence of distinct exponentials $e^{\lambda r}$ on any open interval. Theorem 21.3 adds the interpretation of R_h as the exact dimension of a continuous summary, not a new classification. It concerns evaluation from source data, not the size of conic lifts or barrier parameters.

21.3 A two-moment relative-entropy example

For positive definite Hermitian matrices we use the homogeneous Umegaki relative entropy without trace correction, $D(X\|Y) = \text{tr } X(\log X - \log Y)$. Fix weights w_i and positive scalar scales c_i, d_i , and define

$$A = \sum_i w_i c_i, \quad B = \sum_i w_i c_i \log(c_i/d_i), \quad D_{\text{eff}} = A e^{-B/A}. \quad (21.6)$$

Then

$$\sum_i w_i D(c_i X \| d_i Y) = A D(X\|Y) + B \text{tr } X = D(AX \| D_{\text{eff}} Y). \quad (21.7)$$

Indeed, scaling a matrix by c adds $(\log c)I$ to its logarithm; X and Y need not commute. The identity extends to the positive-semidefinite closure with the usual condition $\text{supp } X \subseteq \text{supp } Y$ for finite relative entropy. If that condition fails, both sides are $+\infty$. If $X = 0$, both sides vanish.

Once A and B are known, every value and derivative evaluation uses one relative-entropy term and no further scenario access. In the scalar case, $-\log r$ has dilation rank two, so Theorem 21.3 shows that two continuous real statistics are necessary for arbitrary positive coefficients and scales. If A is fixed in advance, only B needs to be stored. For $c_i = 1$ and $\sum_i w_i = 1$, $D_{\text{eff}} = \prod_i d_i^{w_i}$.

The convention matters. For $D_+(X\|Y) = D(X\|Y) + \text{tr } Y - \text{tr } X$ and $C = \sum_i w_i d_i$, the exact identity instead is

$$\sum_i w_i D_+(c_i X \| d_i Y) = D_+(AX \| D_{\text{eff}} Y) + (C - D_{\text{eff}}) \text{tr } Y.$$

Identity (21.7) applies to a weighted sum inside one objective or epigraph constraint, not to separate scenario inequalities or separate outputs. If the scales depend on the optimization variables, it no longer allows evaluation without source access.

The relative-entropy epigraph barrier of parameter $2n + 1$ for $n \times n$ matrices is established by Fawzi and Saunderson [35]. Positive-map composition and removal of redundant input log determinants are treated more generally by He et al. [50]. Equation (21.7) is an elementary scaling identity, not a new barrier construction; it makes an exact source summary explicit.

Computing the moments is a separate task. For a uniform average of bounded coherently accessible entries, quantum amplitude estimation uses $O(\min\{N, \eta^{-1}\})$ queries for constant-success additive error η , compared with the randomized upper bound $O(\min\{N, \eta^{-2}\})$ [18]. This assumes that the access model includes preparing the uniform index state and computing an entry; arbitrary implicit weights are not free. For Boolean entries, computing the mean exactly or with error strictly

below $1/(2N)$ recovers the Hamming weight and requires $\Omega(N)$ quantum queries [67]. If A is known exactly, replacing B by an estimate $\hat{B}$ shifts the function by exactly $(\hat{B} - B) \text{tr} X$. Since $X \succeq 0$, an upper bound on B gives an inner approximation of the epigraph and a lower bound an outer one, both globally valid.

A small barrier parameter alone does not allow evaluation without source access. For $z \in \{0, 1\}^N$, let

$$\phi_z(x_1, x_2) = \sum_{i=1}^N D(x_1 \| e^{-z_i} x_2) = Nx_1 \log(x_1/x_2) + |z|x_1.$$

The epigraph on $x_1, x_2 > 0$ is a linear image of a single scalar relative-entropy cone: use $D(Nx_1 \| Nx_2) = Nx_1 \log(x_1/x_2)$ and shift the epigraph coordinate by $|z|x_1$. Thus

$$F_z(t, x_1, x_2) = -\log(t - \phi_z(x_1, x_2)) - \log x_1 - \log x_2$$

is a three-parameter logarithmically homogeneous self-concordant barrier, up to an additive constant [35]. Under the promise $|z| \in \{0, 1\}$, the values of F_z at the interior point $(2, 1, 1)$ differ by exactly $\log 2$ between $|z| = 0$ and $|z| = 1$. An oracle returning that barrier value with error less than $\log 2/3$ therefore decides whether $|z| = 0$ or $|z| = 1$ and requires $\Omega(\sqrt{N})$ quantum or $\Omega(N)$ randomized source queries. This lower bound concerns initial source access; it does not apply once $|z|$ is stored.

21.4 Exponential features versus exponential-cone factors

We use the coordinate convention

$$K_{\text{exp}} = \text{cl}\{(a, b, c) : b > 0, c \geq be^{a/b}\}.$$

In particular, $(a, 1, c) \in K_{\text{exp}}$ means $c \geq e^a$. An exact *signed exponential representation* of a function on an open set is an identity $f(z) = \sum_{j=1}^M \beta_j e^{\langle \lambda_j, z \rangle}$ with real nodes λ_j and real coefficients β_j . The *signed exponential rank* of f is the minimum number of nonzero terms over such representations with arbitrary nodes.

Proposition 21.5 (Unbounded signed exponential rank with three cones). *For every integer $m \geq 1$, the function*

$$F_m(z) = \left(\frac{1 + e^{z/m}}{2} \right)^m$$

has signed exponential rank $m + 1$ on every nonempty open interval, but its full epigraph has an exact lift using three exponential cones and one scalar linear inequality:

$$(z/m - s, 1, u), (-s, 1, v), (ms, 1, t) \in K_{\text{exp}}, \quad u + v \leq 2. \quad (21.8)$$

All affine coefficients have $O(\log(m + 1))$ -bit rational encodings.

Proof. The first two cones and the inequality admit u, v exactly when $s \geq \log((1 + e^{z/m})/2)$. The last cone then projects exactly to $t \geq F_m(z)$, with equality attainable. On the other hand,

$$F_m(z) = 2^{-m} \sum_{j=0}^m \binom{m}{j} e^{jz/m}.$$

Distinct real exponentials are linearly independent on an open interval: differentiating a finite zero combination at any interior point gives a Vandermonde system. Comparing any competing representation with this one therefore forces all $m + 1$ displayed nodes to remain. $\square$

Here F_m is the moment-generating function of $m^{-1} \sum_{j=1}^m B_j$ for independent Bernoulli(1/2) variables. Hence the number of cones does not bound the signed exponential rank, even for moment-generating functions of probability measures on $[0, 1]$. The log-sum-exp and outer-exponential operations used in (21.8) are standard modeling constructions [64, Sections 5.2.5–5.2.6]. The gap arises because the lift composes exponentials, whereas a signed exponential representation is a sum of exponentials of linear functions.

Theorem 21.6 (Bounds for exact exponential lifts of a product-cosh graph). *Let $r \geq 1$, $K > 0$, and $f_r(z) = \prod_{i=1}^r \cosh z_i$. Its signed exponential rank on every nonempty open subset of $\mathbb{R}^r$ is 2^r . Its epigraph has a lift with $2r + 1$ exponential-cone factors and r scalar inequalities. Conversely, any exact affine lift of its epigraph over $[-K, K]^r$ through $K_{\text{exp}}^R \times \mathbb{R}_+^P$, with arbitrary affine projections and free auxiliary variables, satisfies*

$$R \geq r. \quad (21.9)$$

Consequently, if scalar inequalities are not counted, the minimum number of exponential-cone factors lies between r and $2r + 1$. For any such original ambient product, every self-concordant barrier has parameter at least $3R + P \geq 3r + P$, and the optimal ambient parameter is $3R + P$.

Proof. The expansion $f_r(z) = 2^{-r} \sum_{\epsilon \in \{-1, 1\}^r} e^{\langle \epsilon, z \rangle}$ has 2^r distinct nodes. To prove independence against any competing finite node set, choose a direction not perpendicular to any pairwise node difference, and restrict the identity to a short line segment inside the given open set. The one-dimensional Vandermonde argument then applies.

For the upper bound impose, for each i ,

$$(z_i - s_i, 1, u_i), (-z_i - s_i, 1, v_i) \in K_{\text{exp}}, \quad u_i + v_i \leq 2, \quad \left(\sum_i s_i, 1, t \right) \in K_{\text{exp}}.$$

Eliminating u_i, v_i gives $s_i \geq \log \cosh z_i$, and eliminating s gives exactly $t \geq f_r(z)$. Restricting to the cube adds only scalar inequalities.

For the lower bound choose $T > 1$ and intersect the given epigraph with $t \leq T$. Its image is the compact convex body

$$C = \{(z, t) : z \in [-K, K]^r, f_r(z) \leq t \leq T\} \subset \mathbb{R}^{r+1}.$$

It has nonempty interior. In a neighborhood of $(0, 1)$ its lower boundary is the graph of f_r , and

$$\nabla^2 f_r(z) = f_r(z) \left[\text{diag}(\text{sech}^2 z_i) + (\tanh z)(\tanh z)^T \right] \succ 0. \quad (21.10)$$

Its second fundamental form is therefore positive definite in all r tangent directions. The intersection adds one nonnegative-ray factor and does not add an exponential factor. After translation to put an interior point at the origin, Lemma 2.1 eliminates free variables and passes to the minimal face. These sets are definable in the real exponential structure, so Lemma 2.2 and Theorem 3.2 apply. Each retained face of an exponential cone has dimension at most three and thus contributes at most one unit to $\sum_i \max\{\dim K_i - 2, 0\}$; every ray contributes zero. The r -dimensional positive-curvature patch gives $r \leq R$. Finally, Theorem 19.2 gives the exact ambient parameter $3R + P$ for the *original* product. The extra ray from the truncation $t \leq T$ is used only in the curvature argument and is not counted in that original product. $\square$

The same proof gives $R \geq r$ for any definable convex epigraph with an r -dimensional C^2 graph patch having positive definite Hessian, provided that a compact polyhedral truncation retains that patch. The argument must keep the epigraph coordinate: a level set of f_r alone has only $r - 1$ curved directions. We do not claim that the constant 2 in the upper bound of Theorem 21.6 is optimal. Its lower bound comes from curvature, independently of the signed exponential rank.

Proposition 21.7 (Polyhedral relative approximation of convex functions). *Let $D \subset \mathbb{R}^r$ be a compact polytope and let $f > 0$ be convex and C^2 on a neighborhood of D . For every $0 < \varepsilon < 1$, a finite piecewise-affine convex function p satisfies*

$$(1 - \varepsilon)f \leq p \leq f \quad \text{on } D.$$

In particular,

$$\text{epi}(p/(1 - \varepsilon)) \subseteq \text{epi } f \subseteq \text{epi } p \quad \text{over } D,$$

and the two approximating epigraphs use only scalar linear inequalities.

Proof. Let $m_f = \min_D f > 0$ and $L_f = \sup_D \|\nabla^2 f\|_{\text{op}}$. If $L_f = 0$, one tangent is exact on D . Otherwise choose a finite δ -net $\mathcal{N} \subset D$, with $\delta^2 \leq 2\varepsilon m_f / L_f$, and set

$$p(z) = \max_{x \in \mathcal{N}} \{f(x) + \langle \nabla f(x), z - x \rangle\}.$$

Convexity gives $p \leq f$. Taylor's theorem on the segment between z and a net point within δ gives $f(z) - p(z) \leq L_f \delta^2 / 2 \leq \varepsilon f(z)$. The epigraph of a finite maximum of affine functions is their halfspace intersection, giving the claim. $\square$

Thus, if arbitrarily many scalar inequalities are free, uniform relative approximation on a fixed cube needs no exponential cones. The question changes if those inequalities are counted, or if the factor conditions of Section 20 are imposed. Neither the exact signed exponential rank nor the exact curvature count alone is a lower bound for approximation. In each of the three compiler models, the conclusion depends on the stated restriction on the representation or data access; none gives a lower bound for all optimization algorithms.

22 Projection, sufficient statistics, and limits of local compilation

In a problem with first-stage variables x and recourse variables for N scenarios, an exact reformulation may eliminate the recourse variables and optimize over the projection onto x . This changes what an algorithm can access: a small-parameter barrier on x does not by itself produce the recourse variables, and a small-parameter barrier known only to exist need not come with a cheap reusable oracle. Farkas elimination, quadratic epigraph barriers, and quantum spectral approximation are established tools; we combine them under explicit access assumptions. We also prove parameter lower bounds for two compilation rules: a scalar-rescaled convolution construction and a weighted sum of local logarithmic barriers. We do not claim that a small-parameter projected barrier preserves the geometry of the full product of slacks.

22.1 Polyhedral and ordered recourse

Let $W \in \mathbb{R}^{q \times p}$ and $T \in \mathbb{R}^{q \times k}$ be public and consider

$$x \in X_0, \quad W y_i = h_i - T x, \quad y_i \geq 0 \quad (1 \leq i \leq N). \quad (22.1)$$

Write $H = \text{range } W$ and $C = \{z \in H : W^T z \geq 0\}$. This polyhedral cone is pointed in H : if $z, -z \in C$, then $W^T z = 0$ and $z \in H$, whence $z = 0$. Let $z_1, \dots, z_L$ generate its extreme rays; $L = 0$ is allowed.

Proposition 22.1 (Shared recourse directions). *The projection of (22.1) is exactly*

$$x \in X_0, \quad h_i - Tx \in H \ (i \leq N), \quad a_\ell^T x \geq \beta_\ell \ (\ell \leq L), \quad (22.2)$$

where $a_\ell = -T^T z_\ell$ and $\beta_\ell = \max_i (-z_\ell^T h_i)$. On the affine space defined by the displayed equalities, suppose X_0 has a ν_0 -barrier G , and remove inequalities that are identically satisfied. If the remaining inequalities have a common strict feasible point in $\text{ri } X_0$, then

$$G(x) - \sum_{\ell=1}^L \log(a_\ell^T x - \beta_\ell)$$

is a barrier of parameter at most $\nu_0 + L$ on that relative domain.

Proof. Farkas' lemma gives $Wy = h_i - Tx$, $y \geq 0$, precisely when $z^T(h_i - Tx) \geq 0$ for every z satisfying $W^T z \geq 0$. The lineality space of this dual cone is $H^\perp$; it imposes exactly $h_i - Tx \in H$. In H , the extreme rays generate C , so only their inequalities remain. Taking the maximum over i gives (22.2). The barrier statement follows by affine restriction and addition of logarithmic half-space barriers. $\square$

If W has full row rank, the equalities $h_i - Tx \in H$ hold automatically. Assume a coherent oracle for the blocks h_i , exact comparisons at the given data precision, and public rays. Then quantum maximum finding computes all thresholds β_ℓ , jointly correct with constant probability, using $\tilde{O}(L\sqrt{N})$ block queries [28]. Here and below, the tilde hides logarithmic factors from success amplification. Derivatives of the compiled barrier then need no scenario queries. An iteration bound for path following also needs an initial point and depends as usual on the target accuracy and the initial gap; it does not follow from the barrier parameter alone. Without full row rank, the compatibility equalities must also be supplied or checked.

The following block-by-block reconstruction rule requires strict feasibility. Suppose $h_i - Tx \in \text{ri}(W\mathbb{R}_+^p)$, equivalently there exists $y > 0$ with $Wy = h_i - Tx$. Then

$$y_i^c(x) = \underset{Wy=h_i-Tx, y>0}{\text{argmin}} \left\{ \frac{1}{2} \|y\|^2 - \sum_{j=1}^p \log y_j \right\} \quad (22.3)$$

exists uniquely. Indeed, the objective is coercive at infinity, diverges at the coordinate boundary, and has positive-definite Hessian. Let x vary in the affine compatibility space $\mathcal{A}_i = \{x : h_i - Tx \in \text{range } W\}$. For a direction h tangent to $\mathcal{A}_i$, after reducing W to independent rows and applying the same row operations to T , the derivative of the KKT conditions is

$$\begin{bmatrix} I + \text{diag}(y_j^{-2}) & W^T \\ W & 0 \end{bmatrix} \begin{bmatrix} Dy[h] \\ D\lambda[h] \end{bmatrix} = \begin{bmatrix} 0 \\ -Th \end{bmatrix}.$$

The matrix is invertible by positive definiteness and full row rank. Thus y_i^c is smooth on the relatively open subset of $\mathcal{A}_i$ where strictly positive feasible recourse exists, and depends only on block i and x . Computing it requires solving a convex problem for that block, not one arithmetic operation. Nonnegative feasibility alone does not suffice for (22.3); at a boundary point one must instead solve a local feasibility problem or first determine its minimal face.

For varying $W_i \in \mathbb{R}^{q \times p}$, suppose first that each has full row rank. Its pointed dual cone has at most $\binom{p}{q-1}$ extreme rays. An extreme ray has $q-1$ independent active inequalities: otherwise its minimal face has dimension at least two. Each such subset specifies a line, and pointedness

permits at most one orientation of that line. This proves the count, including $q = 1$. For rank $r_i < q$, apply the argument in range W_i , obtaining $\binom{p}{r_i-1}$ rays when $r_i > 0$, and retain the orthogonal compatibility equalities separately. A rank-zero block gives only equalities. Consequently, for fixed q , one block query and polynomial-time local arithmetic enumerate the inequalities defining the projection of that block.

This has a conditional algorithmic consequence. Suppose these inequalities and any constraints for X_0 define a full-dimensional LP in k effective variables with finite optimum. Assume coherent row access, together with the data-precision assumptions and an initial point close to the central path that are required by Apers and Gribling [5, Theorems 1.3 and 6.1]. Their algorithm returns an explicit feasible ϵ -optimal first-stage vector. With fixed q and a polynomial number of public constraints for X_0 , simulating each row query locally gives $\tilde{O}(\sqrt{N} \text{poly}(p, k))$ block queries. This is only a reduction to that algorithm; initialization, facial reduction, and output of the full recourse are separate tasks.

Finally, if the recourse constraints have the ordered form $h_0 + \alpha_i e - Tx \in K$, where K is a proper cone and $e \in K$, they are equivalent to the single constraint with $\alpha_* = \min_i \alpha_i$, because $(\alpha_i - \alpha_*)e \in K$ and the minimum is attained. For a symmetric cone of rank r , its standard barrier adds at most r to the first-stage parameter. The threshold again takes $\tilde{O}(\sqrt{N})$ queries. The scalar family $x \geq z_i$, $z_i \in \{0, 1\}$, gives the standard OR lower bounds $\Omega(\sqrt{N})$ quantum and $\Omega(N)$ randomized [14] for computing the threshold with error below $1/2$. These statements concern ordered offsets only; arbitrary offsets have no such single-threshold reduction.

22.2 Additive quadratic and matrix sufficient statistics

Consider the closed formulation

$$r_i = A_i x - b_i, \quad t_i \geq \|r_i\|^2, \quad \tau \geq \sum_i t_i, \quad x \in X_0. \quad (22.4)$$

It uses N rotated Lorentz factors and one ray. With the canonical barriers, its ambient parameter is $2N + 1$, excluding a barrier for X_0 . Its exact projection is $\tau \geq q(x) = \sum_i \|A_i x - b_i\|^2$.

Proposition 22.2 (Quadratic compilation). *Set $C_i = [A_i, -b_i]$, $\xi = (x, 1)$, and*

$$M = \sum_i C_i^T C_i, \quad Q = \sum_i A_i^T A_i, \quad g = \sum_i A_i^T b_i.$$

Then $q(x) = \xi^T M \xi$. On the quotient by $\ker Q$, the function $f(x, \tau) = -\log(\tau - q(x))$ is a barrier of exact parameter one. On unquotiented variables, if G is a ν_0 -barrier for X_0 , then $G + f$ is a barrier of parameter at most $\nu_0 + 1$. With $s = \tau - q(x)$ and $d = 2(Qx - g)$, the derivatives of f are

$$\nabla f = (d, -1)/s, \quad \nabla^2 f = \begin{bmatrix} 2Q/s & 0 \\ 0 & 0 \end{bmatrix} + (d, -1)(d, -1)^T/s^2. \quad (22.5)$$

Proof. Expansion gives the sufficient statistic and the derivatives. Since $\ker Q = \bigcap_i \ker A_i$, q is constant along the deleted directions. For a line direction h , put $a = Ds[h]$ and $b = -D^2s[h, h] \geq 0$. Direct differentiation gives

$$Df[h] = -a/s, \quad D^2f[h, h] = a^2/s^2 + b/s, \quad D^3f[h, h, h] = -2a^3/s^3 - 3ab/s^2.$$

For $u, v \geq 0$, $2u^3 + 3uv \leq 2(u^2 + v)^{3/2}$, as is seen by squaring and subtracting, leaving $3u^2v^2 + 4v^3 \geq 0$. Apply this with $u = |a|/s$, $v = b/s$ to obtain self-concordance. Also $(Df[h])^2 \leq D^2f[h, h]$. The

Hessian is positive definite on the quotient, and the function diverges at the epigraph boundary. The direction that varies only τ attains equality in the gradient inequality, so the parameter is exactly one. Addition of G removes any remaining flat directions and gives the asserted parameter bound. $\square$

This is the standard quadratic epigraph barrier, not a new formula. It replaces the N product factors exactly by one statistic while keeping local access to the recourse variables. Querying block i recovers $r_i = A_i x - b_i$ and $t_i = \|r_i\|^2$. These values extend every feasible point (x, τ) of the projection to a feasible point of (22.4); strict feasibility follows by distributing part of the positive coupling slack $\tau - q(x)$ among the t_i . Materializing all N recourse scalars requires $\Omega(N)$ output operations, whereas one requested block uses one block query plus local arithmetic.

The statistic can also be approximated without losing feasibility. Let C be the stack of all C_i , with m rows and $k + 1$ columns. The quantum spectral approximation theorem of Apers and Gribling [5, Theorem 3.1] gives, with high probability, $\widetilde{M}$ satisfying

$$(1 - \delta)\widetilde{M} \preceq M \preceq (1 + \delta)\widetilde{M}$$

in $\widetilde{O}(\sqrt{m(k+1)}/\delta)$ row queries, for $0 < \delta \leq 1$. Row sparsity affects arithmetic cost separately; rank deficiency is allowed. Taking $\widehat{M} = (1 + \delta)\widetilde{M}$ yields

$$M \preceq \widehat{M} \preceq \frac{1 + \delta}{1 - \delta} M. \quad (22.6)$$

Thus $\tau \geq \xi^T \widehat{M} \xi$ implies original feasibility. For pure least-squares minimization over a fixed X_0 , an exact minimizer of the inflated objective has original objective at most $(1 + \delta)/(1 - \delta)$ times the original infimum, whenever a minimizer exists; an additive solve error adds the same error to this bound. Subsequent optimization uses only the explicit statistic. When $m = Np$, the cited bound is $\widetilde{O}(\sqrt{Np(k+1)}/\delta)$ row queries; it bounds block queries only if the given block oracle simulates one row query at constant cost. The case $A_i = 0$, $b_i \in \{0, 1\}$, with zero or one nonzero b_i , shows that approximating the objective with error below $1/2$ computes OR, which gives the corresponding quantum and randomized lower bounds on scenario queries.

The same exact compilation holds for matrix residuals. Let $R_i(x) = B_{i0} + \sum_{j=1}^k x_j B_{ij} \in \mathbb{R}^{d_i \times s}$ and let $R(x)$ stack the $R_i(x)$. We quotient x by the directions h with $DR[h] = 0$; the remaining coordinates and Y are the effective variables.

Proposition 22.3 (Matrix quadratic compilation). *On $Y - R(x)^T R(x) \succ 0$ in the effective variables,*

$$f(x, Y) = -\log \det(Y - R(x)^T R(x))$$

is a barrier of exact parameter s . No barrier for this domain has parameter below s . Without quotienting, addition of a ν_0 -barrier for X_0 gives a barrier of parameter at most $\nu_0 + s$. All data enter through the positive-semidefinite block Gram matrix

$$\mathcal{M}_{j\ell} = \sum_i B_{ij}^T B_{i\ell}, \quad 0 \leq j, \ell \leq k,$$

of order $(k + 1)s$.

Proof. Put $S = Y - R^T R$. Along a direction, set

$$A = S^{-1/2} D S^{-1/2}, \quad B = 2S^{-1/2} (DR)^T D R S^{-1/2} \succeq 0.$$

Differentiation gives

$$Df = -\operatorname{tr} A, \quad D^2f = \operatorname{tr} A^2 + \operatorname{tr} B, \quad D^3f = -2\operatorname{tr} A^3 - 3\operatorname{tr}(AB).$$

The bounds $|\operatorname{tr} A^3| \leq \|A\|_F^3$ and $|\operatorname{tr}(AB)| \leq \|A\|_F \operatorname{tr} B$, together with the scalar inequality in Proposition 22.2, prove self-concordance. Moreover $(\operatorname{tr} A)^2 \leq s \operatorname{tr} A^2 \leq s D^2f$. The Hessian vanishes only if $DR = 0$ and $DY = 0$, so is positive definite in the effective variables. The boundary divergence is immediate. Varying only Y in direction S gives decrement squared s . For the lower bound against any barrier, fix x and restrict $Y = R(x)^T R(x) + \operatorname{diag}(u_1, \dots, u_s)$. This is the positive orthant, with its coordinate boundary in the boundary of the full domain, so its parameter lower bound is s . Finally, expanding $R^T R$ gives $\sum_{j,\ell} \xi_j \xi_\ell \mathcal{M}_{j\ell}$, with $\xi_0 = 1$. $\square$

Equivalently, by a Schur complement, the domain is a slice of a PSD cone with one diagonal block fixed to the identity; the parameter s does not count that block. Applying spectral approximation to the stack $[B_{i0}, \dots, B_{ik}]$ and using (22.6) gives a uniform Loewner upper bound on $R(x)^T R(x)$, by congruence with $\xi \otimes I_s$. Thus the scalar feasibility argument also compiles a matrix epigraph. It does not by itself give a relative objective guarantee for an arbitrary SDP objective.

22.3 A scalar normalization cannot repair every convolution barrier

The preceding reduction uses an additive quadratic statistic. For general blocks, a natural alternative convolves measures associated with the blocks and takes the Legendre transform of the log-Laplace function of the result. The next proposition concerns exactly this construction, multiplied by a scalar that depends only on the number of blocks.

Proposition 22.4 (Convolution normalization obstruction). *Let μ be the uniform probability measure on $[0, 1]$, with log-Laplace function f and conjugate $F = f^*$. For N independent variables, consider the conjugate of the log-Laplace function of their average, multiplied by a positive scalar a_N . Suppose this rule must be a standard self-concordant barrier for both of the following collections: N copies of μ , and one copy of μ with $N - 1$ point masses at zero. Then $a_N \geq 1$. Its barrier parameter on the replicated collection is therefore at least N .*

Proof. Independence gives $f_{\text{av}}(\theta) = Nf(\theta/N)$ for identical copies, hence $f_{\text{av}}^*(x) = NF(x)$. For the second collection it gives $f_{\text{av}}(\theta) = f(\theta/N)$, hence $f_{\text{av}}^*(x) = F(Nx)$. For $\theta < 0$,

$$\begin{aligned} f(\theta) &= \log(1 - e^\theta) - \log(-\theta), & f'(\theta) &= -1/\theta + O(e^\theta), \\ f''(\theta) &= \theta^{-2} + O(e^\theta), & f'''(\theta) &= -2\theta^{-3} + O(e^\theta). \end{aligned}$$

As $x = f'(\theta) \downarrow 0$, Legendre differentiation gives $F' = \theta$, $F'' = 1/f''(\theta)$ and $F''' = -f'''(\theta)/(f''(\theta))^3$. Therefore

$$\frac{(F')^2}{F''} \longrightarrow 1, \quad \frac{|F'''}{2(F'')^{3/2}} \longrightarrow 1.$$

Affine rescaling $x \mapsto Nx$ does not change either ratio. Multiplying a function by a_N divides the second ratio by $\sqrt{a_N}$, so self-concordance for the second collection forces $a_N \geq 1$. On the first collection, the first ratio tends to $a_N N$, which is a lower bound on its barrier parameter. $\square$

The hypothesis requires a barrier also for the collection with $N - 1$ point masses at zero. The proposition does not rule out a rule that uses the individual distributions, and gives no lower bound for all barriers of the projected interval; in particular, it does not exclude a nonfactorized entropic or universal barrier. We claim only the stated impossibility for scalar normalizations; the conjugacy identities used are classical.

22.4 Two obstructions to reusable exact summaries

Quadratic sufficient statistics have bounded dimension for fixed first-stage dimension. For sums of exponentials this fails, even with one first-stage variable.

Proposition 22.5 (Continuous exact summaries). *Fix distinct real $a_1, \dots, a_N$ and let $q_w(x) = \sum_i w_i e^{a_i x}$, with $w \in \mathbb{R}_{++}^N$. Suppose a continuous statistic $S : \mathbb{R}_{++}^N \rightarrow \mathbb{R}^d$ determines the entire strict epigraph of q_w . Then $d \geq N$. The decoder need not be continuous.*

Proof. Equal statistics give equal epigraphs and hence equal functions. Their first N derivatives at zero give the Vandermonde system $\sum_i (w_i - v_i) a_i^j = 0$, $0 \leq j < N$, so $w = v$. Thus S is injective. If $d < N$, composing with the coordinate inclusion $\mathbb{R}^d \hookrightarrow \mathbb{R}^N$ contradicts invariance of domain, since its image lies in a subspace with empty interior. $\square$

The next statement counts preprocessing queries instead of bounding the dimension of a continuous statistic. An offline summary, classical or quantum, is *reusable* if, after preprocessing, it answers all the stated requests with joint success probability at least $2/3$ and makes no further input queries. This excludes treating a single quantum state, which measurement consumes, as an oracle for arbitrarily many later answers.

Proposition 22.6 (Reusable boundary oracles). *Let $z \in \{0, 1\}^N$, $\gamma = 1/4$, and define the planar epigraph*

$$E_z = \{(x, t) : t \geq \varphi_z(x)\}, \quad \varphi_z(x) = x^2 + \max_{1 \leq i \leq N} (2ix - i^2 + \gamma z_i). \quad (22.7)$$

Each of its N defining inequalities has a constant-size representation with a rotated Lorentz cone. A reusable summary giving all values $\varphi_z(i)$ to absolute error less than $\gamma/3$ requires $\Omega(N)$ quantum preprocessing queries to the bit oracle for z . The same lower bound holds for a reusable summary that answers membership queries at the public points $(i, 2i^2 + \gamma/2)$, each at vertical distance $\gamma/2$ from the boundary of E_z .

Proof. At $x = i$, the i th affine term exceeds term $j \neq i$ by $(i - j)^2 + \gamma(z_i - z_j) \geq 3/4$. Thus $\varphi_z(i) = 2i^2 + \gamma z_i$. Either oracle recovers every bit. For completeness, a T -query quantum algorithm on an N -bit input has final state in the span of at most $D_T = \sum_{j=0}^T \binom{N}{j}$ fixed vectors: expand its amplitudes in the characters of the input signs, whose degree increases by at most one per query. Purification handles intermediate measurements and randomness. An ensemble of 2^N states in a space of dimension at most D_T can be identified, on a uniform input, with probability at most $D_T/2^N$, because each density matrix is bounded above by the identity on that space and the decoding POVM sums to that identity. For $T \leq N/4$, the binomial entropy bound makes this probability tend to zero. Constant successful recovery therefore requires $T = \Omega(N)$, as in oracle interrogation [84]. $\square$

Rescaling the exposed locations to $x = i/N$ and the affine terms to $2Nix - i^2$ preserves the gap. Replacing the common quadratic by x^r on $x > 0$, for fixed rational $r > 1$, or by e^x , gives the same recovery proof with constant-size power or exponential constraints. The locations then stay in a bounded interval, so the common exponential term needs no high precision. The lower bound applies only to reusable preprocessing: a single requested boundary value can still be computed with $\tilde{O}(\sqrt{N})$ fresh maximum-finding queries. The bound also does not rule out approximate summaries with error in a weaker norm, or summaries that need to succeed only under an input distribution for which the recovery argument fails.

22.5 The parameter cost of additive local logarithms

The same epigraph E_z forces a large parameter for any barrier that is a weighted sum of the local logarithms of the slacks, even though E_z is two-dimensional.

Proposition 22.7 (Additive local barrier cost). *For the slacks $s_i(x, t) = t - x^2 - 2ix + i^2 - \gamma z_i$ in (22.7), suppose*

$$F(x, t) = - \sum_{i=1}^N w_i \log s_i(x, t), \quad w_i > 0,$$

is a standard self-concordant barrier with parameter ν on $\text{int } E_z$. Then $w_i \geq 1$ for every i , and $\nu \geq \sum_i w_i \geq N$. Nevertheless, this line-free planar domain admits a barrier of parameter two.

Proof. At $(i, \varphi_z(i) + \epsilon)$, the i th slack is ϵ and all other slacks stay at least $3/4$. In direction $h = (0, 1)$,

$$D^2 F[h, h] = w_i \epsilon^{-2} + O(1), \quad |D^3 F[h, h, h]| = 2w_i \epsilon^{-3} + O(1).$$

Hence the self-concordance ratio tends to $1/\sqrt{w_i}$ as $\epsilon \rightarrow 0$, forcing $w_i \geq 1$. For fixed x and $t \rightarrow \infty$,

$$\frac{(DF[h])^2}{D^2 F[h, h]} = \frac{(\sum_i w_i / s_i)^2}{\sum_i w_i / s_i^2} \rightarrow \sum_i w_i.$$

The gradient inequality implies the parameter bound. The epigraph is full-dimensional and contains no affine line, because its boundary grows quadratically in either horizontal direction and is bounded below on each vertical line. The universal-barrier theorem in dimension two therefore supplies a two-parameter barrier [60]. $\square$

The lower and upper bounds concern different classes of barriers: a barrier that combines the scenarios through a global construction can have parameter independent of N ; a weighted sum of these local logarithms cannot. Together with the lower bound for reusable oracles, this shows that three requirements are distinct: existence of a small-parameter projected barrier, of an explicit sufficient statistic, and of a cheap oracle.

Part V

Movement in barrier metrics

23 Certificate ranks and movement in a fixed barrier metric

This section bounds primal movement: the distance, in the Hessian metric of a fixed barrier, from a reference point to the set of ϵ -accurate points. A lower bound on the barrier parameter does not by itself bound this distance; one also needs to know which part of the cone the objective forces toward its boundary. A certificate for the objective describes this part, and the distance bound grows with its total Jordan rank. The weighted exposed-minor estimate (Theorem 23.1) is also proved in the companion manuscript [21]; we include its proof to keep this paper self-contained. The Jordan identities are classical [31], as is the conversion from bounded local steps to Riemannian distance [72]. What is new is their combination with the certificate-rank bounds of Sections 4–7 for a fixed cone dictionary, using the same certificate in both bounds.

For a barrier F with positive-definite Hessian on its effective affine domain Ω , write

$$\|h\|_{F,x}^2 = D^2F(x)[h, h], \quad d_F(x, y) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\|_{F,\gamma(t)} dt.$$

The infimum is over piecewise C^1 curves in Ω joining the two points. For a set $T \subset \Omega$, put $d_F(x, T) = \inf_{y \in T} d_F(x, y)$, with value $+\infty$ if T is empty. Any free affine direction along which the slack is constant is removed first.

23.1 The weighted exposed-minor estimate

Let $K = \prod_i K_i$ be a product of symmetric cones with their Jordan trace inner products. Let $\Omega = \text{int } K \cap \mathcal{A} \neq \emptyset$ for an affine subspace $\mathcal{A}$. Suppose an affine objective ℓ has finite supremum ℓ_* and a genuine certificate $s = (s_i) \in K$ satisfying

$$g(x) := \ell_* - \ell(x) = \sum_i \langle s_i, x_i \rangle \quad (x \in \Omega). \quad (23.1)$$

This identity must hold on the entire feasible affine slice. For each nonzero s_i , let $c_i = \text{supp } s_i$, $q_i = \text{rank}_J s_i$, and let $\det_{c_i}$ denote the determinant in the Peirce algebra with unit c_i . The expressions below omit factors with $q_i = 0$.

Theorem 23.1 (Weighted exposed-minor distance). *Use the restricted barrier*

$$F(x) = - \sum_i \alpha_i \log \det_i x_i, \quad \alpha_i \geq 1, \quad Q_\alpha = \sum_i \alpha_i q_i > 0.$$

For a fixed reference $x^0 \in \Omega$, define

$$\Delta_\alpha = Q_\alpha \left[\prod_{i: q_i > 0} \frac{\{\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^0)\}^{\alpha_i}}{\alpha_i^{\alpha_i q_i}} \right]^{1/Q_\alpha}. \quad (23.2)$$

Then $\Delta_\alpha > 0$ and, for every $\epsilon > 0$,

$$d_F(x^0, \{x : g(x) \leq \epsilon\}) \geq \sqrt{Q_\alpha} [\log(\Delta_\alpha/\epsilon)]_+. \quad (23.3)$$

Here $[t]_+ = \max(t, 0)$. The bound applies to every feasible path in Ω , including paths that change coordinates not exposed by s or additional lift variables such as completion variables. Neither strict complementarity nor boundedness of those variables is assumed.

Proof. For an idempotent c of rank q , put $a = P(c)x$ and $\phi_c(x) = -\log \det_c a$. The element a is interior to its Peirce cone: $x \succeq \delta e$ implies $a \succeq \delta c$. The gradient and inverse Hessian of $-\log \det x$ are $-x^{-1}$ and $P(x)$, respectively. Since $P(c)$ is self-adjoint and $a^{-1} \in V(c, 1)$,

$$D\phi_c(x)[h] = -\langle a^{-1}, h \rangle, \quad \|D\phi_c(x)\|_{-\log \det, *}^2 = \langle a^{-1}, P(x)a^{-1} \rangle = q.$$

The last equality follows from the fundamental identity

$$P(c)P(x)P(c) = P(P(c)x) = P(a), \quad P(a)a^{-1} = a, \quad \langle a^{-1}, a \rangle = q.$$

These are Jordan identities, so they also hold in the exceptional algebra. For $\Phi(x) = \sum_{i:q_i>0} \alpha_i \phi_{c_i}(x_i)$, the full product metric consequently gives $\|D\Phi\|_{F,*}^2 = Q_\alpha$. Restricting a covector and the metric to an affine tangent subspace can only decrease its dual norm. Hence

$$|D\Phi(x)[h]| \leq \sqrt{Q_\alpha} \|h\|_{F,x} \quad (h \in T_x \Omega). \quad (23.4)$$

For a feasible endpoint x , set $g_i = \langle s_i, x_i \rangle > 0$ on the active factors. In $V(c_i, 1)$, the positive element $P(s_i^{1/2})P(c_i)x_i$ has trace g_i and determinant $\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i)$. Spectral AM–GM therefore gives

$$\det_{c_i}(P(c_i)x_i) \leq \det_{c_i}(s_i)^{-1} (g_i/q_i)^{q_i}.$$

If $\sum_i g_i \leq \epsilon$, maximizing $\sum_i \alpha_i q_i \log(g_i/q_i)$ over these positive g_i gives $g_i = \epsilon \alpha_i q_i / Q_\alpha$. It follows that

$$\prod_i \det_{c_i}(P(c_i)x_i)^{\alpha_i} \leq (\epsilon/Q_\alpha)^{Q_\alpha} \prod_i \alpha_i^{\alpha_i q_i} \det_{c_i}(s_i)^{-\alpha_i}.$$

Comparison with the reference point yields $\Phi(x) - \Phi(x^0) \geq Q_\alpha \log(\Delta_\alpha/\epsilon)$. Integrating (23.4) along any feasible curve proves the result. $\square$

Corollary 23.2 (Bounded-move consequence). *Suppose a procedure starts at x^{in} with $d_F(x^{\text{in}}, x^0) \leq D_0$ and makes T rounds, each with metric displacement at most $B_0 > 0$, before attaining gap at most ϵ . Then*

$$T \geq \frac{[\sqrt{Q_\alpha} \log(\Delta_\alpha/\epsilon) - D_0]_+}{B_0}. \quad (23.5)$$

In particular, for an integer $m \geq 1$ and $0 < R < 1$, if each round consists of at most m feasible forward chords $x + th$, $0 \leq t \leq 1$, with starting local norm $\|h\|_{F,x} \leq R$, one may take $B_0 = -m \log(1 - R)$. If $\|x^{\text{in}} - x^0\|_{F,x^0} \leq \rho < 1$, one may take $D_0 = -\log(1 - \rho)$. More generally, if the procedure makes $M \geq 1$ chords with starting local norms $r_j < 1$ and $D = [\sqrt{Q_\alpha} \log(\Delta_\alpha/\epsilon) - D_0]_+$, then

$$\sum_{j=1}^M -\log(1 - r_j) \geq D, \quad \max_j r_j \geq 1 - e^{-D/M}. \quad (23.6)$$

Proof. The triangle inequality proves (23.5). For a chord $x + th$, the standard self-concordant Hessian comparison gives $\|h\|_{F,x+th} \leq \|h\|_{F,x}/(1 - t\|h\|_{F,x})$. Its integral is at most $-\log(1 - \|h\|_{F,x})$. Summing these bounds proves the variable-chord assertion as well; see also [72, Lemma 3.2 and Corollary 3.2]. $\square$

The square-root coefficient is sharp on the full product cone. A quadratic-representation isometry sends x^0 to the unit. In a Jordan frame diagonalizing the transformed s_i , multiply exactly its q_i

positive-support coordinates by t and keep all remaining coordinates equal to one. This curve has gap $t \sum_i \langle s_i, e_i \rangle$ and length $\sqrt{Q_\alpha} \log(1/t)$ from the unit. Thus, on the full cone,

$$\lim_{\epsilon \downarrow 0} \frac{d_F(x^0, \{g \leq \epsilon\})}{\log(1/\epsilon)} = \sqrt{Q_\alpha}. \quad (23.7)$$

In this statement the certificate is transformed with the coordinates; its pairing, and hence the gap, is unchanged. An affine slice can exclude the displayed curve, so (23.7) does not assert a matching upper bound for every slice.

Hauser and Güler's classification [49, Theorem 5.5] says that every self-scaled barrier on a symmetric cone is, up to an additive constant, of the displayed form on its irreducible factors, with $\alpha_i \geq 1$. Consequently the theorem covers all self-scaled barriers on a fixed symmetric cone. Changing the weights changes both Q_α and Δ_α ; the bound must use Δ_α , not its unit-weight value. The classification gives no analogous assertion for arbitrary coupled barriers on an affine slice.

23.2 Movement bounds from certificate ranks

For unit weights, put $Q = \sum_i q_i$ and write Δ for (23.2). For a fixed lift, reference, objective, and certificate, Theorem 23.1 gives

$$\liminf_{\epsilon \downarrow 0} \frac{d_F(x^0, \{g \leq \epsilon\})}{\log(1/\epsilon)} \geq \sqrt{Q}. \quad (23.8)$$

The following corollary applies this bound to the earlier rank results.

Corollary 23.3 (Dictionary consequences). *Let the fixed symmetric-cone dictionary have capacity B from (4.1), and use its standard product barrier on the relative-Slater lift. For each lift of $\mathbb{B}_2^s$, on a dense open full-measure set of supports, the right-hand side of (23.8) is at least $\sqrt{\lceil (s-1)/B \rceil}$. For every lift of $\prod_a \mathbb{B}_2^{s_a}$, there exist directions u_a and genuine pure-row certificates Y^a such that every fixed choice of positive weights λ_a gives an objective $\ell(x) = \sum_a \lambda_a \langle u_a, \pi_a x \rangle$ for which the right-hand side is at least*

$$\sqrt{\sum_a \left\lceil \frac{s_a - 1}{B} \right\rceil}.$$

Under the globally bi- C^1 genuine-selection hypotheses of Theorem 7.7, the right-hand side is instead at least $\sqrt{\kappa_{\mathcal{X}}(s)}$ for some support of a single ball.

Proof. For one ball choose any genuine certificate at a support in Theorem 4.1's regular set. For the product take exactly the certificate $s_i = \sum_a \lambda_a Y_i^a$ supplied by Theorem 5.2. It represents the full gap $\sum_a \lambda_a (1 - \langle u_a, \pi_a x \rangle)$ and has the asserted aggregate rank. In the smooth case use the genuine selected certificate whose rank is bounded below in Theorem 7.7. Apply (23.8) in each case. $\square$

The product assertion uses only the existence of one such aggregate certificate; it does not say that every aggregate certificate has this rank, and it does not minimize over the whole dual fiber. One high-rank certificate suffices because its determinant estimate holds at every accurate primal endpoint. For self-scaled barriers the same statements hold with the weighted $Q_\alpha \geq Q$. Since Δ_α can depend on the lift, objective, reference, and certificate, the corollary is an asymptotic statement for a fixed instance, not a finite-accuracy estimate uniform over all lifts. In particular, this argument does not allow replacing Q by a larger ambient or restricted barrier parameter.

23.3 Affine PSD pencils and spectral scales

A second form of the estimate compresses an affine matrix pencil L to a subspace. It is useful when an exposed minor $\det(R^*LR)$ is known directly, including in completion formulations. The convexity used is classical matrix-fractional convexity [17, Example 3.4]. The matrix-ball specialization and its rank-profile interpretation also appear in [21].

Proposition 23.4 (Principal-minor contraction). *Let $L(x) = L_0 + Ax \succ 0$ be a real symmetric or complex Hermitian affine pencil and $F = -\log \det L$ on an effective affine domain with positive-definite Hessian. For an injective matrix R with $r \geq 1$ columns put $\psi_R(x) = -\log \det(R^*L(x)R)$. Then*

$$D^2\psi_R \preceq D^2F, \quad |D\psi_R(x)[h]| \leq \sqrt{r}\|h\|_{F,x}. \quad (23.9)$$

If the compressed function has the sharper restricted gradient bound $|D\psi_R[h]| \leq \sqrt{\vartheta} (D^2\psi_R[h, h])^{1/2}$ for $\vartheta > 0$, the second constant can be replaced by $\sqrt{\vartheta}$. Consequently

$$d_F(x^0, x) \geq \frac{1}{\sqrt{\vartheta}} \left[\log \frac{\det(R^*L(x^0)R)}{\det(R^*L(x)R)} \right]_+, \quad (23.10)$$

with $\vartheta = r$ always permitted.

Proof. First take R isometric and complete it to a unitary basis. In that basis write $L = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$, where $A = R^*LR$. If R is square there is nothing to prove. Otherwise

$$F - \psi_R = -\log \det(C - B^*A^{-1}B).$$

The Schur complement is matrix concave because the matrix-fractional map $B^*A^{-1}B$ is matrix convex: test the scalar convex function $(Bv)^*A^{-1}(Bv)$ against each fixed vector v . The scalar matrix-fractional convexity follows directly from its Schur-complement epigraph, over either field. Composition with the convex, Loewner-decreasing function $-\log \det$ is convex. This proves the Hessian inequality. Also, with $T = A^{-1/2}(DA[h])A^{-1/2}$,

$$D\psi_R[h] = -\operatorname{tr} T, \quad D^2\psi_R[h, h] = \operatorname{tr}(T^2), \quad |\operatorname{tr} T| \leq \sqrt{r} \sqrt{\operatorname{tr}(T^2)}.$$

A thin QR factorization of an arbitrary injective R changes ψ_R only by a constant, proving the general case. Integration proves (23.10), also with any stated sharper constant. $\square$

For example, for the matrix-ball pencil $L(X) = \begin{pmatrix} I_p & X \\ X^* & I_q \end{pmatrix}$, compression to $1 \leq r \leq \min(p, q)$ chosen left coordinates and all right coordinates has determinant $\det(I_r - U^*XX^*U)$, where U is the p by r matrix of the selected orthonormal left coordinate vectors. Its restricted gradient parameter is r , although the compressed pencil has order $r + q$, so one should use $\vartheta = r$. The gradient bound remains valid when this compressed Hessian is singular in some directions.

Theorem 23.5 (Exposed spectral profile). *In Proposition 23.4, let $W \succeq 0$ have rank $r \geq 1$, and suppose the target satisfies $\operatorname{tr}(WL(x)) \leq \epsilon$. Write $W = QDQ^*$ with $Q^*Q = I_r$ and $D \succ 0$, and set*

$$Z_0 = D^{1/2}Q^*L(x^0)QD^{1/2}, \quad d_1 \geq \dots \geq d_r > 0$$

for its eigenvalues. Then

$$d_F(x^0, x) \geq \max_{1 \leq m \leq r} \sqrt{m} \left[\log \frac{m(d_1 \dots d_m)^{1/m}}{\epsilon} \right]_+. \quad (23.11)$$

The same bound divided by $-\log(1 - R_0)$ is a lower bound on the number of feasible forward chords of local norm at most $0 < R_0 < 1$.

Proof. Take top- m orthonormal eigenvectors U_m of Z_0 and use the injective compression $R_m = QD^{1/2}U_m$. Its reference determinant is $\prod_{j \leq m} d_j$. At the endpoint,

$$\mathrm{tr}(R_m^* L(x) R_m) \leq \mathrm{tr}(W L(x)) \leq \epsilon, \quad \det(R_m^* L(x) R_m) \leq (\epsilon/m)^m.$$

Apply Proposition 23.4 with $\vartheta = m$, then maximize over m . The chord statement follows from Corollary 23.2's chord estimate. $\square$

For $m = r$ this is the unweighted exposed-minor estimate. Taking $m < r$ can avoid a large loss when some positive eigenvalues are tiny. The quantities d_i include both the exposing matrix and the reference slack; if $L(x^0) = I$, they are precisely the positive eigenvalues of W .

Denote the right side of (23.11) by $\mathcal{E}_r(\epsilon)$ and put $\Lambda = \log(1/\epsilon)$. For two eigenvalue profiles we give asymptotics of this lower bound; we do not claim they equal the exact distance for an arbitrary pencil. For $d_i = e^{-a(i-1)}$, $a > 0$, and any fixed $\delta > 0$, if $r \geq (2/(3a) + \delta)\Lambda$ for all sufficiently small ϵ , then

$$\mathcal{E}_r(\epsilon) = \left(\frac{2}{3} \sqrt{\frac{2}{3a}} + o(1) \right) \Lambda^{3/2}. \quad (23.12)$$

For $d_i = i^{-\beta}$, $\beta > 1$, put $C_\beta = e^{(2-\beta)/(\beta-1)}$. If $r \geq (C_\beta + \delta)\epsilon^{-1/(\beta-1)}$, then

$$\mathcal{E}_r(\epsilon) = \left(2(\beta-1) \sqrt{C_\beta} + o(1) \right) \epsilon^{-1/[2(\beta-1)]}. \quad (23.13)$$

To verify (23.12), maximize $\sqrt{m}[\Lambda + \log m - a(m-1)/2]_+$ over positive real m . Its positive maximizer satisfies

$$m_* = \frac{2}{3a}(\Lambda + \log m_* + a/2 + 2) = \frac{2\Lambda}{3a} + O_a(\log \Lambda).$$

The bracket there is $am_* - 2$. Rounding m_* to an integer gives (23.12). For the polynomial case, Stirling's estimate gives

$$\log \frac{m(m!)^{-\beta/m}}{\epsilon} = \Lambda - (\beta-1) \log m + \beta + O_\beta(\log(m+1)/m).$$

The error after multiplication by $\sqrt{m}$ is bounded uniformly in $m \geq 1$. The corresponding continuous expression is maximized at $m_* = C_\beta \epsilon^{-1/(\beta-1)}$, with bracket $2(\beta-1)$, proving (23.13). In both cases integer rounding is a lower-order error. If the rank r is smaller than this scale, the maximum over $m \leq r$ in (23.11) is the applicable bound. In particular, for fixed finite r , its leading behavior as $\epsilon \downarrow 0$ is $\sqrt{r} \log(1/\epsilon)$.

The orders of these lower bounds are attained by particular diagonal pencils. For each finite list d_i , take

$$L(q) = \mathrm{Diag}(q_1, \dots, q_r, 2 - q_1, \dots, 2 - q_r), \quad q \in (0, 2)^r, \quad q^0 = \mathbf{1}, \quad W = \mathrm{Diag}(d_1, \dots, d_r, 0, \dots, 0).$$

The target is $\sum_i d_i q_i \leq \epsilon$. On $0 < q \leq 1$ define

$$\varrho(q) = \int_q^1 \sqrt{t^{-2} + (2-t)^{-2}} dt, \quad \log(1/q) \leq \varrho(q) \leq \sqrt{2} \log(1/q).$$

The coordinates $\varrho(q_i)$ turn the product metric into the Euclidean metric. Set $q_i^* = \min\{1, \epsilon/(rd_i)\}$. This endpoint has gap at most ϵ , and a straight segment in the ϱ coordinates gives a feasible path of length at most

$$\sqrt{2} \left(\sum_{i=1}^r [\log(rd_i/\epsilon)]_+^2 \right)^{1/2}. \quad (23.14)$$

Choose $r = \lceil c\Lambda \rceil$ with fixed $c > 2/(3a)$ in the geometric case. Every summand in (23.14) is $O_a(\Lambda^2)$, so the upper order is $\Lambda^{3/2}$. In the polynomial case take $r = \lceil c\epsilon^{-1/(\beta-1)} \rceil$, $c > C_\beta$. Then $\log(rd_i/\epsilon)$ is a bounded constant minus $\beta \log(i/r)$, whose positive-part square has a finite integral on $(0, 1]$. The sum is $O_\beta(r)$, giving the matching order $\epsilon^{-1/[2(\beta-1)]}$. Constants here may depend on the fixed c and decay parameter. These examples use arbitrary feasible paths, not central paths, in specially chosen box spectrahedra whose rank grows at the specified rate; they give no matching upper bound for general affine pencils.

24 Arbitrary barriers and the distinction between primal and dual movement

The movement bounds of Section 23 depend on certificate ranks and use a specified symmetric-cone metric. In the primal–dual product metric, a lower bound depending only on the barrier parameter holds for every logarithmically homogeneous barrier. We state this classical argument and combine it with the earlier formulation bounds. The dimension bound and the objective-weight calculations below also appear in the companion manuscript [21, Sections 5, 9, and 10]; we include them to make precise the metric and target set of each bound.

24.1 The feasible gap set in the product metric

Let F be a ν -logarithmically homogeneous standard self-concordant barrier on a proper cone K . Define the conjugate barrier F_* (with pairing $-\langle s, x \rangle$) and the product barrier $\hat{F}$ by

$$F_*(s) = \sup_{x \in \text{int } K} \{-\langle s, x \rangle - F(x)\}, \quad \hat{F}(x, s) = F(x) + F_*(s).$$

Consider the strictly feasible conic pair

$$\min\{\langle c, x \rangle : Ax = b, x \in K\}, \quad \max\{\langle b, y \rangle : A^T y + s = c, s \in K^*\}.$$

Write $z(\eta) = (x(\eta), s(\eta))$ for its exact centers, with $s(\eta) = -\eta^{-1}F'(x(\eta))$ and gap ν/η . Distances $d_{\hat{F}}$ allow all curves in $\text{int } K \times \text{int } K^*$; requiring them to satisfy the affine equations can only increase distances. A dot denotes differentiation in $\lambda = \log \eta$.

Proposition 24.1 (Primal and dual parts of the central velocity). *Put $H = F''(x)$, $g = H^{-1/2}F'(x)$, and let Π project orthogonally onto $H^{1/2} \ker A$. Then*

$$\begin{aligned} \nu_P(\eta) &:= \|\dot{x}\|_{F,x}^2 = \|\Pi g\|^2, & \nu_D(\eta) &:= \|\dot{s}\|_{F^*,s}^2 = \|(I - \Pi)g\|^2, \\ \nu_P(\eta) + \nu_D(\eta) &= \nu. \end{aligned} \tag{24.1}$$

After removing redundant equality rows,

$$\nu_P = \eta^2 c^T M c, \quad M = H^{-1} - H^{-1} A^T (A H^{-1} A^T)^{-1} A H^{-1}.$$

For $p(\eta) = \langle c, x(\eta) \rangle$ and $d(\eta) = \langle b, y(\eta) \rangle$,

$$-\eta^2 p'(\eta) = \nu_P(\eta), \quad \eta^2 d'(\eta) = \nu_D(\eta). \tag{24.2}$$

If both objectives converge to the finite optimum p^* , then

$$p(\eta) - p^* = \int_\eta^\infty \frac{\nu_P(u)}{u^2} du, \quad p^* - d(\eta) = \int_\eta^\infty \frac{\nu_D(u)}{u^2} du.$$

Proof. Homogeneity and conjugacy give $Hx = -F'(x)$, $\|g\|^2 = \nu$, and $F''(s) = \eta^2 H^{-1}$. Differentiation gives $\eta \dot{s} = F'(x) - H\dot{x}$. Thus $v = H^{1/2}\dot{x}$ and $d_0 = \eta H^{-1/2}\dot{s}$ add to g . Primal and dual feasibility put these vectors in $H^{1/2} \ker A$ and its orthogonal complement, respectively. This proves the speed formulas. Differentiating primal stationarity gives $\dot{x} = -\eta M c$; $MHM = M$ proves the displayed Schur formula and the first identity in (24.2). Differentiate $p - d = \nu/\eta$ for the second, and integrate under the stated convergence assumption. $\square$

This classical orthogonal decomposition underlies the constant-speed primal–dual path of Nesterov and Todd [72, Theorems 4.1 and 5.2].

Theorem 24.2 (Classical distance to a feasible gap sublevel). *Let $z_0 = z(\eta_0)$ have gap $\Delta_0 = \nu/\eta_0$, and let $\mathcal{Z}_\epsilon$ be the strictly primal–dual feasible pairs with gap at most $0 < \epsilon < \Delta_0$. Then*

$$\sqrt{\nu/2} \log(\Delta_0/\epsilon) \leq d_{\hat{F}}(z_0, \mathcal{Z}_\epsilon) \leq \sqrt{\nu} \log(\Delta_0/\epsilon). \quad (24.3)$$

Consequently, let $m \geq 1$ be an integer and $0 < R < 1$. If a trajectory from z_0 to $\mathcal{Z}_\epsilon$ consists of T rounds, each of at most m chords from z to $z + h$ with $\|h\|_z \leq R$, where $\|h\|_z^2 = \hat{F}''(z)[h, h]$, then

$$T \geq \frac{\sqrt{\nu/2} \log(\Delta_0/\epsilon)}{m \log(1/(1-R))}. \quad (24.4)$$

Intermediate chord endpoints need not satisfy the affine equations.

Proof. At $z_c = z(\eta)$, conjugacy gives $F'_*(s_c) = -\eta x_c$ and $\hat{F}(z_c) = \nu \log \eta - \nu$. For any feasible $z = (x, s)$, $\langle x - x_c, s - s_c \rangle = 0$, since the first difference belongs to $\ker A$ and the second to $\text{range } A^T$. Hence

$$D\hat{F}(z_c)[z - z_c] = -\eta\{\langle s_c, x - x_c \rangle + \langle x_c, s - s_c \rangle\} = -\eta\{\langle x, s \rangle - \nu/\eta\}.$$

At $\eta = \nu/\epsilon$, convexity shows that z_c minimizes $\hat{F}$ on $\mathcal{Z}_\epsilon$. Every target therefore has barrier increase at least $\nu \log(\Delta_0/\epsilon)$ from z_0 . The ambient dual norm of $D\hat{F}$ is exactly $\sqrt{2\nu}$; integration along any joining curve proves the lower bound. The central path has constant speed $\sqrt{\nu}$ by Proposition 24.1, proving the upper bound. For a chord $z + th$ with $\|h\|_z = r < 1$, self-concordant Hessian comparison bounds its speed by $r/(1-tr)$; integration gives length at most $-\log(1-r)$. Sum over the chords. $\square$

This argument is Theorem 5.1(c) of Nesterov and Todd; the chord comparison and the primal–dual geometry are also theirs [72, Section 3 and Theorems 5.1–5.2]. By the triangle inequality, an initial metric error D_0 subtracts D_0 from the numerator’s distance bound. The theorem says nothing about unrestricted long moves, primal-only target sets, or query counts.

24.2 A dimension bound for arbitrary cone dictionaries

Theorem 24.3 (Barrier parameter bound from dimensions alone). *Let a compact full-dimensional body $C \subset \mathbb{R}^D$, $D \geq 1$, have an exact affine lift over a product of proper cones. After minimal-face reduction (Lemma 2.1) and deletion of zero factors, suppose all factors have dimension at most an integer $d \geq 2$. Define*

$$\Psi_d(h) = 2\lfloor h/d \rfloor + \min\{h \bmod d, 2\}.$$

Every possibly coupled logarithmically homogeneous standard self-concordant barrier on this reduced product has

$$\nu \geq \Psi_d(D+1) \geq 2(D+1)/d. \quad (24.5)$$

Thus the coefficient $\sqrt{\nu/2}$ in (24.4) is at least $\sqrt{\Psi_d(D+1)/2}$.

Proof. The total dimension M of the reduced product is at least $D + 1$, by Theorem 13.4. This also follows from the ordinary rank $D + 1$ of the body's affine slack operator after the reductions of Lemma 2.1. Let r_0 factors be rays and q_0 have dimension at least two. In each nonray factor a two-dimensional linear section through an interior point is a proper planar cone, hence an orthant after a linear change of coordinates. Their product with all the rays is an interior orthant section of dimension $r_0 + 2q_0$. Restriction of the barrier preserves its homogeneity degree and self-concordance, so the sharp orthant lower bound gives $\nu \geq r_0 + 2q_0$ [53, 70]. The argument permits arbitrary coupling. The cap d gives $r_0 + dq_0 \geq M \geq D + 1$. For $h = ad + b$, $0 \leq b < d$, the minimum of $r + 2q$ over nonnegative integers with $r + dq \geq h$ is $2a + \min(b, 2)$: for $q \leq a$ its minimum is $h - (d - 2)q$, while $q \geq a + 1$ costs at least $2a + 2$. Finally $\min(b, 2) \geq 2b/d$. $\square$

The function Ψ_d is the exact integer minimum implied by these two bounds; we do not claim that some lift attains it for every body. No definability or contact regularity is needed. For a product of balls it uses $D = \sum_a s_a$, whereas on a positively curved D -body with definable factors, Theorem 3.2 gives the usually stronger bound

$$\nu \geq 2 \left\lceil \frac{D - 1}{d - 2} \right\rceil \quad (d \geq 3).$$

Under the hypotheses of Theorem 6.2, the exact minimum number $\sum_a h_a$ of Lorentz factors instead gives $\nu \geq 2 \sum_a h_a$ and coefficient $\sqrt{\sum_a h_a}$. For the single shared product-ball cone below, the optimal parameter is $k + 1$ and the coefficient $\sqrt{(k + 1)/2}$. All these bounds concern barriers on the ambient cone and its conjugate, not the restricted standard barrier on a fixed-scale slice.

24.3 Primal and dual speeds on shared product balls

For positive dimensions s_a , consider

$$\mathcal{H} = \{(t, y_1, \dots, y_k) : t \geq \|y_a\|_2 \ (1 \leq a \leq k)\}.$$

Theorem 17.3, specialized to vector rows, gives the optimal $(k + 1)$ -logarithmically homogeneous barrier

$$F(t, y) = - \sum_a \log(t^2 - \|y_a\|^2) + (k - 1) \log t. \quad (24.6)$$

Its fixed-scale restriction is $f(y) = - \sum_a \log(1 - \|y_a\|^2)$. For unit vectors u_a , weights $w_a \geq 0$, and objective $-\sum_a w_a \langle u_a, y_a \rangle$ on $t = 1$, the centers are

$$y_a(\eta) = r(\eta w_a) u_a, \quad r(z) = \frac{z}{\sqrt{1 + z^2 + 1}}. \quad (24.7)$$

Indeed $2r/(1 - r^2) = \eta w_a$. Differentiation in $\log \eta$ gives $\dot{r} = r(1 - r^2)/(1 + r^2)$; the radial Hessian is $2(1 + r^2)/(1 - r^2)^2$. Consequently

$$\nu_P(\eta) = q_{\text{eff}}(\eta) := \sum_a \left(1 - \frac{1}{\sqrt{1 + (\eta w_a)^2}} \right), \quad \nu_D(\eta) = k + 1 - q_{\text{eff}}(\eta). \quad (24.8)$$

Thus q_{eff} smoothly counts the factors whose weight is significant at scale $1/\eta$; we call w_a visible at η when $\eta w_a \geq 1$. Let $L_P(\eta_0, \eta_1)$ and $L_D(\eta_0, \eta_1)$ be the primal and dual central-path lengths between η_0 and η_1 , in the metrics of F and F_* . With $w_1 = 1$ and all other weights zero, any interval $0 < \eta_0 < \eta_1$ satisfies

$$L_P \leq \log(\eta_1/\eta_0), \quad L_D \geq \sqrt{k} \log(\eta_1/\eta_0), \quad d_{\widehat{F}}(z(\eta_0), z(\eta_1)) \geq \sqrt{(k + 1)/2} \log(\eta_1/\eta_0).$$

From the primal analytic center $\eta = 0$ to $\eta_1 \geq 1$, the primal length is at most $1/\sqrt{2} + \log \eta_1$, since $q_{\text{eff}}(\eta) \leq \eta^2/2$ near zero. A variant has a unique optimizer. For fixed $0 < \epsilon \leq 1$, set $\nu = k + 1$, $w_1 = 1$, $w_a = \epsilon^2/\nu^2$ for $a > 1$, and $\eta_1 = \nu/\epsilon$. Then all weights are positive, the final gap ν/η_1 is ϵ , and throughout $1 \leq \eta \leq \eta_1$,

$$q_{\text{eff}}(\eta) \leq 1 + \frac{\epsilon^2}{2\nu}, \quad L_P(1, \eta_1) \leq \sqrt{1 + \frac{\epsilon^2}{2\nu}} \log(\nu/\epsilon).$$

Thus, even with a unique optimizer, the barrier parameter alone gives no lower bound on primal distance uniform over objectives. The weights here depend on ϵ ; for any fixed instance with all $w_a > 0$, $q_{\text{eff}}(\eta) \rightarrow k$ as $\eta \rightarrow \infty$. The general distinction between primal and product geometry is classical [69, 72]; the explicit specializations above are also recorded in [21, Section 9].

24.4 Accuracy and movement from the objective-weight distribution

Conversely, the objective-weight distribution determines primal accuracy and primal central length up to constant factors. Write

$$Q_j(\eta) = \sum_a \min\{(\eta w_a)^j, 1\} \quad (j = 1, 2), \quad E(\eta) = \sum_a w_a [1 - r(\eta w_a)].$$

Here $E(\eta)$ is the primal objective error of the center. For $z \geq 0$, direct scalar calculation gives

$$(1 - 1/\sqrt{2}) \min(z^2, 1) \leq 1 - (1 + z^2)^{-1/2} \leq \min(z^2/2, 1), \\ (2 - \sqrt{2}) \min(z, 1) \leq z[1 - r(z)] \leq \min(z, 1).$$

For the first lower bound, use $[1 - (1 + z^2)^{-1/2}]/z^2 = [\sqrt{1 + z^2}(1 + \sqrt{1 + z^2})]^{-1}$ on $(0, 1]$, and monotonicity on $[1, \infty)$. For the second, use monotonicity of $1 - r(z)$ below one and of $1 + z - \sqrt{1 + z^2}$ above one. Both lower constants are attained at $z = 1$. Summation and integration give

$$(2 - \sqrt{2})Q_1(\eta)/\eta \leq E(\eta) \leq Q_1(\eta)/\eta, \quad (24.9)$$

$$\sqrt{1 - 1/\sqrt{2}} \mathcal{J}(\eta_0, \eta_1) \leq L_P(\eta_0, \eta_1) \leq \mathcal{J}(\eta_0, \eta_1), \quad \mathcal{J} = \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta}. \quad (24.10)$$

The integral converges at $\eta_0 = 0$. After sorting the positive weights, on an interval with j weights satisfying $\eta w_a \geq 1$, write $B_j^2 = \sum_{a>j} w_a^2$. Then $Q_2 = j + \eta^2 B_j^2$ and, when $j, B_j > 0$, an antiderivative is

$$\sqrt{j + \eta^2 B_j^2} + \sqrt{j} \log \frac{\eta B_j}{\sqrt{j} + \sqrt{j + \eta^2 B_j^2}}.$$

The cases $j = 0$ and $B_j = 0$ give ηB_0 and $\sqrt{j} \log \eta$, respectively.

Proposition 24.4 (Head and tail estimates). *Assume $1 \geq w_1 \geq \dots \geq w_k \geq 0$, $0 < \epsilon \leq 1$, and $1 \leq m \leq k$. If $\sum_{a>m} w_a \leq \epsilon/2$, the center at $\eta_f = 2m/\epsilon$ is primal ϵ -accurate and*

$$L_P(0, \eta_f) \leq \sqrt{(m + \epsilon^2/4)/2} + \sqrt{2m} \log(2m/\epsilon). \quad (24.11)$$

Alternatively, if $\nu = k + 1$ and $\sum_{a>m} w_a^2 \leq \epsilon^2 m/\nu^2$, then at $\eta_g = \nu/\epsilon$ the central primal-dual gap is ϵ and

$$L_P(0, \eta_g) \leq \sqrt{(m + m\epsilon^2/\nu^2)/2} + \sqrt{3m/2} \log(\nu/\epsilon). \quad (24.12)$$

Proof. The head error is at most m/η_f , the tail error at most its weight sum. For $\eta \leq 1$, speed is at most $\eta\|w\|_2/\sqrt{2}$, contributing $\|w\|_2/\sqrt{2}$ to length. For $1 \leq \eta \leq \eta_f$ use $q_{\text{eff}} \leq Q_1 \leq m + \eta \sum_{a>m} w_a \leq 2m$, and $\|w\|_2^2 \leq m + \epsilon^2/4$. For the second claim, throughout $0 < \eta \leq \eta_g$, $q_{\text{eff}} \leq m + (\eta^2/2) \sum_{a>m} w_a^2 \leq 3m/2$; the same early-time integration and $\|w\|_2^2 \leq m + m\epsilon^2/\nu^2$ prove the result. $\square$

The second estimate bounds only the primal part of the central path to gap ϵ ; by (24.8), the remaining squared speed $k + 1 - q_{\text{eff}}$ is dual. All tail weights can be positive in either estimate. If at least m weights are visible throughout an interval, (24.10) gives the matching lower bound $\sqrt{(1 - 1/\sqrt{2})m \log(\eta_1/\eta_0)}$ on arc length.

These arc estimates also give an explicit bounded-move schedule. Set $\mathcal{L}(\eta) = \int_0^\eta \sqrt{q_{\text{eff}}(u)} du/u$ and choose successive centers at increments of at most $\log(1 + R)$ in $\mathcal{L}$, where $0 < R < 1$. For a nonzero objective this scalar function is strictly increasing, so the parameters are uniquely specified. Consecutive centers are at distance at most the arc length between them, so the standard self-concordant comparison $\log(1 + \|y - x\|_x) \leq d_f(x, y)$ [72, Section 3] bounds each chord's starting Dikin norm by R . The number of chords is at most $\lceil L_P / \log(1 + R) \rceil$. Evaluating and numerically inverting $\mathcal{L}$, reading the weights, and computing the centers have separate costs; the schedule bounds movement, not running time.

As an example with separated scales, take $w_a = 2^{-(a-1)}$ and $m = \lceil \log_2(4/\epsilon) \rceil \leq k$. The first estimate gives $L_P(0, 2m/\epsilon) = O(\log^{3/2}(1/\epsilon))$ as $\epsilon \downarrow 0$. This is also the order of this central arc: on $2^{j-1} \leq \eta \leq 2^j$, $1 \leq j \leq m$,

$$j \leq Q_2(\eta) \leq j + 4/3.$$

Each interval contributes $\Theta(\sqrt{j})$, giving $\Theta(m^{3/2})$ up to 2^m ; the remaining interval to $2m/\epsilon$ contributes $O(\sqrt{m} \log m)$ because $q_{\text{eff}} \leq 2m$ there. This conclusion concerns central arc length and requires $k \geq m$, so k grows as $\epsilon \downarrow 0$; for fixed finite k , all positive weights eventually become visible. These distribution estimates and their spectral versions are companion results [21, Section 5]. Here they show that neither the ambient barrier parameter nor the factor count can replace the primal movement of a given formulation.

Remark 24.5 (A pointwise obstruction to a parameter-only argument). Dikin-ellipsoid containment, determinant growth, and the gradient inequality alone do not give a primal distance bound depending only on the parameter. In the cube at $x_\delta = (1 - \delta)\mathbf{1}$, $0 < \delta \leq 1$, put $P = \mathbf{1}\mathbf{1}^T/k$ and

$$H_\delta = \frac{2}{k\delta^2}P + \frac{2}{\delta^2}(I - P), \quad g_\delta = \frac{\mathbf{1}}{k\delta}.$$

Then $(H_\delta^{-1})_{ii} = \delta^2(1 - 1/(2k))$, so its unit ellipsoid fits inside every cube facet, while $\|\mathbf{1}\|_{H_\delta} = \sqrt{2}/\delta$, $\det H_\delta = 2^k/(k\delta^{2k})$, and $g_\delta^T H_\delta^{-1} g_\delta = 1/2$. We do not claim that these pointwise data come from a global barrier. The product-ball examples above already rule out such a bound uniform over objectives; these matrices show that the weaker pointwise tests cannot restore it. Neither construction decides whether such a bound holds for all barriers when objectives are uniformly nondegenerate.

25 Movement in concrete affine formulations

We compute distances in the restricted barrier metric for grouped balls, norm trees, balance slices, and shared spectral cones; the number of cones in the ambient product does not determine this distance. The barrier-height argument and the grouped, packed, and unweighted tree central paths also appear in the companion manuscript [21]; we include the proofs to make the comparisons self-contained. Section 26 gives the exact distance coefficient for norm trees with a fixed objective.

Throughout, distance means the infimum of lengths of piecewise C^1 curves in the indicated open affine slice, with its displayed Hessian metric; the distance to a target set is the infimum over its points. We write $[u]_+ = \max\{u, 0\}$.

25.1 Barrier height and simultaneous slack contraction

Lemma 25.1 (Two endpoint estimates). *Let F be a barrier whose restricted gradient satisfies $\|DF\|_{F,*} \leq \sqrt{\nu}$. For every reference x^0 and target set E ,*

$$d_F(x^0, E) \geq \frac{[\inf_{y \in E} F(y) - F(x^0)]_+}{\sqrt{\nu}}. \quad (25.1)$$

If, in addition, $F = -\sum_{i=1}^H \log q_i$ for positive C^2 concave functions on a convex domain and the Hessian is positive definite, then

$$d_F(x, y) \geq \left(\sum_i \left| \log \frac{q_i(y)}{q_i(x)} \right|^2 \right)^{1/2}. \quad (25.2)$$

In particular, if $\prod_i q_i(y) \leq (A\epsilon/H)^H$ throughout E_ϵ and $\bar{q}_0 = (\prod_i q_i(x^0))^{1/H}$, then

$$d_F(x^0, E_\epsilon) \geq \sqrt{H} \left[\log \frac{H\bar{q}_0}{A\epsilon} \right]_+.$$

Proof. Integrate $|DF[\dot{x}]| \leq \sqrt{\nu} \|\dot{x}\|_F$ along a curve for the first assertion. Concavity gives

$$D^2(-\log q_i)[u, u] = \frac{Dq_i[u]^2}{q_i^2} - \frac{D^2 q_i[u, u]}{q_i} \geq (D \log q_i[u])^2.$$

Thus the map $x \mapsto (\log q_i(x))_i$ into Euclidean space does not increase lengths. Integration and Cauchy–Schwarz prove the remaining assertions. $\square$

Both estimates concern the whole accurate set, not the length of a particular central path. We record the resulting bound on bounded moves. A feasible chord of starting local norm $r < 1$ has length at most $-\log(1-r)$, by integrating the self-concordant Hessian comparison. Hence if a trajectory from x^0 , with $\|x^0 - \bar{x}\|_{\bar{x}} \leq \rho < 1$, reaches E_ϵ by chords of starting local norms $r_j < 1$, and D_ϵ is a lower bound on $d_F(\bar{x}, E_\epsilon)$, then

$$\sum_j -\log(1-r_j) \geq [D_\epsilon + \log(1-\rho)]_+. \quad (25.3)$$

In particular, for an integer $m \geq 1$ and $0 < R < 1$, rounds of at most m chords of starting local norm at most R require at least $[D_\epsilon + \log(1-\rho)]_+ / [-m \log(1-R)]$ rounds. With M chords in total, where $M \geq 1$ is an integer, some r_j is at least $1 - \exp(-[D_\epsilon + \log(1-\rho)]_+ / M)$. These statements restrict only the geometry of the primal trajectory and allow arbitrary computation between the counted moves.

25.2 Grouped balls and unrestricted PSD completion

For each of k balls, partition the coordinates into h nonempty groups, and use the grouped domain from (9.6):

$$q_{aG} = s_{aG} - \|w_{aG}\|^2 > 0, \quad \sum_G s_{aG} = 1, \quad F = -\sum_{a,G} \log q_{aG}.$$

Choose unit directions c_a , maximize $\ell = \sum_a \langle c_a, w_a \rangle$, and put $H = kh$. The optimum is k . A PSD packing puts these H group columns into arbitrary real matrix blocks W_j , pads shorter columns by zeros, and replaces the scalar residuals by

$$D_j = S_j - W_j^T W_j \succ 0, \quad F = - \sum_j \log \det D_j, \quad \sum_{G \text{ of } a} (S_{j(a,G)})_{GG} = 1.$$

All off-diagonal entries of S_j are free completion variables. Each column occurs once and contains precisely its assigned group coordinates. This includes the one-factor packing of Section 9.

Proposition 25.2 (Completion variables do not shorten the distance). *For both formulations the analytic center has $w = 0$ and all residual eigenvalues equal to $1/h$. The central path minimizing $F - \tau \ell$ is given by*

$$D(\tau) = \sqrt{h^2 + \tau^2}, \quad q(\tau) = \frac{2}{D(\tau) + h}, \quad r(\tau) = \frac{\tau}{D(\tau) + h}, \quad (25.4)$$

with $w_a = r(\tau)c_a$ and $q_{aG} = q(\tau)$ or $D_j = q(\tau)I$. From every central reference $z(\tau_0)$,

$$d_F(z(\tau_0), \{k - \ell \leq \epsilon\}) \geq \sqrt{H} \left[\log \frac{Hq(\tau_0)}{2\epsilon} \right]_+. \quad (25.5)$$

The bound allows arbitrary motion in the completion variables.

Proof. For the grouped barrier, allocation stationarity makes all slacks of one ball equal, and leaf stationarity gives $2w_{aG}/q = \tau c_{aG}$. The root allocation identity is $hq + \|w_a\|^2 = 1$. Solving it gives (25.4). For packing, stationarity in off-diagonal entries of S_j makes D_j^{-1} diagonal; diagonal allocation stationarity equates its entries within each ball. The remaining equations are the same. Strict convexity on the stated affine coordinates gives uniqueness.

Let $\delta_a = 1 - \langle c_a, w_a \rangle > 0$. In either representation the sum of residual diagonal entries belonging to ball a is $1 - \|w_a\|^2 \leq 2\delta_a$. Hadamard's determinant inequality for packing, followed by AM–GM within and across balls, gives

$$\prod_{a,G} q_{aG} \leq (2\epsilon/H)^H, \quad \text{or} \quad \prod_j \det D_j \leq (2\epsilon/H)^H \quad \text{if} \quad \sum_a \delta_a \leq \epsilon.$$

The grouped barrier has parameter H . So does the unrestricted matrix epigraph barrier before allocation restriction: the shear calculation in the proof of Theorem 9.6 gives squared gradient norm equal to the number of residual columns, and restriction can only decrease it. At $z(\tau_0)$ the residual determinant product is $q(\tau_0)^H$. Lemma 25.1 proves the result, with no constraint on the completion variables along the path. $\square$

A blockwise version of this estimate retains more information than the total barrier height. If c_j is the number of columns in block j , differentiation gives

$$D^2 F_j[(\dot{S}_j, \dot{W}_j), (\dot{S}_j, \dot{W}_j)] = \text{tr}((D_j^{-1/2} \dot{D}_j D_j^{-1/2})^2) + 2 \text{tr}(D_j^{-1} \dot{W}_j^T \dot{W}_j), \quad \dot{D}_j = \dot{S}_j - \dot{W}_j^T W_j - W_j^T \dot{W}_j.$$

Consequently every path from x to y has length at least

$$\left(\sum_j \frac{1}{c_j} \left| \log \frac{\det D_j(y)}{\det D_j(x)} \right|^2 \right)^{1/2}. \quad (25.6)$$

Indeed, trace Cauchy–Schwarz bounds each squared logarithmic derivative by c_j times its block metric, and integration gives the assertion. No off-diagonal entry is fixed in this argument.

For different group counts $h_a \geq 1$ and positive objective weights λ_a , set $H = \sum_a h_a$, $p_a = h_a/H$, and $\Delta_{\text{eff}} = \prod_a (\lambda_a/p_a)^{p_a}$. In both grouped and packed formulations, the residual diagonal entries of source a at the analytic center $\bar{z}$ equal $1/h_a$, the off-diagonal residuals are zero, and

$$d_F \left(\bar{z}, \left\{ \sum_a \lambda_a (1 - \langle c_a, w_a \rangle) \leq \epsilon \right\} \right) \geq \sqrt{H} \left[\log \frac{\Delta_{\text{eff}}}{2\epsilon} \right]_+. \quad (25.7)$$

To verify it, put $e_a = \lambda_a(1 - \langle c_a, w_a \rangle)$. Hadamard’s inequality and AM–GM within each source bound the residual determinant product by $\prod_a (2e_a/(\lambda_a h_a))^{h_a}$; its maximum for $\sum_a e_a \leq \epsilon$ occurs at $e_a = \epsilon p_a$. Compare with the center product $\prod_a h_a^{-h_a}$ and use Lemma 25.1. In particular, the one-block lift of order $s + k$ with one length- s column per ball has $H = k$, despite the unrestricted completion variables in its single PSD factor. For the one-column-per-ball Hermitian packing of (9.9), these endpoint and central-path statements also hold over $\mathbb{C}$ and $\mathbb{H}$, with arbitrary nonzero real column subspaces and unit support directions c_a in them. Use the real trace and, over $\mathbb{H}$, the Jordan determinant. Hermitian Hadamard gives the same diagonal product bound; Theorem 9.6 gives parameter k ; and stationarity in every real component of the free off-diagonal entries forces D^{-1} to be diagonal. Column stationarity in its real subspace then gives $w_a = r c_a$, as above. Thus the arc and tube formulas below apply to this Hermitian specialization with $h = 1$ as well.

25.3 Norm trees: uniform bounds and weighted exact central paths

Let T be a finite rooted tree whose internal nodes have at least two children. Its leaves index the coordinates of a Euclidean unit ball. Write $I(v)$ and $J(v)$ for the internal and leaf children of v , respectively, and set

$$q_v = t_v^2 - \sum_{u \in I(v)} t_u^2 - \sum_{j \in J(v)} w_j^2 > 0, \quad t_v > 0, \quad t_{\text{root}} = 1. \quad (25.8)$$

All blocks must be on the positive Lorentz sheet. Assign fixed weights $\omega_v \geq 1$ and use $F_T = -\sum_v \omega_v \log q_v$. Put $W = \sum_v \omega_v$, and let Ω_v be the sum of the weights of internal nodes in the subtree rooted at v , including v . For a unit leaf direction c , put $\alpha_v = \|c|_{\text{leaves below } v}\|$, so $\alpha_{\text{root}} = 1$.

Proposition 25.3 (Weighted central profile). *The unique minimizer of $F_T - \tau \langle c, w \rangle$ has*

$$\begin{aligned} q_v &= \omega_v q, & q &= \frac{2}{\sqrt{W^2 + \tau^2} + W}, & r &= \frac{\tau}{\sqrt{W^2 + \tau^2} + W}, \\ w &= r c, & t_v^2 &= \Omega_v q + r^2 \alpha_v^2. \end{aligned} \quad (25.9)$$

For $\tau > 0$ its squared speed per unit $\log \tau$ is

$$\left\| \frac{dz}{d \log \tau} \right\|_{F_T}^2 = W \left(1 - \frac{W}{\sqrt{W^2 + \tau^2}} \right). \quad (25.10)$$

The formulas allow zero leaf coefficients and zero subtree norms.

Proof. Internal-coordinate stationarity is $-2\omega_v t_v/q_v + 2\omega_{p(v)} t_v/q_{p(v)} = 0$. Since $t_v > 0$, all q_v/ω_v are equal. Leaf stationarity is $2w_j/q = \tau c_j$. The determinant identities telescope to $\sum_v q_v = 1 - \|w\|^2$, giving $Wq + r^2 = 1$ and the stated solution. The subtree identities determine the positive t_v uniquely. The weighted barrier is strictly convex on the slice $t_{\text{root}} = 1$, so the stationary point is the unique minimizer. Differentiating restricted central stationarity gives $H_\tau z'(\tau) = \nabla \ell$; therefore squared logarithmic speed is $\tau^2 r'(\tau)$, which is (25.10). $\square$

For k independent trees with the same total weight W , unit objective weights, and $L_\omega = kW$, all profiles agree and their squared speeds add. The same expression holds for grouped and packed balls with $W = h$ and $L_\omega = H$. Since restricted central stationarity gives $\nabla F = \tau \nabla \ell$, their squared restricted gradient norm along the central path tends to H . Thus the standard grouped and packed barriers have exact parameter H , not just the upper bound used in Proposition 25.2. Define

$$\mathcal{A}(r) = \sqrt{2} \left[\sqrt{2} \operatorname{artanh} \frac{\sqrt{2}r}{\sqrt{1+r^2}} - \operatorname{artanh} \frac{r}{\sqrt{1+r^2}} \right], \quad 0 \leq r < 1. \quad (25.11)$$

The exact central length from the analytic center to profile r is

$$\sqrt{L_\omega} \mathcal{A}(r), \quad \mathcal{A}(r) = \log \frac{1}{1-r} + O(1) \quad (r \uparrow 1). \quad (25.12)$$

Indeed, $\tau = 2Wr/(1-r^2)$ transforms the length element into $\sqrt{2L_\omega} \sqrt{1+r^2} dr/(1-r^2)$; the substitution $y = r/\sqrt{1+r^2}$ gives (25.11). For $0 < \epsilon \leq k$, the central path to gap ϵ uses $r = 1 - \epsilon/k$. At $\epsilon = k$ its length is zero; larger tolerances are already met at the analytic center. An arc of length a ends within local norm $e^a - 1$ of its start, measured at the start: after arclength s , self-concordance bounds the ratio of local norms by e^s , and integration gives the estimate. Partitioning the central path into arcs of length $\log(1+R)$, where $0 < R < 1$, therefore gives at most

$$\left\lceil \frac{\sqrt{L_\omega} \mathcal{A}(1 - \epsilon/k)}{\log(1+R)} \right\rceil \quad (25.13)$$

feasible chords of starting local norm at most R .

The next estimate holds uniformly at finite accuracy and allows different tree weights and objective weights.

Proposition 25.4 (Entropy scale for different trees). *For source a , let $W_a = \sum_v \omega_{av}$ and let $\lambda_a > 0$ multiply its unit support objective. Put*

$$L_\omega = \sum_a W_a, \quad p_a = W_a/L_\omega, \quad \Delta_{\text{eff}} = \prod_a (\lambda_a/p_a)^{p_a}.$$

If the displayed restricted barrier $F = -\sum_{a,v} \omega_{av} \log q_{av}$ satisfies $\|DF\|_{F,} \leq \sqrt{\nu}$, then from its analytic center $\bar{z}$,*

$$d_F \left(\bar{z}, \left\{ \sum_a \lambda_a (1 - \langle c_a, w_a \rangle) \leq \epsilon \right\} \right) \geq \frac{L_\omega}{\sqrt{\nu}} \left[\log \frac{\Delta_{\text{eff}}}{2\epsilon} \right]_+. \quad (25.14)$$

The product-cone value $\nu = 2L_\omega$ is always valid. For unit barrier weights, write b_a for the number of internal nodes. One may use the exact parameter $\nu = \sum_a \nu_{T_a}$ from Theorem 9.1, where $\nu_{T_a} = 2b_a - 2$ if the root has only internal children and $\nu_{T_a} = 2b_a - 1$ otherwise.

Proof. At the center $q_{av}^0 = \omega_{av}/W_a$. Write $e_a = \lambda_a(1 - \langle c_a, w_a \rangle)$. Telescoping gives $\sum_v q_{av} = 1 - \|w_a\|^2 \leq 2e_a/\lambda_a$. Weighted AM–GM within a tree gives

$$\prod_v \left(\frac{q_{av}}{q_{av}^0} \right)^{\omega_{av}} \leq (2e_a/\lambda_a)^{W_a}.$$

Under $\sum_a e_a \leq \epsilon$, the product on the right is maximized at $e_a = \epsilon p_a$. Thus $F(y) - F(\bar{z}) \geq L_\omega \log(\Delta_{\text{eff}}/(2\epsilon))$. Apply Lemma 25.1. $\square$

For unit objective weights, $\Delta_{\text{eff}} = \exp(H(p))$, where $H(p) = -\sum_a p_a \log p_a$ is Shannon entropy in natural units. In general $\Delta_{\text{eff}} \leq \sum_a \lambda_a$, with equality exactly when the objective weights are proportional to W_a . For k unweighted identical trees with b internal nodes, $L = L_\omega = kb$ and the bound is $(L/\sqrt{\nu})[\log(k/(2\epsilon))]_+$. Since $L \leq \nu < 2L$, this bound and (25.13) show that reaching the target takes $\Theta_R(\sqrt{L} \log(k/\epsilon))$ chords of starting local norm at most R for $0 < \epsilon \leq k/4$, with constants independent of the tree. The lower estimate applies to arbitrary feasible paths. Section 26 determines the exact leading coefficient for a fixed objective, which can be smaller than the central-path coefficient.

25.4 A separate control on central-path parameter changes

The endpoint bounds above hold whether or not accepted iterates stay near the central path. If they do, each increase of the path parameter is also bounded. This refines the short-step analysis for grouped balls, and the proof applies to the tree profile as well.

Proposition 25.5 (Tube parameter increments). *Consider a profile above with k sources of unit objective weight and common scalar parameter $W \geq 1$ (grouped or packed balls with $W = h$, or trees of total weight W), $L_\omega = kW$, and objective optimum k . Fix $a > 0$, $0 < R < 1$, an integer $m \geq 1$, and*

$$\gamma_a = 1 + a - \sqrt{1 + a^2}, \quad b_a = \sqrt{1 - (1 + a^2)^{-1/2}}, \quad 0 \leq \rho < \gamma_a.$$

Suppose $aW \leq \tau_0 \leq \dots \leq \tau_T$, accepted points satisfy $\|x_j - z(\tau_j)\|_{z(\tau_j)} \leq \rho$, and each round, from x_j to x_{j+1} , uses at most m feasible chords of starting norm at most R . Put

$$B_\rho = \frac{\rho}{(1 - \rho)^2}, \quad A_{\rho,R,m} = (1 - R)^{-m} - 1 + B_\rho(1 + (1 - R)^{-m}).$$

Then every round satisfies

$$\log \frac{\tau_{j+1}}{\tau_j} \leq \frac{A_{\rho,R,m}}{(1 - \rho)b_a \sqrt{L_\omega}}. \quad (25.15)$$

If the final objective gap is at most ϵ , and $\Delta = L_\omega/\tau_0$, then

$$T \geq \frac{(1 - \rho)b_a \sqrt{L_\omega}}{A_{\rho,R,m}} \left[\log \frac{(\gamma_a - \rho)\Delta}{\epsilon} \right]_+. \quad (25.16)$$

Proof. All covectors are restricted to the affine hull. Hessian comparison and integration from $z(\tau_j)$ to x_j give

$$\|\nabla F(x_j) - \tau_j \nabla \ell\|_{x_j,*} \leq B_\rho.$$

For one chord of starting norm at most R , the gradient change in its starting dual norm is at most $R/(1 - R)$; transporting a dual norm backwards across that chord costs at most $(1 - R)^{-1}$. A geometric sum over at most m chords, including both endpoint residuals, consequently gives

$$(\tau_{j+1} - \tau_j) \|\nabla \ell\|_{x_j,*} \leq A_{\rho,R,m}.$$

The exact speed formula and the tube comparison give $\tau_j \|\nabla \ell\|_{x_j,*} \geq (1 - \rho)b_a \sqrt{L_\omega}$. Use $\log u \leq u - 1$ to prove (25.15). For $u = \tau/W$, the central gap g satisfies $\tau g/L_\omega = 1 + u - \sqrt{1 + u^2} \geq \gamma_a$. The objective change from a center to a point within local norm ρ of it, measured at the center, is at most $\rho \sqrt{L_\omega}/\tau \leq \rho L_\omega/\tau$. Thus accuracy forces $\tau_T \geq (\gamma_a - \rho)L_\omega/\epsilon$. Sum (25.15) to finish. $\square$

25.5 The balance slice attains the scale set by its certificate rank

Return to Z in (15.1), with total Jordan rank $\rho = \sum_i r_i$, and use the restricted standard barrier $\Phi = -\sum_i \log \det X_i$. Choose the frame primitive $c = c_{11}$ and put $S = (e_1 - c, e_2, \dots, e_L)$ and $Q = \rho - 1$. The gap $g(X) = \langle S, X \rangle = 1 - \langle c, X_1 \rangle$ vanishes only at $(c, 0, \dots, 0)$: positivity and vanishing pairing force every block into the corresponding complementary face, and trace one fixes its coefficient. This point is feasible because all balance moments vanish on the chosen frame.

Proposition 25.6 (Exact leading distance on the balance slice). *Let $X_i^0 = e_i/\rho$ and $g_0 = Q/\rho$. For $0 < \epsilon < g_0$,*

$$\sqrt{Q} \log \frac{g_0}{\epsilon} \leq d_\Phi(X^0, \{g \leq \epsilon\}) \leq \sqrt{Q} \log \frac{g_0}{\epsilon} + \log(\rho(1 - \epsilon)). \quad (25.17)$$

Thus, for this fixed instance, the leading coefficient of the distance in $\log(1/\epsilon)$ is exactly $\sqrt{\rho - 1}$.

Proof. The certificate S has total Jordan rank Q , determinant one on its support, and exposed determinant ρ^{-Q} at X^0 . Theorem 23.1 therefore gives scale $\Delta = Q/\rho = g_0$ and the lower bound. More explicitly, the negative logarithm of the exposed determinant has dual gradient norm at most $\sqrt{Q}$ on the slice, and spectral AM–GM bounds that determinant at gap ϵ by $(\epsilon/Q)^Q$.

The central path minimizing $\Phi + \tau g$ has

$$X_1 = ac + b(e_1 - c), \quad X_i = be_i \ (i > 1), \quad a + Qb = 1, \quad \frac{1}{b} = \tau + \frac{1}{a}.$$

This point is stationary on the whole slice: its balance moments vanish, and $\nabla \Phi + \tau S$ is the total-trace normal $-e/a$. Strict convexity gives uniqueness and also proves the center assertion at $\tau = 0$. Writing $g = Qb$ gives

$$\tau = \frac{Q - (Q + 1)g}{g(1 - g)}, \quad g(0) = g_0, \quad g(\tau) \sim Q/\tau.$$

The frame-diagonal metric has central length

$$\int_\epsilon^{g_0} \sqrt{\frac{Q}{g^2} + \frac{1}{(1 - g)^2}} dg.$$

Bound its integrand above by $\sqrt{Q}/g + 1/(1 - g)$ to obtain the upper bound in (25.17). $\square$

Theorem 15.1 independently proves that every barrier on this body has parameter at least $\rho - 1$. The distance statement above concerns only the displayed standard barrier. Its squared coefficient equals this parameter bound, but that does not prove the same distance bound for every optimal coupled barrier.

25.6 Sharing spectral cones does not change the reduced movement

Use the real or complex matrix balls and orientations of Section 17, with $r_a \leq c_a$, and put $R_0 = \sum_a r_a$. Their common reduced barrier is

$$\Phi(X) = -\sum_a \log \det(I_{r_a} - X_a X_a^*).$$

Let $C_a = (I_{r_a} \ 0)$, and maximize $\ell(X) = \sum_a \operatorname{Re} \operatorname{tr}(C_a^* X_a)$, whose optimum is R_0 .

Proposition 25.7 (Grouping-independent spectral distance). *For every shared-cone grouping and its fixed-identity chordal block-star realization,*

$$d_\Phi(0, \{R_0 - \ell \leq \epsilon\}) \geq \sqrt{R_0} \left[\log \frac{R_0}{2\epsilon} \right]_+. \quad (25.18)$$

The exact central path is $X_a(\tau) = r(\tau)C_a$, where $r(\tau) = \tau/(\sqrt{1+\tau^2}+1)$. Its length to gap $0 < \epsilon < R_0$ is $\sqrt{R_0}\mathcal{A}(1 - \epsilon/R_0)$. Consequently, for this fixed instance, the leading coefficient of the distance in $\log(1/\epsilon)$ is exactly $\sqrt{R_0}$.

Proof. Let $\sigma_{ai} < 1$ be the singular values of X_a . Von Neumann's trace inequality gives $\sum_{a,i}(1 - \sigma_{ai}) \leq R_0 - \ell(X)$. Since $1 + \sigma_{ai} \leq 2$, AM–GM gives

$$\prod_a \det(I - X_a X_a^*) \leq (2\epsilon/R_0)^{R_0} \quad \text{when } R_0 - \ell(X) \leq \epsilon.$$

Theorem 17.3 gives the restricted parameter R_0 , and $\Phi(0) = 0$; apply Lemma 25.1. The barrier gradient at rC_a is $2rC_a/(1 - r^2) = \tau C_a$, so the path is stationary, hence optimal by convexity, in all variables. Along the path the barrier is $-R_0 \log(1 - r^2)$, and its length is exactly the scalar arc already computed. The two bounds have the same leading coefficient. Finally, Proposition 17.5 identifies the restricted barrier literally with Φ , so the metric and every displayed distance and path formula are identical in those formulations. $\square$

For vector rows these bodies are products of Euclidean balls. Maximal sharing can therefore reduce the number of cones without changing this reduced path geometry. Equations (25.3) and (25.18) give the same bounded-chord lower bound for all these descriptions. This conclusion depends only on their common barrier after affine elimination; it gives neither a query lower bound nor the cost of evaluating the ambient Newton system.

26 Exact distance coefficients for weighted norm trees

Proposition 25.4 gives a finite-accuracy lower bound uniform over objectives. For a fixed objective, we now determine the leading coefficient of the distance exactly. An internal subtree on which the objective vanishes contributes half of its barrier weight to the squared coefficient. The rank of an exposing certificate does not detect this contribution, because its dual blocks on such a subtree are zero. The contribution is nevertheless unavoidable: as the gap tends to zero, the subtree must shrink toward the vertex of its Lorentz cone.

We use the norm-tree domain Ω_T of Section 9. Write V for its internal nodes, $b = |V|$, and w for its vector of leaf coordinates. Fix weights $\omega_v \geq 1$ and the displayed barrier

$$F_\omega = - \sum_{v \in V} \omega_v \log q_v, \quad W = \sum_{v \in V} \omega_v.$$

It is a self-concordant barrier with positive definite restricted Hessian. All distances in this section refer to this Hessian, with the root fixed to one and all linking equations imposed. Fix a unit vector c in the leaf space, put $g(w) = 1 - c^T w$, and define

$$\alpha_v = \|c\|_{\text{leaves below } v}, \quad V_+ = \{v : \alpha_v > 0\}, \quad V_0 = V \setminus V_+, \quad K_c = \sum_{v \in V_+} \omega_v + \frac{1}{2} \sum_{v \in V_0} \omega_v.$$

We call the nodes of V_+ active and those of V_0 inactive. The root is active, and V_0 is a union of complete subtrees.

Theorem 26.1 (Weighted norm-tree distance). *For every fixed $x_0 \in \Omega_T$, as $\epsilon \downarrow 0$,*

$$d_{F_\omega}(x_0, \{x \in \Omega_T : g(w) \leq \epsilon\}) = \sqrt{K_c} \log(1/\epsilon) + O(\log \log(1/\epsilon)). \quad (26.1)$$

The lower bound holds with an $O(1)$ remainder; the $O(\log \log(1/\epsilon))$ remainder comes only from our upper-bound construction. The constants may depend on the fixed tree, weights, objective, and reference.

Proof. We first construct a potential that bounds every feasible path. For the primal Lorentz block $x_v = (t_v, y_v)$, define the Euclidean dual vector

$$s_v = (\alpha_v, -(\alpha_u)_{u \in I(v)}, -(c_j)_{j \in J(v)}).$$

Here $I(v)$ and $J(v)$ are the internal-child and leaf-child sets. The identity $\alpha_v^2 = \sum_{u \in I(v)} \alpha_u^2 + \sum_{j \in J(v)} c_j^2$ says that s_v is a nonzero boundary vector of the Lorentz cone for $v \in V_+$ and is zero otherwise. Consequently

$$\beta_v = \langle s_v, x_v \rangle > 0 \quad (v \in V_+), \quad \sum_{v \in V_+} \beta_v = 1 - c^T w = g(w).$$

The sum telescopes, including all internal links. Thus each $\beta_v \leq g$.

There is also a uniform bound $q_z \leq C_c g$ for every inactive node $z \in V_0$. To see this, first let $u \in V_0$ have an active parent v . Write $e_v = ((\alpha_u)_{u \in I(v)}, (c_j)_{j \in J(v)})/\alpha_v$, which is a unit vector. The coordinate occupied by t_u is orthogonal to e_v . Since $\|y_v\| < t_v$ and $t_v \leq 1$,

$$t_u^2 \leq \|y_v - (e_v^T y_v) e_v\|^2 \leq t_v^2 - (e_v^T y_v)^2 \leq 2t_v(t_v - e_v^T y_v) = \frac{2t_v \beta_v}{\alpha_v} \leq \frac{2g}{\alpha_v}.$$

Every internal node z below u , including u , has $q_z \leq t_z^2 \leq t_u^2$. Taking the maximum of $2/\alpha_v$ over the finitely many active parents of inactive subtrees proves the assertion.

The needed dual-norm identities can be checked directly, without Jordan normalization. For $f(x) = -\log(x^T J x)$ on a Lorentz cone, with $J = \text{Diag}(1, -I)$ and $q = x^T J x$, direct inversion gives

$$(D^2 f(x))^{-1} = x x^T - \frac{q}{2} J.$$

If $s \neq 0$ lies on the boundary of the Lorentz cone, then $s^T J s = 0$ and $\|D \log(s^T x)\|_{(D^2 f)^{-1}}^2 = 1$. Logarithmic homogeneity likewise gives $\|D f\|_{(D^2 f)^{-1}}^2 = 2$. It follows that the potential

$$\Psi(x) = - \sum_{v \in V_+} \omega_v \log \beta_v - \frac{1}{2} \sum_{v \in V_0} \omega_v \log q_v$$

has squared dual gradient norm exactly K_c in the ambient product metric $D^2 F_\omega$. Restriction to the slice can only decrease this norm: its value is the supremum over fewer tangent directions. At every point with $g \leq \epsilon$, the preceding determinant and pairing bounds imply

$$\Psi(x) - \Psi(x_0) \geq K_c \log(1/\epsilon) - O(1).$$

Integrating $|D\Psi[h]| \leq \sqrt{K_c} \|h\|_{F_\omega, x}$ along an arbitrary path proves the claimed lower bound.

For the upper bound put $A = |V_+|$, $Z = |V_0|$, and, for large λ , set $h = e^{-\lambda}$ and prescribe

$$q_v = \begin{cases} h, & v \in V_+, \\ h/\lambda^2, & v \in V_0, \end{cases} \quad r^2 = 1 - Ah - Zh/\lambda^2, \quad w = rc, \quad t_v^2 = r^2 \alpha_v^2 + \sum_{z \in V \text{ below } v} q_z. \quad (26.2)$$

The last sum includes v . Choose the positive square roots. The subtree identities give precisely the prescribed determinants, and the root equation gives $t_{\text{root}} = 1$. Thus this is an interior feasible curve. Its gap is $1 - r = Ah/2 + O(h/\lambda^2 + h^2)$, since $A \geq 1$.

A dot will denote differentiation with respect to λ . The Lorentz metric has the identity

$$D^2 f(x)[\dot{x}, \dot{x}] = \left(\frac{\dot{q}}{q}\right)^2 - \frac{2(\dot{t}^2 - \|\dot{y}\|^2)}{q}. \quad (26.3)$$

At an active node, t_v tends to $\alpha_v > 0$ and $\dot{t}_v = O(h)$. The same holds for active child axes and for leaf-coordinate derivatives; inactive child derivatives are $O(\sqrt{h}/\lambda)$. Since $q_v = h$, this block contributes $1 + O(h + \lambda^{-2})$ to the unweighted squared speed. At an inactive node, all its leaf coordinates vanish and all its descendants are inactive. Its whole block is a fixed interior Lorentz vector multiplied by $\sqrt{h}/\lambda$. Its squared speed is therefore exactly $2(1/2 + 1/\lambda)^2$. Adding the weighted contributions gives

$$\|\dot{x}\|_{F_\omega, x}^2 = K_c + O(1/\lambda).$$

Starting the curve at any sufficiently large fixed λ_0 , its length to λ is $\sqrt{K_c}\lambda + O(\log \lambda)$. Connect x_0 to the starting point by an interior line segment, whose length is finite. Along the curve, $\lambda = \log(1/g) + O(1)$, which proves the upper bound. $\square$

Corollary 26.2 (Central-path excess and independent products). *For the weighted tree, the central path from its analytic center to gap ϵ has length $\sqrt{W} \log(1/\epsilon) + O(1)$. Its ratio to the shortest distance to that target tends to $\sqrt{W/K_c}$. The central path has the optimal leading coefficient exactly when c is nonzero on the leaves below every internal node, that is, when V_0 is empty.*

For independent trees and the objective $\sum_a \lambda_a c_a^T w_a$, with fixed $\lambda_a > 0$ and unit c_a , the squared distance coefficient is $\sum_a K_{c_a}$. Additional independent trees that do not appear in the objective contribute zero.

Proof. The weighted central-path formulas of the preceding section give the first length. The ratio follows from Theorem 26.1; $K_c = W$ holds exactly when V_0 is empty. For products, sum the lower potentials. Each individual gap is at most ϵ/λ_a , which changes only the fixed remainder, and the squared dual norms add. For the upper bound synchronize the parameter λ in (26.2) across all positively weighted trees. The total objective gap is a fixed positive constant times $e^{-\lambda}(1 + o(1))$, and squared speeds add. Keep the remaining independent factors at their reference points. $\square$

For example, take a tree with two internal nodes, the root and one internal child, and let c be supported on a leaf child of the root. For unit weights, the root is active and its internal child is inactive. The shortest-distance coefficient is $\sqrt{3/2}$, whereas the central coefficient is $\sqrt{2}$. If the objective also has nonzero support in the child subtree, both coefficients are $\sqrt{2}$. Thus the coefficient can jump when a component of the objective tends to zero; the asymptotics hold with the objective fixed while the accuracy tends to zero.

This calculation builds on the classical study of barrier metrics and central-path distances [69, 72] and on the uniform norm-tree estimates of the companion manuscript [21]. Our contribution is the explicit weighted coefficient K_c , including the contribution of objective-free subtrees, for this fixed family of linked Lorentz barriers. It does not identify the best barrier on the tree domain, and it does not give an iteration lower bound unless the metric distance traversed per accepted step is restricted.

Part VI

Resource counts and computational costs

27 Exact resource counts and their optimization

The number of cone factors, the cone dimension, the ambient and restricted barrier parameters, and the number of free affine coordinates measure different aspects of a formulation. This section determines exact integer tradeoffs among them. Each optimization statement concerns only the resource model it names and does not by itself bound running time. The lifts and slack factorizations are those of Sections 5–9, interpreted in the factorization framework of Gouveia et al. [43]; the balancing and integer optimization arguments are elementary consequences of the curvature and nullity bounds proved here.

27.1 Real PSD order caps

Let $N \geq 4$ and $R \geq 2$ be integers, and consider exact lifts of $\mathbb{B}_2^N$ through real PSD blocks of orders $2 \leq r_i \leq R$ with globally bi- C^1 full-slack contact selections. Put

$$c_r = \lfloor r^2/4 \rfloor, \quad d_r = r(r+1)/2, \quad M = \sum_i d_{r_i}, \quad \rho = \sum_i r_i.$$

Here c_r is the capacity of an order- r block. Optional ray factors add resources but no curvature, so we omit them from the lower bounds. The ambient product cone has optimal barrier parameter ρ , even among coupled barriers; this does not determine the parameter after the affine equations are imposed. By Corollary 7.6,

$$\sum_i c_{r_i} \geq N. \tag{27.1}$$

Indeed, the local lower bound is $N - 1$, and equality would require the single matching spin factor Q_{N+1} , which is not among these real PSD cones when $N \geq 4$.

Proposition 27.1 (Exact integer consequences). *Define $K_R(N), V_R(N), F_R(N)$ as the minima of, respectively, $\sum_r a_r$, $\sum_r r a_r$, and $\sum_r d_r a_r$ over nonnegative integers $a_2, \dots, a_R$ satisfying $\sum_r c_r a_r \geq N$. Every lift in the stated class, with L PSD blocks, satisfies*

$$L \geq K_R(N), \quad \rho \geq V_R(N), \quad M \geq F_R(N).$$

Write $N = qc_R + t$ with $0 \leq t < c_R$. Then

$$K_R(N) = \lceil N/c_R \rceil, \tag{27.2}$$

$$V_R(N) = qR + \begin{cases} 0, & t = 0, \\ \lceil 2\sqrt{t} \rceil, & t > 0. \end{cases} \tag{27.3}$$

$$F_R(0) = 0, \quad F_R(n) = \min_{2 \leq r \leq R} \{d_r + F_R((n - c_r)_+)\}. \tag{27.4}$$

In particular,

$$\begin{aligned} \rho &\geq \max\{\lceil RN/c_R \rceil, \lceil 2\sqrt{N} \rceil\}, \\ M &\geq \max\{\lceil (2 + 1/\lfloor R/2 \rfloor)N \rceil, 2N + \lceil \sqrt{N} \rceil\}. \end{aligned} \tag{27.5}$$

Two explicit cases are

$$F_4(N) = 10\lfloor N/4 \rfloor + 3(N \bmod 4), \quad F_5(N) = \lceil 5N/2 \rceil \quad (N \geq 4).$$

These values are exact for the integer problems; for $R \geq 4$, we do not claim that lifts attain them.

Proof. Formula (27.2) is immediate. Extend c_r to nonnegative integer r . Its increments $c_r - c_{r-1} = \lfloor r/2 \rfloor$ are nondecreasing. Moving one unit from a smaller positive order to a larger order below R therefore increases or preserves capacity. Repeating, a list of total order $S = \ell R + j$, $0 \leq j < R$, has capacity at most $\ell c_R + c_j$. This relaxation permits zero and one as orders, which only strengthens its use as an upper bound on capacity. The smallest order with $c_r \geq t > 0$ is $\lceil 2\sqrt{t} \rceil$. Maximal-order blocks and this last block attain (27.3); the capacity bound just proved excludes every smaller total order. Selecting the first block of an optimal multiset proves both inequalities in (27.4).

For $r = 2j$ or $2j + 1$ one has $d_r/c_r = 2 + 1/j$, whereas c_r/r increases with r . These give the cap-dependent bounds in (27.5). Also $N \leq \sum_i r_i^2/4 \leq \rho^2/4$ and $d_r = 2c_r + \lceil r/2 \rceil$, proving the other two bounds. For $R = 4$, orders two and three cost three per unit of capacity; order four costs ten for four units. Using the maximal possible number of complete four-unit blocks and paying three per remaining unit is optimal; overshooting by another four-unit block costs ten rather than at most nine. For $R = 5$, the ratio lower bound is $5/2$. Every even integer at least four is a nonnegative combination of four and six, realized by orders four and five at that ratio. For odd $N \geq 5$, add one order-two block to a realization of $N - 1$. This attains the ceiling. $\square$

27.2 Grouped Schur constructions and exact minima for small orders

Partition $[N]$ into nonempty groups G_j of sizes $g_j \leq R - 1$. The formulation

$$\begin{pmatrix} s_j & x_{G_j}^T \\ x_{G_j} & I_{g_j} \end{pmatrix} \succeq 0, \quad \sum_{j=1}^k s_j = 1 \quad (27.6)$$

projects exactly onto $\mathbb{B}_2^N$. At a boundary point the allocation is $s_j = \|x_{G_j}\|^2$. Polynomial slack factors are

$$A_j(x) = \begin{pmatrix} \|x_{G_j}\|^2 & x_{G_j}^T \\ x_{G_j} & I \end{pmatrix}, \quad B_j(y) = \frac{1}{2}(1, -y_{G_j})^T(1, -y_{G_j}),$$

since $\sum_j \text{tr}(A_j(x)B_j(y)) = \frac{1}{2}\|x - y\|^2 = 1 - x^T y$ on the unit sphere. Thus this construction meets the global hypotheses. For a prescribed $\lceil N/(R - 1) \rceil \leq k \leq N$, write $N = ak + t$ with $0 \leq t < k$. Balancing the group sizes minimizes the ambient dimension within this construction and gives

$$(L, \rho, M, \nu_{\text{slice}}) = (k, N + k, (k - t)d_{a+1} + td_{a+2}, k). \quad (27.7)$$

Here $\nu_{\text{slice}} = k$ is also the intrinsic barrier parameter of this fixed affine domain, by Theorem 9.4. Strict convexity of d_{g+1} in g proves balancing. The minimum count is $\lceil N/(R - 1) \rceil$, whereas the minimum dimension is $3N$:

$$d_{g+1} - 3g = (g - 1)(g - 2)/2 \geq 0,$$

with equality for groups of size one or two. Eliminating the constant identity blocks leaves $N + k$ variables and one equality. In the unreduced cone encoding there are $1 + \sum_j d_{g_j}$ affine equations, giving the same affine dimension $N + k - 1$.

The lower bounds above and this construction give the following exact minima, attained simultaneously by one lift, in the global contact-selection class:

real PSD dictionary	$L_{\min}$	$M_{\min}$	$\rho_{\min}$
$r \leq 2$	N	$3N$	$2N$
$r \leq 3$	$\lceil N/2 \rceil$	$3N$	$\lceil 3N/2 \rceil$.

(27.8)

For the fixed complex order-two dictionary, the corresponding triple is

$$(\lceil N/2 \rceil, 4\lceil N/2 \rceil, 2\lceil N/2 \rceil) \quad (N \geq 4); \quad (27.9)$$

for the fixed quaternionic order-two dictionary it is

$$(\lceil N/4 \rceil, 6\lceil N/4 \rceil, 2\lceil N/4 \rceil) \quad (N \geq 6).$$

The exceptions are $(1, 4, 2)$ for complex $N = 3$ and $(1, 6, 2)$ for quaternionic $N = 5$. Indeed, these cones are Q_4, Q_6 , with capacities two and four; the claims follow from the strict capacity increment and the fixed-tail construction of Section 7. Dimensions count full dictionary cones, even for a partially filled final group.

27.3 Tradeoffs for packing columns into shared PSD blocks

Let $C = (\mathbb{B}_2^s)^b$, where $b \geq 2$ and $s \geq 3$ are integers. Choose an integer $1 \leq p \leq s$ and split each source into $h = \lceil s/p \rceil$ nonempty groups of at most p coordinates. Pad columns by zeros and pack the $H = bh$ columns into ℓ nonempty blocks. Block j has a $p \times c_j$ matrix W_j and a free symmetric matrix S_j :

$$\begin{pmatrix} S_j & W_j^T \\ W_j & I_p \end{pmatrix} \succeq 0, \quad \sum_{\gamma \text{ belonging to } a} (S_{j(\gamma)})_{\gamma\gamma} = 1 \quad (a = 1, \dots, b). \quad (27.10)$$

Every column belongs to one source group and appears once.

Proposition 27.2 (Exact resource counts of finite packings). *Write $H = q\ell + t$ with $0 \leq t < \ell$. For fixed p, ℓ , balanced column counts minimize the maximum order, the ambient cone dimension M , and the number V of free coordinates, giving*

$$\begin{aligned} L &= \ell, & r_{\max} &= p + \lceil H/\ell \rceil, & \rho &= p\ell + H, \\ M &= (\ell - t)d_{p+q} + td_{p+q+1}, & V &= bs + (\ell - t)d_q + td_{q+1}, & \nu_{\text{slice}} &= H. \end{aligned} \quad (27.11)$$

The last value is the exact parameter of the restricted standard barrier; in fact, the fixed affine domain has intrinsic parameter H . The count V is taken before the b independent allocation equations are imposed. Under an order cap R , all undominated resource vectors of this fixed-width family are obtained by enumerating

$$1 \leq p \leq \min(s, R - 1), \quad \lceil H/(R - p) \rceil \leq \ell \leq H \quad (27.12)$$

and discarding dominated vectors.

Proof. Schur complementation gives $D_j = S_j - W_j^T W_j \succeq 0$. The allocation identities bound the squared norm of each source vector by one. Conversely, any source points in the balls admit diagonal residuals with the required nonnegative sums. This proves exactness. Moving a column from a block with at least two more columns to a smaller block decreases both sums of d_{p+c} and of d_c and does not increase the maximum order. This proves balancing and all counts except the barrier equality.

The barrier $-\sum_j \log \det D_j$ has parameter at most $\sum_j c_j = H$ by the matrix-epigraph calculation in Theorem 9.6. Project the closed slice onto all W_j and all diagonal entries of S_j . Its image is exactly the grouped domain of Theorem 9.4, with H nonempty groups. Fibers are bounded: $|D_{uv}|^2 \leq D_{uu}D_{vv}$ and each diagonal residual is at most one. There is a strictly feasible point at $W = 0$ and positive allocation to every column. Lemma 9.5 therefore gives an intrinsic lower bound H , proving equality. Finally, balanced maximum order is at most R exactly in (27.12); widths $p > s$ only add zero padding and resources. $\square$

For example, the exact minimum factor count and ambient rank in this family are

$$\min_p \left\lceil \frac{H_p}{R-p} \right\rceil, \quad \min_p \left\{ H_p + p \left\lceil \frac{H_p}{R-p} \right\rceil \right\}, \quad H_p = b \lceil s/p \rceil.$$

Minimizing M , or a combined cost such as $M\sqrt{H_p}$, requires the finite enumeration over ℓ ; a continuous balanced aspect ratio would ignore the overshoot of the final block. The answer changes if only free affine coordinates are counted. For every $\theta \geq 0$, put $p_* = \min(s, R-1)$ and $H_* = b \lceil s/p_* \rceil$. Then

$$\min V \nu_{\text{slice}}^\theta = (bs + H_*)H_*^\theta. \quad (27.13)$$

Indeed splitting a column block of sizes $c_1 + c_2$ saves $c_1 c_2$ free coordinates, so for fixed p the unique optimal column allocation is $\ell = H_p$. The function $(bs + H)H^\theta$ increases with $H > 0$. Thus the largest feasible p is optimal; a smaller width with the same value of $\lceil s/p \rceil$ ties it. With one column per block, the optimizers are the unshared grouped Schur blocks.

Without a cap, direct Lorentz lifting has $(L, \rho, M, \nu_{\text{slice}}) = (b, 2b, b(s+1), b)$. Every packing instead satisfies

$$M \geq (2p+1)H_p \geq b(2s+1), \quad H_p \geq b, \quad (27.14)$$

since $d_{p+c} = (2p+1)c + (p-c)(p-c+1)/2$ and the last term is nonnegative for integer $p-c$. Thus the direct Lorentz formulation has strictly smaller cone dimension and no larger restricted parameter, with equality for $p = s$. Nevertheless, a packing can use a single PSD factor, of minimum order within this family

$$r_{\text{one}} = \min_{1 \leq p \leq s} \{p + b \lceil s/p \rceil\}, \quad r_{\text{one}} < 2b \iff b > s.$$

If $\lceil s/p \rceil \geq 2$, the order exceeds $2b$; otherwise $p = s$ and the order is $s+b$. This proves the equivalence.

Now cap both the Lorentz dimension and the PSD order by an integer $3 \leq R < s+1$. Grouped Lorentz blocks use $h_L = \lceil s/(R-2) \rceil$ groups per source and have

$$(L, \rho, M, \nu_{\text{slice}}, V) = (bh_L, 2bh_L, bs + 2bh_L, bh_L, bs + bh_L).$$

Grouped PSD Schur blocks instead use $h_S = \lceil s/(R-1) \rceil$, with $(L, \rho, \nu_{\text{slice}}, V) = (bh_S, bs + bh_S, bh_S, bs + bh_S)$ and balanced dimension from (27.7), summed over sources. The PSD blocks have no larger free-coordinate and restricted-parameter counts, although a PSD cone of order R has d_R scalar coordinates and a Lorentz cone of dimension R only R . Under a cap of D scalar coordinates per cone, the PSD order is only about $\sqrt{2D}$. When source-boundary and last-block effects are negligible, grouped Lorentz has $(L, \nu_{\text{slice}}, \rho, M) \sim (bs/D, bs/D, 2bs/D, bs)$, whereas balanced PSD packing has $(2bs/D, \sqrt{2bs}/\sqrt{D}, 2\sqrt{2bs}/\sqrt{D}, 2bs)$. In this comparison, the advantage of packing comes from sharing blocks across sources, not from using fewer scalar coordinates.

27.4 Maximal capacity for a nullity budget and a general resource inequality

Return to globally bi- C^1 full product-slack factorizations over L real PSD blocks of orders at most R . Write $n = b(s-1)$ and let $\nu = \nu_{\text{std}, \text{slice}}$. At a simultaneous contact, let $c_i = \text{nullity } X_i$. The mixed derivative has rank n , so

$$n \leq \sum_i c_i(r_i - c_i), \quad \sum_i c_i \leq \nu. \quad (27.15)$$

The second inequality is Lemma 8.1. For integers $z \geq 0$, define the largest capacity allowed by a nullity budget z as

$$C_R(L, z) = \max \left\{ \sum_i c_i(R - c_i) : 0 \leq c_i \leq R, c_i \in \mathbb{Z}, \sum_i c_i \leq z \right\}.$$

Put $m = \lfloor R/2 \rfloor$ and $z' = \min(z, Lm)$, and write $z' = qL + t$ with $0 \leq t < L$. Concavity and balancing give the exact expression

$$C_R(L, z) = (L - t)q(R - q) + t(q + 1)(R - q - 1). \quad (27.16)$$

Indeed, replacing $c_i > m$ by $R - c_i$ preserves capacity and uses less budget, and below m all marginal increments are positive; so using the budget up to Lm and balancing is optimal. Together with the sourcewise bound on selected ranks in Theorem 7.8, this proves the joint necessary conditions

$$\lfloor \nu \rfloor \geq b \lfloor s/(R-1) \rfloor, \quad C_R(L, \lfloor \nu \rfloor) \geq n. \quad (27.17)$$

Grouped Schur lifts attain the first bound. We do not claim that every integer point satisfying these conditions is realized by a lift. If the total nullity is $z \leq LR/2$, Cauchy–Schwarz also gives $n \leq Rz - z^2/L$. Here $n > 0$ forces $z > 0$ and $Rz - n \geq z^2/L > 0$, so $L \geq z^2/(Rz - n)$.

Theorem 27.3 (Resource product bound). *Let $J_\omega = \sum_i r_i^\omega$. Under the preceding contact and standard-barrier hypotheses, for $\theta \geq 0$ and $0 \leq \omega \leq 2 + \theta$,*

$$J_\omega \nu^\theta \geq C_\theta \frac{n^{1+\theta}}{R^{2+\theta-\omega}}, \quad C_\theta = \frac{(2+\theta)^{2+\theta}}{(1+\theta)^{1+\theta}}. \quad (27.18)$$

In the continuous relaxation of the capacity bound, equality requires $p : c = (1 + \theta) : 1$ in each block, where $p = r - c$.

Proof. For $c_i > 0$, write $p_i = r_i - c_i = x_i r_i$. Maximizing $x(1-x)^{1/(1+\theta)}$ on $[0, 1]$ gives

$$p_i c_i \leq K_\theta R^{(2+\theta-\omega)/(1+\theta)} r_i^{\omega/(1+\theta)} c_i^{\theta/(1+\theta)}, \quad K_\theta = \frac{1+\theta}{2+\theta} (2+\theta)^{-1/(1+\theta)}.$$

Zero-capacity terms can be discarded. For $\theta > 0$, sum and apply Hölder with exponents $1 + \theta$ and $(1 + \theta)/\theta$. Use (27.15) and $\sum c_i \leq \nu$, then raise to power $1 + \theta$. Since $K_\theta^{-(1+\theta)} = C_\theta$, the result follows. At $\theta = 0$ it is simply the sum of $p_i c_i \leq R^{2-\omega} r_i^\omega / 4$. The shape maximum is unique at $x = (1 + \theta)/(2 + \theta)$. For $\omega < 2 + \theta$ equality also requires full order R in every contributing block; at the endpoint $\omega = 2 + \theta$ the scale does not matter. $\square$

For $\theta = 1/2$, $C_{1/2} = (25/6)\sqrt{5/3}$. Since $M \geq (1 + 1/R)J_2/2$, the theorem gives

$$M\sqrt{\nu} \geq \frac{25}{12} \sqrt{\frac{5}{3}} (1 + 1/R) \frac{n^{3/2}}{\sqrt{R}}. \quad (27.19)$$

Divisible packings with $p + c = R$, $p : c = (1 + \theta) : 1$ and $H = bs/p$, $L = H/c$ attain (27.18) with bs in place of $b(s-1)$. Their ratio to the lower bound is $(s/(s-1))^{1+\theta}$. The cone-coordinate comparison has exactly the same ratio because all blocks have full order. Thus these divisible families attain the relaxation asymptotically as $s \rightarrow \infty$. When $b\lceil s/(R-1) \rceil$ constrains the optimizer, or divisibility fails, use (27.17) and the finite enumeration instead.

The same ratio $p : c$ follows directly by optimizing a uniform full-order packing: $J_\omega H^\theta = (bs)^{1+\theta} R^\omega / (p^{1+\theta} c)$. For static resources, $\theta = 0$, the ratio is $1 : 1$; for the common iteration-count factor $\sqrt{\nu}$ it is $3 : 2$. If $\omega > 2 + \theta$, larger blocks increase $J_\omega H^\theta$, so the cap R need not bind. No ratio $p : c$ is universal once constraints on source width, minimum block size, and integrality apply. In particular, $3 : 2$ need not be optimal for a cost model based on dense cubic linear algebra, $\omega = 3$.

Unshared PSD formulations have a different optimum. For divisible Schur groups of common width g and $A = bs$,

$$L = \nu = A/g, \quad \rho = A(1 + 1/g), \quad M\sqrt{\nu} = A^{3/2} \frac{(g+1)(g+2)}{2g^{3/2}}.$$

The first two resources times $\sqrt{\nu}$ decrease with g , whereas the last has its unique integer minimum $15A^{3/2}/8$ at $g = 4$, if this width is available. Indeed, consecutive values decrease exactly when $(g+1)^5 - g^3(g+3)^2 = -g^4 + g^3 + 10g^2 + 5g + 1 \geq 0$; this holds for $g = 1, 2, 3$ and fails thereafter. The polynomial is negative at four and has negative derivative thereafter. Heterogeneous groups cannot improve the value when $4 \mid s$: at fixed mean width u , convexity balances widths between two neighboring integers. The interpolated cost is $au + b'$ with $a > 0$ and $b' \leq 0$, so the derivative of $(au + b')u^{-3/2}$ changes sign, if at all, from positive to negative. There is no interior minimum. Its minimum is therefore at an integer width, where the preceding comparison applies. If four is unavailable, the largest allowed width below four is optimal in the divisible model. The exact integer counts above handle all remaining cases.

28 When movement lower bounds imply work lower bounds

The resource inequalities of Section 27 imply lower bounds on computational work only under an explicit assumption about the cost of each round. We prove that the certificate rank carrying the boundary curvature also controls the movement lower bound; a larger ambient or restricted barrier parameter cannot replace that rank.

28.1 The same aggregate certificate in both estimates

Consider the full rows of $C = \prod_{a=1}^b \mathbb{B}_2^{s_a}$, with $s_a \geq 2$, and globally bi- C^1 selections as in (5.2). Suppose each dual row is a genuine certificate on the whole feasible affine slice. Choose positive weights λ_a and a simultaneous contact u . The objective and its certificate are

$$\ell(x) = \sum_a \lambda_a u_a^T x_a, \quad \ell_* = \sum_a \lambda_a, \quad S_i = \sum_a \lambda_a Y_i^a(u_a).$$

Write $n = \sum_a (s_a - 1)$, $p_i = \text{rank}_J X_i(u)$, $q_i = \text{rank}_J S_i$ and $Q = \sum_i q_i$. For simple factors with rank r_i and Peirce constant a_i , the mixed-channel bound gives

$$n \leq \sum_{i \text{ non-ray}} a_i p_i q_i, \quad p_i + q_i \leq r_i. \quad (28.1)$$

To see this directly: the weighted slack has mixed derivative $-\sum_a \lambda_a \langle h_a, k_a \rangle$, a nondegenerate form on an n -dimensional tangent space. At contact, positivity makes $X_i(u)$ and S_i complementary.

The differentiated mixed pairing factors through the cross Peirce space between their supports, of dimension $a_i p_i q_i$. The ranks add to at most r_i by complementarity; rank subadditivity proves (28.1). In real PSD notation this is the space $\text{Hom}(\text{Ran } S_i, \text{Ran } X_i)$; its transpose is not a second independent channel. Because the certificate identity holds on the whole slice, Theorem 23.1 applies with exactly this Q .

The bound (28.1) also bounds Q below in terms of the number of factors. Consider a common Jordan family with Peirce constant $a > 0$, orders at most an integer $R \geq 2$, and at most an integer $L \geq 1$ non-ray factors. With $\mathcal{C}_R$ as in (27.16), set

$$Q_{\min}(n; L, a, R) = \min\{z \in \mathbb{Z}_{\geq 0} : a\mathcal{C}_R(L, z) \geq n\}. \quad (28.2)$$

It exists only if $n \leq aL\lfloor R^2/4 \rfloor$; otherwise no such factorization exists. In the feasible case every aggregate certificate above has $Q \geq Q_{\min}$. Optional ray factors can add to Q but not to capacity, and do not affect this bound. For the non-ray rank Q_0 , Cauchy–Schwarz gives $n \leq a(RQ_0 - Q_0^2/L)$, and hence

$$Q \geq \frac{L}{2} \left(R - \sqrt{R^2 - \frac{4n}{aL}} \right). \quad (28.3)$$

Thus, when n is a constant fraction of its maximum $aL\lfloor R^2/4 \rfloor$, the factor $\sqrt{Q}$ in the movement bound is of order $\sqrt{LR}$, although there are only L factors. For a fixed non-ray rank $Q_0 \leq LR$, the untruncated maximum capacity is $a[(L-t)u(R-u) + t(u+1)(R-u-1)]$ with $Q_0 = uL + t$. For a rank *budget*, the truncation in $\mathcal{C}_R$ is necessary: capacity is maximized by stopping at $L\lfloor R/2 \rfloor$. All these bounds are integer consequences of the curvature bound; we do not claim that every balanced rank vector is realized by a lift.

These bounds can also be imposed at every stage of the sequential compression in Theorem 7.8. For a dictionary $\mathcal{K}$ whose non-ray faces retain the common Peirce constant a and order bound R , the same proof selects a simultaneous contact with

$$Q \geq \sum_{a'=1}^b \max\{\kappa_{\mathcal{K}}(s_{a'}), Q_{\min}(s_{a'} - 1; L, a, R)\}. \quad (28.4)$$

Indeed, there are at most L non-ray faces at every stage, each compressed full row has mixed rank $s_{a'} - 1$, and newly exposed ranks add by Lemma 5.1. The bound applies to each fixed classical Hermitian family and to a fixed spin family; non-ray faces of the Albert cone also keep Peirce constant eight. If the factors also have real dimension at most d , the bound $h_d(s_{a'})$ of Section 7 is an alternative one-row bound and can be combined with $Q_{\min}$ in the same way. For mixed families, use the adaptive face ranks and constants of that section instead of an artificial common constant.

28.2 A conditional total-work theorem

Specialize to real PSD blocks $2 \leq r_i \leq R$ and the standard restricted barrier $F = -\sum_i \log \det X_i$. A method starts at a fixed interior point x^0 and proceeds in rounds, each moving intrinsic distance at most $B_0 > 0$; $\mathcal{W}$ denotes its total work. We assume that each round costs at least $a_0 J_\omega$ work, where $a_0 > 0$ and $J_\omega = \sum_i r_i^\omega$. The assumption concerns what each round must process or output; it does not follow from access to sparse entries or quantum oracles.

Theorem 28.1 (Total work when every round has a minimum cost). *For the support objective and certificate above, let $\Delta > 0$ be (23.2) with all weights $\alpha_i = 1$. If $0 < \epsilon < \Delta$ and $0 \leq \omega \leq 5/2$, then every such method that reaches objective gap at most ϵ has total work*

$$\mathcal{W} \geq \frac{a_0}{B_0} \frac{25}{6} \sqrt{\frac{5}{3}} \frac{n^{3/2}}{R^{(5-2\omega)/2}} \log \frac{\Delta}{\epsilon}. \quad (28.5)$$

If each round instead costs at least $a_0 M$, where $M = \sum_i d_{r_i}$, then

$$\mathcal{W} \geq \frac{a_0}{B_0} \frac{25}{12} \sqrt{\frac{5}{3}} (1 + 1/R) \frac{n^{3/2}}{\sqrt{R}} \log \frac{\Delta}{\epsilon}. \quad (28.6)$$

A round of at most m feasible chords, where $m \geq 1$ is an integer and each chord has starting local norm at most $0 < \eta < 1$, satisfies the distance assumption with $B_0 = m \log(1/(1 - \eta))$.

Proof. Apply the algebraic proof of Theorem 27.3 to $p_i \leq r_i - q_i$ and (28.1), taking $\theta = 1/2$. It gives

$$J_\omega \sqrt{Q} \geq \frac{25}{6} \sqrt{\frac{5}{3}} \frac{n^{3/2}}{R^{(5-2\omega)/2}}.$$

Theorem 23.1 bounds the endpoint distance below by $\sqrt{Q} \log(\Delta/\epsilon)$. The triangle inequality therefore requires at least this distance divided by B_0 rounds. Multiplying by the assumed cost per round proves (28.5). The inequality $M \geq (1 + 1/R) J_2/2$ proves the coordinate version. The chord estimate is the classical self-concordant Hessian comparison already used in Corollary 23.2. $\square$

28.3 Matching upper bounds in specified output models

For a common source dimension s , suppose that $p = 3R/5$, $c = 2R/5$, $h = s/p$ and $L = bh/c$ are integers. The packed lift has $H = bh$ columns and full order- R blocks. For unit source directions v_a , maximize $\sum_a v_a^T x_a$. The central path of Proposition 25.2, whose auxiliary variables are at the fiber center, has

$$x_a = r(\tau) v_a, \quad r(\tau) = \frac{\tau}{\sqrt{h^2 + \tau^2} + h}, \quad D_j = q(\tau) I_c, \quad q(\tau) = \frac{2}{\sqrt{h^2 + \tau^2} + h}.$$

At its support contact, the c dual column vectors in each block are linearly independent because their allocation coordinates are different basis vectors. Hence $q_i = c$ and $Q = H$. Its resource product is exactly

$$J_\omega \sqrt{H} = \frac{25}{6} \sqrt{\frac{5}{3}} \frac{(bs)^{3/2}}{R^{(5-2\omega)/2}}. \quad (28.7)$$

The central arc has length $O(\sqrt{H} \log(b/\epsilon))$ for $0 < \epsilon \leq b/4$, and the distance from the analytic center to the points of objective gap at most ϵ is at least $\sqrt{H} [\log(b/(2\epsilon))]_+$. Thus checkpoints with bounded distance between consecutive ones attain the order of the movement lower bound, and the resource product exceeds that in (28.5) only by the factor $(s/(s-1))^{3/2}$. The reference scale Δ of that theorem depends on the instance; we claim no scale uniform over all lifts.

For an explicit upper bound on dense output cost, preprocess the fixed objective columns V_j and their Gram matrices $G_j = V_j^T V_j$ once. At each checkpoint the entire lifted point is

$$W_j = r V_j, \quad S_j = r^2 G_j + q I_c.$$

Writing all ambient cone entries therefore takes $O(M)$ exact real arithmetic operations and writes per checkpoint, including the fixed identity blocks if the output format requires them. This proves a matching upper bound on *postprocessing and writing the output* along this path, with preprocessing and scalar evaluation counted separately; no Gram matrix is recomputed. An explicit schedule of bounded-distance checkpoints also avoids a hidden optimization subproblem: from $\tau = 0$ to h , use increments $O(h/\sqrt{H})$; subsequently use constant increments $O(1/\sqrt{H})$ in $\log \tau$ until $\tau = H/\epsilon$. The squared local-norm speed in $\log \tau$ is $H(1 - h/\sqrt{h^2 + \tau^2}) \leq H$, and before h the ordinary speed is at

most $\sqrt{H/2}/h$. Also $b(1 - r(\tau)) \leq H/\tau$. These estimates verify the step bound and the endpoint accuracy. Square roots and exponentials, exact or to a chosen precision, are separate scalar costs; no dense linear solve is hidden.

Counting one output record per factor (cost J_0), or r_i records per factor (cost J_1), is a different output model. If those records consist of the public scalars and references to preprocessed fixed directions and Gram matrices, they can be refreshed in $O(J_0)$ or $O(J_1)$ word operations. For this representation, (28.7) shows that the record work matches the lower bound; it does not follow that all bs projected coordinates, or all dense factors, can be produced at that cost. The full-coordinate output model, by contrast, does require writing every output entry anew in every round. If only free coordinates are counted, the relevant optimization is (27.13), whose optimizer is grouped Schur rather than 3 : 2 packing.

Finally, none of these per-round costs follows from one-shot query complexity. Fixed input data can be preprocessed and reused, and public constant cone blocks can be added without changing the problem. For example, let $v_{aj} = \sigma_{aj}/\sqrt{s}$ with hidden signs $\sigma_{aj} \in \{-1, 1\}$. One phase-oracle query prepares

$$\frac{1}{\sqrt{bs}} \sum_{a,j} \sigma_{aj} |a, j\rangle.$$

Every nonzero projected center and projected predictor on the displayed path is a public scalar multiple of this same vector. Preparing it anew costs one oracle query per checkpoint, independently of the number of PSD factors, not $\Omega(L)$ queries. This says nothing about auxiliary Gram-state preparation, final classical readout, or other solver subroutines. It shows that a one-shot lower bound cannot be multiplied by a movement count without a further assumption forcing new work in every round or a specific output.

29 Newton systems: elimination, packing, and access

The cone resources of a formulation do not determine the cost of its Newton equation; the projected variables, the auxiliary iterate, the chosen elimination, and the requested output also matter. We show this for the grouped and column-packed formulations of products of balls. Both have the same projected Newton equation at the fiber center (the minimizer of the barrier over the auxiliary variables for fixed x). Away from it, exact auxiliary allocations make the comparison with the centered Hessian depend linearly, not quadratically, on the relative spectral bounds μ, L of the residuals; this is sharp. We also bound the cost of exact recentering, with lower bounds on specific promise families.

29.1 The common marginal barrier and its Newton equation

Let $C = \prod_{a=1}^b B_2^{s_a}$, let $n = \sum_a s_a$, and partition each source vector x_a into $h_a \geq 1$ nonempty coordinate groups. Pad a group by public zero coordinates when necessary, and arrange the resulting columns in real matrices $W_\ell \in \mathbb{R}^{p_\ell \times c_\ell}$. Every projected coordinate occurs once. Let E_a^ℓ be the diagonal projector onto the columns of block ℓ assigned to source a . These projectors are orthogonal and sum to the identity in each block; source a has h_a columns in total. The packed formulation is

$$Z_\ell = \begin{pmatrix} S_\ell & W_\ell^T \\ W_\ell & I_{p_\ell} \end{pmatrix} \succ 0, \quad \sum_\ell \text{tr}(E_a^\ell S_\ell) = 1, \quad F = - \sum_\ell \log \det Z_\ell. \quad (29.1)$$

All additional constraints and the objective in this subsection depend only on x . Write $D_\ell = S_\ell - W_\ell^T W_\ell$ and $\rho_a = 1 - \|x_a\|^2 > 0$.

Proposition 29.1 (Packing-independent marginal and centered system). *For fixed $x \in \text{int } C$, F has a unique minimizer, the fiber center, on the auxiliary fiber of x (the auxiliary variables feasible in (29.1) for this x). Its residuals D_ℓ^0 are diagonal, with entries*

$$(D_\ell^0)_{\gamma\gamma} = d_a := \rho_a/h_a \quad (\gamma \text{ assigned to } a). \quad (29.2)$$

The marginal barrier Φ (the minimum value of F on the fiber) and its Hessian are

$$\Phi(x) = - \sum_a h_a \log \rho_a + \sum_a h_a \log h_a, \quad (29.3)$$

$$H_0(x) = \bigoplus_a \left(\frac{2h_a}{\rho_a} I_{s_a} + \frac{4h_a}{\rho_a^2} x_a x_a^T \right). \quad (29.4)$$

Its self-concordant barrier parameter is exactly $\sum_a h_a$. The grouped paraboloid formulation with the same groups has the same marginal. At their fiber centers, both formulations give the same projected Newton matrix and right-hand side for every common affine constraint $Ax = d$, linear objective, multiplier, and projected iterate.

Proof. The Schur determinant identity gives $\det Z_\ell = \det D_\ell$. For a fixed diagonal, Hadamard's inequality shows that the determinant is uniquely maximized by a positive diagonal matrix. The allocation equations become $\sum_{\gamma \in \Gamma_a} d_\gamma = \rho_a$; AM–GM uniquely maximizes their product at $d_\gamma = \rho_a/h_a$. The same AM–GM calculation applies to the grouped slacks $t_\gamma - \|x_\gamma\|^2$. Differentiation gives (29.4). Each summand is h_a times the standard unit-ball barrier, and its squared dual gradient norm is $2h_a\|x_a\|^2/(1 + \|x_a\|^2)$. This tends to h_a at the boundary. Additivity and simultaneous approach of all sources to the boundary prove exactness of the parameter.

The equality of Newton systems also holds for the uncondensed residual equations. For a direction $(A_\ell, U_\ell) = (\delta S_\ell, \delta W_\ell)$ put $B_\ell = A_\ell - W_\ell^T U_\ell - U_\ell^T W_\ell$. Direct differentiation along this affine direction gives

$$D^2 F[(A, U)]^2 = \sum_\ell \left[\text{tr}(D_\ell^{-1} B_\ell D_\ell^{-1} B_\ell) + 2 \text{tr}(D_\ell^{-1} U_\ell^T U_\ell) \right]. \quad (29.5)$$

At the diagonal center D^0 , the off-diagonal entries of B enter as independent positive quadratic terms with zero gradient, objective, and constraint coefficients. The diagonal modes $r_\gamma = B_{\gamma\gamma}$ obey

$$\sum_\gamma \frac{r_\gamma^2}{d_\gamma^2} + 2 \sum_\gamma \frac{\|u_\gamma\|^2}{d_\gamma}, \quad \sum_{\gamma \in \Gamma_a} (r_\gamma + 2x_\gamma^T u_\gamma) = 0. \quad (29.6)$$

These are precisely the grouped residual equations, including their linear terms $-\sum_\gamma r_\gamma/d_\gamma$ and the common objective. Additional rows $Au = 0$ coincide. Eliminating r and the allocation multipliers therefore gives the same matrix and right-hand side. $\square$

With no additional affine coupling, the graph of (29.6), with one allocation multiplier per source, is a disjoint union of stars. Solving these stars, or applying the diagonal-plus-rank-one inverse in (29.4), costs $O(n + \sum_a h_a) = O(n)$ arithmetic. The graph of a fully materialized generic Hessian is instead a union of s_a -cliques; the two graphs describe different representations of the same operator. Let G be the graph on the projected variables, the source hubs (the allocation multipliers), and one constraint hub per row of A . Attaching the leaves r_γ gives width $\max\{1, \text{tw}(G)\}$; this graph, not the PSD packing layout, determines the cost of the corresponding sparse solve.

Corollary 29.2 (Matched projected-oracle equivalence). *Supply both formulations with the same complete canonical oracles for their common projected data and reduced Newton equation, at the same precision, where “same” includes their action on unused input and ancilla subspaces. Require the same projected direction, or its normalized quantum state when the direction is nonzero. Then their minimum classical and quantum query complexities agree exactly. This equivalence also holds along any adaptive procedure that uses only these projected oracles and recenters the auxiliary fibers before each Newton call.*

Proof. Proposition 29.1 identifies both the operator and the right-hand side, in the same coordinates. Forward each query to the identical oracle; retain all other operations, measurements, and stopping decisions. The simulation works in both directions. Induction on calls gives the adaptive statement. Building both oracle sets from the same raw data by the same charged procedure gives identical preparation costs. $\square$

Equality of the whole unitary is needed: if only an encoded matrix block were equal, the rest of the unitary could reveal extra information. The corollary does not construct these oracles or make them free. Under its hypothesis, normalization, spectrum, right-hand-side norm, solution overlap, and output accuracy agree. In residual coordinates, the centered PSD KKT system is the direct sum of the grouped system and a positive diagonal block, with zero right-hand side, for the off-diagonal residual entries; with public zero padding, the solution states also agree. The original δS output is different: reconstructing it requires $W^T U + U^T W$, which the projected equivalence does not cover. If all $h_a = h$, changing h multiplies the unconstrained marginal Hessian by h ; this changes its metric scale, but not its condition number or, for a fixed right-hand side, its normalized inverse state. Hence a smaller number of cone factors does not by itself show an advantage in preparing Newton states.

29.2 A sharp comparison away from the fiber center

Theorem 29.3 (Exact quotient and allocation-aware comparison). *At a strictly feasible point of (29.1), define*

$$z_a(U) = \sum_{\ell} \text{tr}(E_a^{\ell} W_{\ell}^T U_{\ell}), \quad M_{ab}(D) = \sum_{\ell} \text{tr}(E_a^{\ell} D_{\ell} E_b^{\ell} D_{\ell}). \quad (29.7)$$

Minimizing the Hessian form of F over auxiliary directions that satisfy the linearized allocation constraints gives

$$Q_D(U) = 2 \sum_{\ell} \text{tr}(D_{\ell}^{-1} U_{\ell}^T U_{\ell}) + 4z(U)^T M(D)^{-1} z(U). \quad (29.8)$$

Let H_D denote this form on the projected coordinates. If $\mu D_{\ell}^0 \preceq D_{\ell} \preceq L D_{\ell}^0$ for all ℓ , where $0 < \mu \leq 1 \leq L$, then

$$L^{-1} H_0 \preceq H_D \preceq \mu^{-1} H_0, \quad \kappa(H_0^{-1/2} H_D H_0^{-1/2}) \leq L/\mu. \quad (29.9)$$

The first-power dependence on L and μ and the condition bound are sharp. The comparison also holds after restriction to any common projected tangent space. For any full-row-rank matrix A , the dual normal matrices satisfy

$$\mu A H_0^{-1} A^T \preceq A H_D^{-1} A^T \preceq L A H_0^{-1} A^T. \quad (29.10)$$

Proof. Equip symmetric residual directions with $\langle B, C \rangle_D = \sum_\ell \text{tr}(D_\ell^{-1} B_\ell D_\ell^{-1} C_\ell)$. The allocation functional $B \mapsto \sum_\ell \text{tr}(E_a^\ell B_\ell)$ has Riesz representative $(D_\ell E_a^\ell D_\ell)_\ell$. The Gram matrix of these representatives is $M(D)$. It is positive definite: for $E_\ell(\theta) = \sum_a \theta_a E_a^\ell$,

$$\theta^T M(D) \theta = \sum_\ell \|D_\ell^{1/2} E_\ell(\theta) D_\ell^{1/2}\|_F^2,$$

and its vanishing forces every $\theta_a = 0$, since each source has a column. The tangent allocation constraint is $\sum_\ell \text{tr}(E_a^\ell B_\ell) = -2z_a(U)$. Its minimum squared D -norm is $4z^T M^{-1} z$, proving (29.8) from (29.5).

The first-power comparison uses the exact allocations; the order of $D_\ell \otimes D_\ell$ alone would give the factors μ^2 and L^2 . Although $E_\ell(\theta)$ may be indefinite, $E_\ell(\theta) D_\ell E_\ell(\theta) \succeq 0$. Positive trace pairing therefore gives

$$\mu \sum_\ell \text{tr}(D_\ell^0 E_\ell D_\ell E_\ell) \leq \theta^T M(D) \theta \leq L \sum_\ell \text{tr}(D_\ell^0 E_\ell D_\ell E_\ell).$$

The diagonal matrices D_ℓ^0 and E_ℓ commute. Exact source allocations, $\sum_\ell \text{tr}(E_a^\ell D_\ell) = \rho_a$, yield

$$\sum_\ell \text{tr}(D_\ell^0 E_\ell D_\ell E_\ell) = \sum_a \theta_a^2 d_a \rho_a = \theta^T M_0 \theta, \quad M_0 = \text{diag}(\rho_a^2 / h_a). \quad (29.11)$$

Hence $\mu M_0 \preceq M(D) \preceq L M_0$. Both terms in (29.8) now obey the first-power bounds L^{-1} and μ^{-1} . At the same W , their vector $z(U)$ is unchanged, so these inequalities prove (29.9). Restriction preserves order; inversion followed by congruence proves (29.10). Sharpness is established in Example 29.4 below. $\square$

This Hessian elimination differs from first minimizing the barrier value over the fiber, which always replaces D by D^0 , and it must keep the gradient of the eliminated variables. For minimizing $F + \eta \langle c, x \rangle$ at a feasible packed point, the reduced Newton gradient is the linear functional

$$g_D[U] = 2\rho^T M(D)^{-1} z(U) + \eta \langle c, U \rangle. \quad (29.12)$$

Indeed, the Newton quadratic before auxiliary elimination is

$$\frac{1}{2} \|B\|_D^2 - \sum_\ell \text{tr}(D_\ell^{-1} B_\ell) + \sum_\ell \text{tr}(D_\ell^{-1} U_\ell^T U_\ell) + \eta \langle c, U \rangle.$$

Completing the square replaces B by $B - D$ and its allocation by $-2z(U) - \rho$. Its minimum is $\frac{1}{2}(2z + \rho)^T M^{-1}(2z + \rho)$, plus the displayed U terms and a constant. This proves (29.12). At D^0 it is $\sum_a 2h_a \langle x_a, U_a \rangle / \rho_a + \eta \langle c, U \rangle$, the marginal gradient. Away from the center, a spectral comparison of the matrices alone does not determine Newton directions or their normalized states. Additional homogeneous tangent constraints are the same in both systems; violated affine constraints add right-hand-side terms.

Example 29.4 (Sharpness and fixed-projection obstruction). Pack two sources as two columns of one block, take $W = 0$, and put

$$D = \begin{pmatrix} 1 & \xi \\ \xi & 1 \end{pmatrix}, \quad D^0 = I_2, \quad |\xi| < 1.$$

The source equations hold exactly. Formula (29.8) reduces to

$$Q_D(U) = \frac{2}{1 - \xi^2} (\|u_1\|^2 + \|u_2\|^2 - 2\xi u_1^T u_2).$$

The relative eigenvalues are $1/(1 + \xi)$ and $1/(1 - \xi)$. Thus $\mu = 1 - |\xi|$, $L = 1 + |\xi|$, and the relative condition number is exactly L/μ . For the objective with columns $(v, 0)$, where $\|v\| = 1$, the projected Newton directions are $-\eta(v, \xi v)/2$ and $-\eta(v, 0)/2$ at D and D^0 . Their normalized states have trace distance $|\xi|/\sqrt{1 + \xi^2}$. Replacing ξ by $-\xi$ leaves all projected input records unchanged, but the two resulting states have trace distance $2|\xi|/(1 + \xi^2) \rightarrow 1$. Consequently, projected data alone cannot uniformly recover unrestricted off-center Newton states. The obstruction concerns a single iterate; it gives no lower bound on the number of outer iterations.

Corollary 29.5 (Approximate allocations). *Keep W and its exact center D^0 fixed, but allow positive residuals $\hat{D}$ with allocations*

$$(1 - \zeta)\rho_a \leq \sum_{\ell} \text{tr}(E_a^{\ell} \hat{D}_{\ell}) \leq (1 + \zeta)\rho_a, \quad 0 \leq \zeta < 1.$$

If $\mu D^0 \preceq \hat{D} \preceq L D^0$, with $0 < \mu \leq 1 \leq L$, the Hessian quotient on homogeneous tangent directions obeys

$$\frac{H_0}{L(1 + \zeta)} \preceq H_{\hat{D}} \preceq \frac{H_0}{\mu(1 - \zeta)}. \quad (29.13)$$

In particular, a relative residual error $\|(D_{\ell}^0)^{-1/2}(\hat{D}_{\ell} - D_{\ell}^0)(D_{\ell}^0)^{-1/2}\|_2 \leq \delta < 1$ gives a relative condition number of at most $((1 + \delta)/(1 - \delta))^2$ without exact allocations, and at most $(1 + \delta)/(1 - \delta)$ when the allocations are exact.

Proof. The middle trace in (29.11) lies between $(1 - \zeta)\theta^T M_0 \theta$ and $(1 + \zeta)\theta^T M_0 \theta$. Repeat the proof and use the weaker of the resulting bounds for the two nonnegative terms of Q . Spectral error δ implies allocation error at most δ by positive trace pairing with each projector. Set $\mu = 1 - \delta$, $L = 1 + \delta$, and $\zeta = \delta$ or 0 , respectively. $\square$

The first-power improvement depends on the feasible allocation structure; otherwise the proof of Theorem 29.3 uses only elementary trace inequalities for positive matrices; we do not claim a new general matrix-order principle. Neither the theorem nor the corollary provides access to D^{-1} , $M(D)^{-1}$, or $H_0^{-1/2}$; constructing them and any encoding normalization are separate costs.

29.3 What exact recentering requires

The fiber center has a cheap implicit representation: store (W, d) , which represents $S_{\ell} = W_{\ell}^T W_{\ell} + \text{Diag}(d_{\ell})$. We call the slack ρ_a the scale of source a . Computing all squared source norms and all diagonal entries $d_{\gamma} = \rho_a/h_a$ uses $O(n)$ arithmetic and storage, independently of packing, because $\sum_a h_a \leq n$. The source equations hold exactly by $\|x_a\|^2 + h_a d_a = 1$. For B -bit dyadic coordinates with a common denominator, the squared norms and the entries d_{γ} are exact rationals of $O(B + \log n)$ bits. Fast integer arithmetic gives the conservative bound $\tilde{O}(nM(B + \log n))$, where $M(t)$ is the cost of multiplying t -bit integers. This is the exact center of the stored rational interior iterate, obtained without subtracting rounded Gram matrices. Coordinates without a common denominator, evaluation of inverses, and the requested numerical accuracy add further bit costs.

A reversible value oracle for ρ_a similarly supplies d_{γ} with one scale query and reversible division by the public h_a . Together with blockwise access to the direction states, it permits the common rank-one compiler for (29.4); radii and certified normalization bounds must be supplied or computed. The normalized state of x_a alone does not provide this scale: the same complete state-preparation unitary prepares the normalized states of both $r_0 v$ and $r_1 v$, while their slacks $1 - r_j^2$ can differ by arbitrarily large factors, so no algorithm with only this unitary can return both scales to a fixed relative accuracy.

Proposition 29.6 (Cost of missing radial metadata). *In coordinate-bit access, where each query returns one bit determining one coordinate of a source of dimension $s \geq 2$, computing a classical center scale to relative error $\delta < 1/3$ can require $\Omega(\sqrt{s})$ quantum or $\Omega(s)$ randomized queries. For b independent sources, returning all scales with joint success at least $2/3$ requires $\Omega(b\sqrt{s})$ and $\Omega(bs)$, respectively. These orders are attained on the promise used in the proof. For $k \mid s$, $1 \leq k \leq s/2$, the one-source promise can instead have smaller slack $\rho = k/s$, with tight costs $\Theta(\rho^{-1/2})$ quantum and $\Theta(\rho^{-1})$ randomized.*

Proof. Take $x_i = z_i/\sqrt{s}$ with $z_1 = 0$ public and all other bits one, except for at most one additional hidden zero. The slack is $1/s$ or $2/s$. Their relative- δ intervals are disjoint precisely when $\delta < 1/3$, so recovering the scale solves promised unstructured search. The standard search bounds apply [14, 18].

The quantum direct-sum bound has a short adversary proof. Index one source's promise by the no-mark input and its $s - 1$ single-mark inputs. The star adjacency matrix Γ has norm $\sqrt{s - 1}$, and filtering by any queried bit leaves norm one. For b sources use $\sum_{a=1}^b I^{\otimes(a-1)} \otimes \Gamma \otimes I^{\otimes(b-a)}$. Its norm is $b\sqrt{s - 1}$; filtering a query in one source retains only that summand. Its entries vanish between inputs having the same output vector. The adversary lower bound for function evaluation [59] gives the claim. For randomized algorithms, give each source an independent fair no-mark/single-uniform-mark promise. Joint success implies constant average success for each output, and the usual search indistinguishability argument requires a constant fraction of the $s - 1$ expected queries per such source; summing proves $\Omega(bs)$. On this promise, exact amplitude amplification for the known single-mark case followed by verification distinguishes zero from one mark with $O(\sqrt{s})$ queries and no error. Applied sourcewise it attains joint success without an amplification logarithm. Scanning attains the randomized order.

Finally, partition the coordinates into blocks of size k . One public block is zero; all candidate blocks are one except possibly one hidden zero block. The two slacks are k/s and $2k/s$. A representative coordinate is an oracle for its block mark, reducing to search on $s/k - 1$ labels. This number is within a constant factor of s/k in the stated range. $\square$

The joint-output bound also applies to a standalone compiled value interface whose values can be read on all labels with the same joint correctness guarantee and no further queries to the coordinate input (raw queries). It does not apply to a single online coherent evaluation, which can run the promised search controlled by the source label with $O(\sqrt{s})$ raw queries, charged again on every invocation. Under the following output requirement, we prove a lower bound linear in the number T of epochs, each with a fresh hidden update.

A *compile-and-commit maintainer* returns, before the next epoch, either a classical record of all source scales or a complete reusable unitary for their encoded numerical values. The latter must allow every basis-label value to be copied out, using the unitary and its inverse, with no further raw queries and with the entire backing workspace restored exactly, including correlations with other retained registers. Each committed object has fixed value strings v_a ; compute-copy-uncompute on label a acts as

$$|a, z\rangle|\psi\rangle \longmapsto |a, z \oplus v_a\rangle|\psi\rangle$$

on its backing state and any retained reference. The strings encode numbers $\hat{d}_a$. Correctness means that, with joint probability at least $2/3$, all committed values at all epochs obey $|\hat{d}_a - d_a| \leq \delta d_a$. A state consumed or disturbed by reading does not meet this requirement. Rational encodings are exact, even for $\delta = 0$; fixed-point encodings need enough bits for the requested positive error.

Theorem 29.7 (Fresh-update scale maintenance). *Let $b, T \geq 1$, $s \geq 2$, and $0 \leq \delta < 1/3$. For the packing (29.1) with b sources of dimension s and public column counts h_a , there is a family of T*

fresh coordinate-bit update batches, each zero or one-sparse per source, on which compile-and-commit maintenance has tight worst-case raw-query costs

$$Q = \Theta(Tb\sqrt{s}), \quad R = \Theta(Tbs) \quad (29.14)$$

for quantum and randomized algorithms, respectively. Each epoch starts from the same public interior base and is reset after commitment. For $k \mid s$, $1 \leq k \leq s/2$, a replicated-block version has k -sparse updates and smaller slack $\rho_* = k/s$, with costs

$$Q = \Theta(Tb\rho_*^{-1/2}), \quad R = \Theta(Tb\rho_*^{-1}). \quad (29.15)$$

The lower bounds hold even if the algorithm has all epoch oracles from the start, makes queries in superposition over epoch, source, and coordinate, and keeps arbitrary workspace across epochs. The upper bounds have zero error on these promises.

Proof. Use $\bar{x} = (0, 1, \dots, 1)/\sqrt{s}$. Independently for each pair (t, a) , the update is either $u_{t,a} = 0$ or $u_{t,a} = -e_j/\sqrt{s}$ at one hidden $j \in \{2, \dots, s\}$. The raw oracle returns the corresponding mark bit at a queried coordinate. After the update,

$$\rho_{t,a} = 1 - \|\bar{x} + u_{t,a}\|^2 \in \{1/s, 2/s\}, \quad d_{t,a} = \rho_{t,a}/h_a.$$

Both cases are strictly interior, and Proposition 29.1 gives these exact diagonal entries of the PSD fiber center. Their relative-error intervals are disjoint because $1 + \delta < 2(1 - \delta)$. Undo each batch after commitment; inverse-update access uses the same bits and reveals no new information. A public reset only strengthens the model.

Read every committed value before resuming the maintainer, restoring its workspace in the quantum case. Thresholding recovers $m = Tb$ independent presence bits with joint success $2/3$ and no extra raw queries. Apply the star adversary from Proposition 29.6, with tensor factors now indexed by (t, a) . Its Kronecker sum has norm $m\sqrt{s-1}$ and every coordinate filter has norm one. It is entrywise nonnegative and vanishes between inputs with the same output vector, so the finite-output spectral adversary bound applies [2, Theorems 3–4]. Because the adversary matrix covers all epochs at once, the bound already allows access to all epoch oracles from the start and queries scheduled across epochs.

For the randomized bound, independently choose no mark with probability $1/2$ and a uniform single mark otherwise. Fix the randomness and all other pairs, and let L count distinct candidates queried when this pair has no mark. The no-mark and random-mark transcripts coincide unless the mark lies in these L locations. Averaging over the fixed data, the total variation distance is at most $\mathbb{E}L/(s-1)$. Marginal success at least $2/3$ forces $\mathbb{E}L \geq (s-1)/3$. Since the no-mark case has probability $1/2$, this pair contributes at least $(s-1)/6$ expected queries under the product distribution. Summing over the m pairs proves the worst-case randomized bound.

Exact amplitude amplification designed for one marked coordinate, followed by verification, decides zero versus one mark in $O(\sqrt{s})$ queries [18, Theorems 4 and 16]. Verification always rejects in the zero-mark case. Search each pair and store its exact rational scale; no joint-success amplification logarithm is needed. Scanning gives the classical upper bound.

For the margin version, partition coordinates into s/k blocks of size k , with one public zero block in the base. An update zeros either no candidate block or one whole candidate block. The slacks are k/s and $2k/s$. A coordinate query reveals only its block's mark bit, and one representative coordinate per block suffices. Replace $s-1$ throughout by $s/k-1 = \Theta(s/k)$, which is at least one in the stated range. $\square$

Corollary 29.8 (Universal invocation versus compilation tradeoff). *On the same promises, suppose a service must support any adaptive client making at most r_t value invocations at epoch t , including coherent label inputs in the quantum case. Invocations implement the reusable value interface above, but may query the raw input; all preprocessing and invocation queries are charged. Require joint correctness with probability at least $2/3$ over the entire interaction. For prescribed nonnegative integer budgets r_t , the tight worst-case costs are*

$$Q = \Theta \left(\sqrt{s} \sum_{t=1}^T \min\{r_t, b\} \right), \quad R = \Theta \left(s \sum_{t=1}^T \min\{r_t, b\} \right). \quad (29.16)$$

In the margin version replace $\sqrt{s}, s$ by $\rho_^{-1/2}, \rho_*^{-1}$, respectively.*

Proof. Choose the permitted client that reads $\min\{r_t, b\}$ distinct basis labels at each epoch. Fix all other update pairs to zero. Its transcript recovers $\sum_t \min\{r_t, b\}$ independent search bits, so the same adversary and product-distribution proofs give the lower bounds. For the upper bound, if $r_t < b$, run the exact promised search controlled by the source label on each invocation. Coherently verify, copy the deterministic presence bit, and uncompute the search workspace; this is a clean value unitary at $O(\sqrt{s})$ queries per call. If $r_t \geq b$, search all sources once and keep a classical table for query-free reversible lookup. Classical scans give the analogous strategies. Apply the same choice to candidate-block representatives for the margin version. $\square$

Serving every permitted client is stronger than serving one downstream algorithm's structured calls. A single online coherent evaluation costs no factor b ; a service that must support at least b arbitrary calls per epoch costs as much as compilation. Neither result concerns arbitrary one-shot state output or finite-precision gate complexity.

For an explicit sparse list, the exact identity

$$\rho_a(x + \alpha u) = \rho_a(x) - 2\alpha x_a^T u_a - \alpha^2 \|u_a\|^2 \quad (29.17)$$

instead maintains scales in $O(\sum_a |\text{supp } u_a|)$ arithmetic, given access to the current coordinates and a list of affected sources. In particular one-sparse lists cost $O(b)$ per epoch. The lower bound measures the cost of discovering a hidden support, not of handling sparse updates. Likewise, a snapshot lower bound does not imply a factor T for one fixed oracle: if every round requests the same promised OR bit, an algorithm can compute and cache it once with $\Theta(\sqrt{s})$ quantum or $\Theta(s)$ randomized queries, and all T requests then use no further raw queries.

Proposition 29.9 (Fixed central-path scale caching). *For the uncoupled product of balls, minimize $\Phi(x) + \eta \sum_a c_a^T x_a$ with fixed c_a and $\eta \geq 0$. Put $q_a = \|c_a\|$ and $\tau_a = \eta q_a / h_a$. The unique center is*

$$\rho_a(\eta) = \frac{2}{1 + \sqrt{1 + \tau_a^2}}, \quad x_a(\eta) = -\frac{\eta \rho_a(\eta)}{2h_a} c_a, \quad d_a(\eta) = \frac{\rho_a(\eta)}{h_a}. \quad (29.18)$$

For $s_a \geq 2$, its source Hessian has radial-to-tangential eigenvalue ratio $2/\rho_a(\eta) - 1 = \sqrt{1 + \tau_a^2}$.

Proof. Stationarity is $2h_a x_a / \rho_a + \eta c_a = 0$. Substitution in $\rho_a = 1 - \|x_a\|^2$ gives $\rho_a + \tau_a^2 \rho_a^2 / 4 = 1$; its positive solution is (29.18), also when $q_a = 0$ or $\eta = 0$. It is interior, and strict convexity of Φ gives uniqueness. Equation (29.4) gives eigenvalues $2h_a / \rho_a$ and $2h_a / \rho_a + 4h_a(1 - \rho_a) / \rho_a^2$, proving the ratio (equal to one at $x_a = 0$). $\square$

Thus reversible access to the fixed norms q_a gives a requested scale with $O(1)$ norm queries and charged scalar arithmetic; uncomputing norm scratch may require an inverse query. Explicit objective data allows all norms to be acquired once in $O(n)$ real arithmetic, including square roots, and reused for every η . Requested precision and reversible arithmetic still have their own costs. Additional affine constraints or inexact neighborhood iterates need not satisfy this stationarity equation. We have not embedded the fresh batches of Theorem 29.7 in the Newton trajectory of one fixed conic program. Multiplying a query lower bound by an iteration count would require such an embedding and an output requirement under which the iterates reveal the independent hidden bits of every epoch. The closed-form uncoupled path only shows reuse of fixed data and gives no such embedding.

Explicit dense auxiliary matrices cost $\Omega(\sum_\ell c_\ell^2)$ scalar writes, and ordinary Gram formation costs $O(\sum_\ell p_\ell c_\ell^2)$. An implicit ambient encoding can avoid these writes: put $K_\ell = (W_\ell \ I_{p_\ell})$ and write $Z_\ell = K_\ell^T K_\ell + \text{Diag}(d_\ell, 0)$. Products of a supplied α_K -normalized encoding of K_ℓ and its adjoint, followed by a positive linear combination with an α_d -normalized diagonal encoding, give normalization $\alpha_K^2 + \alpha_d$ with a constant number of supplied encoding calls. Constructing these encodings, their norms, and an inverse is still charged, and nothing follows about preparing the flattened Gram matrix as a state.

29.4 Sparse augmented Lorentz systems

Low-rank augmentation of cone Hessians is classical, including in ECOS and nonsymmetric-cone implementations [23, 27]. We use it to give an explicit classical baseline for the formulations above. Below, the graph of a system is the fixed structural graph of its augmented scalar matrix; entries that vanish at special numerical values can only remove edges. The treewidth of a graph is the minimum, over its tree decompositions, of the largest bag size minus one.

For a Lorentz block $x = (t, z)$, let $q = t^2 - \|z\|^2 > 0$ and $J = \text{diag}(1, -I)$. Its standard Hessian is

$$H_x = -\frac{2J}{q} + \frac{4(Jx)(Jx)^T}{q^2} = D_x + w_x w_x^T. \quad (29.19)$$

Thus the equality-form Newton matrix and its one-hub augmentation, with one extra hub variable per cone carrying $w_x w_x^T$, are

$$K_0 = \begin{pmatrix} H & A^T \\ A & 0 \end{pmatrix}, \quad \widetilde{K}_0 = \begin{pmatrix} D & W & A^T \\ W^T & -I & 0 \\ A & 0 & 0 \end{pmatrix}. \quad (29.20)$$

Here W has one embedded w_x column per cone. Eliminating the hub variables (the $-I$ block) gives K_0 , and the augmented right-hand side is $(g_x, 0, g_y)$. We do not assume that D has a positive diagonal.

Proposition 29.10 (Exact field-operation comparison). *Suppose $A \in \mathbb{R}^{p \times M}$ has full row rank, there are k Lorentz blocks, and a compact width- τ tree decomposition of the structural graph of $\widetilde{K}_0$ is supplied. Then a Newton equation can be solved in $O((M + p + k)(\tau + 1)^2)$ exact field operations, after $O(M + \text{nnz } A)$ coefficient formation. Compactness here means $O(M + p + k)$ bags; processing a larger supplied decomposition is an additional input cost.*

Proof. Positive definiteness of H and full row rank of A imply nonsingularity of both matrices in (29.20). Replace each vertex of each bag by its row and column copies. The resulting decomposition of the matrix's row-column bipartite graph has width at most $2\tau + 1$: every off-diagonal entry

has both endpoints in some original bag, each diagonal entry is covered, and connectedness of each copy's bags is preserved. Apply the exact low-treewidth linear-system algorithm of Fürer et al. [39, Corollary 3]. Its bound is $O((m+n)(w+1)^2)$ for a supplied compact width- w bipartite decomposition. It allows nonzero pivots chosen during elimination and algebraic cancellations, so the zero or negative diagonal entries here cause no difficulty. Formula (29.19) uses only field operations and gives the formation cost. $\square$

A positive-diagonal version is useful when a reusable symmetric factorization is wanted. If $r = \|z\| > 0$ and $e = z/r$, put

$$D = \frac{2}{q}I, \quad a = \frac{\sqrt{2r(t+r)}}{q} \begin{pmatrix} 1 \\ -e \end{pmatrix}, \quad v = \frac{\sqrt{2r(t-r)}}{q} \begin{pmatrix} 1 \\ e \end{pmatrix}. \quad H_x = D + aa^T - vv^T. \quad (29.21)$$

At $r = 0$ take $a = v = 0$. The identity follows by restricting $H_x - D$ to the span of $(1, 0)$ and $(0, e)$, where its matrix is $4q^{-2} \begin{pmatrix} r^2 & -tr \\ -tr & r^2 \end{pmatrix}$. Moreover $v^T D^{-1} v = 2r/(t+r) < 1$, so $D - vv^T \succ 0$. With embedded update and downdate columns denoted by U, V , respectively, the regularized augmentation is

$$\begin{pmatrix} D & V & U & A^T \\ V^T & I & 0 & 0 \\ U^T & 0 & -I & 0 \\ A & 0 & 0 & -\varpi I \end{pmatrix}, \quad \varpi > 0. \quad (29.22)$$

Grouping (x, V) as positive variables and (U, y) as negative variables shows symmetric quasidefiniteness: the first diagonal block has Schur complement $D - VV^T \succ 0$, and the second is negative definite. Every symmetric permutation therefore admits an exact pivot-free LDL^T factorization [86]. A supplied width- τ elimination order gives $O((M+p+k)(\tau+1)^2)$ real arithmetic, $O((M+p+k)(\tau+1))$ storage, and width-linear subsequent solves. Unlike Proposition 29.10, this uses the square roots in (29.21), so it is not a field-operation result.

This is not an unconditional numerical-stability statement. Writing $K_\varpi = K_0 - \varpi \text{diag}(0, I)$, the resolvent identity gives, for $d_0 = K_0^{-1}g \neq 0$ and $\varpi \|K_0^{-1}\|_2 < 1$,

$$\frac{\|K_\varpi^{-1}g - d_0\|}{\|d_0\|} \leq \frac{\varpi \|K_0^{-1}\|_2}{1 - \varpi \|K_0^{-1}\|_2}. \quad (29.23)$$

Working precision, small pivots, growth, and residual certification still need bounds. Quasidefiniteness rules out exact zero-pivot breakdown, not arbitrarily small pivots. Also, two hubs cannot generically be reduced to one while retaining a positive diagonal base: if the Lorentz block has dimension $m \geq 3$ and every coordinate of Jx is nonzero, then off-diagonal consistency in $H = D_+ + \epsilon aa^T$, with D_+ diagonal and $\epsilon = \pm 1$, forces $\epsilon = 1$ and $a_i^2 = 4(Jx)_i^2/q^2$. The axial entry of D_+ is then $-2/q$. This proves minimality only within the class of diagonal matrices plus one signed rank-one term.

To bound the width of the two-hub graph, form a coarse incidence graph with one vertex for each cone and each equality row, joining them when the row uses a coordinate of that cone. If this graph has width w and each scalar column of A has at most one nonzero, the two-hub graph has width at most $\max\{2w+1, 3\}$. Replace every cone vertex in a coarse bag by its two hubs. For each used coordinate attach a bag consisting of that coordinate, its two hubs, and its single row, to a bag covering the corresponding coarse edge. For unused coordinates omit the row. These bags cover all edges and preserve the connected-bag condition. Thus a coarse forest gives width at most three regardless of cone dimension.

29.5 Exact graph counts for the ball constructions

For N projected coordinates grouped into k nonempty groups, set $q_j = s_j - \|w_j\|^2$ and impose $\sum_j s_j = 1$. The reduced barrier has Hessian blocks

$$\text{diag}(0, 2I/q_j) + q_j^{-2}(1, -2w_j)(1, -2w_j)^T. \quad (29.24)$$

Let λ be the multiplier of $\sum_j s_j = 1$, and let h_j be the hub of group j (not the group count h_a used above). The one-hub graph has the paths $\lambda - s_j - h_j$ and the leaves $h_j - w_{ji}$. It is a tree with $N + 2k + 1$ vertices and $N + 2k$ edges. Because the s_j diagonal entries are zero, scalar leaf pivoting can fail; Proposition 29.10 gives the exact solve instead. Restricting (29.21) by $(s, w) \mapsto ((1 + s)/2, (1 - s)/2, w)$ gives positive base $\text{diag}(1/q, 2I/q)$ and a positive downdated base by congruence. The two-hub reduced graph has $N + 3k + 1$ vertices, $2N + 3k$ edges, and width two. Bags containing the two hubs and one coordinate, together with bags $\{\lambda, s_j\}$, prove the upper bound; each nonempty group supplies a cycle for the lower bound.

For comparison, retain all raw Lorentz coordinates (u_j, v_j, w_j) and equations $u_j + v_j = 1$, $\sum_j(u_j - v_j) = 1$. The exact structural counts are

representation	vertices	edges	treewidth
raw grouped, one hub	$N + 4k + 1$	$N + 6k$	2
raw grouped, two hubs	$N + 5k + 1$	$2N + 8k$	3
reduced grouped, one hub	$N + 2k + 1$	$N + 2k$	1
reduced grouped, two hubs	$N + 3k + 1$	$2N + 3k$	2

To verify the raw widths, the one-hub graph has $K_{2,3}$ on axial coordinates u_j, v_j and the hub, local row, and global row. Bags $\{u_j, v_j, a\}$ for these three vertices a , plus hub-coordinate leaf bags, give width two. The two-hub graph has a subdivision of K_4 : its branch vertices are the two hubs and u_j, v_j ; a row connects the two axial vertices and one w coordinate connects the two hubs. A width-three decomposition uses the four branch vertices as a core, the two axial vertices with the local and global rows as another bag, and bags of the two hubs with each w coordinate. Different groups join through the global row. Counting incidences gives the edge totals.

The direct ball formulation $(t, x) \in Q_{N+1}$, $t = 1$, has a one-hub tree with $N + 3$ vertices; eliminating t gives a star with $N + 1$ vertices. A binary norm tree with $L = N - 1$ internal nodes has $3L$ raw cone coordinates, L hubs, $L - 1$ link rows, and one root row. It is connected with $5L$ vertices and $3L + 2(L - 1) + 1 = 5L - 1$ edges, hence a tree. Its rows are independent because each link or root equality uses its own cone-coordinate slots. The smooth coordinatewise construction ($k = N$) thus has raw width two but reduced width one; the width two comes from keeping the raw Lorentz coordinates, not from the smooth contact selection.

For a single ball with no external affine coupling, all these Newton equations can be solved exactly in $O(N)$ arithmetic operations, while the ambient and restricted barrier parameters and the movement bounds of Section 25 differ between these formulations. Thus replacing a large cone by many small cones can increase the standard short-step iteration bound and the movement cost of steps with bounded Dikin norm without making this linear solve cheaper. For products of cone epigraphs, fixing their axial coordinates to one reduces to products of balls; the additive curvature bound of Theorem 5.3 forces at least $k(s - 1)$ copies of Q_3 to represent Q_{s+1}^k , even under joint reformulation. Their ambient product barrier has parameter at least $2k(s - 1)$ by its orthant section. Separate binary trees attain the count. This compares the barrier parameter in the generic ambient short-step bound with the parameter $2k$ of the direct product, not the iteration counts of all algorithms.

29.6 The full-output comparison and its boundary

Suppose that an outer algorithm requests a classical direction or updated iterate in every round, its convergence theorem accepts any direction satisfying a stated Newton-residual condition, and the current coefficients and right-hand side are classically evaluable at the cost charged to its quantum access model. If fixed compact decompositions of widths τ_t are supplied for its augmented systems, replacing the quantum Newton module by Proposition 29.10 costs

$$\sum_t O((M_t + p_t + k_t)(\tau_t + 1)^2) \quad (29.25)$$

exact field operations, plus coefficient formation and any decomposition input cost. The outer guarantee is preserved because the exact solution satisfies the residual condition. This need not reproduce the quantum algorithm's random trajectory; the comparison requires the outer guarantee and the structural width bound for every admitted trajectory. Dense materialization already requires $\Omega(M_t + p_t)$ scalar writes per round. Hence, if both modules output full classical vectors under the same access model with exact arithmetic, $p_t + k_t = O(M_t)$, and $\tau_t = M_t^{o(1)}$, a quantum Newton module gives no polynomial saving in the dimension. The centered packed family and the grouped or norm-tree ball family have $O(n)$ or $O(N)$ replacements. Dense external affine coupling can invalidate their width bounds. A finite-precision replacement additionally needs a stable implementation and certification at the prescribed tolerance.

There is a complementary classical route for sparse well-conditioned systems. For a consistent full-column-rank system $Kd = r$ and zero initial guess, conjugate gradients on its normal equations, implemented by matrix–vector products as CGLS or LSQR [74], has the exact-arithmetic residual estimate

$$\|Kd_j - r\|_2 \leq 2 \left(\frac{\kappa_2(K) - 1}{\kappa_2(K) + 1} \right)^j \|r\|_2.$$

This follows from the usual CG energy-norm bound, because that norm is $\|K(d_j - d_*)\|_2$ and $\sqrt{\kappa_2(K^T K)} = \kappa_2(K)$. If one pair of products by K, K^T costs $\mathcal{M}$, including current entry evaluation, a relative residual $0 < \epsilon_{\text{lin}} < 1$ costs $O(\mathcal{M}\kappa_2(K) \log(2/\epsilon_{\text{lin}}))$. The case $\kappa_2(K) = 1$ terminates after one step. No normal matrix need be materialized. Sparse bounded-size conic blocks fit this comparison only when their actual augmented entries are cheaply available. Quantum-only encodings and tall implicit Hessian formation are different access models; in particular this argument does not replace the row-sampling work in Apers and Gribbling [5].

The classical algorithms, signed low-rank augmentations, and search bounds used here are known. The results specific to these formulations are the packing-independent centered Newton equation, its sharp off-center comparison at feasible auxiliary points, and the exact access and output conditions under which the same classical Newton work applies. Published quantum SDP and portfolio IPM analyses explicitly account for conditioning, data access, and classical readout [7, 26]; our comparison covers only the structured families displayed here. In a block-diagonal SDP embedding, the matrix order is the sum of the block orders, not their maximum, and a generic solver theorem must use that sum. Outputs that are states, samples, scalars, or coherent iterates can have different costs, as the query results in the next section show.

30 Query access and classical output

A formulation can have few cone factors, a simple Newton system, and an inexpensive normalized quantum state, while its requested classical output still requires reading most of the input. We show

this on the instances used for the geometric results. The query tools are standard: the polynomial method, search, approximate counting, and quantum information bounds [11, 14, 18, 66, 67]. What is new is their application to specified conic lifts, barriers, and output requirements; we do not claim new Boolean query bounds.

30.1 Access conventions and an information bound

The input is a string $b \in \{0, 1\}^n$, with $\sigma_j = (-1)^{b_j}$, accessed through the clean indexed oracle

$$O_b|j, z\rangle = |j, z \oplus b_j\rangle.$$

Quantum and randomized query complexities below have worst-case success probability at least $2/3$. *Public* means independent of b . Public arithmetic, magnitude preparation, index routing, numerical precision, and writing the output are counted separately. An explicit classical vector means a list of real coordinates at the specified accuracy, not a circuit, a sampling data structure, or an amplitude state. For a nonzero real vector v , its amplitude state is $|v\rangle = \|v\|^{-1} \sum_j v_j |j\rangle$ in an ordinary complex Hilbert space.

Sample-and-query access supplies coordinate queries, squared-coordinate samples, and the specified global norm. Coherent preparation access is a fixed unitary completion $U_b = O_\sigma U_{\text{pub}}$ of the preparation, where U_{pub} is public and O_σ is diagonal: it multiplies specified basis states (slots) by their hidden signs and fixes all other slots. Both U_b and its inverse cost a constant number of queries to O_b . A reference slot on which the oracle acts trivially may be added, so controlled phase queries and clean bit queries are equivalent up to a constant factor. The model excludes free norms of hidden-dependent projections, free postselection probabilities, and arbitrary hidden-dependent unitary completions.

Lemma 30.1 (Explicit approximate sign recovery). *Suppose a T -query algorithm, on each $\sigma \in \{-1, 1\}^n$, returns a classical string with at most αn errors with probability at least $2/3$, where $0 \leq \alpha < 1/4$ is fixed. Then $T = \Omega_\alpha(n)$.*

Proof. Purify the algorithm and defer its measurements. Induction on its clean bit queries gives

$$|\psi_\sigma\rangle = \sum_{S \subseteq [n], |S| \leq T} \left(\prod_{j \in S} \sigma_j \right) |a_S\rangle.$$

Indeed a bit query, in its target's $|+\rangle, |-\rangle$ basis, either preserves a coefficient or multiplies it by one σ_j ; input-independent unitaries preserve degree. Thus the ensemble of final states is supported on a subspace of dimension $D_T \leq \sum_{j=0}^T \binom{n}{j}$. For uniform σ , the expected fraction of erroneous decoded bits is at most $d = (1 + 2\alpha)/3 < 1/2$. If $\hat{\sigma}$ is the decoded string, elementary conditional-entropy bounds and concavity of the binary entropy H_2 give

$$H(\sigma | \hat{\sigma}) \leq \sum_j H(\sigma_j | \hat{\sigma}_j) \leq nH_2(d).$$

Holevo's information bound therefore implies

$$n[1 - H_2(d)] \leq I(\sigma; \hat{\sigma}) \leq \log_2 D_T.$$

For $T \leq n/2$, the right-hand side is at most $nH_2(T/n)$, proving a linear lower bound; larger T is trivial. These are standard entropy arguments underlying quantum random access bounds [66]; the Fourier span also appears in oracle interrogation [84]. $\square$

We also use the standard bounded-error complexities of estimating the mean of n bits to additive error δ :

$$Q(n, \delta) = \Theta(\min\{n, \delta^{-1}\}), \quad R(n, \delta) = \Theta(\min\{n, \delta^{-2}\}) \quad (0 < \delta \leq c_0), \quad (30.1)$$

for a sufficiently small universal $c_0 > 0$. Amplitude estimation, or reading all bits, gives the quantum upper bound; Nayak and Wu's bound for separated central weights gives the lower bound [18, 67]. For the randomized lower bound, use Bernoulli inputs of means $1/2 \pm C\delta$ when $\delta \geq C'/\sqrt{n}$. By concentration, an accurate mean estimate distinguishes the two distributions, while the divergence between T -query transcripts is $O(T\delta^2)$. This gives $\Omega(\delta^{-2})$. Applying this at $\delta = C'/\sqrt{n}$ and monotonicity in δ give $\Omega(n)$ at finer accuracy; the constants absorb finitely many small n . At error strictly below half the gap between adjacent means, parity gives a direct linear lower bound [11]; under the promise of zero or one marked bit, search instead gives $\Theta(\sqrt{n})$ quantum and $\Theta(n)$ randomized queries [14, 18].

30.2 Certificate rank, movement, and sign output on one lift

Fix a simple Euclidean Jordan algebra of rank $r \geq 2$ attaining the dictionary capacity $B = a(r-1) > 0$. Let $n = qB$, $s = n + 1$, $q \geq 2$, and use full half-Peirce groups in the lift (4.9). Put

$$g_\sigma = n^{-1/2}(\sigma_1, \dots, \sigma_n), \quad v_\sigma = (g_\sigma, 0), \quad X_G^0 = c + q^{-1}d.$$

The reference point is strictly feasible; it need not be the analytic center.

Theorem 30.2 (One instance, different costs for different outputs). *For every hidden string, the minimum total Jordan rank of a genuine certificate for v_σ on this lift is q . In the restricted standard Jordan log-determinant metric,*

$$d_F(X^0, \{X : 1 - \langle v_\sigma, \pi X \rangle \leq \epsilon\}) \geq \sqrt{q} \log(1/\epsilon), \quad 0 < \epsilon < 1. \quad (30.2)$$

One sign query prepares the amplitude state of the projected optimizer. It also prepares the real-coordinate amplitude states of the structured lifted optimizer and of the standard-barrier central point for every positive multiplier, provided their public magnitude states can be prepared. For every fixed $0 < \epsilon < 1/8$, outputting an explicit feasible ϵ -optimal point, projected or lifted, requires $\Theta_\epsilon(n)$ quantum and randomized queries.

Proof. Every group has $\|g_G\|^2 = 1/q$. The unique projected optimizer has $w_G = g_G$, $b = 0$, and feasibility forces $z_G = 1/q$. Set

$$A = \frac{1}{2q}, \quad Y_G = Ac - \frac{g_G}{\sqrt{2}} + \frac{P(g_G)c}{2A}.$$

Lemma 4.2 gives a positive rank-one element with tail trace $1/2$. Direct pairing gives $\sum_G \langle Y_G, X_G \rangle = 1 - g_\sigma^T w$ on the whole affine slice. Every certificate has nonzero half-Peirce coefficient in each full group, so every block is nonzero, proving the exact minimum rank q . The positive eigenvalue of Y_G is $(q+1)/(2q)$. Its primitive support idempotent u_G has c -mass $1/(q+1)$ and d -mass $q/(q+1)$, hence

$$\det_{u_G} Y_G = \frac{q+1}{2q}, \quad \det_{u_G} (P(u_G)X_G^0) = \frac{2}{q+1}.$$

Thus the scale in Theorem 23.1 is $\Delta = q[(1/q)^q]^{1/q} = 1$, proving (30.2).

The optimizer state is $n^{-1/2} \sum_j \sigma_j |j\rangle$. For central points, the determinant formula (4.7) and stationarity in w_G give $w_G = t_G g_G$ with $t_G > 0$. The objective and barrier depend on group

orientations only through these inner products and norms. Equal group norms and strict convexity imply common public values $t_G = t$ and $z_G = z$, and a public b . All remaining coordinates are public. Phase kickback therefore supplies the hidden signs in one query after public magnitude preparation. For the Albert algebra, this statement encodes its 27 real coordinates; it assumes neither octonionic quantum amplitudes nor an efficient Jordan-product circuit.

For any feasible projected point x of support gap at most ϵ ,

$$\|x - v_\sigma\|^2 \leq 2(1 - v_\sigma^T x) \leq 2\epsilon.$$

Each incorrectly decoded sign costs at least $1/n$ in this squared error, so at most $2\epsilon n$ signs are wrong. Lemma 30.1 applies with $\alpha = 2\epsilon < 1/4$. Reading all signs constructs the optimizer and its structured lift, proving the upper bound. Projection extends the lower bound to explicit lifted output. $\square$

The movement theorem implies at least $\sqrt{q} \log(1/\epsilon)/[-\log(1 - R)]$ feasible chords of starting local norm at most $0 < R < 1$. This bound and the query lower bound hold simultaneously but do not multiply: the query bound allows reuse of all information learned at earlier points.

To see what the quantifier reversal (4.11) costs in the query model, let $u = (n^{-1/2}\mathbf{1}, 0)$, choose a public orthogonal U_0 sending u to the pole, and set

$$U_\sigma = U_0(D_\sigma \oplus 1).$$

Then $U_\sigma v_\sigma$ is the rank-one pole support. One coherent use of this objective-dependent rotation costs one sign query plus public gates. An explicit classical matrix $\tilde{U}_\sigma$ approximating U_σ with operator-norm error strictly below $1/2$ reveals every sign through the diagonal of $U_0^T \tilde{U}_\sigma$, so producing it costs $\Theta(n)$ queries. Thus the cost of a reformulation depends on whether it is used coherently or written out; the geometric quantifier reversal alone does not determine it.

30.3 Well-conditioned disk checkpoints with hard scalar readout

For $N \geq 2$, minimize $\sum_i c_i^T z_i$ over $(\mathbb{B}_2^2)^N$, where $c_i = e_1$ if $b_i = 0$ and $c_i = -e_2$ if $b_i = 1$. For

$$F(z) = -\sum_i \log(1 - \|z_i\|^2), \quad r(a) = \frac{a}{\sqrt{1 + a^2} + 1},$$

the central point is $z_i(\eta) = -r(\eta)c_i$. The Hessian of one block is

$$H(z) = \frac{2}{1 - \|z\|^2} I + \frac{4}{(1 - \|z\|^2)^2} z z^T. \quad (30.3)$$

Its radial-to-transverse eigenvalue ratio is $\sqrt{1 + \eta^2}$, so $\kappa(F''(z(2))) = \sqrt{5}$, independently of N . At this checkpoint the scalar Hessian is diagonal; its structural graph has only disjoint constant-size components. With $r_0 = r(2)$ and Hamming weight $w = |b|$, the fixed affine observable

$$Nr_0 + \sum_i (z_i)_1 = r_0 w \quad (30.4)$$

requires $\Theta(N)$ quantum and randomized queries to estimate to any fixed error strictly below $r_0/2$. Under the promise $w \in \{0, 1\}$ the complexities are instead $\Theta(\sqrt{N})$ and $\Theta(N)$. These bounds follow from parity and search; reading all bits gives the first upper bound.

The optimal value itself is public and equals $-N$; the observable (30.4) reads out the solution, not the value. An explicit feasible point with total objective gap below $1/8$ also needs linearly many queries, since

$$\|z_i + c_i\|^2 \leq 2(1 + c_i^T z_i)$$

places each block within distance $1/2$ of its optimizer, and the two possible block optimizers $-e_1, e_2$ are at distance $\sqrt{2}$. In contrast, one clean bit query prepares $|z(\eta)\rangle$ for any $\eta > 0$: query a uniform index into an axis register and apply the corresponding public sign. The norm $\sqrt{N}r(\eta)$ is public. Ordinary sample-and-query access to the raw objective adds no information: a sample is a uniformly random block, and one query reveals its axis. Access to norms of arbitrary coordinate subvectors would be stronger; for example, the first-coordinate part of the objective has norm $\sqrt{N-w}$.

The affine functional in (30.4) has Euclidean norm $\sqrt{N}$. Normalizing it to norm one changes its required error to $O(N^{-1/2})$. Likewise, a point within constant local-norm distance of the checkpoint need not have the fixed scalar accuracy: the functional has dual local norm $\Theta(\sqrt{N})$, so a generic local-norm error guarantee must be $O(N^{-1/2})$. The lower bound concerns the specified scalar output.

30.4 Query cost versus precision with public magnitudes

Impose $y_i = x_i$ in each disk and use the raw objective pairs

$$\tilde{c}_i = \frac{\Gamma}{\sqrt{10N}}(1, 3\sigma_i), \quad \Gamma > 0.$$

All raw magnitudes and the raw global norm Γ are public. In the orthonormal coordinates $(x_i, y_i) = t_i(1, 1)/\sqrt{2}$, the reduced domain is $[-1, 1]^N$ and the reduced objective coefficients are

$$\beta_i = \frac{\Gamma}{\sqrt{5N}} \begin{cases} 2, & b_i = 0, \\ -1, & b_i = 1. \end{cases}$$

Hence

$$\text{OPT}(b) = -\frac{2\Gamma\sqrt{N}}{\sqrt{5}} + \frac{\Gamma}{\sqrt{5N}}w. \quad (30.5)$$

Theorem 30.3 (Query cost versus precision on the product domain). *For $0 < \epsilon \leq c_0\Gamma\sqrt{N}$, with $c_0 > 0$ a sufficiently small universal constant, estimating (30.5) to additive error ϵ has worst-case query complexities*

$$Q = \Theta\left(\min\left\{N, \frac{\Gamma\sqrt{N}}{\epsilon}\right\}\right), \quad R = \Theta\left(\min\left\{N, \frac{\Gamma^2 N}{\epsilon^2}\right\}\right). \quad (30.6)$$

The same bounds hold for the objective value at the exact central point of multiplier $\eta_\Gamma = 2\sqrt{5N}/\Gamma$. The reduced Hessian has condition number at most four along the entire central path. One query prepares the normalized optimizer state. The normalized projected states of the checkpoint and of the central predictor there have heralded exact preparation with $O(1)$ expected queries, or constant-error preparation with $O(1)$ worst-case queries. An explicit feasible point with objective gap below $\Gamma/\sqrt{5N}$ requires $\Theta(N)$ queries.

Proof. Equation (30.5) is an affine rescaling of w/N with scale $\Gamma\sqrt{N}/5$. Equation (30.1) proves (30.6), with constants absorbed into c_0 . The reduced barrier is $-\sum_i \log(1 - t_i^2)$ and its central point is $t_i = -\text{sign}(\beta_i)r(\eta|\beta_i|)$. At force $a = \eta|\beta_i|$, the scalar Hessian of the reduced barrier is

$$h(a) = \sqrt{1 + a^2}(\sqrt{1 + a^2} + 1).$$

For $a > 0$, $h(a)/a^2 = s/(s-1)$, $s = \sqrt{1+a^2}$, is decreasing; thus $h(2a)/h(a) \leq 4$, and continuity handles $a = 0$. At η_Γ , the forces are four and two. Subtracting the public central objective value for $b = 0$ gives

$$L_b(\eta_\Gamma) + \frac{2\Gamma\sqrt{N}}{\sqrt{5}}r(4) = \frac{\Gamma}{\sqrt{N}}\Delta_{\text{cp}}w, \quad \Delta_{\text{cp}} = \frac{\sqrt{17}-\sqrt{5}}{2\sqrt{5}} > 0. \quad (30.7)$$

This is another affine rescaling of w/N by $\Theta(\Gamma\sqrt{N})$.

The optimizer coordinates are $-\sigma_i$, so phase kickback gives their normalized state. Checkpoint magnitudes are the two public constants $r(4), r(2)$. Query a uniform index, rotate a flag with amplitude proportional to the desired signed magnitude, uncompute the input bit, and postselect the flag. The success probability is uniformly bounded below by $(r(2)/r(4))^2 > 0$. Repetition gives heralded exact preparation with constant expected queries; a fixed number of repetitions gives constant error. The central predictor is $\dot{t}_i = -\beta_i/h(\eta|\beta_i|)$, so the same argument at the specified checkpoint applies to its two nonzero public magnitudes. This concerns this particular right-hand side, not arbitrary Newton systems. Finally, any wrong coordinate sign incurs gap at least $\min_i |\beta_i| = \Gamma/\sqrt{5N}$. Below that threshold every sign is correct and all bits are recovered; parity supplies the lower bound. $\square$

Squared-coordinate sampling of the raw objective picks a uniform i , then one of its two coordinates with probabilities $1/10, 9/10$; a returned sign costs one query. The canonical coherent completion applies the sign phase only to the second coordinate of a public magnitude preparation. The theorem holds under these forms of access, but not if $\|\beta\|^2 = \Gamma^2(4N - 3w)/(5N)$ is free, since that reveals the answer.

Moving the signs into the constraints. Equivalently, use the public objective $\Gamma(1, 3)/\sqrt{10N}$ in every disk and impose the hidden equalities $y_i = \sigma_i x_i$. Their constraint matrix E has rows $(-\sigma_i, 1)$ with disjoint supports, hence $EE^T = 2I$. Row norms, singular values, column degrees, and squared-sampling distributions are all public. The hidden orthonormal coordinates $(x_i, y_i) = t_i(1, \sigma_i)/\sqrt{2}$ give exactly the preceding coefficients and query bounds. The canonical coherent matrix completion applies the sign phases to the first-coordinate slots of a public preparation; each call costs $O(1)$ queries. A coherent use of the hidden basis costs $O(1)$ queries; writing it out as a classical matrix costs $\Theta(N)$. The ambient optimizer pairs are $(-\sigma_i, -1)/\sqrt{2}$, so one query after a public coordinate split prepares their amplitude state. At the checkpoint they are $(-r(4), -r(4))/\sqrt{2}$ or $(r(2), -r(2))/\sqrt{2}$, and the heralded preparation just proved applies. These statements concern the projected variables; they provide neither a free hidden-vector norm nor an oracle for the full lifted state.

Comparison with a single direct ball. Keep the same raw objective and public equalities $y_i = x_i$, but replace the product of disks by the one ball $\mathbb{B}_2^{2N}$. The reduced domain is $\mathbb{B}_2^N$, its standard restricted barrier has parameter one, and

$$\text{OPT}_{\text{ball}}(b) = -\Gamma\phi(w/N), \quad \phi(u) = \sqrt{(4-3u)/5}.$$

Since $3\sqrt{5}/20 \leq |\phi'(u)| \leq 3\sqrt{5}/10$, the query complexities of the optimal value become

$$Q = \Theta(\min\{N, \Gamma/\epsilon\}), \quad R = \Theta(\min\{N, \Gamma^2/\epsilon^2\}) \quad (0 < \epsilon \leq c_0\Gamma). \quad (30.8)$$

At multiplier $1/\Gamma$, the central value is $-\Gamma\phi(u)r(\phi(u)) = -\Gamma(\sqrt{1+\phi(u)^2} - 1)$. Its derivative in u is $3\Gamma/[10\sqrt{1+\phi(u)^2}]$, bounded above and below by positive multiples of Γ , so the same bounds hold. By (30.3), the reduced Hessian has condition number at most $3/\sqrt{5}$; it is diagonal plus rank

one. The normalized optimizer and central states all equal $|\beta\rangle$, prepared by projecting the raw objective state onto the public equality subspace. The success probability is $\|\beta\|^2/\Gamma^2 \in [1/5, 4/5]$, so heralded exact preparation uses $O(1)$ expected queries. An explicit feasible point with gap below $\Gamma/(8\sqrt{5}N)$ determines all signs: its squared distance to $-\beta/\|\beta\|$ is at most $2\epsilon/\|\beta\|$, whereas every optimizer coordinate has magnitude at least $1/(2\sqrt{N})$. The same holds for the public-objective hidden-equality encoding.

For this direct ball, $[\log(\Gamma/(2\sqrt{5}\epsilon))]_+$ is a lower bound on the barrier distance from zero to the ϵ -accurate points. Indeed, if $d = -\beta/\|\beta\|$ and gap $g = \|\beta\|(1 - d^T t)$, then $1 - \|t\|^2 \leq 2g/\|\beta\|$, and the barrier gradient has dual norm at most one. Thus the logarithmic movement bound and the value query bound hold on the same direct-ball instances. The half-power difference between (30.6) and (30.8) compares two different feasible bodies; it does not separate two formulations of one optimization problem. Moreover, the sliced disk product is a box, so the factor-minimality results for the full disk product do not apply to it.

30.5 Reading out several independent search bits

An intermediate output between one scalar and a dense vector is one bit for each of several independent search problems. Let k source balls have even dimension $s \geq 4$, and let each have one hidden marked index $j_a \in [s]$. Define $\sigma_{ai} = 1 - 2\mathbf{1}_{i=j_a}$ and minimize $\sum_{a,i} \sigma_{ai} w_{ai}$. Use the grouped Lorentz formulation (6.4), globally smooth and optimal under a cone-dimension cap: for an integer cap $3 \leq d < s + 1$, put $\ell = \lceil s/(d-2) \rceil$, partition each ball into balanced nonempty groups, and use

$$q_{aG} = t_{aG} - \|w_{aG}\|^2 > 0, \quad \sum_G t_{aG} = 1, \quad F = - \sum_{a,G} \log q_{aG}.$$

For $d \geq s + 1$, take $\ell = 1$ and the direct ball barrier. Stationarity at multiplier $\eta_0 = 4\ell/(3\sqrt{s})$ gives exactly

$$q_{aG} = \frac{3}{4\ell}, \quad w_{ai} = -\frac{\sigma_{ai}}{2\sqrt{s}}, \quad t_{aG} = \frac{3}{4\ell} + \frac{|G|}{4s}. \quad (30.9)$$

Indeed $w = -\eta_0 q \sigma / 2$ and $\ell q + s(\eta_0 q / 2)^2 = 1$. For the public first half $H \subset [s]$, the affine readout $p_a = \sqrt{s} \sum_{i \in H} w_{ai} + |H|/2$ equals $\mathbf{1}_{j_a \in H}$.

Returning all k bits p_a with error less than $1/3$ requires $\Theta(k\sqrt{s})$ quantum and $\Theta(ks)$ randomized queries, whereas one phase query prepares the projected central state. For the quantum lower bound, on one block use the adversary matrix joining all indices in H to all in its complement. Its norm is $s/2$; masking by a single query gives norm $\sqrt{s}/2$. For k blocks, sum its tensor-factor copies. They share a top eigenvector, giving numerator $ks/2$; a query masks all terms except its own block, giving the same denominator. The spectral adversary bound [56] proves the claim. The matching upper bound finds all k marks among ks positions: knowing the number remaining, exact amplitude amplification uses $O(\sqrt{ks/j})$ queries at the j -remaining stage, whose sum is $O(k\sqrt{s})$ [18]. This avoids the cost of making k separate searches all succeed. For randomized algorithms under independent uniform marks, a randomly chosen block, with all other blocks simulated for free, has expected query cost T/k . Deciding whether its mark lies in H requires $\Omega(s)$ expected queries, hence $T = \Omega(ks)$. Scanning gives the upper bound.

The checkpoint is also well conditioned after an explicit public scaling. Write $z = \sqrt{\ell} \delta t$, $y = \delta w$, and $A = 2\sqrt{\ell} w_G$. Dividing the Hessian quadratic form by ℓ gives

$$\frac{16}{9}(z - A^T y)^2 + \frac{8}{3}\|y\|^2, \quad \|A\|^2 = \ell|G|/s < 2.$$

The coupled two-dimensional matrix, multiplied by nine, has determinant 384 and trace below 72, so all blocks have common eigenvalue bounds $384/(72 \cdot 9)$ and $72/9$. The condition number is at most $27/2$, also after restriction to $\sum_G z_{aG} = 0$. The direct-ball case has condition number $5/3$. This bound concerns the scaled restricted Hessian, not an arbitrary augmented KKT matrix. Writing a dense classical vector still takes $\Omega(k)$ writes, whereas a list of the k marked indices keeps the search advantage. These different output requirements explain why the linear arithmetic cost of the Newton solves does not settle the query comparison.

30.6 The same conclusions with one cone factor

The full disk family has the one-factor PSD formulation

$$\begin{pmatrix} S & W^T \\ W & I_2 \end{pmatrix} \succeq 0, \quad S_{ii} = 1, \quad D = S - W^T W, \quad F = -\log \det D, \quad (30.10)$$

where the columns of W are the disk variables. The cone order is $N + 2$. For fixed W , the fiber center is $D = \text{diag}(1 - \|w_i\|^2)$ by Hadamard's inequality, so the marginal barrier is exactly the disk barrier, with slice parameter N . Consequently, the projected central states and readouts above are unchanged. The statement on inexpensive state preparation does not cover the dense auxiliary Gram coordinates $S = W^T W + D$.

Starting at $W = 0, D = I$, any feasible endpoint of gap at most ϵ in the full disk problem satisfies

$$\det D \leq \prod_i (1 - \|w_i\|^2) \leq (2\epsilon/N)^N.$$

The height estimate of Lemma 25.1, with parameter N , therefore gives

$$d_F \geq \sqrt{N} [\log(N/(2\epsilon))]_+.$$

For the public-objective hidden-equality version, let $a_i = |\beta_i|$ and $g_i = a_i(1 - t_i t_i^*)$. Since $a_i \geq \Gamma/\sqrt{5N}$ and $1 - t_i^2 \leq 2g_i/a_i$, the corresponding bound is

$$d_F \geq \sqrt{N} \left[\log \frac{\Gamma\sqrt{N}}{2\sqrt{5}\epsilon} \right]_+. \quad (30.11)$$

Both bounds allow arbitrary feasible off-diagonal completions along the path, not just fiber centers. If the initial point is within local norm $0 \leq \rho < 1$ of the center, subtract $-\log(1 - \rho)$ from the lower bound before taking the positive part. If each round has at most $m \geq 1$ feasible chords (m an integer), each of starting local norm at most $0 < R < 1$, dividing by $m[-\log(1 - R)]$ gives a lower bound on the number of rounds.

Alternatively, the disk product is the slice $t = 1$ of the single cone

$$\mathcal{H}_{N,2} = \{(t, z) : \|z_i\| \leq t \ (1 \leq i \leq N)\}.$$

By Theorem 17.3, the optimal parameter of an LHSC barrier on this cone is $N + 1$, attained by

$$F_{\mathcal{H}}(t, z) = - \sum_i \log(t^2 - \|z_i\|^2) + (N - 1) \log t.$$

At $t = 1$ its restriction is again the disk barrier. The public-objective hidden-equality problem is strictly primal feasible at $t = 1, z = 0$ and strictly dual feasible: the dual cone consists of (M, v) with $M \geq \sum_i \|v_i\|$, so take a sufficiently large positive M and the public objective blocks as v_i .

Therefore, for *any* ambient LHSC barrier, Theorem 24.2 bounds the number of rounds from an exact primal–dual central point of gap Δ_0 to a strictly feasible point of gap $0 < \epsilon < \Delta_0$:

$$T \geq \frac{\sqrt{(N+1)/2} \log(\Delta_0/\epsilon)}{m[-\log(1-R)]}. \quad (30.12)$$

Chords are measured in the primal–dual product metric, and intermediate points need not be affinely feasible. If the start is at the public multiplier $\eta_0 = 2\sqrt{5}$ and $\epsilon < (N+1)/\eta_0$, the numerator can be replaced by $\sqrt{(N+1)/2} \log((N+1)/(\eta_0\epsilon))$. Indeed $\Delta_0 = \nu/\eta_0$, $\nu \geq N+1$, and $\sqrt{\nu} \log(\nu/(\eta_0\epsilon))$ is increasing on this range. The inexpensive-state and diagonal-Hessian statements hold for the displayed barrier, whereas (30.12) holds for every ambient LHSC barrier with its own metric and central start.

Thus, at $\Gamma = \sqrt{N}$ and a fixed, sufficiently small error, these explicit formulations have one cone factor, need $\Theta(N)$ queries for scalar or explicit output, have projected states preparable with $O(1)$ queries, and need $\Omega(\sqrt{N} \log N)$ bounded-movement rounds in the stated metrics. A cost model counting both queries and rounds has their maximum as a lower bound. Multiplying them would require a separate theorem showing that every round needs new queries; these fixed-input families provide no such theorem. This is consistent with quantum optimization analyses that count data access, linear solving, and classical output separately [5].

31 Acquiring an exact smaller formulation

The preceding formulation bounds assume a known representation. When constraints depend on the input, removing redundant ones can require queries. We call the query cost of computing an exact smaller formulation its acquisition cost. For explicit families we compute this cost and compare barrier parameters and central-path lengths before and after compilation. The search bounds are standard query-complexity results; our contribution is their application to exact conic reformulations. In particular, a barrier with a small parameter may exist without being cheap to evaluate from the supplied data.

31.1 Output and access conventions

A source query returns one input symbol, classically or coherently through the usual reversible oracle. Public arithmetic and the preparation of index registers are not source queries; their gate and bit costs are counted separately. In every bounded-error statement, the complete output must be correct with probability at least $2/3$ on every promised input. An *exact reusable compiler* outputs a finite classical description of the exact projected body that is usable without further source queries, for example the retained inequalities and a barrier for their intersection. The lower bounds also cover any classical description that supports exact membership or boundary evaluation without further source queries; they do not count the cost of computing with it. A solution is *coordinate-readable* if its coordinates are output classically.

Lemma 31.1 (Acquisition of independent marked positions). *Let $b \geq 1$ and $m \geq 2$. In each of b public blocks of m bits, exactly one bit is one. Returning all marked positions has quantum query complexity $\Theta(b\sqrt{m})$ and randomized query complexity $\Theta(bm)$. If exactly k of N bits are marked, with k known, all marks can be found with certainty in $O(\sqrt{Nk})$ quantum queries.*

Proof. Exact amplitude amplification with a known number of marked positions finds a mark in $O(\sqrt{n/t})$ queries when $t > 0$ of n positions are marked [18, Theorem 4]. Apply this separately to

the blocks. For known total cardinality, suppress previously found indices and sum $O(\sqrt{N/t})$ for $t = k, k-1, \dots, 1$. There is no accumulation of failure probabilities in these exact-query upper bounds.

For completeness, index block inputs by $r \in [m]^b$ and use the nonnegative adversary matrix

$$\Gamma = \sum_{j=1}^b I^{\otimes(j-1)} \otimes (J_m - I_m) \otimes I^{\otimes(b-j)}.$$

Its norm is $b(m-1)$. A query to position ℓ of block j retains only the j th summand, replacing $J_m - I_m$ by the adjacency matrix of the star joining ℓ to the other positions. Its norm is $\sqrt{m-1}$. The spectral adversary bound therefore gives $\Omega(b\sqrt{m-1}) = \Omega(b\sqrt{m})$; this is the standard direct-sum argument [2, Theorems 3–4]. Classically, under independent uniform marked positions, success probability $2/3$ for all outputs implies that probability for each individual position. Revealing the other blocks in advance cannot make a uniform marked position in an unread block less uncertain. A successful search in that block requires $\Omega(m)$ expected queries, even allowing a final unqueried guess. Summing over blocks gives $\Omega(bm)$; scanning gives the matching upper bound. $\square$

The exact quantum upper bounds use the standard query model, which allows the public rotations that exact amplitude amplification needs. Approximate gate implementations must account for their precision separately.

31.2 Nested constraints and consensus

Let $d \geq 1$ be fixed and let $r_i = 2 - b_i$, where $b \in \{0, 1\}^N$ has weight zero or one. Consider

$$\max e_1^T z_1, \quad (r_i, z_i) \in Q_{d+1}, \quad z_i = z_{i+1} \quad (1 \leq i < N). \quad (31.1)$$

We call (31.1) the consensus system. Its projection is exactly $\{z : \|z\| \leq r_*\}$, where $r_* = \min_i r_i$, and its optimum is r_* . The origin is strictly feasible. The public path equalities have constant incidence; only the radii r_i are hidden.

Proposition 31.2 (Exact radius compilation). *The exact projected ball and an optimizer of (31.1) can be acquired with $\Theta(\sqrt{N})$ quantum queries or $\Theta(N)$ randomized queries. The same lower bounds hold for an optimum-value estimate with additive error below $1/3$. After acquisition,*

$$f_{r_*}(z) = -\log(r_*^2 - \|z\|^2)$$

is a barrier with exact parameter one. On the all-radius-two input, the natural barrier restricted to the consensus equalities is Nf_2 and has exact parameter N . Its central path from $z = e_1$ to $z = (2 - \epsilon)e_1$, $0 < \epsilon \leq 1/2$, has length $\Theta(\sqrt{N} \log(1/\epsilon))$, compared with $\Theta(\log(1/\epsilon))$ for f_2 .

Proof. The acquisition problem is the zero-versus-one search promise. Exact amplitude amplification, followed by verification of the returned index, decides it in $O(\sqrt{N})$ queries [18, Theorem 16]. The search lower bound is [14]; the uniform single-mark distribution gives the classical lower bound. The optimum values are separated by one. The fixed ball barrier has parameter one by the ball calculation in Section 9. Alternatively, direct inversion of its Hessian gives gradient squared dual norm $2\|z\|^2/(r_*^2 + \|z\|^2) < 1$, with boundary limit one; self-concordance follows from its Lorentz restriction. Multiplication by the integer N preserves self-concordance and multiplies this exact parameter by N . On the axis,

$$f_2''(a) = \frac{2(4 + a^2)}{(4 - a^2)^2},$$

whose square root is comparable to $(2 - a)^{-1}$ on $[1, 2)$. The length follows by integration. Reflection symmetry and strict convexity put every center for objective e_1 on this axis. $\square$

Every move of local norm at most $R \in (0, 1)$ has metric length at most $-\log(1 - R)$. Hence following the central path of Nf_2 with such moves requires $\Omega_R(\sqrt{N} \log(1/\epsilon))$ moves; this concerns only that barrier and such local schedules. Dividing the natural barrier $\sum_i f_{r_i}$ by N does not give a barrier valid for all inputs. If exactly one radius is one and $N > 1$, the divided barrier restricted to the normal line near the tighter boundary is $-N^{-1} \log(1 - a) + O(1)$ with smooth bounded corrections. For $-c \log t$, the self-concordance ratio $|f'''|/[2(f'')^{3/2}]$ equals $c^{-1/2}$. The divided barrier therefore violates self-concordance near that boundary.

Without elimination of the copy equalities, the consensus system also has a conditioning problem. At its analytic center and in one coordinate, put $D = \text{Diag}(2/r_i^2)$ and let A be the path incidence matrix. Then

$$K = \begin{pmatrix} D & A^T \\ A & 0 \end{pmatrix}, \quad \kappa(AD^{-1}A^T) = \Theta(N^2), \quad \kappa(K) = \Theta(N^2). \quad (31.2)$$

Here condition numbers use singular values and $N \geq 2$. Indeed, $I/2 \preceq D \preceq 2I$, whereas the extreme eigenvalues of AA^T have orders N^{-2} and one. For a negative eigenvalue $-t$ of K , eliminating its primal component gives $A(D + tI)^{-1}A^T y = ty$. The smallest positive solution has order N^{-2} by these same comparisons and the min-max principle; all positive eigenvalues are bounded below by $1/2$, and all eigenvalues have bounded magnitude. Thus K has the stated condition number although its sparsity pattern is a forest. Eliminating the public consensus equalities removes this factor, so (31.2) is not a lower bound on the cost of a linear-system algorithm that may perform that elimination.

More generally, suppose that N constraints are partitioned into k given chains under inclusion, of lengths $n_1, \dots, n_k$. Each constraint has a queried key from an ordered set. The keys are unsorted, and within each chain their order identifies the smallest member. Suppose that this member, once known, has a specified barrier of parameter ν_j , that an additional public barrier of parameter ν_0 is present, and that common strict feasibility and positive definiteness hold on the effective affine space. Minimum finding in each chain [28], amplified to joint success, uses

$$O\left(\log(2k) \sum_{j=1}^k \sqrt{n_j}\right) \leq O\left(\sqrt{Nk} \log(2k)\right) \quad (31.3)$$

queries and produces a compiled barrier with parameter $\nu_0 + \sum_j \nu_j$. This covers homothetic copies $r_i C$ of a public compact convex body containing zero: the copies are nested, so only the minimum radius matters.

The dependence $\sqrt{Nk}$ is necessary in general. For $N = km$, take k independent interval chains $|z_j| \leq 2 - b_{j\ell}$, with each chain containing zero or one mark. Returning all optimal coordinates of $\max \sum_j z_j$ within $1/3$ determines all k OR values. The direct sum of the k star adversaries for OR has norm $k\sqrt{m}$, and any single query filters it to norm one. The quantum lower bound is $\Omega(k\sqrt{m})$ and the classical bound is $\Omega(km)$. For this promised family the upper bound is also $O(k\sqrt{m})$, by the exact promised search in the preceding proof; no logarithmic amplification is needed. On the all-zero input the natural and compiled barriers are respectively $m \sum_j f_2(z_j)$ and $\sum_j f_2(z_j)$. Their common diagonal path from zero to total objective gap ϵ , for $0 < \epsilon \leq k/2$, has lengths

$$\Theta\left(\sqrt{N} \log(k/\epsilon)\right), \quad \Theta\left(\sqrt{k} \log(k/\epsilon)\right).$$

If additional public constraints couple the chains, the compiled problem may not be separable; its solution cost comes on top of (31.3), which bounds only acquisition.

31.3 A two-dimensional family with incomparable active factors

For $v > 0$ write

$$q(v) = (1 + v, 3 - v^2), \quad E(q) = \{z \in \mathbb{R}^2 : q_1 z_1^2 + q_2 z_2^2 \leq 1\}, \quad u_i = \frac{i}{2(N+1)}.$$

All parameters used below lie in $(0, 1]$, so both coefficients of $q(v)$ are positive. The public factor $E(q(1)) = E(2, 2)$ is the sentinel. The input is a string of N trits; the trit at position i selects $E(q(1))$, $E(q(u_i))$, or $E(q_i^-)$ with $q_i^- = (1 + u_i, 1)$, according as it is 0, +, or $-$. Let S be the set of plus positions. Since the sentinel lies inside every minus factor, the projected body is

$$D_S = E(q(1)) \cap \bigcap_{i \in S} E(q(u_i)). \quad (31.4)$$

Each factor is a transformed slice of the three-dimensional Lorentz cone. Copies of (z_1, z_2) , one per factor and linked by equalities along a path, give a lift with constant incidence whose scalar KKT graph has bounded treewidth: a bag can contain the two coordinates of consecutive copies, their equality multipliers, and one constant-size local cone block. Consecutive bags share only a copy. The origin is strictly feasible.

Proposition 31.3 (Pareto compilation). *If $1 \leq k \leq N/2$ and $|S| = k$ is promised, the $k + 1$ factors of (31.4) are essential in that intersection. Exact reusable compilation has quantum query complexity $\Theta(\sqrt{Nk})$ and randomized query complexity $\Theta(N)$. These lower bounds concern an exact description of the body, not solving for one fixed objective or the minimum number of factors over all possible lifts.*

Proof. For each retained v , put $s_v = (2v, 1)$. The identity

$$\langle q(v) - q(w), s_v \rangle = (v - w)^2$$

shows that squared coordinates $s_v / \langle q(v), s_v \rangle$ define a point on factor v and strictly inside all the others. Thus removal of v enlarges the body: a small outward perturbation still satisfies the other inequalities. It also follows that different sets of retained parameters give different bodies. Hence an exact source-free membership description determines the retained set: evaluate it at the finitely many public points obtained by perturbing the witness point of every candidate parameter slightly outward. The perturbations may be chosen so that each perturbed point violates only its own factor among all candidates.

The upper bound is the known-cardinality search of Lemma 31.1. For the lower bound restrict to one plus position in each of k public blocks of size $m = \lfloor N/k \rfloor \geq 2$, with all other positions zero. The reusable description determines every marked position, so the lemma applies. Since $km \geq N/2$, its bounds have the claimed orders. $\square$

The retained constraints can be simultaneously active. If $v < w$ are consecutive retained parameters and $s = (v + w, 1)$, then

$$\langle q(r) - q(v), s \rangle = (r - v)(w - r).$$

Thus the point with squared coordinates $s / \langle q(v), s \rangle$ makes exactly the v and w factors tight. Their outward normals are linearly independent, and a positive combination of them exposes this point, so every objective in the interior of its normal cone is optimized where both factors are tight. The compiled problem therefore does not reduce to independent scalar minimum problems.

The dimension stays two, but compilation can reduce the barrier parameter greatly. Set $F_q(z) = -\log(1 - z^T \text{Diag}(q)z)$. On the hard inputs of the proof, with only zeros and pluses, the natural and compiled barriers are

$$F_o = MF_{q(1)} + \sum_{i \in S} F_{q(u_i)}, \quad F_c = F_{q(1)} + \sum_{i \in S} F_{q(u_i)}, \quad M = N - k + 1. \quad (31.5)$$

Their exact parameters satisfy $M \leq \nu(F_o) \leq N + 1$ and $\nu(F_c) \leq k + 1$. For the lower bound, approach the sentinel boundary along the positive first axis. The other factors and all their derivatives stay bounded, and the squared gradient dual norm tends to M . The upper bounds follow by adding the parameter-one ellipsoid barriers. The planar body itself admits a barrier of parameter two [60]; the M lower bound is specific to F_o .

For the objective $\max z_1$, both central paths lie on the axis $z = (a, 0)$, $0 \leq a < 2^{-1/2}$. If $f_A(a) = -\log(1 - Aa^2)$, then

$$f_A''(a) = \frac{2A(1 + Aa^2)}{(1 - Aa^2)^2}.$$

For marked factors, $1 < A < 3/2$, so f_A'' is bounded above and below by positive universal constants on this whole interval. The square root of the sentinel Hessian is comparable to $(2^{-1/2} - a)^{-1}$ near its endpoint. Integrating the square root of the Hessians in (31.5), and using $\max(\sqrt{x}, \sqrt{y}) \leq \sqrt{x + y} \leq \sqrt{x} + \sqrt{y}$, gives, uniformly for $0 < \epsilon \leq 1/4$,

$$L_o(\epsilon) = \Theta(\sqrt{M} \log(1/\epsilon)), \quad L_c(\epsilon) = \Theta(\sqrt{k} + \log(1/\epsilon)). \quad (31.6)$$

Lengths are measured from the origin to $a = 2^{-1/2} - \epsilon$. The assumption $k \leq N/2$ ensures $M \geq k$. The endpoint estimate of Lemma 25.1 also gives an $\Omega(\sqrt{M} \log(1/\epsilon))$ distance lower bound for F_o at sufficiently small universal ϵ : its barrier increase is at least $M \log(c/\epsilon)$, while $\nu(F_o) \leq M + k \leq 2M$. Thus any method with bounded local steps needs $\Omega(\sqrt{M} \log(1/\epsilon))$ steps, even if its path leaves the central path.

31.4 A fixed objective that reveals the input when the dimension grows

If the projected dimension may grow, a single fixed objective can be as hard as exact compilation: in the family below, both have quantum query complexity $\Theta(\sqrt{Nk})$ and randomized query complexity $\Theta(N)$. Let $N = km$, $m \geq 2$, and $u_\ell = \ell/[2(m+1)]$. In block j exactly one position ℓ_j is marked. In coordinates (y, p_j, q_j) retain the public sentinel and the marked inequality

$$4(y^2 + p_j^2 + q_j^2) \leq 1, \quad y^2 + (1 + u_{\ell_j})^2 p_j^2 + (3 - u_{\ell_j})^2 q_j^2 \leq 1. \quad (31.7)$$

Before compilation each block has m hidden factors, each either the sentinel (the left inequality) or the marked factor, and one more public copy of the sentinel. The coefficient matrices are rational with $O(\log m)$ bits, and all factors are direct Q_4 slices. The shared coordinate y is the only coupling between blocks.

Proposition 31.4 (Fixed-objective acquisition and metric separation). *The body (31.7) has dimension $2k + 1$ and $2k$ essential retained inequalities. It has a lift with $\Theta(N)$ constant-size cone factors, bounded row and column incidence, and bounded scalar KKT treewidth, all with bounds independent of k, m . Exact reusable compilation, and a coordinate-readable solution of $\max \sum_j q_j$ with coordinate error below $1/[40(m+1)]$, each have quantum query complexity $\Theta(\sqrt{Nk})$ and randomized query complexity $\Theta(N)$. The same lower bounds hold for an exactly feasible coordinate-readable solution with total objective gap below $1/[40(m+1)]$.*

For the separate objective $\max y$, the natural and compiled explicit barriers have parameters of orders N and k , respectively. Their central paths from their common center to $y = 1/2 - \epsilon$ have lengths

$$\Theta\left(\sqrt{N} \log(1/\epsilon)\right), \quad \Theta\left(\sqrt{k} \log(1/\epsilon)\right) \quad (0 < \epsilon \leq 1/4).$$

Proof. The origin is strictly feasible. To make the marked factor of block j alone tight, set $y = p_j = 0$, $q_j = (3 - u_{\ell_j})^{-1}$, and all other coordinates zero. Its sentinel is strict since $3 - u_{\ell_j} > 5/2$. To make its sentinel alone tight, take $y = q_j = 0$, $p_j = 1/2$; the marked factor is strict since $1 + u_{\ell_j} < 3/2$. These points also prove essentiality after small outward perturbations. For bounded incidence, give every slot a triple copy and join triples along paths within blocks. Connect representative y copies between blocks along a path. A constant-size bag for consecutive copies, their multipliers, and each local cone block gives a tree decomposition; at block junctions only a constant number of representative coordinates need be retained. This also proves the assertion for the scalar KKT graph, including dense local cone Hessians.

For the decoding objective, every feasible point has $q_j \leq (3 - u_{\ell_j})^{-1}$. All these bounds are attained simultaneously only at

$$y = 0, \quad p_j = 0, \quad q_j^* = (3 - u_{\ell_j})^{-1}.$$

The optimum is unique. Neighboring candidate q_j^* values differ by more than $1/[18(m+1)]$. A coordinate error below $1/[40(m+1)]$ is less than half that spacing, so rounding recovers every mark. For a feasible point the deficits $q_j^* - q_j$ are nonnegative; a total objective gap below the same threshold bounds each deficit. Lemma 31.1 proves the lower bounds and the acquisition upper bounds. Once the marks are known, the exact optimizer takes $O(k)$ arithmetic operations. The conclusion is for readable coordinates, not the objective value alone or a normalized quantum state.

For $\max y$, reflection symmetry gives $p_j = q_j = 0$ on both central paths. Write $f_A(y) = -\log(1 - Ay^2)$. Their restrictions are exactly

$$F_o(y) = Nf_4(y) + kf_1(y), \quad F_c(y) = kf_4(y) + kf_1(y).$$

The term f_1'' is uniformly bounded on $[0, 1/2)$, whereas $\sqrt{f_4''}$ has order $(1/2 - y)^{-1}$ near the endpoint. Integration proves the length assertions. The squared gradient dual norm on this axis tends to N or k , respectively, while additivity of barrier parameters gives the upper bounds $N + k$ and $2k$. The diagonal symmetry removes mixed Hessian entries on the axis, so these are also lower bounds for the full barriers. $\square$

The decoding objective is easy once the hidden constraints are known, so the quantum advantage lies in acquisition; it does not use a quantum linear-system solver.

31.5 Blockwise relative-entropy compilation

The same acquisition cost arises for the nonsymmetric relative-entropy cone. Use $D(U\|V) = U \log(U/V)$, with its closed perspective extension. Let $N = Bm$, $B \geq 1$, $m \geq 2$, and choose public values $\gamma_1 < \dots < \gamma_m$ in $[1, 2]$ with minimum spacing $h \asymp m^{-1}$. Each block has exactly one marked position r_j . Set $d_{ji} = 1$ except $d_{jr_j} = e^{-\gamma_{r_j}}$. An oracle query returns the mark bit; the coefficient at a marked position is a public function of that position. The coefficients $e^{-\gamma_{r_j}}$ can be kept symbolic, so this query convention hides no cost of writing transcendental numbers exactly. All source terms in block j share U_j, V_j .

The elementary identity (21.7) becomes

$$\sum_{i=1}^m D(U_j \| d_{ji} V_j) = m U_j \log(U_j/V_j) + \gamma_{r_j} U_j = D(m U_j \| m e^{-\gamma_{r_j}/m} V_j). \quad (31.8)$$

One can equivalently retain the rational/public coefficient γ_{r_j} in the middle expression. Under $t_j = \sum_i t_{ji}$, the projection of the m separate epigraphs is exactly the single epigraph of (31.8). One inclusion follows by summing the inequalities. Conversely, distribute the nonnegative aggregate slack among the m epigraph coordinates. This argument includes the closed perspective boundary whenever the aggregate value is finite; if it is infinite, neither epigraph has a point over those U_j, V_j .

Proposition 31.5 (Entropy acquisition and path lengths at equal gap). *An exact reusable compiler for these B aggregate epigraphs has query complexities*

$$Q = \Theta(B\sqrt{m}) = \Theta(\sqrt{NB}), \quad R = \Theta(Bm) = \Theta(N).$$

The same lower bounds apply to a coordinate-readable, exactly feasible solution of the slice $U_j = V_j = 1$ minimizing $\sum_j t_j$, with objective excess below $h/3$. On that slice, at matched positive total gap Δ , the natural and compiled projected central points coincide. Between gaps $\Delta_0 > \epsilon > 0$ their exact barrier-metric lengths are

$$\sqrt{N} \log(\Delta_0/\epsilon), \quad \sqrt{B} \log(\Delta_0/\epsilon), \quad (31.9)$$

in the metrics of their respective product barriers.

Proof. The projected epigraph in block j , evaluated at $U_j = V_j = 1$, has lower boundary $t_j = \gamma_{r_j}$. Hence an exact reusable description determines all r_j ; exact membership at the public midpoints between candidate boundary values suffices. Lemma 31.1 gives both bounds and the upper algorithm. For the solution task, feasibility gives $e_j = t_j - \gamma_{r_j} \geq 0$. Total excess below $h/3$ implies $e_j < h/3$ for every j , and nearest-value rounding recovers all marks.

With $U_j = V_j = 1$, the usual relative-entropy cone barrier restricts to $-\log(t_{ji} - D(1 \| d_{ji}))$, up to a constant. At multiplier $\eta > 0$, all N source slacks equal $1/\eta$, so their total gap is N/η and each projected block slack is m/η . The compiled barrier has B slacks, each $1/\hat{\eta}$, total gap $B/\hat{\eta}$. At $\eta = N/\Delta$ and $\hat{\eta} = B/\Delta$, both project to $t_j = \gamma_{r_j} + \Delta/B$. A positive orthant logarithmic barrier has line element $\sum_i (ds_i/s_i)^2$. Along these curves it is respectively $N(d\Delta/\Delta)^2$ and $B(d\Delta/\Delta)^2$, proving (31.9). $\square$

For the natural formulation the first length is measured in its full source-slack coordinates. It is also the length of the common projected path for the marginal barrier obtained by minimizing over the source slack allocations: that marginal is $-m \sum_j \log(t_j - \gamma_{r_j})$, up to a constant. Equal multipliers do not give equal projected points, so the comparison is at equal gaps. We do not claim it for varying U_j, V_j . If feasibility violations are allowed, the decoding argument must also bound negative deficits, in addition to the objective excess.

The two full ambient exponential-cone products have optimal barrier parameters $3N$ and $3B$, even among coupled barriers, by Theorem 19.2. On the fixed slice in Proposition 31.5, the displayed parameters are N and B . Neither ambient parameter is a lower bound for every barrier on the projected epigraph. The relevant prior barrier results are the small relative-entropy barriers of Fawzi and Saunderson [35] and the positive-map constructions of He et al. [50]. Identity (31.8) is the elementary scalar case of their setting, not a new relative-entropy barrier theorem. Our conclusions here concern acquiring the effective coefficients, decoding readable outputs, and comparing the specified metrics at equal gap.

For scalar variables, the uncompiled formulation can have bounded incidence and bounded KKT treewidth. Arrange each block as a binary tree; carry copies of U_j, V_j down its edges and sum the epigraph variables up the same tree. Each node has only a constant number of variables and equations, and each parent–child separator has constant size. These separators give an explicit bounded-width tree decomposition. For matrix entropy variables the separator size grows with the matrix dimension, so no dimension-free treewidth bound for the scalarized system follows.

31.6 Scope of the oracle advantage

Recovering the consensus copies needs no queries: the projected vector determines every copied coordinate. For a nonzero amplitude-encoded vector, the public map $|z\rangle \mapsto |u_N\rangle|z\rangle$ produces the equal copies. An explicit array of all Nd copied entries still costs $\Omega(Nd)$ output operations. In the entropy example, an implicit online compiler could instead search one requested block with $O(\sqrt{m})$ queries while keeping access to the input; it pays for all blocks only when all are requested.

Nor can an explicit input be loaded for free. In a circuit of fan-in at most $g > 1$, a readable OR output sensitive to m source bits depends on all of them, so the circuit needs $\Omega(m)$ input connections and depth $\Omega(\log_g m)$. For the independent promised OR chains this gives $\Omega(N)$ such connections in total and depth $\Omega(\log_g(N/k))$. Parallel classical reductions match these orders. The query improvements above therefore assume a supplied coherent oracle or an already constructed access structure.

Removing redundant constraints before an interior-point solve is classical presolve [42], and quantum search on partially ordered sets is known [63]. In our chain model the inclusion keys are unsorted, whereas the poset-search models of that reference have their own comparison or ordering promises. Minimum finding, amplitude amplification, and adversary direct sums are standard. Our families combine them with witnesses of essential constraints, sparse conic lifts, reductions from readable solutions to the hidden input, and exact comparisons of central-path lengths. We do not claim a general quantum speedup for optimization, a lower bound over all barriers, or that retaining fewer inequalities always gives the best lift.

32 Summary

A conic reformulation has several distinct costs. Local curvature bounds the number and dimensions of the cone factors. Minimizing over whole certificate fibers determines the least worst-case certificate rank; globally C^1 selections of boundary points and certificates can raise such minima. Affine restriction can lower the standard barrier parameter; choosing another barrier on the same domain is a separate optimization problem. For the cases in (8.6) and for nonsymmetric barrier parameters, we prove only two-sided bounds. The integer tradeoffs are necessary conditions whose minima need not be attained by lifts.

The weighted tree coefficient shows that certificate ranks do not determine distance. In the squared leading coefficient, active nodes contribute their full weight, and inactive nodes, whose subtrees miss the support of the objective, contribute half, although their certificate blocks vanish; along the central path, every node contributes its full weight. The PSD Hessian theorem shows that exact allocation equations improve the spectral comparison from squared to first powers of the residual factors, but changing the auxiliary point can still change the Newton right-hand side and the data it needs.

Geometric conclusions become computational ones only in a specified model of access and output; a short description, a cheap normalized state, a readable solution, and a reusable exact compiler are different outputs. Under the compile-and-commit requirement of Theorem 29.7, on an explicit family,

the query cost of maintaining scales grows linearly in the number of fresh update batches; fixed data can instead be cached. Work bounds here state their access and per-round work assumptions, and metric bounds their objective, endpoint, and barrier.

Appendices

A Further topological restrictions

This appendix records topological restrictions that use less information than the joint covering theorem of Section 10: only the tangent splitting induced by the mixed contact form, or saturation of a single channel.

A.1 What the abstract tangent splitting alone implies

For C^2 factors with $P = D = M$ and ζ the identity, suppose the full mixed form H is positive definite. Each individual form $H_i = -DA_i^*DB_i$ is positive semidefinite: the Hessian of the nonnegative function $\langle A_i(x), B_i(z) \rangle$ at (x, x) is positive semidefinite and annihilates the tangent diagonal. Its diagonal blocks are therefore H_i , and its off-diagonal blocks are $-H_i$. If $\sum_i \text{rank } H_i = n$ everywhere, lower semicontinuity of each rank and constancy of their sum make the individual ranks locally constant. Whitening by H gives positive semidefinite matrices of total rank n summing to the identity. Factoring them as $Z_i Z_i^*$ and concatenating the n columns shows that each whitened matrix is an orthogonal projection. Thus

$$TM = \bigoplus_i E_i, \quad E_i = \text{im}(H^{-1}H_i). \quad (\text{A.1})$$

The following necessary conditions use only the splitting (A.1); at capacity equality with exact globally smooth cone factors, Theorem 10.1 gives more.

Proposition A.1 (Ranks in a sphere tangent splitting). *Suppose $T\mathbb{S}^n = \bigoplus_i E_i$, with positive ranks r_i , and put $s = \rho(n+1) - 1$, where $\rho((2a+1)2^{4b+c}) = 8b + 2^c$ for $0 \leq c \leq 3$. Every subset sum $r_I = \sum_{i \in I} r_i$ satisfies*

$$0 < r_I \leq n/2 \implies r_I \leq s.$$

If $s < n/3$, there is a unique rank greater than $n/2$, and it is at least $n - s$. If all ranks are at most $c \leq n/2$, the splitting is impossible when $s = 0$; otherwise it needs at least $\lceil n/\min\{c, s\} \rceil$ summands. For $n \geq 3$, if a summands have rank one and b have rank two, then $a + 2b \leq s$.

Proof. For completeness, we show by clutching that such a splitting yields independent tangent vector fields. At the endpoint $2r = n$, a splitting would make $e(T\mathbb{S}^n) = e(E)e(F) = 0$, contradicting its Euler number two. For $0 < 2r < n$, both summands in a decomposition $T\mathbb{S}^n = E^r \oplus F^{n-r}$ are orientable. Their clutching maps take values in $SO(r)$ and $SO(n-r)$. If $r \leq n-r$, the two inclusions in $SO(n)$ can be conjugated into the same copy of $SO(n-r)$; the conjugations are homotopic to the identity in $SO(n)$. The sum of the resulting homotopy classes lies in the image of $\pi_{n-1}SO(n-r)$. Hence $T\mathbb{S}^n$ admits a reduction to $SO(n-r)$, which supplies r independent tangent vector fields. This is the classical plane-field argument of Steenrod [83]. Adams's theorem bounds their number by s [1]. Apply it to $\bigoplus_{i \in I} E_i$. For $n = 1$ the subset assertion is empty.

If every rank were at most $n/2$, take the first partial sum exceeding $n/2$. The preceding sum and the last added rank are each at most s . The complementary sum, when nonzero, is also at most s . This gives $n \leq 3s$ (and $n \leq 2s$ for empty complement), a contradiction. The complement of the unique large rank then has rank at most s . The capped-count assertion follows by applying the subset bound to each summand. For $n \geq 3$, rank-one bundles are trivial and oriented rank-two bundles are classified by their Euler class in $H^2(\mathbb{S}^n; \mathbb{Z}) = 0$. These summands jointly supply $a + 2b$ fields. $\square$

The largest-rank bound is sharp for abstract splittings: a maximal s -frame and its orthogonal complement have ranks s and $n-s$. The inequality $s < n/3$ holds for $n+1 \geq 3$ except $n+1 = 4, 8, 16$, as follows directly from the displayed formula for ρ . For even n , a proper splitting is impossible already from $e(E \oplus F) = e(E)e(F)$: both intermediate-degree Euler classes vanish, whereas $\langle e(T\mathbb{S}^n), [\mathbb{S}^n] \rangle = 2$. For higher ranks the precise classical test is the unstable clutching condition

$$\text{blockdiag}_*(\alpha_1, \dots, \alpha_k) = \tau_n \quad \text{in } \pi_{n-1}SO(n), \quad \alpha_i \in \pi_{n-1}SO(r_i),$$

where τ_n classifies $T\mathbb{S}^n$. Stable triviality of the tangent bundle alone does not imply this condition.

A.2 Restrictions from one everywhere saturated channel

The normalization step in the proof of Theorem 10.1 uses only that the chosen channel has positive capacity c_i and rank c_i everywhere. It then gives a submersion $P \rightarrow \mathbb{S}^{c_i}$ for that channel alone, even when the total capacity is larger than $\dim P$. The next proposition restricts such submersions when P is a sphere (a classical result) or a product of equal spheres.

Proposition A.2 (Sphere and equal-sphere-product submersions). *A C^1 submersion $\mathbb{S}^n \rightarrow \mathbb{S}^r$ with $0 < r < n$ is possible only for $(n, r) = (3, 2), (7, 4), (15, 8)$, and each pair occurs. More generally, for $q \geq 2$ and $1 \leq r < kq$, a C^1 submersion $(\mathbb{S}^q)^k \rightarrow \mathbb{S}^r$ requires*

$$r = q \quad \text{or} \quad q = 2r - 1 \quad \text{with } r \text{ even.} \quad (\text{A.2})$$

Projection realizes the first possibility when $k \geq 2$. Projection followed by a Hopf map realizes the three corresponding possibilities in the second case. We do not claim that the other dimension pairs in (A.2) occur.

Proof. A C^1 submersion from a compact source has a uniformly positive surjectivity modulus. Smooth approximation preserves submersivity, so Ehresmann's theorem gives a smooth fiber bundle for purposes of an existence obstruction [29]. A target circle is excluded by lifting to $\mathbb{R}$ and taking a maximum. For a sphere total space and $r \geq 2$, the fiber is connected. Browder's theorem gives its homotopy type as $\mathbb{S}^j$, $j \in \{1, 3, 7\}$ [19, Theorem 1]. The two-row mod-two Serre spectral sequence forces $r = j + 1$: its only possible nonzero differential killing the fiber class has length $j + 1$, and must land in the sole positive base degree r . The top surviving class gives $n = 2j + 1$. The Hopf fibrations attain these pairs.

Now write $E = (\mathbb{S}^q)^k$. If $r \geq 3$, the closed fiber F is simply connected by the homotopy exact sequence. It is orientable because the base and total space are orientable. Both its rational homotopy and its cohomology are finite-dimensional, so it is elliptic. The classical formal-dimension identity [38, Theorem 32.6] is

$$\dim F = \sum_{j \geq 2} w_j d_j(F), \quad w_j = \begin{cases} j, & j \text{ odd,} \\ -(j-1), & j \text{ even,} \end{cases} \quad d_j(X) = \dim_{\mathbb{Q}} \pi_j(X) \otimes \mathbb{Q}.$$

Let a_j denote the rational rank of $\pi_j(E) \rightarrow \pi_j(\mathbb{S}^r)$. Exactness gives $d_j(F) = d_j(E) + d_{j+1}(\mathbb{S}^r) - a_j - a_{j+1}$. Since $w_j + w_{j-1}$ is two for odd j and zero for even j , substitution gives

$$\dim F = \begin{cases} kq - (r-2) - 2a_r, & r \text{ odd,} \\ kq + 2 - r - 2a_{2r-1}, & r \text{ even.} \end{cases}$$

Comparison with $\dim F = kq - r$ forces the displayed rank to equal one. An odd sphere has rational homotopy only in its own degree; an even q -sphere has it in degrees q and $2q - 1$. Thus odd r forces

$r = q$ if q is odd. The other candidate, $r = 2q - 1$ for even q , contradicts $2^k = \chi(E) = \chi(\mathbb{S}^r)\chi(F) = 0$. Even r gives $r = q$ for even q , and $q = 2r - 1$ for odd q .

It remains to treat $r = 2$, where the fiber need not be simply connected. If $q \geq 4$, a minimal Sullivan model of E has no elements in degrees two or three. Hence a model of f sends both generators of $(\Lambda(x_2, y_3), dy_3 = x_2^2)$ to zero. The homotopy-fiber model is then

$$\mathcal{M}_E \otimes \Lambda(u_1) \otimes \mathbb{Q}[v_2], \quad Du_1 = Dv_2 = 0,$$

by the standard fiber-model construction [38, Section 15]. One can obtain it explicitly by adjoining u_1, v_2 to the sphere model with $Du_1 = x_2$ and $Dv_2 = y_3 - u_1x_2$, then pulling back along the zero model map. Its cohomology is infinite-dimensional, contradicting the compact fiber. Consequently $q = 2$ or 3 , exactly the alternatives in (A.2) for $r = 2$. $\square$

In particular, the result for sphere total spaces cannot be applied to a product source to exclude, for example, $(q, r) = (11, 6)$, and the rational argument above does not decide that case. At joint capacity equality, however, Theorem 10.1 already rules out every smaller target, including the Hopf targets.

A.3 Balanced real projector orbits

The preceding sphere-fiber argument also restricts individual support maps into real Grassmannians. It applies when a single symmetric-cone block is saturated but the collection of all blocks is not.

Proposition A.3 (No sphere submersion onto a large balanced real orbit). *Let $B_r = \text{Gr}_{\lfloor r/2 \rfloor}(\mathbb{R}^r)$. For $r \geq 4$ no sphere admits a C^1 submersion onto B_r or any of its connected finite covers. For $r = 3$, the possible source dimensions are exactly two and three. For $r = 2$, the only source dimension is one, and the maps are circle coverings. The $r = 1$ target is a point.*

Proof. For $r \geq 3$, a submersion from $\mathbb{S}^n$ must have $n \geq 2$ and lifts to the simply connected oriented Grassmannian

$$\tilde{B}_r = SO(r)/(SO(p) \times SO(t)), \quad p = \lfloor r/2 \rfloor, \quad t = \lceil r/2 \rceil.$$

The lift is onto because its image is open and compact. Smoothing and Ehresmann give a bundle with connected fiber. If the fiber has positive dimension, Browder's theorem and the two-row Serre sequence give

$$H^*(\tilde{B}_r; \mathbb{F}_2) = \mathbb{F}_2[e]/(e^{a+1}), \quad |e| = h \in \{2, 4, 8\}, \quad pt = ah, \quad \chi(\tilde{B}_r) = pt/h + 1. \quad (\text{A.3})$$

Indeed multiplication by the transgression e is an isomorphism in all base degrees below the total-space dimension, and the fundamental class fixes the last power.

For completeness, we compute the required Euler characteristic directly. The Schubert cells of the real Grassmannian are indexed by p -subsets of $\{1, \dots, r\}$. Summing $(-1)^{\dim}$ over these cells gives

$$\chi(B_r) = \begin{cases} 0, & p, t \text{ both odd,} \\ \binom{\lfloor r/2 \rfloor}{\lfloor p/2 \rfloor}, & \text{otherwise.} \end{cases}$$

The cell decomposition is classical [61, Section 6]. The formula follows by separating the subsets according to whether they contain r , giving the signed Pascal recurrence. The oriented double cover has twice this Euler characteristic. For $r = 2m$, odd m gives zero; even $m \geq 2$ gives $2\binom{m}{m/2} > m^2/2 + 1$, except that $m = 2$ is checked directly as $4 > 3$. For $r = 2m + 1$, $m \geq 4$ gives

$2\binom{m}{\lfloor m/2 \rfloor} > m(m+1)/2 + 1$. The last inequality starts at $m = 4, 5$ and propagates by the two-step binomial recurrence; the even case starts at $m = 4$ and uses the same recurrence. At $m = 3$ the Euler characteristic is six, whereas the admissible values in (A.3) are seven and four. Thus the only numerical candidates are $r = 3$ and $r = 5$, both with $h = 2$.

For $r = 5$, the map $(u, v) \mapsto [u + iv]$ identifies $\tilde{B}_5$ with the smooth three-dimensional complex quadric. In coordinates $z_0 z_4 + z_1 z_3 + z_2^2 = 0$, the loci $z_0 \neq 0$ and $z_0 = 0, z_1 \neq 0$ are respectively $\mathbb{C}^3$ and $\mathbb{C}^2$; the remainder is $\mathbb{CP}^1 = \mathbb{C} \sqcup \{\text{pt}\}$. The resulting closed filtration and compactly supported cohomology sequence give one integral generator in each of degrees 0, 2, 4, 6. Its hyperplane class x is primitive: if $x = kz$ with $|k| \geq 2$, then k^3 would divide its cube integral, which equals two. Poincaré duality therefore supplies a degree-four generator y with $\int xy = 1$. The degree computation gives $x^2 = 2y$. Thus the unique nonzero mod-two degree-two class has square zero, contradicting (A.3) with $a = 3$.

If the fiber is zero-dimensional, the lifted map would identify a sphere with $\tilde{B}_r$. This is impossible for $r = 4$, where the target is $\mathbb{S}^2 \times \mathbb{S}^2$, and for $r = 5$ by the preceding cohomology. For $r \geq 6$, the homogeneous-space homotopy sequence gives

$$\pi_2(\tilde{B}_r) = \ker[\mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow \mathbb{Z}/2] = \mathbb{Z}/2,$$

where the map is addition. This differs from $\pi_2(\mathbb{S}^{pt}) = 0$. Finally $\tilde{B}_3 = \mathbb{S}^2$ gives the two stated dimensions by the sphere-submersion result and the identity. Composing with its double cover realizes the ordinary projective-plane target. The circle case follows from the lifting-to- $\mathbb{R}$ obstruction and circle covering theory. $\square$

The topology used here—plane fields, sphere fiberings, rational formal dimension, and Grassmannian cohomology—is classical. Our contribution is the passage from saturated contact channels of cones to submersions onto normalized cone-base boundaries and their joint covering, together with the consequences for formulations. These statements concern one specified globally C^1 slack factorization. Aubrun, La Piana, and Müller-Hermes [6] instead study factorization of all positive linear maps through sums of Lorentz cones, a statement with different quantifiers.

B Restricted kernels: embeddings, locality, and flat residuals

The full labelled rows of an affine lift carry more information than a single weighted sum of them, which is a contact kernel. This section studies what factorizations of *restricted kernels* can and cannot prove about the number of factors. These are prescribed kernels, such as the contact kernel of a product of spheres or of projective spaces, used in place of the full slack operator. We also give a nullity obstruction that applies when the mixed contact curvature is zero. Each result states its factorization class. A factorization of a restricted kernel need not extend to the full slack operator of a convex body.

B.1 Projective contact forces an embedding

Write $P_x = xx^T$ for $x \in \mathbb{S}^{r-1}$ and let $\mathbb{RP}^{r-1}$ denote the space of such rank-one projectors. The contact kernel on this space is

$$\kappa([x], [y]) = 1 - \text{tr}(P_x P_y) = 1 - (x^T y)^2.$$

This kernel is *strict-diagonal*: it vanishes exactly on the diagonal. A factorization is *nowhere-zero* if every selected primal and dual factor is nonzero at every point, not merely on an open dense subset.

Theorem B.1 (Projective phase embedding). *Let $r \geq 3$, $k \geq 1$, and $\lambda_a > 0$. Suppose that on $M = (\mathbb{RP}^{r-1})^k$ the kernel*

$$\kappa_k(u, v) = \sum_{a=1}^k \lambda_a (1 - (x_a^T y_a)^2) = \sum_{i=1}^L \langle A_i(u), B_i(v) \rangle$$

has a globally C^1 , nowhere-zero factorization with $A_i, B_i \in Q_3$. Then M has a C^1 embedding in $\mathbb{R}^L$. In particular, with $m = \lceil \log_2 r \rceil$,

$$L \geq \max\{k(r-1) + 1, k(2^m - 1)\}. \quad (\text{B.1})$$

For one projective plane ($r = 3, k = 1$), the minimum of L in this class is four. For all r, k there is a smooth nowhere-zero factorization with

$$L = k \left(\frac{r(r+1)}{2} - 2 \right). \quad (\text{B.2})$$

Proof. Nonnegative summands and the diagonal zero force complementarity in every factor pair. Since both factors are nonzero, their normalizations have the form

$$A_i(u) = a_i(u)(1, p_i(u)), \quad B_i(u) = b_i(u)(1, -p_i(u)), \quad a_i, b_i > 0, \quad p_i(u) \in \mathbb{S}^1.$$

The fundamental group of M is finite, so every homomorphism from it to $\pi_1(\mathbb{S}^1) = \mathbb{Z}$ vanishes. The covering lifting criterion therefore gives a global C^1 angle $\theta_i : M \rightarrow \mathbb{R}$ with $p_i = (\cos \theta_i, \sin \theta_i)$. Equivalently, $H^1(M; \mathbb{Z}) = 0$; mod-two first cohomology would not suffice. Taking the negative mixed derivative on the diagonal gives

$$2 \sum_a \lambda_a g_a = \sum_i a_i b_i d\theta_i \otimes d\theta_i,$$

where g_a is the quotient round metric on the a th projective space. Only first derivatives of the factors enter. Since the left side is positive definite, $\Theta = (\theta_1, \dots, \theta_L)$ is an immersion. If $\Theta(u) = \Theta(v)$, every phase agrees and all pairings $\langle A_i(u), B_i(v) \rangle$ vanish. The strict-diagonal property then gives $u = v$. Compactness makes this injective immersion an embedding.

For completeness, we include the characteristic-class calculation behind the numerical bound. Put $n = r - 1$ and let a generate $H^1(\mathbb{RP}^n; \mathbb{F}_2)$. The tangent bundle is $\text{Hom}(\gamma, \gamma^\perp)$, where γ is the canonical line bundle, so $T\mathbb{RP}^n \oplus \varepsilon^1 \cong (n+1)\gamma$. The Whitney product formula [61] gives, for the normal bundle η of an embedding,

$$w(\eta) = (1 + a)^{-r} = \sum_{j=0}^n \binom{n+j}{j} a^j.$$

The coefficient is odd exactly when the binary expansions of n and j have no common nonzero bit. This also follows directly by expanding $(1 + a)^{-r}$ over $\mathbb{F}_2$. The largest such $j \leq n$ is $j_* = 2^m - r$, the complement of the first m bits of n . For the product, $w(\eta) = \prod_a (1 + a_a)^{-r}$ has the nonzero monomial $\prod_a a_a^{j_*}$. Hence the normal rank is at least kj_* and $L \geq k(2^m - 1)$. Independently, a nonempty compact manifold of dimension kn cannot embed in $\mathbb{R}^{kn}$, by invariance of domain. This proves (B.1). The projective plane cannot embed in $\mathbb{R}^3$: a compact embedded hypersurface of Euclidean space is orientable, whereas $\mathbb{RP}^2$ is not. Thus its lower bound is four.

For the construction put $D = r(r+1)/2 - 1$ and use the normalized Veronese map into the unit sphere of traceless symmetric matrices,

$$X([x]) = \sqrt{\frac{r}{r-1}} \left(xx^T - \frac{I}{r} \right) \in \mathbb{S}^{D-1}, \quad \kappa(u, v) = \frac{r-1}{r} (1 - \langle X(u), X(v) \rangle).$$

Its image is a proper compact subset of that sphere. Choose a pole p outside it, set $c(u) = 1 - \langle p, X(u) \rangle > 0$, and let $F(u) = (X(u) - \langle p, X(u) \rangle p)/c(u) \in \mathbb{R}^{D-1}$. The scalar null-paraboloid maps

$$U(t) = \left(\frac{1+t^2}{2}, \frac{1-t^2}{2}, t \right), \quad V(t) = \left(\frac{1+t^2}{2}, -\frac{1-t^2}{2}, -t \right) \quad (\text{B.3})$$

satisfy $\langle U(t), V(w) \rangle = (t-w)^2/2$. The stereographic identity is

$$1 - \langle X(u), X(v) \rangle = \frac{c(u)c(v)}{2} \|F(u) - F(v)\|^2.$$

Thus $\sqrt{(r-1)/r} c(u) U(F_j(u))$ and the corresponding V factors give $D-1$ smooth nowhere-zero channels. Concatenation with scales $\sqrt{\lambda_a}$ proves (B.2), and gives four when $r=3$. $\square$

The bundle obstruction itself is classical. The step specific to factorizations is the passage from positive complementary Lorentz phases to an injective Euclidean phase map. The theorem concerns only the displayed rank-one kernel and its nowhere-zero factorizations. In particular, its finite upper bound is consistent with Fawzi's theorem that the full cone $\mathbb{S}_+^3$ has no finite second-order-cone lift [32].

B.2 What local tests and separated constructions can show

Fix $q \geq 2$, $k \geq 2$, positive weights with $\sum_a \lambda_a = 1$, and

$$M = (\mathbb{S}^q)^k, \quad K(x, z) = \sum_{a=1}^k \lambda_a (1 - x_a^T z_a), \quad n = kq, \quad D = k(q+1). \quad (\text{B.4})$$

For arbitrary globally C^1 factorizations of this joint kernel with Q_3 factors, Proposition 6.3 and the construction after it give $n+1 \leq L \leq n + \lceil k/2 \rceil$. The results below show what several natural attempts to close this gap can and cannot prove.

Proposition B.2 (Exact local saturation). *Every finite collection of primal and dual points of M lies in a product of punctured-sphere charts on which K has a smooth nowhere-zero factorization with exactly n channels. The identity holds on the whole product chart, not merely on the finite samples.*

Proof. Choose poles $p_a \in \mathbb{S}^q$ outside the finite projections of the sample sets. On the punctured sphere set

$$c_a(x_a) = 1 - p_a^T x_a, \quad F_a(x_a) = \frac{x_a - (p_a^T x_a) p_a}{c_a(x_a)} \in \mathbb{R}^q.$$

For each of the q coordinates of F_a , use the two factors

$$\sqrt{\lambda_a} c_a(x_a) U((F_a)_j(x_a)), \quad \sqrt{\lambda_a} c_a(z_a) V((F_a)_j(z_a))$$

from (B.3). The stereographic chord identity gives exactly the a th summand of (B.4). There are kq channels in total. $\square$

Consequently, neither a finite submatrix of the kernel nor finitely many of its jets at sample pairs can distinguish the local count n from the larger global smooth count. This concerns the kernel only; it does not say whether finitely many samples of *prescribed factor maps* determine their possible global extension.

Proposition B.3 (Partition separation). *Suppose the source coordinates are partitioned into nonempty groups $G_1, \dots, G_m$, and the channel labels into groups $I_1, \dots, I_m$, such that both factors of every channel in I_g depend only on the coordinates in G_g . Every global C^1 factorization of K of this form satisfies $L \geq n + m$. Equality is attainable for every partition into singletons and pairs.*

Proof. All diagonal pairings vanish separately. Freeze every group other than G_g at equal primal and dual arguments. The remaining channels then factor exactly $\sum_{a \in G_g} \lambda_a (1 - x_a^T z_a)$. The proof of Proposition 6.3 applies to any product of simply connected spheres: if its $q|G_g|$ -dimensional contact metric were factored with that many Q_3 channels, each channel would have rank one everywhere; the real phase map would be a local diffeomorphism from a compact manifold to $\mathbb{R}^{q|G_g|}$. Therefore $|I_g| \geq q|G_g| + 1$. Summing proves the bound. A singleton uses the $q + 1$ scalar chord maps applied to its ordinary coordinates. A pair uses the $2q + 1$ stereographic channels in Proposition 6.3. Rescale by the sum of its weights when that sum is not one. $\square$

Another restricted class consists of representations by finitely many *common-scale squared-distance kernels*:

$$K(x, z) = \sum_{g=1}^m \frac{r_g(x)r_g(z)}{2} \|F_g(x) - F_g(z)\|^2, \quad r_g > 0, \quad F_g : M \rightarrow \mathbb{R}^{L_g}, \quad L = \sum_g L_g, \quad (\text{B.5})$$

with globally C^1 maps. Each scalar coordinate produces one Q_3 channel by (B.3). Different summands may have the same scale function.

Proposition B.4 (Functional inertia). *Let e be the number of summands in (B.5) for which the image of the feature map F_g is not contained in a Euclidean sphere. Then*

$$L \geq D - e \geq D - m. \quad (\text{B.6})$$

In particular an $L = n + 1$ representation in this class requires $e \geq k - 1$. Within the narrower class in which each summand is itself a positive block-additive kernel

$$\frac{r_g(x)r_g(z)}{2} \|F_g(x) - F_g(z)\|^2 = \sum_{a \in G_g} \mu_{ga} (1 - x_a^T z_a), \quad \mu_{ga} > 0, \quad \sum_{g: a \in G_g} \mu_{ga} = \lambda_a, \quad (\text{B.7})$$

the exact minimum count is $n + \lceil k/2 \rceil$, even when the groups overlap.

Proof. Set $\alpha_g = r_g \|F_g\|^2/2$, $\beta_g = r_g$, and $h_{gj} = r_g (F_g)_j$. The corresponding kernel is

$$\alpha_g(x)\beta_g(z) + \beta_g(x)\alpha_g(z) - \sum_j h_{gj}(x)h_{gj}(z).$$

Its coefficient form has at most $L_g + 1$ negative directions. If the image lies on a sphere, translate the center to zero without changing squared distances. Then α_g is a nonnegative constant multiple of β_g and the bound improves to L_g negative directions. Any pullback to the actual span of functions has no larger negative index. Conversely, the constant function and the D coordinate functions of $X(x) = (\sqrt{\lambda_a} x_a)_a$ are linearly independent, and $K(x, z) = 1 - X(x)^T X(z)$. Choose finitely many

evaluation points whose feature evaluation matrix has full column rank. Their kernel matrix has exactly D negative eigenvalues. Applying the preceding upper bounds to this same finite matrix proves (B.6).

For (B.7), a nonsingleton group contributes at least $|G_g|(q+1) - 1$ channels by the same inertia argument. A singleton contributes at least $q+1$ by the compact phase obstruction. If $I = \sum_g |G_g|$ and t groups are nonsingletons, then $I \geq k$, $t \leq \lfloor I/2 \rfloor$, and

$$L \geq (q+1)I - t \geq (q+1)I - \lfloor I/2 \rfloor \geq (q+1)k - \lfloor k/2 \rfloor.$$

Disjoint pairs and, if needed, one singleton attain this value. $\square$

In the paired construction, the image of each stereographic pair lies in no Euclidean sphere. Indeed, the inverse image of a Euclidean sphere under stereographic projection is contained in a hyperplane section of the ambient unit sphere, whereas a weighted product of two full spheres affinely spans its entire ambient space. The image of a singleton lies in a sphere. Thus this construction also attains (B.6), with $e = \lfloor k/2 \rfloor$. In the general class (B.5), m and e can be larger; the paired construction is proved optimal only in the class (B.7).

Example B.5 (A channel can depend on every source). Choose a pole $p = (p_a)_a \in \mathbb{S}^{D-1}$ with every $p_a \neq 0$ and $(\|p_a\|)_a \neq (\sqrt{\lambda_a})_a$. Then

$$\max_{x \in M} p^T X(x) = \sum_a \sqrt{\lambda_a} \|p_a\| < 1.$$

Global stereographic projection of $X(M)$ from p therefore gives $D-1$ smooth nowhere-zero channels. To verify dependence on all sources, choose a unit $v \in p^\perp$ such that each $v_a \neq 0$ and $v_a \perp p_a$. This is possible because each block has dimension at least three. One coordinate is $f = v^T X/c$, where $c = 1 - p^T X$. The time component of its primal factor is $T = c(1 + f^2)/2$. Fix a source a . On an open subset of its sphere there are tangent variations with $d(v^T X) = 0$ and $dc \neq 0$, because v_a and p_a are independent. At points with $f \neq \pm 1$ these give $dT = (1 - f^2)dc/2 \neq 0$. Such points exist: the equation $v^T X = \pm c$ cannot hold on an open subset of the product, since its two sides would give a nontrivial affine relation among the constant and coordinate functions. This argument applies to every a . Thus one channel can depend nontrivially on all k sources. Positivity alone therefore does not let a proof for unrestricted factorizations assume that each channel depends on one source or one pair of sources.

B.3 Column ranks detect residuals with zero contact curvature

The next theorem uses finite-dimensional linear algebra instead of contact differentiation, and it needs no topology. For a kernel $F_a : X_a \times X_a \rightarrow \mathbb{R}$, its *column rank* is the dimension of the span of the functions $F_a(\cdot, v)$ on X_a .

Theorem B.6 (A column-rank nullity criterion). *Let $X_1, \dots, X_b$ be nonempty sets and $M = \prod_a X_a$. Suppose nonnegative kernels F_a vanish on the diagonal, have finite nonzero column ranks ρ_a , and admit the labelled real-PSD factorization*

$$F_a(x_a, v) = \sum_{i=1}^L \text{tr}(A_i(x) B_i^a(v)), \quad A_i(x), B_i^a(v) \in \mathbb{S}_+^{r_i}, \quad r_i \leq R. \quad (\text{B.8})$$

If, for every nonempty $J \subseteq \{1, \dots, b\}$,

$$1 + \sum_{a \in J} (\rho_a - 1) > \frac{R+1}{2} (|J| - 1), \quad (\text{B.9})$$

then some $x \in M$ satisfies $\sum_i \text{nullity } A_i(x) \geq b$. A sufficient uniform condition, when $\rho_a \geq \rho > 0$, is

$$(R+3)(b-1) < 2b\rho. \quad (\text{B.10})$$

For $b = 1$ both conditions hold for every R .

Proof. Assume $\sum_i \text{nullity } A_i(x) \leq b-1$ for every $x \in M$. In $\mathcal{H} = \bigoplus_i \mathbb{R}^{r_i}$ write $A = \bigoplus_i A_i$ and $B^a = \bigoplus_i B_i^a$. Diagonal contact implies $\text{Ran } B^a(x_a) \subseteq \ker A(x)$. For each a let S_a be the union of the nonzero vectors in the ranges of all $B^a(v)$. It is nonempty because $\rho_a > 0$. If there were linearly independent vectors $w_a \in S_a$, then choosing the corresponding v_a and setting $x = (v_a)_a$ would put b independent vectors in $\ker A(x)$.

The independent-transversal criterion of Rado [76] therefore gives a nonempty J , with $j = |J|$, such that

$$\dim \text{span} \bigcup_{a \in J} S_a \leq j-1.$$

We give a direct proof that also covers infinite sets S_a . For finite-dimensional subspaces $V_a = \text{span } S_a$, independent vectors $v_a \in V_a$ exist if $\dim \sum_{a \in J} V_a \geq |J|$ for every J . Prove this by induction. If a nonempty proper subset is tight, first choose its independent vectors and then apply induction to the remaining subspaces in the quotient by its sum. Otherwise every nonempty proper subset has dimension at least its cardinality plus one; select a nonzero vector in the last subspace and quotient by its span, preserving the inductive inequalities for the others. Finally, multilinearity of the exterior product expands independent choices from the V_a into at least one independent choice from the generating sets S_a . This proves the claimed criterion without a finiteness assumption on those sets.

Put $V_i = \text{span}_{a \in J, v} \text{Ran } B_i^a(v)$ and $m_i = \dim V_i$. The range of a block-diagonal matrix contains each block-supported range vector. Consequently the span of $\bigcup_{a \in J} S_a$ is exactly $\bigoplus_i V_i$, not merely a subspace of it, and $\sum_i m_i \leq j-1$. All the dual matrices for $a \in J$ lie in $\bigoplus_i \text{Sym}(V_i)$, whose dimension is at most

$$\sum_i \frac{m_i(m_i+1)}{2} \leq \frac{R+1}{2} \sum_i m_i \leq \frac{R+1}{2}(j-1).$$

Pairing these matrices with $A(x)$ bounds by the same number the span of all cylindrical column functions $x \mapsto F_a(x_a, v)$, $a \in J$. On the other hand, a relation $\sum_{a \in J} f_a(x_a) = 0$ makes each f_a constant, by varying one coordinate at a time. Its relation space has dimension at most $j-1$, so the cylindrical span has dimension at least $\sum_{a \in J} \rho_a - (j-1)$. This contradicts (B.9). For the uniform assertion, with ρ in place of each ρ_a , twice the difference of the two sides of (B.9) is $R+3+j(2\rho-R-3)$, an affine function of $1 \leq j \leq b$. It is positive at $j=1$, and at $j=b$ positivity is exactly (B.10). For $b=1$ this condition is automatic. $\square$

Corollary B.7 (Antipodal separation and squared slack). *Let $s \geq 3$, $X_a = \mathbb{S}^{s-1}$, and assume the kernels in (B.8) are continuous. If each $F_a(u, u) = 0$ and $F_a(u, \tau_a(u)) > 0$, where τ_a is topologically conjugate to the antipodal map, then the nullity conclusion of Theorem B.6 holds whenever*

$$(R+3)(b-1) < 2bs. \quad (\text{B.11})$$

For the squared kernels $F_a(u, v) = (1 - u^T v)^2/2$, the column rank is exactly $\rho = s(s+3)/2$. Thus (B.11) can be replaced by the less restrictive sufficient condition

$$(R+3)(b-1) < bs(s+3). \quad (\text{B.12})$$

Without any cap condition, every factorization of these squared kernels satisfies

$$\sum_i \frac{r_i(r_i+1)}{2} \geq 1 + b \left(\frac{s(s+3)}{2} - 1 \right). \quad (\text{B.13})$$

Proof. Choose a basis $f_1, \dots, f_{\rho_a}$ of the column space and write $F_a(u, v) = \sum_{\ell} f_{\ell}(u) c_{\ell}(v)$. The coefficient map $c : \mathbb{S}^{s-1} \rightarrow \mathbb{R}^{\rho_a}$ is continuous: select ρ_a evaluation points for which the basis evaluation matrix is invertible and recover c from those evaluations. If $c(v) = c(\tau_a(v))$, evaluation at $u = v$ contradicts the assumed separation. After conjugating τ_a to the antipodal map, the Borsuk–Ulam theorem [48, Section 2.2] implies $\rho_a \geq s$. Theorem B.6 proves (B.11).

For the squared kernel the odd parts of the columns span all s linear coordinate functions. The even parts span all quadratic forms restricted to the sphere: their coefficient matrices are proportional to $I + vv^T$. A matrix C orthogonal to every $I + vv^T$ obeys $\text{tr } C + v^T C v = 0$ for every unit v , which first makes C scalar and then zero. The restriction of homogeneous quadratic forms to the sphere is injective, and the odd and even spaces are independent. Thus $\rho = s + s(s+1)/2$. This gives (B.12). The constant function belongs to the column span, for example by averaging $F_a(\cdot, v)$ over the sphere. Coordinate column spaces are independent modulo this common one-dimensional space, so the labelled cylindrical column rank is exactly $1 + b(\rho - 1)$. Pairing with the full matrix space $\bigoplus_i \mathbb{S}^{r_i}$ proves (B.13). $\square$

These squared kernels have zero mixed contact curvature, so the contact Hessian estimate cannot recover this lower bound. If the selected primal tuples also lie in the closure of a relative-Slater affine slice, Lemma 8.1 converts the nullity conclusion into $\nu_{\text{std}, \text{slice}} \geq b$ for the standard barrier of that slice.

For example, the identity

$$1 - u^T v = \frac{1 - (u^T v)^2}{2} + \frac{(1 - u^T v)^2}{2}$$

splits the ball slack into a projective term and a flat residual. The first term has the order- s factorization $\text{tr}((I - uu^T)vv^T/2)$. A construction that uses one such separate block for each source has b primal nullities at every contact. If the residual uses blocks of order at most s , Corollary B.7 forces at least another b nullities at some simultaneous contact. Moreover, with $d_s = s(s+1)/2$, (B.13) gives

$$L_{\text{residual}} \geq b + \left\lceil \frac{b(s-1) + 1}{d_s} \right\rceil > b.$$

Thus this prescribed positive splitting needs at least $2b + 1$ factors, whereas the grouped Schur construction for the unsplit full slacks uses $2b$. The rank theorem for unrestricted full slacks does not assume such a splitting. The example shows that a projective summand attaining equality in the curvature bound need not be an economical part of a complete formulation.

Lower bounds on PSD factorization size from ordinary matrix rank are standard [36]. Theorem B.6 adds the combination of the independent-transversal criterion with simultaneous contact and the blockwise bound on matrix-space dimension. Stating the criterion with the actual column ranks keeps information that the weaker antipodal estimate $\rho_a \geq s$ loses. The conclusions concern a simultaneous primal nullity for a prescribed labelled kernel. Without further hypotheses, they bound neither the nullity at every contact nor an arbitrary barrier.

C Global obstructions for conditioned approximate factors

The local rank estimates of Section 20 have global consequences when the factors have fixed block labels and vary smoothly over the whole contact sphere. An arbitrary approximate lift need not satisfy these hypotheses. Without requiring boundary factors, saturation $\sum_i (m_i - 2) = n$ and accurate approximation force a splitting of the cotangent bundle of the sphere. Nonzero boundary factors give stronger conclusions for the channel (mixed derivative form) of each block.

C.1 A robust tangent splitting

Let $n \geq 1$ and suppose proper cones K_i of dimensions $m_i \geq 2$ admit globally C^2 maps $A_i : \mathbb{S}^n \rightarrow K_i$, $B_i : \mathbb{S}^n \rightarrow K_i^*$ satisfying

$$\sum_i \langle A_i(x), B_i(y) \rangle = h_C(y) - \langle x, y \rangle, \quad \mathbb{B}_2^{n+1} \subset C \subset (1 + \epsilon) \mathbb{B}_2^{n+1}. \quad (\text{C.1})$$

Put $r_i = m_i - 2$ and assume $\sum_i r_i = n$. At x write $a_i = A_i(x)$, $b_i = B_i(x)$, $X_i = dA_i(x)$, $Y_i = dB_i(x)$. Assume uniformly that $\|a_i\|, \|b_i\| \geq \mu > 0$ and $\|X_i\|, \|Y_i\| \leq L$. Fix $0 < r_0 < \pi$ and $H > 0$ such that, for every unit $u \in T_x \mathbb{S}^n$ and $|t| \leq r_0$,

$$|\langle (A_i \circ \exp_x(tu))'', b_i(x) \rangle| \leq H, \quad |\langle a_i(x), (B_i \circ \exp_x(tu))'' \rangle| \leq H.$$

The opposite factor is fixed at x .

Theorem C.1 (Approximate saturation gives a bundle splitting). *If $\epsilon < \mu^2$, $\sqrt{2\epsilon/H} \leq r_0$, and $\mathcal{E}_k(\epsilon) < 1$, where k is the number of blocks and $\mathcal{E}_k$ is defined in (20.8), then*

$$T^* \mathbb{S}^n = \bigoplus_{i:r_i > 0} E_i, \quad \text{rank } E_i = r_i.$$

Consequently, the number of blocks with $m_i = 3$ is at most $\rho(n+1) - 1$. If all blocks have dimension three, $n \in \{1, 3, 7\}$. If $n \geq 2$ is even, exactly one block has positive r_i , and that block has dimension $n + 2$.

Proof. Let $\delta_i = \langle a_i, b_i \rangle$. Equation (C.1) gives $\delta_i \geq 0$, $\sum_i \delta_i \leq \epsilon$, and $-\sum_i X_i^* Y_i = I$. Since $\gamma_i = \delta_i / (\|a_i\| \|b_i\|) < 1$, the vectors a_i, b_i are linearly independent. The spaces $W_i(x) = a_i(x)^\perp \cap b_i(x)^\perp$ therefore form continuous bundles of rank r_i . Define the canonical bilinear forms

$$R_i(u, v) = -\langle P_{W_i} X_i u, P_{W_i} Y_i v \rangle.$$

They depend continuously on x and have rank at most r_i . On W_i , both $P_{b_i^\perp} P_{a_i^\perp}$ and P_{W_i} are the identity. On the plane spanned by a_i, b_i , their difference has norm γ_i . Thus the proof of Lemma 20.2 gives $\| -X_i^* Y_i - R_i \| \leq e_i$, with the sharpened error (20.6). It follows that $\| I - \sum_i R_i \| \leq \mathcal{E}_k(\epsilon) < 1$. The sum is invertible and

$$n \leq \sum_i \text{rank } R_i \leq \sum_i r_i = n.$$

All ranks attain their bounds, and their images are in direct sum. Continuity and constant rank give the stated subbundles. For $n \geq 2$, every real line bundle on $\mathbb{S}^n$ is trivial, so rank-one summands give independent vector fields; Adams's bound is $\rho(n+1) - 1$ [1]. For $n = 1$ the tangent bundle is already trivial. A complete rank-one splitting forces the familiar parallelizable dimensions 1, 3, 7. For even n , a splitting into two positive intermediate ranks would give $e(T\mathbb{S}^n) = e(E)e(F) = 0$, since intermediate integral cohomology vanishes. This contradicts Euler number two. $\square$

The stronger subset-rank restrictions of Proposition A.1 also apply. The proof needs the canonical forms R_i : pointwise low-rank approximants chosen independently need not define vector bundles. The hypotheses exclude rays with nonzero selected factors, since for a ray $\delta_i = \|a_i\| \|b_i\| \geq \mu^2$. Selections with a vanishing factor or without the uniform bounds fall outside the hypotheses.

C.2 An index invariant under fixed blockwise coordinate changes

The following version of Theorem C.1 is invariant under fixed linear coordinate changes in each block, although it still depends on the chosen selections and the parameterization of the contact sphere. Keep (C.1) and $\sum_i(m_i - 2)_+ = n$, now allowing rays. For a fixed gauge $S = (S_i)$, $S_i \in GL(m_i)$, transform

$$A_i \mapsto S_i A_i, \quad B_i \mapsto S_i^{-T} B_i, \quad K_i \mapsto S_i K_i.$$

At each center compute $U_i^S, V_i^S, \alpha_i^S, \beta_i^S, \gamma_i^S$ as in Lemma 20.2 for the transformed maps, using geodesics of a fixed radius r_0 . In particular,

$$\alpha_i^S = \frac{\psi_{r_0}(\delta_i, H_{A,i})}{\|S_i^{-T} b_i\|}, \quad \gamma_i^S = \frac{\delta_i}{\|S_i a_i\| \|S_i^{-T} b_i\|}.$$

The bounds $H_{A,i}, H_{B,i}$ on paired second derivatives are unchanged by this transformation. Let e_i^S be (20.6), or $\alpha_i^S \beta_i^S$ for a ray, and set

$$\mathfrak{D} = \inf_{S \in \prod_i GL(m_i)} \max \left\{ \sup_{x,i} \gamma_i^S(x), \sup_x \sum_i e_i^S(x) \right\}. \quad (\text{C.2})$$

If a_i or b_i vanishes somewhere, the maximum in (C.2) is set to $+\infty$ for that gauge.

Corollary C.2 (Gauge-invariant degeneration). *If $T^*\mathbb{S}^n$ admits no splitting with ranks $(m_i - 2)_+$, then $\mathfrak{D} \geq 1$.*

Proof. If $\mathfrak{D} < 1$, some fixed gauge has both suprema below one. Its $\gamma_i < 1$ excludes rays and makes the primal and dual factors nonparallel everywhere. The canonical maps of the preceding proof then have rank at most $m_i - 2$ and an invertible sum, which gives the excluded splitting. If the original coordinates are changed by T_i , the composite transformations $S_i T_i$ range over the same set as before. Thus the infimum is invariant under every fixed blockwise invertible reparameterization. The argument does not need the individual Euclidean projectors to transform covariantly. $\square$

The index $\mathfrak{D}$ is not a lower bound on a barrier Hessian or a Newton-system condition number. The infimum is over fixed linear block gauges only; it includes neither gauges that depend on the contact point nor reparameterizations of the contact sphere.

C.3 Rank-one phase dropout

For a single block Q_3 with nonzero boundary factors, we show that the channel C has norm at most $K\eta + K^2\epsilon$ at some point (notation below). For spheres of dimension three and seven this is stronger than the bound on three-dimensional blocks in Theorem C.1. Let M be a closed, connected, simply connected smooth Riemannian manifold. Let $A, B : M \rightarrow \partial Q_3 \setminus \{0\}$ be C^1 and write

$$A = a(1, p), \quad B = b(1, -q), \quad a, b > 0, \quad p, q : M \rightarrow \mathbb{S}^1.$$

Set $\sigma = \langle A, B \rangle$, $C_x(u, v) = -\langle dA_x u, dB_x v \rangle$ and

$$K_A = \sup_M \|d \log a\|, \quad K_B = \sup_M \|d \log b\|, \quad K = \min\{K_A, K_B\}.$$

Bilinear forms need not be symmetric; all norms are operator norms.

Theorem C.3 (Quantitative phase dropout). *If $0 \leq \sigma \leq \epsilon$ and $\|d\sigma\| \leq \eta$, then there is $x_* \in M$ with*

$$\|C_{x_*}\| \leq K\eta + K^2\epsilon.$$

If every C_x is within δ of a positive rank-one form whose nonzero eigenvalue is at least $\mu > 0$, then

$$\delta + K\eta + K^2\epsilon \geq \mu. \quad (\text{C.3})$$

Proof. Since M is simply connected, p lifts to a real C^1 phase. At a maximum of that phase, $dp = 0$, so $dA = \alpha \otimes A$, where $\alpha = d \log a$. At this point

$$\langle A, dB \rangle = d\sigma - \sigma\alpha, \quad C = -\alpha \otimes (d\sigma - \sigma\alpha).$$

Its norm is at most $K_A\eta + K_A^2\epsilon$. A critical point of the dual phase gives $C = -(d\sigma - \sigma\beta) \otimes \beta$, with $\beta = d \log b$, and the bound using K_B . Taking the smaller bound proves the first assertion. At the corresponding point, any rank-one approximant of norm at least μ has distance at least $\mu - \|C\|$ from C , proving (C.3). $\square$

The hypotheses can be checked using norms of the original factors: $\|A\| = \sqrt{2}a$ and $\|dA(u)\|^2 = 2|da(u)|^2 + a^2\|dp(u)\|^2$, whence $K_A \leq L_A/m_A$ if $\|A\| \geq m_A$ and $\|dA\| \leq L_A$. The dual bound is identical. Multiplying A by a constant $c > 0$ and B by $1/c$ leaves K_A, K_B, σ , and C unchanged.

Corollary C.4 (Saturated approximate phase channels). *Let $M = \mathbb{S}^n$, $n \geq 2$, and let n pairs satisfy the hypotheses of the preceding theorem with constants ϵ_i, η_i, K_i . Suppose*

$$\left\| \sum_i C_i - I \right\| \leq \xi, \quad \|C_i - R_i\| \leq \delta_i,$$

where R_i is positive of rank one at every point; the choices of R_i need not be continuous. Put $\Delta = \sum_i \delta_i$ and $E_i = K_i\eta_i + K_i^2\epsilon_i$. Then, for every i ,

$$\xi + \Delta + \delta_i + E_i \geq 1.$$

In particular, common bounds $\delta_i \leq \delta$ and $E_i \leq E$ give $\xi + (n+1)\delta + E \geq 1$.

Proof. There is nothing to prove if $\xi + \Delta \geq 1$. Otherwise $\|\sum_i R_i - I\| \leq \xi + \Delta < 1$. Write $R_i = \lambda_i \omega_i \otimes \omega_i$ with $\|\omega_i\| = 1$. At any point, choose a unit vector annihilated by the at most $n-1$ covectors ω_j , $j \neq i$. Evaluation on that vector gives $\lambda_i \geq 1 - \xi - \Delta$. Apply Theorem C.3 to the i th channel with $\mu = 1 - \xi - \Delta$. $\square$

The bound η can be obtained from a modulus of continuity of $d\sigma$. For r_0 below the injectivity radius, suppose covectors $d\sigma$ at endpoints of geodesic segments of length $t \leq r_0$ differ by at most $\omega(t)$ after parallel transport. Then

$$\|d\sigma\| \leq \inf_{0 < s \leq r_0} \left\{ \frac{\epsilon}{s} + \frac{1}{s} \int_0^s \omega(t) dt \right\}. \quad (\text{C.4})$$

Indeed, along a unit geodesic in the gradient direction, the integral of the derivative is at least $s\|d\sigma\| - \int_0^s \omega(t)dt$, whereas the change in σ is at most ϵ . If $\omega(t) \leq Ht$, this bound is $\psi_{r_0}(\epsilon, H)$. For $H > 0$ with $\sqrt{2\epsilon/H} \leq r_0$, it gives $\eta \leq \sqrt{2H\epsilon}$. If $H = 0$, $d\sigma$ is locally parallel; its vanishing at a maximum and connectedness give $d\sigma = 0$ everywhere. For a common modulus with $\omega(t) \rightarrow 0$ as $t \rightarrow 0$, the right side of (C.4) tends to zero with ϵ .

In a nonnegative kernel $F(x, y) = \sum_i \langle A_i(x), B_i(y) \rangle$ with $F(x, x) \leq \epsilon$, every defect $\sigma_i = \langle A_i, B_i \rangle$ satisfies $0 \leq \sigma_i \leq \epsilon$, so the preceding inequalities apply to each channel with its own derivative modulus. A bound on the derivative of the sum does not replace these bounds, because derivatives of different channels can cancel.

C.4 Higher-rank dropout from forbidden sphere submersions

For blocks of dimension $r + 2$ the phase becomes a map into the boundary of a cone base, an r -sphere. Outside three exceptional pairs (n, r) , the Hopf pairs, no map $\mathbb{S}^n \rightarrow \mathbb{S}^r$ with $r < n$ is a submersion. Let $1 \leq r < n$, let $K \subset E$ be proper of dimension $r + 2$, and choose $\ell \in \text{int } K^*$. Suppose the normalized base $D = \{z \in K : \ell(z) = 1\}$ has a C^1 boundary J . Let $A : \mathbb{S}^n \rightarrow \partial K \setminus \{0\}$ and $B : \mathbb{S}^n \rightarrow \partial K^* \setminus \{0\}$ be C^1 . Define

$$\begin{aligned} a &= \ell(A), \quad p = A/a : \mathbb{S}^n \rightarrow J, \quad K_A = \sup \|d \log a\|, \\ g_B(v) &= \langle A, dB(v) \rangle, \quad \eta_B = \sup \|g_B\|, \quad C = -dA^* dB. \end{aligned}$$

Write $s_r(C)$ for the r th largest singular value.

Theorem C.5 (Robust submersion obstruction). *If $(n, r) \notin \{(3, 2), (7, 4), (15, 8)\}$, there is a point with*

$$s_r(C) \leq K_A \eta_B.$$

Consequently, if every C is within δ of a rank- r form whose r th singular value is at least $\mu > 0$, then

$$\delta + K_A \eta_B \geq \mu. \tag{C.5}$$

For a Hopf pair, the same conclusion holds if the particular map p has a critical point. If a dual normalized base also has C^1 boundary, the term $K_A \eta_B$ in (C.5) may be replaced by $\min\{K_A \eta_B, K_B \eta_A\}$, with dual definitions.

Proof. Radial projection from a relative interior point of the convex base identifies J C^1 diffeomorphically with $\mathbb{S}^r$. The inverse is C^1 because the supporting normal at each boundary point has strictly positive inner product with its radial vector. Outside the three Hopf pairs, Proposition A.2 rules out an everywhere submersive map p . Hence $\text{rank } dp \leq r - 1$ somewhere. Writing $\alpha = d \log a$, differentiation of $A = ap$ gives

$$C(u, v) = -a \langle dp(u), dB(v) \rangle - \alpha(u) g_B(v).$$

At that point the first term has rank at most $r - 1$ and the second has norm at most $K_A \eta_B$. The singular-value min-max inequality gives $s_r(C) \leq K_A \eta_B$. Singular values change by at most the operator perturbation norm, proving (C.5). Applying the same argument to the normalized dual map supplies the second bound, possibly at a different point; both inequalities hold, so their minimum is valid. $\square$

Here $g_B(x)$ is the one-sided derivative $d_y \langle A(x), B(y) \rangle|_{y=x}$, with x held fixed. It can be bounded using a small contact defect and a modulus of continuity in y : if $F(x, y) = \langle A(x), B(y) \rangle \geq 0$, $F(x, x) \leq \epsilon$, and the second-variable derivative has uniform geodesic modulus ω_B , then

$$\eta_B \leq \inf_{0 < s \leq r_0} \left\{ \frac{\epsilon}{s} + \frac{1}{s} \int_0^s \omega_B(t) dt \right\}. \tag{C.6}$$

Move the second argument from x in the direction opposite to $g_B(x)$; the value at the endpoint is nonnegative, so integration gives this inequality. The first-variable bound for η_A is identical. Lipschitz constants H_A, H_B give the corresponding ψ bounds and, when the optimizing radii are at most r_0 ,

$$\delta + \min\{K_A \sqrt{2H_B \epsilon}, K_B \sqrt{2H_A \epsilon}\} \geq \mu.$$

The finite-radius formula (C.6) remains valid when either Lipschitz constant is zero.

A bound on the derivative of $\sigma = \langle A, B \rangle$ alone does not suffice for this higher-rank argument, since

$$g_B = d\sigma - \sigma d \log a - a dp^* B.$$

At a critical point of a rank-one phase, $dp = 0$, recovering Theorem C.3. For $r > 1$ a critical point only forces rank $dp < r$ and the last term need not vanish. If it is bounded by τ and $\|d\sigma\| \leq \eta$, threshold (C.5) becomes

$$\delta + K_A \eta + K_A^2 \epsilon + K_A \tau \geq \mu.$$

Finally, suppose $\sum_i r_i = n$ with at least two positive ranks, every block satisfies these boundary and base-regularity hypotheses, and its channel is uniformly within δ_i of a rank- r_i form with r_i th singular value at least μ_i . Some pair (n, r_i) is not a Hopf pair: for $n \neq 3, 7, 15$ no pair with $1 \leq r_i < n$ is a Hopf pair, and for $n = 3, 7, 15$ the only Hopf target is 2, 4, 8, respectively, and copies of it cannot sum to n . Thus, under common bounds $\delta_i \leq \delta$, $K_{A,i} \leq K$, $\eta_{B,i} \leq \eta_\partial$, and $\mu_i \geq \mu$, every such profile satisfies

$$\delta + K \eta_\partial \geq \mu.$$

The obstruction does not apply to a single block of full rank n .

These proofs combine elementary perturbation estimates with the classical sphere topology cited in Appendix A. The results quantify how specified global selections of cone contacts must degenerate. They do not follow from the finite slack-matrix approximation framework of [44], and they do not exclude nonsmooth, unlabelled, or vanishing-factor approximations.

D Contact sensitivity and Newton conditioning

The approximation obstructions of Appendix C concern derivatives of the selected contact maps with respect to the contact label (the point of the sphere). A Newton condition number concerns a linear system at one interior point. Two separations show that connecting them needs an additional sensitivity theorem.

D.1 A sparse change of cone coordinates

Suppose $F(Gz) = F(z) + c_G$ for a linear cone automorphism G . In an equality formulation $Ez = b$, put $z' = Gz$, $E' = EG^{-1}$ and $H = \nabla^2 F(z)$. Differentiation gives

$$H' = G^{-T} H G^{-1}, \quad E'(H')^{-1}(E')^T = E H^{-1} E^T. \quad (\text{D.1})$$

The raw augmented system instead changes by congruence:

$$\begin{pmatrix} H' & (E')^T \\ E' & 0 \end{pmatrix} = \begin{pmatrix} G^{-T} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} H & E^T \\ E & 0 \end{pmatrix} \begin{pmatrix} G^{-1} & 0 \\ 0 & I \end{pmatrix}.$$

Thus the Schur matrix is unchanged, whereas the Euclidean spectrum of the augmented matrix may change.

For Q_m , $m \geq 3$, let $a = e_0 + e_1$, $b = e_0 - e_1$, and let G_λ be the identity on transverse coordinates, with first block

$$\frac{1}{2} \begin{pmatrix} \lambda + \lambda^{-1} & \lambda - \lambda^{-1} \\ \lambda - \lambda^{-1} & \lambda + \lambda^{-1} \end{pmatrix}, \quad \lambda > 0.$$

This boost preserves the Lorentz quadratic form and maps Q_m onto itself, and $G_\lambda a = \lambda a$, $G_\lambda^{-T} b = \lambda b$. As $\lambda \rightarrow 0$, both primal and dual Euclidean norms tend to zero while all pairings stay unchanged. For a contact example, the maps

$$A(u) = (\sqrt{1 + \|u\|^2}, 1, u), \quad B(v) = (\sqrt{1 + \|v\|^2}, -1, -v)$$

satisfy $\langle A(u), B(v) \rangle \geq 0$, with equality at $u = v$. At zero their transverse derivatives are isometries unchanged by the boost, while the norms of $A(0) = a$ and $B(0) = b$ scale by λ . The standard Lorentz barrier is invariant, so (D.1) applies. At e_0 , $H = 2I$, but at $G_\lambda e_0$ it is $2G_\lambda^{-2}$, with condition number λ^{-4} for $0 < \lambda < 1$. Therefore a lower bound in terms of unnormalized Euclidean factor norms cannot be read as a lower bound on the Schur condition number. The boost mixes only two columns of E , but it can change their magnitudes, the data-access normalization, and the cost of a solver using the raw augmented system.

D.2 Bounded pointwise geometry and unbounded contact derivatives

Transforming the normalization covector ℓ together with the factors removes this coordinate effect, but a second problem remains. For maps A, B and a strictly positive covector ℓ , the transformations

$$(A, B, \ell) \mapsto (GA, G^{-T}B, G^{-T}\ell)$$

preserve $\langle A, B \rangle$, $-dA^*dB$, A^*dB , and $d \log \ell(A)$. Yet derivatives in the contact label can grow without bound while Hessian conditioning, scaling-point conditioning, and centrality stay bounded at each interior point.

Use the Lorentz Jordan product $(t, z) \circ (s, w) = (ts + z^T w, tw + sz)$ and fix $c_\pm = (1, \pm p_0)/2$ for $p_0 \in S^1$. On S^n , $n \geq 1$, put $a_k(x) = \exp(\sin(kx_1))$ and

$$A_k = a_k c_+, \quad B_k = a_k^{-1} c_-, \quad X_k = a_k(c_+ + \tau c_-), \quad S_k = a_k^{-1}(c_- + \tau c_+), \quad (\text{D.2})$$

where $0 < \tau < 1$ is fixed. The boundary maps are complementary, $e^{-1} \leq a_k \leq e$, and $\|d \log \ell(A_k)\|_\infty = k$ for any positive ℓ . Yet $X_k \circ S_k = \tau e$ exactly. Since $\nabla^2 F(az) = a^{-2} \nabla^2 F(z)$, the primal and dual Hessian norms and inverse norms are uniformly bounded in k . In the common Jordan frame the Nesterov–Todd scaling point has eigenvalues $a_k/\sqrt{\tau}$ and $a_k\sqrt{\tau}$; its spectral ratio is $1/\tau$. Centrality uses the displayed Jordan product; rescaling the trace inner product by a constant does not change the conclusion.

For the second example, fix $0 < \theta < \pi/2$ and put $p_k(x) = (\cos(kx_1), \sin(kx_1))$, $q_k = R_\theta p_k$, and

$$A_k = (1, p_k), \quad B_k = (1, -q_k), \quad X_k = (1 + \tau, p_k), \quad S_k = (1 + \tau, -q_k). \quad (\text{D.3})$$

The diagonal defect is the constant $1 - \cos \theta$, but

$$d\langle A_k, B_k \rangle = 0, \quad A_k^* dB_k = \sin \theta d(kx_1), \quad \|A_k^* dB_k\|_\infty = k \sin \theta.$$

All interior pairs are common spatial rotations of one fixed pair, so Hessian spectra and scaling-point spectra are independent of k . Their relative central residual $\|X_k \circ S_k - \mu_k e\|/\mu_k$, with $\mu_k = \langle X_k, S_k \rangle$ and $e = (1, 0)$, is

$$\frac{2(1 + \tau) \sin(\theta/2)}{2\tau + \tau^2 + 1 - \cos \theta}.$$

For any positive central-neighborhood radius, a sufficiently small fixed $\theta > 0$ puts the entire family inside that neighborhood. Alternatively, $\theta = k^{-1/2}$ gives defect $\Theta(k^{-1})$ and central residual

$O(k^{-1/2})$, while the one-sided derivative grows as $\Theta(\sqrt{k})$. The mixed channel is $\cos \theta d(kx_1) \otimes d(kx_1)$; it vanishes at the critical points of x_1 , consistent with the phase obstruction of Theorem C.3.

Thus pointwise centrality, Hessian conditioning, scaling-point conditioning, and bounded factor scales control neither the radial sensitivity $\|d \log \ell(A)\|$ nor the one-sided sensitivity $\|A^* dB\|$. The pointwise interpretation of self-scaled geometry is classical [71]; the examples separate it from the quantities used here, which compare contacts for different objectives. One may even impose the equality constraint $z = X_k$, with identity constraint matrix, at each label. The assembled saddle matrix $\begin{pmatrix} \mu H & I \\ I & 0 \end{pmatrix}$ and its inverse are uniformly bounded whenever μ stays in a fixed positive compact interval, since its inverse is $\begin{pmatrix} 0 & I \\ I & -\mu H \end{pmatrix}$. These artificial instances have label-dependent right-hand sides, so they are not a single approximate lift of a fixed body.

D.3 What an additional sensitivity hypothesis would give

In the robust submersion inequality (C.5), suppose the finite-radius estimate reduces to

$$\delta + K_A \sqrt{2H_B \epsilon} \geq \mu, \quad \sqrt{2\epsilon/H_B} \leq r_0,$$

where $\delta < \mu$, $\epsilon > 0$ and $H_B > 0$. Then $H_B \geq (\mu - \delta)^2 / (2K_A^2 \epsilon)$. The finite-radius expression $\inf_{0 < s \leq r_0} (\epsilon/s + H_B s/2)$ remains the valid form when $H_B = 0$ or the optimizing radius exceeds r_0 ; one cannot divide by H_B in those cases. If a separately proved sensitivity theorem for a representation with conditioning parameter χ (for example, a Newton condition number) gives $K_A \leq c_K \chi^a$ and $H_B \leq c_H \chi^b$, with positive constants and exponents, substitution yields

$$\chi \geq \left(\frac{\mu - \delta}{c_K \sqrt{2c_H \epsilon}} \right)^{1/(a+b/2)}.$$

Likewise, for coefficient bit length β and $C > 0$, $K_0 > 0$, $q > 0$, $H_B \leq C 2^{q\beta}$ and $K_A \leq K_0$ imply

$$\beta \geq \frac{1}{q} \log_2 \frac{(\mu - \delta)^2}{2CK_0^2 \epsilon}.$$

These consequences are conditional: we do not prove that a Newton condition number or a coefficient bit count satisfies these hypotheses. In particular, differentiating a KKT solution map requires control of both its inverse Jacobian and the variation of the input data with the contact label.

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